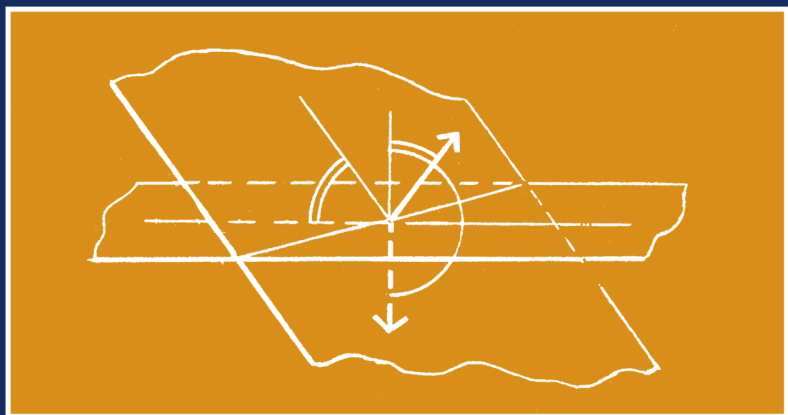


High-School MATHEMATICS

Under the editorship of Professor
____ G. N. YAKOVLEV, D.Sc. ____



____ Part 1 ____

MIR PUBLISHERS · MOSCOW

High-School Mathematics

Пособие по математике для поступающих в вузы

Под редакцией
Г. Н. Яковлева

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High-School Mathematics

Under the editorship of Professor
G. N. YAKOVLEV, D.Sc.

In Two Parts

Part One



MIR PUBLISHERS MOSCOW

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Preface

The contributors to the book are lecturers in the Department of Higher Mathematics of the Moscow Institute of Physics and Technology. The lecturers and students of the Institute do much to spread the knowledge of physics and mathematics among high-school students: they deliver lectures, hold discussions with the students, and give consultations directly and by correspondence. They keep in systematic contact with high-school students from various parts of the Soviet Union: the students are sent the texts of the lectures and test problems and post back their answers and solutions, which are checked up and then sent back to them with the necessary comments.

The present book was written by the lecturers who drew on their long experience as propagandists of knowledge among high-school students.

The book does not contain any systematic exposition of a course in elementary mathematics and cannot replace regular textbooks. Nevertheless, all the principal and most important problems are treated in sufficient detail.

Each chapter of the book includes theoretical material and problems. The elaboration of the theory is followed by a great number of examples of varying difficulty.

The problems of each chapter are divided into two sections, the problems of the first section being followed by com-

plete solutions and those of the second section only by answers.

The book is principally intended for high-school students who are interested in mathematics and wish to obtain a more thorough knowledge. It can be of use to students whose aim is to learn to solve problems in elementary mathematics. The authors also hope that school teachers will find some material in the book which they will use in their everyday work.

The book contains more than two thousand problems, from the simplest to the advanced. About a third of the problems are provided with detailed solutions.

G. N. Yakovlev

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Chapter 1

Sets. A Function and Its Inverse

1.1. A Set. A Subset. A Union and an Intersection of Sets. Number Sets

A Set. The concept of a set is used in mathematics to describe a collection of particular things or objects. It is assumed that these things (objects) can be distinguished from each other and from those not included in that collection. For instance, one can speak of the set of all books from the given library, the set of all vertices of the given polygon, the set of all natural numbers, the set of all points belonging to the given line. The books from the given library, the vertices of the given polygon, natural numbers, the points of the given line are the *members* or *elements* of the corresponding sets.

It is customary to denote sets by capital letters A , B , X , The fact that the object a is an element of the set A is written as $a \in A$ and is read as " a belongs to the set A " or " a enters the set A ". The notation $a \notin A$ (or $a \notin A$) means that a is not an element of the set A .

For example, if we designate as $\mathbb{N}$ the set of natural numbers, then

$$3 \in \mathbb{N}, 20 \in \mathbb{N}, 0 \notin \mathbb{N}, 3/2 \notin \mathbb{N}.$$

We usually give a set by indicating some characteristic property of its elements, that is, a property possessed by all the elements of the set and only by those elements. More often than not such a property is formulated as follows: the set of students of the given class, the set of solutions of the equation $x^2 - 1 = 0$, the set of points of the plane equidistant from the given two points.

It may so happen that the set is defined by a property possessed by neither of the objects. Let us consider, for example, a set of triangles whose sides are equal to 1 cm,

2 cm, and 5 cm. It is known from geometry that a triangle with these sides does not exist, that is, the set so defined does not contain a single element. There is not a single element either in the set of rational numbers whose square is equal to two. The set of solutions of the system of equations

$$\begin{cases} x + y = 1, \\ x + y = 2 \end{cases}$$

does not contain any elements either. Such sets are called *empty* or *null*. An empty set does not contain any elements and is designated as $\emptyset$.

If a set contains a finite number of elements, it is sometimes convenient to define it by enumerating its elements. The elements of such a set are written in a sequence and are enclosed in braces, the order of notation being immaterial. For instance, the set consisting of three letters a , b , c is written as $\{a, b, c\}$.

The sets A and B are said to be *equal* if they contain the same elements; in that case we write $A = B$.

The set of solutions of the equation $x^2 - 5x + 6 = 0$ contains the same elements (the numbers 2 and 3) as the set of prime numbers smaller than five. These two sets are equal.

Number sets. Algebra most often involves sets whose elements are numbers; such sets are *number sets*. The following notation is customarily used for some most frequently encountered number sets; $\mathbb{N}$, the set of natural numbers, $\mathbb{Z}$, the set of integers, $\mathbb{Q}$, the set of rational numbers, $\mathbb{R}$, the set of real numbers.

Real numbers are represented by points of the *coordinate line* (number axis), which is any straight line on which a direction has been identified as positive, a point has been chosen as a reference point and a unit of measurement has been taken whose length is equal to unity. The coordinate line is usually horizontal, with the positive direction shown by an arrow and the reference point designated by the letter O (Fig. 1). The point O divides the coordinate line into two rays or half-lines one of which has the positive direction and is called a *positive ray* and the other, a *negative ray*. The number which the point M represents on the coordinate line is called the *coordinate* of the point M . The

coordinate of the initial point O is zero. The coordinate of any point M belonging to the positive half-line OE is equal to the length of the line segment OM : $x = |OM|$ (in Fig. 2 the coordinate of the point M is equal to 2.5). Now if the point M belongs to the negative half-line, its coordi-

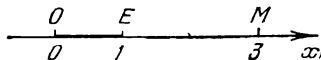


Fig. 1.

nate is equal to the length of the line segment OM taken with the minus sign: $x = -|OM|$ (in Fig. 3 the coordinate of the point M is equal to -1). The whole coordinate line is designated as Ox .

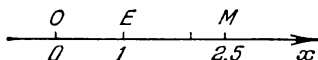


Fig. 2.

Inequalities of real numbers belonging to the coordinate line can be easily interpreted. If $x_1 < x_2$, then the point with the coordinate x_1 lies on the left of the point with the coordinate x_2 . The distance between two points M_1 and M_2

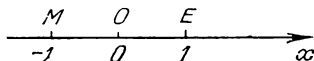


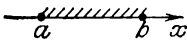
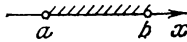
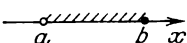
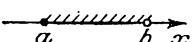
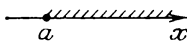
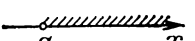
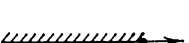
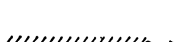
Fig. 3.

of the coordinate line is equal to the absolute value of the difference of their coordinates x_1 and x_2 : $|M_1M_2| = |x_1 - x_2|$.

Suppose a and b are real numbers and $a < b$. Table 1 presents names, definitions and designations of the number sets known as *number intervals*, as well as their graphical representation on the coordinate line. Each of the number intervals is defined as a set of real numbers x satisfying definite inequalities.

Intervals open from the left or from the right (see Table 1) are also known as *semi-open (semi-closed) intervals*, and number half-lines, as *infinite intervals*. A set of real numbers $\mathbb{R}$

TABLE 1. NUMBER INTERVALS

Name	Inequality defining the set	Designation	Representation
Interval from a to b (closed interval)	$a \leq x \leq b$	$[a; b]$	
Interval from a to b (open interval)	$a < x < b$	$(a; b)$	
Interval from a to b open from the left	$a < x \leq b$	$(a; b]$	
Interval from a to b open from the right	$a \leq x < b$	$[a; b)$	
Number half-line from a to $+\infty$	$a \leq x$	$[a; +\infty)$	
Open number half-line from a to $+\infty$	$a < x$	$(a; +\infty)$	
Number half-line from $-\infty$ to a	$x \leq a$	$(-\infty; a]$	
Open half-line from $-\infty$ to a	$x < a$	$(-\infty; a)$	

is also denoted as $(-\infty; +\infty)$ and is called a *number line*. Every coordinate line represents a number line.

When number sets are being considered the word "point" is also used instead of the word "element" or "number". Thus, for instance, instead of saying "the number 1 belongs to the interval $[0; 2]$ " we may say "the point 1 lies on the interval $[0; 2]$ " or "the point 1 belongs to the interval $[0; 2]$ ".

Subsets. Every natural number belongs to the set of integers; any point of the open interval $(a; b)$ is a point of the closed interval $[a; b]$, any regular triangle is an element of the set of all triangles.

If an element of the set A also belongs to the set B , then A is called a *subset* of the set B . This is written as $A \subset B$ or $B \supset A$, and it is said that the set A is contained in the set B or the set B contains the set A . Thus, the set $\mathbb{N}$ of natural numbers is a subset of the set $\mathbb{Z}$ of integers, i.e. $\mathbb{N} \subset \mathbb{Z}$; the open interval $(a; b)$ is a subset of the closed interval $[a; b]$: $(a; b) \subset [a; b]$.

If the set A contains an element which does not belong to the set B , then A is not a subset of B : $A \not\subset B$. For example, the closed interval $[a; b]$ is not a subset of the semi-closed interval $(a; b]$ since $a \in [a, b]$, but $a \notin (a, b]$.

It follows from the definition of a subset that any set is a subset of itself, that is, the statement $A \subset A$ holds true.

It is also assumed that an empty set is a subset of any set, which is only natural because an empty set does not contain any element and, consequently, it does not contain an element which would not belong to any other set.

Let us consider, for example, a set of students in a certain class; we denote the set by X and assume that Y is a set of students in the same class who have got the excellent mark for a history test work. If all the students in the class got an excellent mark for that work, then X and Y are equal sets: $X = Y$. Now if neither of the students got excellent marks, then the set Y is empty, $Y = \emptyset$. But in any case the set Y is a subset of the set X : $Y \subset X$.

Let us now consider an arbitrary set consisting of three elements, which will be denoted by a , b and c , and find all its subsets.

They are: an empty set $\emptyset$; sets containing one element each: $\{a\}$, $\{b\}$, $\{c\}$; sets containing two elements each: $\{a, b\}$, $\{b, c\}$, $\{a, c\}$; the set $\{a, b, c\}$ itself. The number of all these subsets is eight. Thus, any set consisting of three elements has $8 = 2^3$ subsets. It will be established later on (see Ch. 11) that if a set consists of n elements, then the number of all its subsets is equal to 2^n .

Intersection of sets. Let us consider two sets:

$$X = \{0, 1, 3, 5\}, Y = \{1, 2, 3, 4\}.$$

The numbers 1 and 3 and only these numbers simultaneously belong to both sets X and Y . The set $\{1, 3\}$ they form contains all the elements common to the sets X and Y .

The set consisting of all the elements belonging both to the set A and to the set B is called the *intersection* of the sets A and B and is denoted $A \cap B$.

Thus, the set $\{1, 3\}$ is the intersection of the indicated sets X and Y :

$$\{0, 1, 3, 5\} \cap \{1, 2, 3, 4\} = \{1, 3\}.$$

The intersection of the closed interval $[-1; 1]$ and the open interval $(0; 3)$, that is, the set, consisting of elements

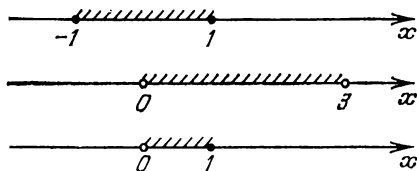


Fig. 4.

common to them, is the semi-open interval $(0; 1]$ (Fig. 4)

The intersection of the set of rectangles and the set of rhombuses is the set of squares. The intersection of the set of final-year students in the given school and the set of

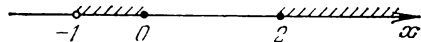


Fig. 5.

students who are members of the chemistry circle in the same school is the set of final-year students who are members of the chemistry circle.

If the sets A and B do not possess any elements in common, then it is said that they are non-intersecting or that their intersection is an empty set, and the notation is $A \cap B = \emptyset$. For instance, the intersection of the set of even numbers and the set of odd numbers is empty, the intersection of the number intervals $(-1; 0]$ and $[2; +\infty)$ is empty as well (Fig. 5).

The intersection of any set A and an empty set is, evidently, an empty set: $A \cap \emptyset = \emptyset$.

Union of sets. Let us take the sets $X = \{0, 1, 3, 5\}$ and $Y = \{1, 2, 3, 4\}$ and the set $\{0, 1, 2, 3, 4, 5\}$. The latter contains all the elements of the set X (the numbers 0, 1, 3, 5) and all the elements of the set Y (the numbers 1, 2, 3, 4), and does not contain any other elements, that is, the set

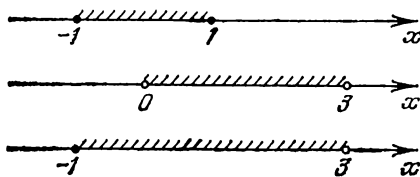


Fig. 6.

$\{0, 1, 2, 3, 4, 5\}$ contains those and only those elements which belong either to X or to Y .

The set consisting of all the elements belonging either to the set A or to the set B is called the *union* of A and B and is designated as $A \cup B$.

Thus we have

$$\{0, 1, 3, 5\} \cup \{1, 2, 3, 4\} = \{0, 1, 2, 3, 4, 5\}.$$

The union of the set of even positive numbers and the set of odd positive numbers is, evidently, the set of natural numbers. The union of the set of pupils younger than twelve and the set of pupils of the same school older than ten is the set of all the pupils of that school.

The union of the number intervals $[-1; 1]$ and $(0; 3)$ is the interval $[-1; 3)$ (Fig. 6).

Note that the union of number intervals cannot always be represented by one number interval. For instance, the union of the sets $(-1; 0]$ and $[2; +\infty)$ is not a number interval (see Fig. 5).

The given definitions of the intersection and union of sets can be vividly illustrated by depicting sets on a plane. Sets can be schematically illustrated by circles or rectangles. For instance, Fig. 7 shows the sets A and B , the set A being hatched horizontally and the set B vertically. Figures 8

and 9 show hatched the sets $A \cap B$ and $A \cup B$ respectively.

It is often necessary to consider a union and an intersection of three and more sets. The union of the sets A , B and C is a set whose each element belongs to at least one of the sets

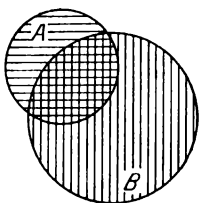


Fig. 7.

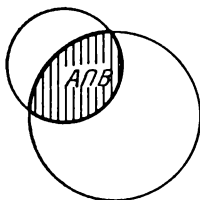


Fig. 8.

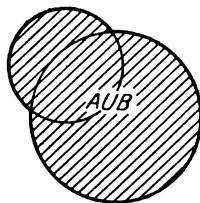


Fig. 9.

A , B and C (Fig. 10). The intersection of the sets A , B and C is the set of all elements belonging to the set A , the set B and the set C (Fig. 11).

For example, the union of the sets of acute, obtuse and right-angled triangles is the set of all the triangles; the uni-

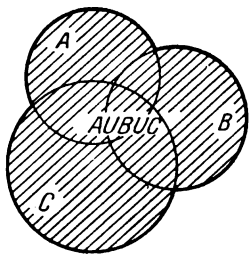


Fig. 10.

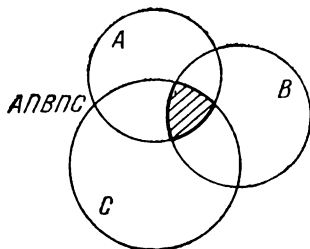


Fig. 11.

on of the sets of the vertices of triangles inscribed in a given circle is the set of points of that circle. If A , B and C are, respectively, the sets of the students in the class who solved a test problem in algebra, a problem in geometry and a problem in trigonometry, then the intersection of those sets is the set of students in that class who solved all the three problems.

A union and an intersection of sets possess many properties similar to those of a sum and a product of numbers, such as commutative, associative, and distributive properties (written on the left are the equalities for sets and on the

right the equalities for numbers):

- | | |
|--|-----------------------------------|
| 1. $A \cup B = B \cup A,$ | $a + b = b + a.$ |
| 2. $A \cap B = B \cap A,$ | $ab = ba.$ |
| 3. $(A \cup B) \cup C = A \cup (B \cup C),$ | $(a + b) + c$
$= a + (b + c).$ |
| 4. $(A \cap B) \cap C = A \cap (B \cap C),$ | $(ab) \cdot c = a \cdot (bc).$ |
| 5. $(A \cup B) \cap C = (A \cap C) \cup (B \cap C),$ | $(a + b)c = ac$
$+ bc.$ |

This similarity, however, is not always valid. For instance, the following equalities are valid for sets (the first of them is illustrated in Fig. 12):

6. $(A \cup C) \cap (B \cup C) = (A \cap B) \cup C.$
7. $A \cup A = A.$
8. $A \cap A = A.$

It is evident that the respective equalities for numbers are not true.

Let us designate by $m(A)$ the number of elements of a finite set A . The number of elements of an empty set is,

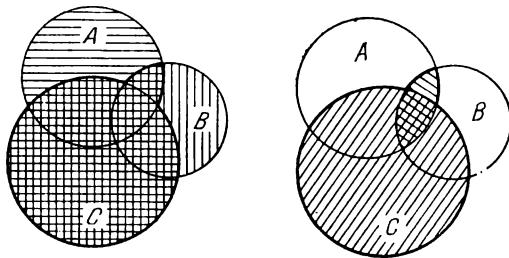


Fig. 12.

evidently, zero. For any finite sets A and B the following equality holds true:

$$m(A \cup B) = m(A) + m(B) - m(A \cap B). \quad (1)$$

Indeed, suppose the sets A and B do not intersect, i.e. $m(A \cap B) = 0$. Their union is obtained by adding all the

elements of one set to the elements of the other set and, therefore,

$$m(A \cup B) = m(A) + m(B).$$

Now if the intersection of A and B is not empty, then the number of their common elements is equal to $m(A \cap B)$. The union of these sets is formed by adding to the elements of A all the elements of B which do not belong to the set A .

The number of such elements is $m(B) - m(A \cap B)$. Thus,

$$m(A \cup B) = m(A) + m(B) - m(A \cap B).$$

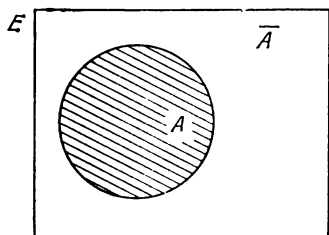


Fig. 13.

Example 1. Out of 250 students taking an exam in mathematics 180 students got a mark lower than the top grade while 210 students passed the exam. How many students got good and satisfactory marks?

△ Assume that A is the set of students having passed the exam, B is the set of students who got marks lower than the highest grade; by the hypothesis, $m(A) = 210$, $m(B) = 180$, $m(A \cup B) = 250$. The students who got good and satisfactory marks form the set $A \cap B$. By formula (1) we find that

$$m(A \cap B) = m(A) + m(B) - m(A \cup B) = 140. \blacktriangle$$

Complement. It is often sufficient to consider various subsets of one and the same set. In that case the set is called *fundamental* or *universal*. Let us denote the fundamental set by the letter E . The following equalities are valid for any set A belonging to the fundamental set E :

$$A \cup E = E, A \cap E = A.$$

The set of the elements of the fundamental set E which do not belong to the set A is called the *complement* of the set A relative to E or simply a complement and is designated as $\overline{A}$. The union of the set A and its complement $\overline{A}$ is a fundamental set: $A \cup \overline{A} = E$. The intersection of the set

and its complement is empty: $A \cap \overline{A} = \emptyset$. The complement of an empty set is a fundamental set: $\overline{\emptyset} = E$, and the complement of a fundamental set is empty: $\overline{E} = \emptyset$. In Fig. 13 the fundamental set E is schematically shown as a

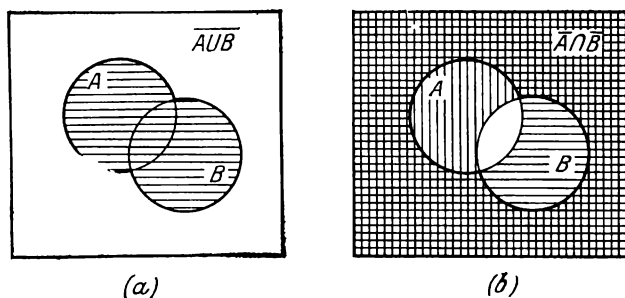


Fig. 14.

rectangle, its subset A is hatched and the complement of A is not hatched.

The following equalities are valid for any subsets A and B of the fundamental set E :

$$\overline{A \cup B} = \overline{A} \cap \overline{B}, \quad (2)$$

$$\overline{A \cap B} = \overline{A} \cup \overline{B}. \quad (3)$$

The first equality is illustrated in Fig. 14a and b. In Fig. 14a the union of A and B is hatched and the complement of that set unhatched, i.e. $\overline{A \cup B}$. In Fig. 14b the set $\overline{A}$ is hatched horizontally and the set $\overline{B}$ vertically, double hatching shows their intersection $\overline{A} \cap \overline{B}$.

A set of real numbers $\mathbb{R}$ can be taken as the fundamental set for number sets. The complement of the set of rational numbers $\mathbb{Q}$ relative to $\mathbb{R}$ is the set of irrational numbers, the complement of the half-line $[0; +\infty)$ is the open half-line $(-\infty; 0)$.

Example 2. There are 1400 students in a school; 1250 of them can ski, 952 can skate, and 60 students can neither ski nor skate. How many students can both ski and skate?

Δ The set of students in the school can be taken as the

fundamental set E ; A and B are the sets of students skiing and skating respectively (Fig. 15). The students unable to ski or skate form the set $\overline{A} \cap \overline{B}$. By the hypothesis, $m(\overline{A} \cap \overline{B}) = 60$, and since we have $\overline{A} \cap \overline{B} = \overline{A \cup B}$ from formula (2), it follows $m(\overline{A \cup B}) = 60$ too. Hence

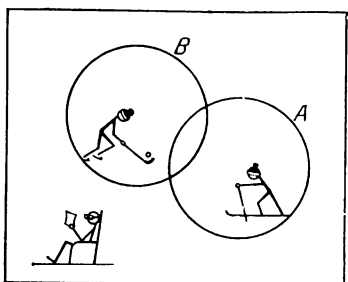


Fig. 15.

$m(A \cup B) = m(E) - m(\overline{A \cup B}) = 1340$. If $m(A)$ and $m(B)$ are known, formula (1) yields

$$\begin{aligned} m(A \cap B) &= m(A) \\ &+ m(B) - m(A \cup B) \\ &= 862. \quad \blacktriangle \end{aligned}$$

1.2. The Concept of a Function

Suppose X is some number set. A *number function* is defined on the set X if a real number is put into correspondence with each element of X . In

this case the set X is called the *domain of definition* of the function. An arbitrary element of the domain of definition of the function is usually denoted by the letter x and is called the *argument* of the function or the *independent variable*. The expression "the argument x runs over the set X " should be understood in the sense that any number from the domain of definition of the function can be taken instead of x .

For example, a number function is defined as follows: every number x belonging to the interval $[0; 1]$ is associated with the number $2x + 1$. The interval $[0; 1]$ being domain of definition, the rule of correspondence can be written as $x \rightarrow 2x + 1$.

Here is another example of a function: every rational number is associated with the number 1 and every irrational number with the number 0. The domain of definition is the whole set of real numbers (this function is known as *Dirichlet's function*).

The rule of correspondence is usually denoted by some letter, say f , and it is said that a function f or $f(x)$ is defined on the set X . A function can be also written in the form $y = f(x)$, where x is the argument and y is the corresponding value of the function, f being the rule of correspondence.

It is sometimes said that the function f associates the value of y with the value of argument x .

The function $x \rightarrow 2x + 1$ with the domain of definition $X = [0; 1]$ can be written in the form $f(x) = 2x + 1$, $x \in [0; 1]$ or $y = 2x + 1$, $x \in [0; 1]$, and the Dirichlet function can be written like this, for instance:

$$f(x) = \begin{cases} 1, & \text{when } x \text{ is rational,} \\ 0, & \text{when } x \text{ is irrational.} \end{cases}$$

It goes without saying that we can take any other letters instead of x , f , y , and write the function as $y = \varphi(t)$ or $z = F(x)$.

If the number a belongs to the domain of definition of the function f , then the function f is said to be *defined at the point a* . To indicate the value of function at the fixed point a , the following notation is used: $f(a)$, $y(a)$, $f(x)|_{x=a}$. Here, for instance, are the values of the function $y = 2x + 1$, $x \in [0; 1]$ at some points:

$$y(0) = 1, y(1/2) = 2, y(3/4) = 5/2, y(1) = 3.$$

The set of all values of the function $f(x)$, the argument x running over the domain of definition of the function, is called the *range* of the function f . For example, the range of the function $y = 2x + 1$, $x \in [0; 1]$, is the interval $[1; 3]$ and the range of the Dirichlet function consists of two numbers, 0 and 1.

The set of values of a function (its range) is a subset of the set $\mathbb{R}$ of real numbers. It is, therefore, sometimes said that the function is the mappings of one subset (domain of definition) into another subset (range) of the set of real numbers.

Two functions are considered to be equal if they have the same domain and associate every number from that domain with one and the same value. For instance, the function $f(x) = x - 1$ with the domain $(-1; +1)$ and the function $g(x) = (x^2 - 1)/(x + 1)$ with the same domain are equal since for any x belonging to the set $(-1; +1)$ there holds the equality

$$\frac{x^2 - 1}{x + 1} = x - 1.$$

A function is often defined by a formula indicating the order of mathematical operations which must be performed on the argument to obtain the value of the function. In this case nothing is said of its domain of definition, for then it is assumed that the function is defined on the set of the values of the argument for which the operations indicated by the formula are feasible. The set of all such values of the argument is called the *natural* domain of definition of the function given by the formula, or the domain of admissible values of the argument. The natural domain of the function f given by a formula is usually denoted by $D(f)$. In case a function is given by a formula, a problem arises of finding its domain (the natural domain of definition).

Example 1. Find the domain of the function

$$y = \sqrt{\frac{x+1}{2-x}}.$$

△ The operations indicated by this formula are feasible for the values of the argument x for which the radicand is nonnegative, i.e. $\frac{x+1}{2-x} \geq 0$. Solving the inequality, we find that it is valid only on the interval $[-1; 2)$, which is precisely the domain of the function being considered. ▲

Example 2. Find the domains of the functions

$$f_1(x) = \sqrt{1-x^2}, \quad f_2(x) = \frac{x-3}{2x+1},$$

$$f_1(x) + f_2(x) = \sqrt{1-x^2} + \frac{x-3}{2x+1}.$$

△ The function $f_1(x) = \sqrt{1-x^2}$ is defined on the set of the values of x for which $1-x^2 \geq 0$. This is the interval $[-1; 1]$. Thus, $D(f_1) = [-1; 1]$.

For the function $f_2(x) = (x-3)/(2x+1)$ the natural domain is the set of all values of the argument for which the denominator of the fraction does not vanish, i.e. $x \neq -1/2$. Thus, $D(f_2) = (-\infty; -1/2) \cup (-1/2; +\infty)$.

The function $f(x) = f_1(x) + f_2(x)$ is defined at a point a if and only if the point a belongs to the domain of the function $f_1(x)$ and to the domain of the function $f_2(x)$. In other words, the natural domain of the function $f(x)$ is the inter-

section of the domains of the functions f_1 and f_2 :

$$D(f) = D(f_1) \cap D(f_2) = [-1; -1/2) \cup (-1/2; 1]. \blacktriangle$$

Let us recall the definitions of some frequently encountered functions.

A *polynomial of degree n ($n \geq 0$) of the standard form* is the function

$$y = a_0 + a_1x + \dots + a_nx^n, \quad a_n \neq 0;$$

the numbers $a_0, a_1, \dots, a_n$ being called the *factors* of this polynomial. The domain of a polynomial is the entire number line. The n th-degree polynomial is also denoted by $P_n(x)$. In particular, the first-degree polynomial

$$y = P_1(x) = a_0 + a_1x, \quad a_1 \neq 0,$$

is called a *linear* function, and the second-degree polynomial

$$y = P_2(x) = a_0 + a_1x + a_2x^2, \quad a_2 \neq 0,$$

is called a *quadratic* function or *quadratic trinomial*.

A *polynomial of degree n* is a function which is defined on the entire number axis and can be reduced to an n th-degree polynomial of standard form. For instance, quadratic functions are the functions defined by the formulas

$$f(x) = 1 - x^3 + (x^2 - 3)(x + 1),$$

$$g(x) = \sqrt{x^4 + 2x^2 + 1}.$$

Indeed, they are both defined on the entire number axis, and for any number x there hold the equalities

$$1 - x^3 + (x^2 - 3)(x + 1) = x^2 - 3x - 2,$$

$$\sqrt{x^4 + 2x^2 + 1} = \sqrt{(x^2 + 1)^2} = x^2 + 1.$$

The function $f(x) = \frac{(x^2 - 1)}{(x + 1)}$ coincides with the first-degree polynomial $y = x - 1$ throughout the domain of definition. However, the function $f(x)$ is not a polynomial since its domain does not coincide with the set $\mathbb{R}$, it is not defined at the point $x = -1$.

A *rational* function is a function which can be represented as the ratio of two standard polynomials

$$y = \frac{P_n(x)}{P_m(x)} = \frac{a_0 + a_1x + \dots + a_nx^n}{b_0 + b_1x + \dots + b_mx^m}.$$

The natural domain of a rational function is the entire number line with the exception of the points at which the denominator vanishes. For example, the function $f(x) = \frac{(x^2-1)}{(x+1)}$ is rational for its domain is the entire number line except for $x = -1$.

1.3. A Coordinate Plane. The Graph of a Function

Two mutually perpendicular coordinate lines with the common reference point form a Cartesian (rectangular) system of coordinates on a plane. Scale units are usually taken to be equal; however sometimes use is made of coordinate lines with different scale units. One of these lines is usually horizontal, it is called *abscissa axis* and is designated as Ox . The other line is drawn vertically and is called the *axis of ordinates*, its designation being Oy (Fig. 16). The common reference point, the point O , is called the *origin of coordinates* or simply *origin*.

A plane with a system of coordinates chosen on it is called a *coordinate plane*. On a coordinate plane, every point is associated with a pair of numbers known as the *coordinates* of that point with respect to the given system of coordinates. Assume that M_x and M_y are the orthogonal projections of the point M on the axis of abscissas and the axis of ordinates respectively (see Fig. 16). The point M_x , being a point of the coordinate line Ox , has the coordinate x , and the point M_y , as a point of the coordinate line Oy , has the coordinate y . The pair of numbers $(x; y)$ (it is called an "ordered" pair in the sense that x occupies the first place and y the second) is called the *coordinates* of the point M and the notation is $M(x; y)$. The coordinates x and y are called the *abscissa* and the *ordinate* of the point M respectively.

Each ordered pair of numbers $(a; b)$ is associated on the coordinate plane with a single point M for which these numbers serve as coordinates: $x = a$, $y = b$. Thus, there is a one-to-one correspondence between the points of the coordinate plane and the ordered pairs of real numbers $(x; y)$. The set of the pairs of real numbers is sometimes called a *number plane*.

The coordinate lines divide the plane into four quadrants which are enumerated as shown in Fig. 16.

The distance between the points $M_1(x_1; y_1)$ and $M_2(x_2; y_2)$ (Fig. 17) is expressed in terms of their coordinates by the formula

$$|M_1M_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}.$$

Suppose a function $y = f(x)$ is given whose domain of definition is X . The *graph* of the function $f(x)$ is the set of

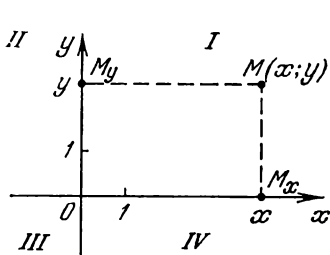


Fig. 16.

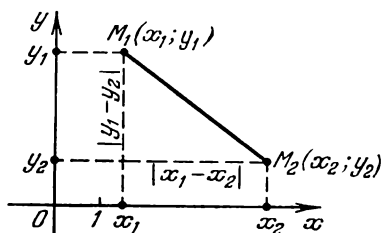


Fig. 17.

points of the coordinate plane with the coordinates $(x; f(x))$, that is, the set of points whose abscissas belong to the set X and the ordinates are equal to the corresponding val-

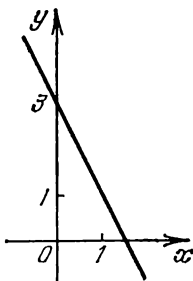


Fig. 18.

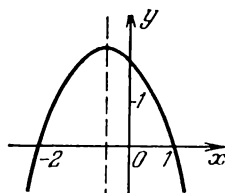


Fig. 19.

ues of the function. Figures 18 and 19 illustrate the graphs of the linear function $y = 3 - 2x$ and of the quadratic function $y = 2 - x - x^2$.

The graph of a function on a coordinate plane is a vivid illustration of the properties and behaviour of the function. The simplest way to construct the graph of the function $y = f(x)$ is to use points. A table is compiled featuring the values of the argument and the corresponding values of the

function:

$$\frac{x_1 | x_2 | x_3 | \dots}{y_1 | y_2 | y_3 | \dots}$$

and the points $M_1(x_1; y_1)$, $M_2(x_2; y_2)$, etc. are plotted on the coordinate plane. These points are connected by a smooth curve which represents, to a certain approximation, the graph of the function $y = f(x)$. This approximation may, however, prove to be rather rough. Therefore, other methods are used, as a rule, to construct the graph of a function. They will be discussed in Chap. 6.

1.4. An Inverse Function

Assume that the function $y = f(x)$, defined on the set X , is such that two distinct values y_1 and y_2 are put into correspondence with two different values of the argument x_1 and

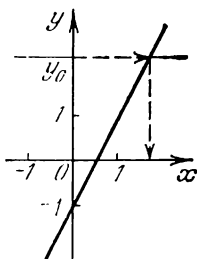


Fig. 20.

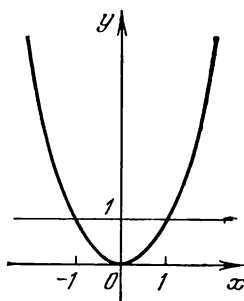


Fig. 21.

x_2 . The function $y = 2x - 1$, for instance, whose graph is shown in Fig. 20, possesses this property. Indeed, if $x_1 \neq x_2$ for it, then $y_1 = f(x_1) \neq f(x_2) = y_2$, since $f(x_1) - f(x_2) = 2(x_1 - x_2)$. This fact admits of the following geometrical interpretation: every straight line which is parallel to the Ox axis and passes through the point y_0 , belonging to the set of values of the function, intersects the graph of the function only once. Conversely, the function $y = x^2$ does not possess this property. For example, the value of the function $y = 1$ corresponds to two values of the argument, $x_1 = -1$ and $x_2 = 1$ (a straight line which is parallel to

the Ox axis and passes through the point $y = 1$ intersects the graph of the function at two points (Fig. 21)).

The function $y = f(x)$ with the domain X which associates different numbers with two different values of the argument establishes a one-to-one correspondence between its domain X and its range Y . Indeed, the function associates every point x from X with a certain y from Y , two distinct values x_1 and x_2 with distinct values $y_1 = f(x_1)$ and $y_2 = f(x_2)$, and every point $y \in Y$ with one of the values of the argument x . In this case, every point y from Y is associated with a single number x such that $y = f(x)$, that is, the correspondence is one-to-one. Thus, a function is defined on the set Y which is said to be an *inverse function* of f and is designated as f^{-1} . The domain of the inverse function f^{-1} is the set of values of the function f . The function having an inverse is called *invertible*.

The function $y = 2x - 1$, for example, is invertible. Its range is entire number line. Every number y is associated with the value $x = \frac{1}{2}y + \frac{1}{2}$ (in Fig. 20 this correspondence for the points y_0 is shown by an arrow). The formula $x = \frac{1}{2}y + \frac{1}{2}$ defines the function $x = f^{-1}(y)$, the inverse of $y = f(x)$.

The graph of the function $y = f(x) = 2x - 1$ is also the graph of its inverse $x = f^{-1}(y) = \frac{1}{2}y + \frac{1}{2}$, but the graph of the inverse function is unusual, the axis of the argument (the Oy axis) is vertical. If we customarily denote the argument of the inverse function by x , that is, represent the function as $y = f^{-1}(x)$, its graph, in the general case, differs from the graph of the function $y = f(x)$. Figure 22 shows the graph of the function $y = f(x) = 2x - 1$ and the graph of its inverse $y = f^{-1}(x) = \frac{1}{2}x + \frac{1}{2}$. They are symmetrical about the bisector of the first and third quadrants.

As a rule, if the function f is invertible, then its argument and the argument of its inverse are denoted by the same letter, that is, $y = f(x)$ and $y = f^{-1}(x)$.

Suppose the function $y = f(x)$ is invertible. Show that the graphs of the function $y = f(x)$ and of its inverse $y = f^{-1}(x)$ are symmetrical about the bisector of the first and

third quadrants, that is, symmetrical about the straight line $y = x$.

□ If the point $M(a; b)$ belongs to the graph of the function $y = f(x)$, then $b = f(a)$ (Fig. 23). Since there exists an inverse function, the equality yields $a = f^{-1}(b)$ and, hence, the point $M'(b; a)$ belongs to the graph of the inverse function $y = f^{-1}(x)$. The converse is also true: if the

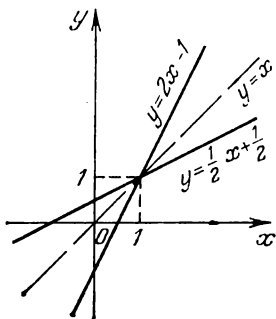


Fig. 22.

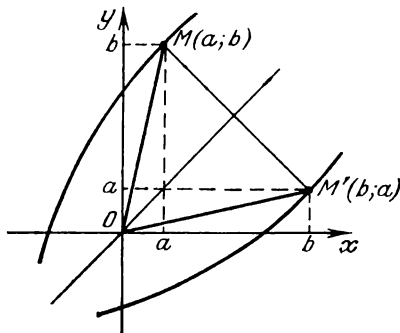


Fig. 23.

point $M'(b; a)$ belongs to the graph of the inverse function $y = f^{-1}(x)$, then $a = f^{-1}(b)$. Consequently, $b = f(a)$, and the point $M(a; b)$ belongs to the graph of the function $y = f(x)$.

Let us prove that the points $M(a; b)$ and $M'(b; a)$ are symmetrical about the bisector of the first and third quadrants. Assume $a \neq b$, that is, M and M' do not coincide. It is easy to see that they are equidistant from the origin. This means that the straight line which passes through the origin and the midpoint of the line segment MM' is perpendicular to that segment, and the points M and M' are symmetrical about that line. Since the midpoint of the segment MM' has the coordinates $((a + b)/2; (a + b)/2)$, the equation of the straight line is $y = x$. ■

It should be emphasized that a function is invertible if and only if this function associates different values with two different values of the argument. The function $f(x) = x^2$, $X = \mathbb{R}$, does not possess this property and, hence, it has no inverse.

Let us discuss a function defined by the same formula $f(x) = x^2$ and having the domain $X = [0; +\infty)$. If x_1 and x_2 belong to the domain of definition and $x_1 \neq x_2$, then $f(x_1) - f(x_2) = (x_1 - x_2)(x_1 + x_2)$. Here $x_1 \neq x_2$ and $x_1 + x_2 > 0$, therefore, $f(x_1) \neq f(x_2)$. Consequently, the function $f(x) = x^2$, $X = [0; +\infty)$, possesses an inverse. To find the inverse function, let us solve the equation $y = x^2$ for x , taking into account that $x \geq 0$. We get $x = \sqrt{y}$. We interchange the places of x and y : $y = \sqrt{x}$. Thus,

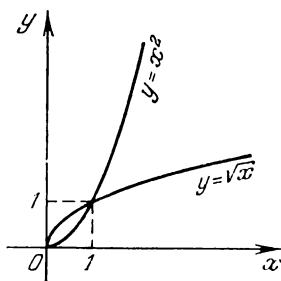


Fig. 24.

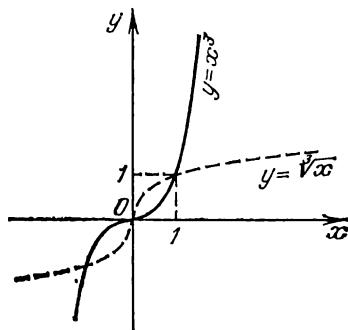


Fig. 25.

the inverse of $f(x) = x^2$, $X = [0; +\infty)$, is of the form $f^{-1}(x) = \sqrt{x}$, its domain of definition being the interval $[0; +\infty)$. The graphs of these functions are illustrated in Fig. 24.

Let us now consider the function $f(x) = x^3$. It is defined on the entire number line, its range also being the whole number line (Fig. 25). Show that the function possesses an inverse. Suppose $x_1 \neq x_2$. We take the difference $f(x_1) - f(x_2)$:

$$f(x_1) - f(x_2) = x_1^3 - x_2^3 = (x_1 - x_2) \left(\left(x_1 + \frac{x_2}{2} \right)^2 + \frac{3}{4} x_2^2 \right).$$

The expression in the first parenthesis is nonzero since $x_1 \neq x_2$, that in the second is also nonzero since it contains the sum of the squares of two summands which by the hypothesis $x_1 \neq x_2$ cannot vanish simultaneously. Thus, different values of the function are associated with different values of the argument from the set $(-\infty; +\infty)$. This means

that there exists an inverse function; it is designated as $y = \sqrt[3]{x}$. Its domain is the entire number axis and its graph is depicted in Fig. 25 by the dash line.

We note in conclusion that the definition of an inverse function implies that there always exists an inverse of the inverse function such that $(f^{-1})^{-1} = f$. Because of this fact, the function f and its inverse f^{-1} (provided it exists) are called mutually inverse functions.

PROBLEMS OF SECTION I

1. Assume that A is the set of the divisors of the number 15, B is the set of prime numbers smaller than 10, C is the set of even numbers smaller than 9. Enumerate the elements of these sets and find $A \cup B$, $A \cap C$, $B \cap C$, $(A \cup C) \cap B$, $A \cap B \cap C$.

2. Assume $A = [-1; 1]$, $B = (-\infty; 0)$, $C = [0, 2)$. Find the following sets: $A \cup C$, $A \cap B$, $A \cup B \cup C$, $(A \cup B) \cap C$, $B \cap C$ and represent them on the coordinate line.

3. Find the subsets X and Y of the set E if for any subset A of E there holds the equality $A \cap X = A \cup Y$.

Represent a set on the coordinate plane the coordinates $(x; y)$ of whose points satisfy the condition indicated in problems 4 through 6:

4. $|x| + |y| = 1$. 5. $|x + y| \leq 1$ and $|x - y| \leq 1$.

6. $x^2 - 2x + y \leq 0$.

7. Seventy tourists out of a group of 100 speak German, 45 speak French and 23 speak both German and French. How many tourists in the group speak neither German nor French?

8. Forty students took part in a mathematics Olympiad. They had to solve one problem in algebra, one in geometry and one in trigonometry. The results are tabulated below:

Problems solved	Number of students having solved the problems
in algebra	20
in geometry	18
in trigonometry	18
in algebra and geometry	7
in algebra and trigonometry	8
in geometry and trigonometry	9

It is also known that three students did not solve a single problem. How many students solved all the three problems? How many students solved two problems?

9. Assume that A is the set of solutions of the equation $f(x) = 0$, B is the set of solutions of the equation $g(x) = 0$. Prove that for any

polynomials $f(x)$ and $g(x)$ the set $A \cup B$ is the set of the roots of the equation $f(x) \cdot g(x) = 0$, and $A \cap B$ is the set of the roots of the equation $(f(x))^2 + (g(x))^2 = 0$.

Is the statement true in the case when $f(x)$ is a polynomial and $g(x)$ is a rational function?

Find the domains of the functions given in problems 10 through 16:

$$10. f(x) = \sqrt{3+x} + \sqrt{3-x}. \quad 11. f(x) = \sqrt[3]{\frac{x}{1-|x|}}.$$

$$12. f(x) = \sqrt{2x-3x}. \quad 13. f(x) = \log_{3+x}(x^2-1).$$

$$14. f(x) = \frac{\sqrt{x+5}}{\log(9-x)}. \quad 15. f(x) = \frac{\log_2 x^3}{\arccos(2x-1)}.$$

$$16. f(x) = \log[(1.25)^{1-x^2} - (0.4096)^{1+x}].$$

Find the range of the functions given in problems 17 and 18:

$$17. f(x) = \sqrt{x^2+2x+2}. \quad 18. f(x) = \frac{2x}{1+x^2}.$$

Find the inverse function $y = f^{-1}(x)$ of the function $y = f(x)$ presented in problems 19-21:

$$19. y = x^2 + 1, x \in (-\infty; 0]. \quad 20. y = 1 + \frac{1}{x}.$$

$$21. y = \frac{x^2+3x}{x^2-5x}.$$

22. Determine at what values of the parameters a and b the linear function $y = ax + b$ possesses an inverse and coincides with it.

PROBLEMS OF SECTION II

Write the sets A , B and C by enumerating their elements and find $A \cup B$, $B \cap C$, $(A \cup B) \cap C$, $A \cap B \cap C$ proceeding from the conditions given in problems 1 and 2:

1. A is the set of the divisors of the number 12; B is the set of the roots of the equation $x^2 - 6x + 5 = 0$; C is the set of odd numbers x such that $3 \leq x \leq 12$.

2. A is the set of even numbers x , $3 < x < 10$; B is the set of the divisors of the number 21; C is the set of prime numbers smaller than 12.

3. Give examples of the number sets A and B such that

(a) $A \cup B = \mathbb{R}$, $A \cap B = \emptyset$; (b) $A \cup B = A$, $A \cap B = B$.

Find $A \cup B$, $A \cap B$, $A \cap C$, $B \cup C$, $A \cap B \cap C$, $(A \cup B) \cap C$ and represent these sets on the coordinate line proceeding from the conditions given in problems 4 through 6:

4. $A = [0; 3]$, $B = (1; 5)$, $C = (-2; 0]$.

5. $A = [-\infty; 1]$, $B = [1; +\infty)$, $C = (0; 1)$.

6. $A = [-3; 1]$, $B = [2; +\infty)$, $C = (-\infty; -2)$.

The sets A and B are subsets of the fundamental set $\mathbb{R}$. Find $A \cup B$, $A \cap B$, $A \cup \bar{B}$, $\bar{A} \cap B$, $\bar{A} \cup \bar{B}$ and represent these sets on the coordinate line given in problems 7 through 10:

7. $A = (-1; 0]$,

$B = [0; 2)$.

8. $A = [0; 3)$,

$B = (-1; +\infty)$.

9. $A = (-\infty; 1]$,

$B = (-\infty; -3)$.

10. $A = (0; +\infty)$,

$B = [-1; 1)$.

11. The sets A and B are subsets of the set E (Fig. 26). Hatch the following sets:

(a) $A \cup \bar{B}$, $\overline{A \cup B}$; (b) $\bar{A} \cup \bar{B}$, $\bar{A} \cap B$;

(c) $A \cap B$, $\overline{A \cup B}$; (d) $\overline{A \cap B}$, $(\bar{A} \cap B) \cup (A \cap \bar{B})$.

12. A is a subset of the set of natural numbers, each element of A being a number which is a multiple of 2, or 3, or 5. Find the number

of elements in A if among them there are: 70 numbers multiple of 2; 60 numbers multiple of 3; 80 numbers multiple of 5; 32 numbers multiple of 6; 35 numbers multiple of 10; 38 numbers multiple of 15; and 20 numbers multiple of 30.

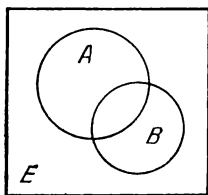


Fig. 26.

13. In a small sweet-shop people usually buy either one cake or one box of chocolates or a cake and a box of chocolates. One day the shop sold 57 cakes and 36 boxes of chocolates. How many cus-

tomers were there that day if 12 people bought both a cake and a box of chocolates?

14. Sixty-five per cent of children in a sports club play football, 70 per cent play volley-ball and 75 per cent play basket-ball. What is the smallest percentage of children playing all the three games?

15. Each of the students in the class twice visited the theatre during the vacation; 25, 12 and 23 students saw the plays A , B and C respectively. How many students are there in the class? How many of them saw the plays A and B , A and C , or B and C ?

16. The films A , B and C were shown in the cinema for a week. Out of 40 students (each of which saw either all the three films, or one of them, 13 students saw film A , 16 students saw film B , and 19 students saw film C . How many students saw all three films?

17. Out of 40 children 30 can swim, 27 can play chess and only five can do neither. How many children can swim and play chess?

18. The literature teacher decided to find out who of the 40 students had read the books A , B and C . The results were the following: 25 students had read the book A , 22 the book B and 22 the book C ; 33 students had read the book A or B , 32 the book A or C , and 31 had read the book B or C ; 10 students had read all three books. How many students had read one book each? How many students had read none of the books?

19. Among the school seniors the top grade was received by 48 students in mathematics, 37 in physics, 42 in chemistry, 75 in mathe-

matics and physics, 76 in mathematics and chemistry, 66 in physics and chemistry, and 4 students in all three subjects. How many students got more than one excellent mark? How many of them got only one excellent mark?

20. Find the intersection of the number sets A and B if every element of A has the form $4n + 2$, $n \in \mathbb{N}$, and every element of B has the form $3n$, $n \in \mathbb{N}$.

Represent a set on the coordinate plane the coordinates $(x; y)$ of whose points satisfy the conditions stated in problems 21 through 28:

$$21. |x + y| = x + y.$$

$$22. y - x^2 \geq 0.$$

$$23. |y| - |x| \geq 0.$$

$$24. x^2 + y^2 - 4y \leq 0.$$

$$25. x^2 + y^2 \leq 2|x| + 2|y|.$$

$$26. |x - y| = |x - y + 1|.$$

$$27. \log_{1/2}(2 - x^2 - y^2) < 0.$$

$$28. (x - |x|)^2 + (y - |y|)^2 \leq 4.$$

Find the domains of the functions given in problems 29 through 38:

$$29. f(x) = \sqrt{3 - 5x - 2x^2}.$$

$$30. f(x) = \sqrt{\frac{x}{1-x}}. \quad 31. f(x) = \sqrt[3]{\frac{x+2}{\log \cos x}}.$$

$$32. f(x) = \frac{\sqrt{4-x^2}}{\arcsin(2-x)}. \quad 33. f(x) = \sqrt{\log \frac{3x-x^2}{x-1}}.$$

$$34. f(x) = \log_3 \log_{1/2} x. \quad 35. f(x) = \sqrt{x^2 - |x| - 2}.$$

$$36. f(x) = \log_{x+1}(x^2 - 3x + 2).$$

$$37. f(x) = \frac{\sqrt{\cos x - \frac{1}{2}}}{\sqrt{6 - 35x - 6x^2}}.$$

$$38. f(x) = \sqrt[4]{\log_{0.2}^3 x + (\log_{0.2} x^3)(\log_{0.2} 0.0016x) + 36}.$$

Find the range of the functions given in problems 39 through 44:

$$39. f(x) = \frac{1-x^2}{1+x^2} \quad 40. f(x) = \sqrt{2x-x^2-1}.$$

$$41. f(x) = \sin x - 5 \cos x. \quad 42. f(x) = \log_3 x + \log_x 3.$$

$$43. f(x) = (\sin x + \cos x)^2. \quad 44. f(x) = 4^x - 2^x + 1.$$

45. Prove that an even function possesses no inverse.

Find the inverse function $y = f^{-1}(x)$ of the function $y = f(x)$ given in problems (46-50):

$$46. y = x^2 - 1, X = (-\infty; 0]. \quad 47. y = \frac{x-1}{x+1}.$$

$$48. y = \sqrt[4]{x}. \quad 49. y = \frac{1}{1+x^3}. \quad 50. y = x|x| + 2x.$$

Establish whether there exists an inverse function of each of the functions presented in problems 51-54, and find the inverse in the case when it exists.

$$51. y = 2 + x - x^2, X = [1/2, +\infty). \quad 52. y = x^3 - x.$$

53. $y = x^2 - 2x - 3$, $X = [0; 3]$. 54. $y = x |x| - 2x - 8$.

55. Prove that for all the values of the parameters a and b such that $ab \neq -4$, the function $y = \frac{2x + a}{bx - 2}$ coincides with its inverse.

56. What conditions must be satisfied by the numbers a , b , c and d ($c \neq 0$) for the function $y = \frac{ax + b}{cx + d}$ to coincide with its inverse?

57. At what values of α does the function $y = x^\alpha$ coincide with its inverse?

Chapter 2

Elements of Logic. Mutually Converse and Mutually Opposite Theorems. Principle of Mathematical Induction

2.1. Sentences. Operations on Sentences

A *sentence* is a statement about which it is worth while deciding whether it is *true* or *false*. We can cite the following examples of sentences:

1. Moscow is the capital of the Soviet Union.
2. Anatoly Karpov is the twelfth world chess champion.
3. The number 221 is prime.
4. $13 < 17$.

Sentences 1, 2 and 4 are certainly true. Sentence 3 is false since $221 = 13 \cdot 17$.

Thus, every sentence is either true or false; it cannot be both true and false simultaneously.

Sentences can be derived from words or symbols, but far from every collection of words or symbols (even making sense) is a sentence.

For example, the statements

- (a) it is easy to enter Moscow University,
- (b) $x > 0$

are not sentences since it is impossible to decide whether they are true or false.

Sentences are usually designated by capital Latin letters A , B , C etc., for example,

$$A \equiv \{6 < 7\}, \quad B \equiv \{\text{the number 6 is prime}\}.$$

This means that sentence B contains the statement that the number 6 is prime, and sentence A contains the statement that $6 < 7$. The " $\equiv$ " sign stands for the words "is the sentence".

Sentences A and B are simple. By means of logical connectives (the conjunctions "and", "or", the words "if ..., then...", "if and only if ...") we can form new sentences from

simple ones. For instance, using logical connectives, we can form from the sentences

$$A \equiv \{6 < 7\} \text{ and } B \equiv \{\text{the number 6 is prime}\}$$

the following composite sentences:

$$C \equiv \{6 < 7 \text{ or the number 6 is prime}\},$$

$$D \equiv \{6 < 7 \text{ and the number 6 is prime}\},$$

$$E \equiv \{6 < 7 \text{ if and only if the number 6 is prime}\},$$

$$F \equiv \{\text{if } 6 < 7, \text{ then the number 6 is prime}\}.$$

Note that new sentences can be formed even from sentences which have nothing in common. For instance, the sentence $G \equiv \{\text{if an elephant is an insect, then the Antarctic continent is covered with tropical forests}\}$

is composed by means of the logical connective "if..., then..." from two sentences which are not connected semantically.

In mathematical logic, the truth or the falsity of composite sentences formed by means of logical connectives is established irrespective of the meaning of the constituent simple sentences. Whether a composite sentence is true or false can be completely determined, first, by the logical connectives used for deriving the composite sentence and, second, by establishing which of the simple constituent sentences are true and which are false. For that purpose, operations are introduced in logic to be performed on the sentences which correspond to the connectives used for forming composite sentences. In this way, an exact unambiguous meaning is assigned to the conjunctions "and", "or" and to the words "if and only if...", "if..., then...".

If a sentence is formed from two sentences by connecting them with the word "or", then it is said to be a logical *sum* (*disjunction*) of those sentences.

A sentence formed from two sentences by connecting them with the word "and" is called a *logical product* (*conjunction*) of those sentences.

Sentences formed from two sentences by connecting them with the words "if and only if...", "if..., then..." are called an *equivalence* and an *implication* respectively.

Somewhat earlier we used logical connectives to form sentences C , D , E , and F from simple sentences

$$A \equiv \{6 < 7\} \text{ and } B \equiv \{\text{the number 6 is prime}\}.$$

Now we can say that those sentences are a disjunction, conjunction, equivalence and implication of the sentences A and B .

The sum (disjunction) of any sentences A and B is written in the form $A + B$ (or $A \vee B$), the product (conjunction), in the form AB (or $A \wedge B$). The sign $\Leftrightarrow$ (or $\sim$) is used for equivalence, i.e. we write $A \Leftrightarrow B$ (or $A \sim B$). Implication is written as $A \Rightarrow B$ (read: " B follows from A " or "if A , then B ").

For any two sentences, the truth or falsity of their disjunction, conjunction, equivalence and implication is determined as follows:

the disjunction $A + B$ of sentences is a true sentence if and only if at least one of the summands is true;

the conjunction AB of sentences is true if and only if both factors are true;

the equivalence $A \Leftrightarrow B$ is a false sentence if one of the sentences is true and the other false; otherwise, that is, if both sentences are true or they are both false, the equivalence is a true sentence;

the implication $A \Rightarrow B$ is a false sentence if and only if A is true and B is false; in all other cases, namely, if

- (a) A is true and B is true,
- (b) A is false but B is true,
- (c) A is false and B is false,

the sentence $A \Rightarrow B$ is taken to be true.

Example 1. Given two sentences

$$A \equiv \{2 = 3\} \text{ and } B \equiv \{2 < 3\}.$$

What is the meaning of the sentences $A + B$, AB , $A \Leftrightarrow B$, $A \Rightarrow B$? Which of these sentences are true and which are false?

$\triangle$ The sentence

$$A + B = \{2 = 3 \text{ or } 2 < 3\}$$

is true since one of the summands is a true sentence. The sentence $A + B$ can be written as one true nonstrict inequality: $2 \leq 3$.

The sentence

$$AB \equiv \{2 = 3 \text{ and } 2 < 3\}$$

is, evidently, false. For the conjunction of two sentences to be true, it is necessary that both sentences be true.

The equivalence

$$A \Leftrightarrow B \equiv \{2 = 3 \text{ if and only if } 2 < 3\}$$

is a false sentence since A is false and B is true.

The implication

$$A \Rightarrow B \equiv \{\text{if } 2 = 3, \text{ then } 2 < 3\}$$

is a true sentence. Indeed, by the definition, the implication $A \Rightarrow B$ is false only if A is true and B is false. ▲

Thus, the truth or falsity of a sentence formed from some sentences by means of operations of summation, multiplication, equivalence and implication depends only on the distribution of truth and falsity between the sentences on which logical operations are performed. It is convenient to write this dependence in the form of the following four tables, which are known as the *tables of truth of logical operations* (the letter T means that the corresponding sentence is true, the letter F means that it is false).

The tables of truth of logical operations make it possible to determine the truth or falsity of *any* sentences having

TABLE I

A	B	$A + B$
T	T	T
T	F	T
F	T	T
F	F	F

TABLE II

A	B	AB
T	T	T
T	F	F
F	T	F
F	F	F

TABLE III

A	B	$A \Leftrightarrow B$
T	T	T
T	F	F
F	T	F
F	F	T

TABLE IV

A	B	$A \Rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

the form $A + B$, AB , $A \Leftrightarrow B$, $A \Rightarrow B$, and so these tables are also known as the *tables of truth of the sentences* $A + B$, AB , $A \Leftrightarrow B$, $A \Rightarrow B$.

For our further discussion, the implication $A \Rightarrow B$ is the most important of all the logical operations we have considered. The first term of the implication $A \Rightarrow B$, the sentence A , is called a *premise*, the second term B is called a *conclusion*. As distinct from Tables I-III, the truth table for implication changes upon a commutation of the columns for A and B . It should also be noted that implication does not completely correspond to the ordinary understanding of the words "if..., then..." and "it follows that". The third and fourth rows of Table IV imply that if A is false, then, whatever B is, the sentence $A \Rightarrow B$ is considered to be true. Thus, a false statement may imply anything. For instance, the statements "if 6 is a prime number, then $7 < 6$ " and "if $7 < 6$, then there exist witches" are true. The sentence "if an elephant is an insect, then the Antarctic continent is covered with tropical forests" stated earlier is true.

In addition to the four logical operations just considered, mathematics employs one more simple but very significant operation, that of negation. This operation corresponds to the logical connective "not". Every sentence A can be associated with a statement consisting in the fact that the sentence A is false. Such a statement is either true or false and is, consequently, a sentence itself, and a true sentence at that if A is false and false if A is true. This new sentence is designated as $\bar{A}$ and is called a negation or denial of A . For instance, a negation for

$$A \equiv \{\text{the number 6 is prime}\}$$

can be constructed as follows:

$$\bar{A} \equiv \{\text{the number 6 is not prime}\}$$

or

$$\bar{A} \equiv \{\text{it is not true that the number 6 is prime}\}$$

or

$$\bar{A} \equiv \{\text{the number 6 is composite}\}.$$

In the given case the original sentence is false and so its negation is true.

The truth table for negation has the form

TABLE V	
A	$\bar{A}$
T	F
F	T

Let us consider the sentence

$A \equiv \{\text{the New-York city is the capital of the USA}\}.$

Its negation is

$\bar{A} \equiv \{\text{the New-York city is not the capital of the USA}\}.$

Since the sentence A is false, the sentence $\bar{A}$ is true. Note that it would be erroneous to consider the sentence

$B \equiv \{\text{Washington is the capital of the USA}\}$

to be a negation of the sentence A .

New sentences can be formed with the aid of several, even all five, logical operations, and each operation can be performed several times. We can determine whether a composite sentence is true or false, depending on the truth or falsity of the constituent sentences, by compiling the truth table of the composite sentence with a consecutive use of the truth tables of logical operations.

Let us compile, for example, a truth table for sentences having the form $\bar{B} \Rightarrow \bar{A}$. Successive use of Tables V and IV yields a table

A	B	$\bar{B}$	$\bar{A}$	$\bar{B} \Rightarrow \bar{A}$
T	T	F	F	T
T	F	T	F	F
F	T	F	T	T
F	F	T	T	T

The third and fourth columns are auxiliary. Thus we have got the following truth table for the sentence of the form $\bar{B} \Rightarrow \bar{A}$:

A	B	$\bar{B} \Rightarrow \bar{A}$
T	T	T
T	F	F
F	T	T
F	F	T

Let us compare the table obtained with Table IV for the implication $A \Rightarrow B$. The tables coincide, that is, the truth tables for the sentences $A \Rightarrow B$ and $\bar{B} \Rightarrow \bar{A}$ are identical. Such sentences are called *equivalent* and are connected by the equality sign.

Consequently, for any sentences A and B we can write

$$A \Rightarrow B = \bar{B} \Rightarrow \bar{A}.$$

The equivalence of the sentences of the forms $A \Rightarrow B$ and $\bar{B} \Rightarrow \bar{A}$ means that whatever the sentences A and B , the sentences $A \Rightarrow B$ and $\bar{B} \Rightarrow \bar{A}$ are either both true or both false. The truth or falsity of one of them correspondingly implies the truth or falsity of the other.

Identically true and identically false sentences play an important part in logic. *Identically true* sentences are always true, irrespective of whether the sentences constituting them are true or false. Here are some examples of identically true sentences: $A + \bar{A}$, $A \Leftrightarrow \bar{\bar{A}}$, $(A \Rightarrow B) \Leftrightarrow (\bar{B} \Rightarrow \bar{A})$. We shall denote identically true sentences by the letter I .

Identically false sentences are always false, irrespective of the truth or falsity of the sentences constituting them. We shall denote them by the letter L . Here are some examples of identically false sentences:

$$A\bar{A}, (BA) \bar{A}, AL.$$

In mathematics itself and in its applications we have often to decide whether various kinds of sentences are equivalent.

When composite sentences are formed from a small number of simple sentences, we can establish equivalence or nonequivalence by compiling the truth tables of the composite sentences and comparing them. But it should be borne in mind that the truth table of a sentence, formed from n simple sentences, consists of 2^n rows. Therefore, another method is usually used to establish equivalence: a certain number of fundamental equivalences (laws of the algebra of sentences) are verified by the truth tables and the equalities obtained are used in proving other equalities in the same way as algebraic laws are used in elementary algebra in identity transformations, e.g.

$$a + b = b + a \text{ (commutative),}$$

$$a + (b + c) = (a + b) + c \text{ (associative),}$$

$$a(b + c) = ab + ac \text{ (distributive).}$$

The following equivalences are easy to verify:

$$1. A + B = B + A.$$

$$2. AB = BA.$$

$$3. A + (B + C) = (A + B) + C.$$

$$4. A(BC) = (AB)C.$$

$$5. A(B + C) = AB + AC.$$

$$6. (A + B)(A + C) = A + BC.$$

$$7. \overline{A + B} = \overline{A}\overline{B}.$$

$$8. \overline{\overline{A}\overline{B}} = \overline{A} + \overline{B}.$$

$$9. \overline{\overline{A}} = A, A + A = A, AA = A.$$

$$10. A + \overline{A} = I, A\overline{A} = L, \\ A + I = I, AI = A, \\ A + L = A, AL = L.$$

The first five laws written above are similar to the laws of ordinary algebra of numbers. The rest have no analogy in elementary algebra.

Let us solve the following problem using the laws of summation and multiplication of sentences.

Example 2. Brown, Johns and Smith are charged with a bank robbery. The thieves escaped in a car that was waiting for them. At the inquest Brown stated that the criminals had escaped in a blue Buick; Johns stated that it had been black

Chevrolet and Smith said that it had been Ford Granada and by no means blue. It turned out that wishing to confuse the Court, each of them only indicated correctly either the make of the car or only its colour. What colour was the car and of what make?

△ Let us discuss the sentences

$$\begin{aligned} A &\equiv \{\text{the car is blue}\}, \\ B &\equiv \{\text{the car is a Buick}\}, \\ C &\equiv \{\text{the car is black}\}, \\ D &\equiv \{\text{the car is a Chevrolet}\}, \\ E &\equiv \{\text{the car is a Ford Granada}\}. \end{aligned}$$

Since each of the criminals gave a correct answer either as to the colour of the car or as to its make, it follows from Brown's statement that the sentence $A + B$ is true. Johns' evidence means that the sentence $C + D$ is true. Smith asserted the truth of the sentence $\bar{A} + E$.

Since the sentences $A + B$, $C + D$ and $\bar{A} + E$ are true, their product is also true:

$$P = (A + B)(C + D)(\bar{A} + E).$$

Removing the brackets, we obtain

$$\begin{aligned} P &= (AC + AD + BC + BD)(\bar{A} + E) \\ &= AC\bar{A} + ACE + AD\bar{A} + ADE + BC\bar{A} + BCE + BD\bar{A} + BDE. \end{aligned}$$

It readily follows from the hypothesis that seven out of the eight summands obtained (all except the fifth) are false. Therefore,

$$\begin{aligned} P &= L + L + L + L + BC\bar{A} + L + L + L \\ &= BC\bar{A} + L = BC\bar{A}, \end{aligned}$$

that is, the sentence $BC\bar{A}$ is true and this means that the criminals escaped in a black Buick.

2.2. Propositions Dependent on a Variable

When studying mathematics, we encounter various propositions (statements) dependent on one or several variables. For instance, the proposition "the number 5 is a divisor of the number n " depends on the variable n assuming natural

values. It is true for every $n = 5k$, $k \in \mathbb{N}$, and false for every $n \neq 5k$, $k \in \mathbb{N}$. Equations and inequalities are also propositions of that kind. The inequality $x - 2 > 0$, for example, can be regarded as a proposition dependent on the variable x . Its truth or falsity depends on the value x assumes. If, for instance, $x = 3$, the proposition is true, now if we assume that $x = 0$, it is false. The equation $x + y = 1$ is a proposition dependent on two real variables, x and y . For $x = y = 1/2$, the proposition $x + y = 1$ is true, for $x = y = 0$ it is, evidently, false.

It is not only in mathematics that propositions depend on a variable. For example, the proposition "a player of the Soviet hockey team Spartak has scored 400 goals" is defined on the set of all players of the team. As is known, this proposition is true for Starshinov and false for all the others.

Propositions dependent on variables are designated as $A(n)$, $B(x)$, $C(x; y)$, etc. For each proposition it must be indicated, in one way or another, on what set it is being considered or, as it is customary to say, on what set it is defined or given. The proposition $A(x)$, $x \in M$, is not, in general, a sentence and it is, therefore, meaningless to verify whether it is true or false. There is no answer to that question (except for the case when the set M consists of one element). But as soon as a certain element $x_0 \in M$ has been chosen, or fixed as it is customary to say, it is assumed that $A(x_0)$ is either true or false, that is, the proposition becomes a sentence. The set M on which the proposition $A(x)$ is defined can be divided into two subsets. One of the subsets contains those and only those elements of M for which $A(x)$ is true. It is called a truth set of the proposition $A(x)$. The other subset contains those and only those elements of M for which $A(x)$ is false. If we designate the first subset as A , we should designate the second as $\bar{A}$ since it is the complement of A with respect to M .

For instance, for the proposition

$$A(x) \equiv \{x^2 - 5x + 6 < 0\}, \quad x \in \mathbb{R},$$

the truth set $\bar{A}$ is the interval $(2; 3)$ and the set $\bar{A}$ is the union of the intervals $(-\infty; 2]$ and $[3; +\infty)$.

Two propositions $A(x)$ and $B(x)$ defined on the same set are *equivalent* if their truth sets coincide.

For example, the propositions

$$A(x) \equiv \{x^2 - 5x + 6 < 0\}, \quad x \in \mathbb{R}$$

and

$$B(x) \equiv \{x^4 - 5x^3 + 6x^2 < 0\}, \quad x \in \mathbb{R}$$

are equivalent since the truth set of each of them is the interval (2; 3).

All logical operations introduced in 2.1 are naturally extended to propositions dependent on variables.

The *negation* (or *denial*) of the proposition $A(x)$, $x \in M$, is the proposition $\overline{A}(x)$ defined on the same set M and turn-

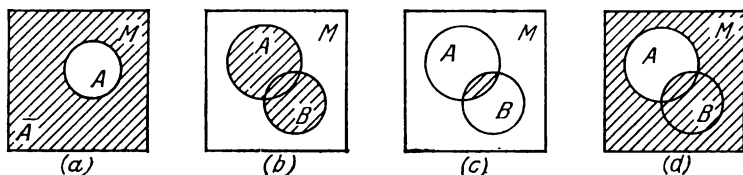


Fig. 27.

ing into a true sentence for *those and only those* elements of M for which $A(x)$ is a false sentence.

It follows from the definition that if A is a truth set of the proposition $A(x)$, then the truth set of the negation $\overline{A}(x)$ is the complement $\overline{A}$ of A with respect to M . Figure 27a shows schematically the sets M , A and $\overline{A}$. The truth set of the negation $\overline{A}(x)$ is hatched.

Let us consider the operation of summation, multiplication and implication for propositions defined on the same set M .

The sum $A(x) + B(x)$ turns into a true sentence for those (and only for those elements) of the set M for which at least one of the summands turns into a true sentence. Two circles in Fig. 27b represent schematically the truth sets A and B for $A(x)$ and $B(x)$ respectively. The truth set of the sum $A(x) + B(x)$ is, evidently, the union $A \cup B$ of A and B . It is hatched in the figure.

The product $A(x)B(x)$ turns into a true sentence for those and only those elements of M for which both factors

turn into true sentences. The truth set of the product $A(x) B(x)$ is the intersection $A \cap B$ of the sets A and B . It is hatched in Fig. 27c.

The implication $A(x) \Rightarrow B(x)$ becomes a false sentence for those and only those elements of M for which the premise $A(x)$ is true and the conclusion $B(x)$ is false. The truth set of the implication $A(x) \Rightarrow B(x)$ is shown hatched in Fig. 27d.

Example 1. Assume that

$$A(x) \equiv \{x - 2 > 0\} \text{ and } B(x) \equiv \{x + 2 \geq 0\}$$

are two propositions dependent on the variable x , $x \in \mathbb{R}$. What is the meaning of the propositions

- (a) $A(x) + B(x)$, (b) $A(x) B(x)$,
 (c) $A(x) \Rightarrow B(x)$, (d) $B(x) \Rightarrow A(x)$,
 (e) $A(x) \overline{B(x)}$, (f) $\overline{B(x)} \Rightarrow \overline{A(x)}$?

What are their truth sets?

Δ (a) The proposition $A(x) + B(x)$ consists in the fact that at least one of the equations

$$x - 2 > 0 \text{ and } x + 2 \geq 0$$

holds true. The truth set of $A(x) + B(x)$ is, evidently, the interval $[-2, +\infty)$.

(b) The product of the propositions $A(x)$ and $B(x)$ is a proposition which states that both equations $x - 2 > 0$ and $x + 2 \geq 0$ are true, that is, the variable x satisfies the system of inequalities

$$\begin{cases} x - 2 > 0, \\ x + 2 \geq 0. \end{cases}$$

The truth set is the interval $(2; +\infty)$.

(c) For implication $A(x) \Rightarrow B(x)$ ("if $x - 2 > 0$, then $x + 2 \geq 0$ ") the truth set consists of all real numbers.

(d) For the implication $B(x) \Rightarrow A(x)$ ("if $x + 2 \geq 0$, then $x - 2 > 0$ ") the truth set is the union of two intervals, namely, $(-\infty; -2) \cup (2; +\infty)$. For every number x belonging to the interval $[-2; 2]$, the implication $B(x) \Rightarrow A(x)$ becomes a false sentence.

(e) For the product $A(x) \overline{B(x)}$, the truth set is the set of solutions of the system of inequalities

$$\begin{cases} x - 2 > 0, \\ x + 2 < 0, \end{cases}$$

that is, an empty set.

(f) For the implication $\overline{B(x)} \Rightarrow \overline{A(x)}$ ("if $x + 2 < 0$, then $x - 2 \leq 0$ "), the truth set is the set of all real numbers. $\blacktriangle$

Two kinds of frequently encountered assertions are related to propositions dependent on variables.

(1) The proposition $A(x)$, $x \in M$, becomes a true sentence for *all* elements of the set M .

(2) The proposition $A(x)$, $x \in M$, becomes a true sentence for *at least one* element of the set M ; in other words, there is an element $x_0 \in M$ for which $A(x_0)$ is a true sentence.

It is customary in mathematics to write such assertions briefly, using special notation: the sign $\forall$ meaning "for all" (the letter A turned upside down, from the word "All"), and the sign $\exists$ meaning "there exists" (the letter E looking left). In statements and formulations, the sign $\forall$ stands for the words "all, every, each, any". The existence sign $\exists$ is used instead of the words "at least one, there is, there exists".

Using these signs, statements 1 and 2 can be written as follows:

$$(1) (\forall x) A(x), (2) (\exists x) A(x); x \in M.$$

Note that the first assertion is equivalent to the following statement: the truth set A of the proposition $A(x)$ is the set M , i.e. $A = M$. The second assertion is equivalent to the following: the truth set of the proposition $A(x)$ is nonempty, i.e. $A \neq \emptyset$.

Each of the statements discussed is either true or false and, consequently, is a sentence.

For instance, using the introduced signs, we can construct the following two assertions for the proposition

$$A(x) \equiv \{x^2 > 0\}, x \in \mathbb{R}:$$

$$(1) (\forall x) A(x) \equiv \{\text{for all real numbers } x^2 > 0\},$$

$$(2) (\exists x) A(x) \equiv \{\text{there exists a real number for which } x^2 > 0\}.$$

Each of these statements is a sentence. The first sentence is, evidently, false, the second, true.

To make sure that the sentence $(\forall x) A(x)$, $x \in M$ is true, it is necessary to verify the truth of the proposition $A(x)$ for all elements of M .

If the elements of the set M are few in number, we can consider them one-by-one and verify the truth of the statement $A(x)$ for each of them. Now if the set M is infinite or, although finite, contains many elements, we can make sure of the truth of the sentence only by way of logical reasoning.

It is very important to understand that to refute the sentence $(\forall x) A(x)$, it is sufficient to indicate at least one element $x \in M$ for which $A(x)$ is false. The element x from the set M for which the proposition $A(x)$ is false is called a *counter-example* for the sentence $(\forall x) A(x)$.

Thus, to make sure of the falsity of the sentence $(\forall x) A(x)$, it is sufficient to find (or to construct, as they say) one counter-example. Let us consider the proposition

$A(n) \equiv \{\text{the number } n^2 + n + 41 \text{ is prime}\}, n \in \mathbb{N}.$

For the sentence

$(\forall n) A(n) \equiv \{\text{for every natural } n$
 $\text{the number } n^2 + n + 41 \text{ is prime}\}$

the element $n = 40$ is a counter-example. Indeed, the number $40^2 + 40 + 41 = 40 \cdot 41 + 41$ can be divided by 41 and, consequently, $A(40)$ is a false sentence.

It is remarkable that for all $n < 40$ the proposition $A(n)$ is true.

2.3. Mutually Converse and Mutually Opposite Theorems. Necessary and Sufficient Conditions

Most of the theorems encountered in school mathematics are formulated (or can be formulated) as follows: for any element x belonging to the set M the proposition $A(x)$ implies a proposition $B(x)$. Using the notation introduced in 2.1 and 2.2, every theorem of this kind can be written in the form $(\forall x) A(x) \Rightarrow B(x)$, $x \in M$. Let us consider, for example, the following four theorems.

Theorem 1. *If the sum of the digits in a natural number is divisible by 3, then the number is also divisible by 3.*

Theorem 2. *If a quadrilateral is a rectangle, then its diagonals are congruent.*

Theorem 3. *If at least one of two numbers is divisible by 7, then their sum is also divisible by 7.*

Theorem 4. *The diagonals of the rhombus are mutually perpendicular.*

Each of these theorems can be understood as a sentence of the form

$$(\forall x) A(x) \Rightarrow B(x), x \in M,$$

and, consequently, all the four theorems have the same structure.

Theorem 1 deals with two propositions,

$$A(n) \equiv \{\text{the sum of the digits in number } n \\ \text{is divisible by 3}\},$$

$$B(n) \equiv \{\text{the number } n \text{ is divisible by 3}\},$$

dependent on the variable $n \in \mathbb{N}$.

The theorem states that the following assertion is true for every natural number: if the sum of the digits forming a number is divisible by 3, then the number is also divisible by 3, that is, the truth of the sentence

$$(\forall n) A(n) \Rightarrow B(n), n \in \mathbb{N}$$

is being asserted. It is known from arithmetic that this sentence is true, that is, Theorem 1 is true.

Theorem 2 is concerned with the set Q of all quadrilaterals q and with the following two propositions on that set:

$$A(q) = \{\text{a quadrilateral } q \text{ is a rectangle}\},$$

$$B(q) = \{\text{the diagonals of the quadrilateral } q \\ \text{are congruent}\}.$$

The theorem states that the following assertion is true for any quadrilateral: if a quadrilateral is a rectangle, then its diagonals are congruent. Thus, Theorem 2 is understood as the sentence

$$(\forall q) A(q) \Rightarrow B(q), q \in Q.$$

It is proved in geometry that this theorem is true.

Theorem 3 contains two propositions:

$$A(n; m) \equiv \{n \text{ or } m \text{ is divisible by } 7\},$$

$$B(n; m) \equiv \{n + m \text{ is divisible by } 7\},$$

defined on the set of all pairs $(n; m)$ of natural numbers.

The theorem asserts the truth of the sentence

$$(\forall n; m) A(n; m) \Rightarrow B(n; m), \quad n, m \in \mathbb{N},$$

which is, evidently, false and, consequently, Theorem 3 is false. Of importance for us here is the fact that the logical structure of this theorem is the same as that of Theorems 1 and 2.

Because of its brevity, Theorem 4 on the diagonals of a rhombus may seem to be of the form other than the form $(\forall x) A(x) \Rightarrow B(x)$. But it is only an apparent difference. The exact formulation of this theorem is this (recall that a rhombus is a parallelogram whose all sides are equal).

Theorem 4. *Suppose P is the set of all parallelograms and assume that*

$$A(p) \equiv \{\text{the parallelogram } p \text{ is a rhombus}\},$$

$$B(p) \equiv \{\text{the diagonals of the parallelogram } p \text{ are mutually perpendicular}\}$$

are two propositions defined on the set P .

Then

$$(\forall p) A(p) \Rightarrow B(p), \quad p \in P$$

that is, the following assertion is true for any parallelogram: if a parallelogram is a rhombus, then its diagonals are mutually perpendicular.

In the formulation of every theorem having the form

$$(\forall x) A(x) \Rightarrow B(x), \quad x \in M,$$

we can isolate the hypothesis (the proposition $A(x)$) and the conclusion (the proposition $B(x)$).

Theorems differing from one another either by the hypothesis or by the conclusion are *different* theorems.

Let us consider one more example.

Theorem 5. *Suppose Q is the set of all quadrilaterals and assume that*

$A(q) \equiv \{\text{the quadrilateral } q \text{ is a rhombus}\},$

$B(q) \equiv \{\text{the diagonals of the quadrilateral } q \text{ are mutually perpendicular}\}$

are two propositions defined on the set Q .

Then

$$(\forall q) A(q) \Rightarrow B(q), q \in Q,$$

that is, the following assertion is true for every quadrilateral: if a quadrilateral is a rhombus, then its diagonal is a rhombus, then its diagonals are mutually perpendicular.

Theorem 5 is not identical to Theorem 4, they are two *different* theorems, since the propositions $A(q)$ and $B(q)$ are defined on the set of all parallelograms in the first case and on the set of all quadrilaterals in the other.

A clear and unambiguous isolation of a hypothesis and a conclusion in a theorem makes it possible to introduce *unambiguously* the concepts of a *converse* and an *opposite* theorem.

The theorems

$$(\forall x) A(x) \Rightarrow B(x), x \in M,$$

and

$$(\forall x) B(x) \Rightarrow A(x), x \in M$$

are called *mutually converse*.

One of these theorems is sometimes called *direct*, then the other is called its *converse*.

It is clear that *any* of the two mutually converse theorems can be taken to be *direct*.

It can be seen from the definition that by interchanging the hypothesis and the conclusion of a theorem, we obtain its *converse*.

It should be emphasized that for a pair of mutually converse theorems three possibilities exist, namely,

- (1) both theorems are correct,
- (2) one of theorems is correct, the other is incorrect,
- (3) both theorems are incorrect.

To show that the first case is possible, it is sufficient to consider Theorem 1 and its converse (if a number is divisible by 3, then the sum of the digits constituting the number is divisible by 3).

Theorem 4 on the diagonals of a rhombus and its converse (the following assertion is true for any parallelogram: if the diagonals of a parallelogram are mutually perpendicular, then the parallelogram is a rhombus) are also true.

The converse of Theorem 2 is formulated as follows: if the diagonals of a quadrilateral are congruent, then the quadrilateral is a rectangle. This theorem is incorrect. As a counter-example we can take a quadrilateral shown in Fig. 28; thus, one of the mutually converse theorems may be true and the other false.

By analogy, Theorem 5 and the following theorem are mutually converse: if the diagonals of a quadrilateral are

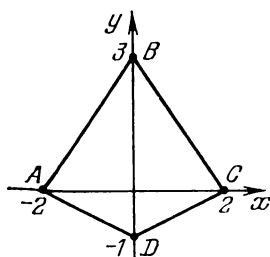


Fig. 28.

mutually perpendicular, then the quadrilateral is a rhombus. Theorem 5 is known to be true and its converse, false. As a counter-example, we can again take the quadrilateral shown in Fig. 28.

The third case, when both theorems, the direct theorem and its converse, are false, can also occur. The converse of Theorem 3 is: if the sum of two numbers is divisible by 7, then

at least one of the summands is divisible by 7. It is evident that the theorem is false, as well as its converse. Let us turn again to Theorems 4 and 5 on the diagonals of a rhombus. We have indicated that these theorems are different; if we made no distinction between them, the converse theorem would lose unambiguity, the theorem would have two converse theorems, one of which would be true (the converse of Theorem 4) and the other false (the converse of Theorem 5).

The words "necessary", "sufficient", "necessary and sufficient" and the like are often used when formulating a direct theorem and its converse.

If the theorem $(\forall x) A(x) \Rightarrow B(x)$ is true, then the proposition $A(x)$ is called a *sufficient condition* for $B(x)$, and the proposition $B(x)$, a *necessary condition* for $A(x)$.

Let us return to Theorem 2:

$$(\forall q) A(q) \Rightarrow B(q), \quad q \in Q,$$

where

$A(q) = \{\text{the quadrilateral } q \text{ is a rectangle}\},$

$B(q) = \{\text{the diagonals of the quadrilateral } q \text{ are congruent}\},$

and Q is the set of all quadrilaterals. This theorem is true and, consequently, $A(q)$ is the *sufficient* condition for $B(q)$, that is, for the diagonals of a quadrilateral to be congruent, it is *sufficient* that the quadrilateral be a rectangle.

The proposition $B(q)$ is the *necessary* condition for $A(q)$, that is, for a quadrilateral to be a rectangle, it is *necessary* that the diagonals of the quadrilateral be congruent.

If not only the theorem

$$(\forall x) A(x) \Rightarrow B(x), \quad x \in M$$

is true, but also its converse

$$(\forall x) B(x) \Rightarrow A(x), \quad x \in M,$$

then $A(x)$ is the necessary and sufficient condition for $B(x)$, and $B(x)$ is the necessary and sufficient condition for $A(x)$.

For instance, at the beginning of the section we presented Theorem 1: If the sum of the digits in a natural number is divisible by 3, then the number is divisible by 3. This theorem is true. Its converse is also true. Therefore, we can say that for the number to be divisible by 3, it is *necessary* and *sufficient* that the sum of the digits in the number be divisible by 3.

It should be remembered that when the formulation of a theorem contains the words "necessary and sufficient", the proof must necessarily consist of two parts: proving the necessity and proving the sufficiency for such a formulation actually comprises the formulations of two theorems, a direct theorem and its converse. Each of them has to be proved since the validity of one of them by no means yields the validity of the other.

It should be pointed out that sometimes the words "if and only if", "in that and only that case", and the like are used instead of the words "it is necessary and sufficient".

Theorem 1 and its converse, for instance, can be formulated as follows: the natural number n can be divided by 3 if and only if the sum of the digits in the number n is divisible by 3. The word "condition" is often replaced by the word

"criterion" and then the formulation of a theorem includes the words a "*necessary criterion*" or a "*sufficient criterion*" or, finally, a "*necessary and sufficient criterion*".

Let us now discuss an opposite theorem.

The theorems

$$(\forall x) A(x) \Rightarrow B(x), x \in M$$

and

$$(\forall x) \overline{A(x)} \Rightarrow \overline{B(x)}, x \in M$$

are called *mutually opposite*.

Consequently, if we replace, in the formulation of some theorem, the hypothesis and conclusion by their negations, we obtain the formulation of a theorem which is opposite to the original theorem.

Every theorem

$$(\forall x) A(x) \Rightarrow B(x)$$

thus generates three more theorems:

its *converse* $(\forall x) B(x) \Rightarrow A(x)$,

its *opposite* $(\forall x) \overline{A(x)} \Rightarrow \overline{B(x)}$,

the *opposite of the converse* $(\forall x) \overline{B(x)} \rightarrow \overline{A(x)}$.

Having taken, as an example, Theorem 5 discussed above as an original theorem, we shall have (for an arbitrary quadrilateral):

(1) the *original theorem*: if a quadrilateral is a rhombus, then its diagonals are mutually perpendicular (the theorem is true);

(2) the *converse theorem*: if the diagonals of a quadrilateral are mutually perpendicular, then the quadrilateral is a rhombus (the theorem is false);

(3) the *opposite theorem*: if a quadrilateral is not a rhombus, then its diagonals are not perpendicular (the theorem is false);

(4) the *theorem opposite to the converse theorem*: if the diagonals of a quadrilateral are not mutually perpendicular, then the quadrilateral is not a rhombus (the theorem is true).

In the example we have considered, the direct theorem and the theorem opposite to the converse theorem have turned out to be true and the converse and opposite theorems, false. This is not an accidental coincidence. The direct

theorem and the theorem opposite to the converse theorem are either both true or both false, that is, there holds the equivalence

$$(\forall x) A(x) \Rightarrow B(x) = (\forall x) \overline{B(x)} \Rightarrow \overline{A(x)}, \quad x \in M.$$

If the proof of one of these theorems turns out to be difficult, then you should try to prove the other one.

The known method of proving by contradiction consists precisely in the fact that the theorem opposite to the converse theorem is proved instead of the original theorem.

2.4. The Method of Mathematical Induction

In many branches of arithmetic, algebra, geometry, analysis it turns out to be necessary to prove the truth of propositions $A(n)$, dependent on a natural variable, for all the values of that variable. This proof is often carried out by a method known as mathematical induction and based on the following principle.

The proposition $A(n)$ is assumed to be true for all the natural values of the variable, provided the following two conditions are fulfilled:

1. The proposition $A(n)$ is true for $n = 1$.
2. From the supposition that $A(n)$ is true for $n = k$ (where k is any natural number) it follows that it is also true for the value $n = k + 1$ next in order.

This principle is known as axioms of *mathematical induction*. It is usually chosen as one of the axioms defining a natural series of numbers and is, consequently, assumed without proof.

Mathematical induction is the following method of establishing the truth: to prove the truth of the proposition $A(n)$ for all natural n , we must verify the truth of the sentence $A(1)$, and then, assuming that the sentence $A(k)$ is true, try to prove the truth of the sentence $A(k + 1)$. If the latter is true and this remains valid for *every* natural value of k , then, in accordance with the principle of mathematical induction, the proposition $A(n)$ is true for all values of n .

Mathematical induction is widely used when proving theorems, identities, inequalities, when solving problems that involve divisibility, when solving geometrical and many other problems.

Example 1. Prove the truth of the proposition

$$A(n) \equiv \{\text{number } 5 \cdot 2^{3n-2} + 3^{3n-1} \text{ is multiple of } 19\},$$

$$n \in \mathbb{N}$$

$\triangle$ 1. The sentence $A(1) \equiv \{\text{the number } 5 \cdot 2 + 3^2 \text{ is a multiple of } 19\}$ is true.

2. Let us assume that for a certain value $n = k$ the proposition

$A(k) \equiv \{\text{number } 5 \cdot 2^{3k-2} + 3^{3k-1} \text{ is a multiple of } 19\}$ is true. Then, since

$$\begin{aligned} 5 \cdot 2^{3(k+1)-2} + 3^{3(k+1)-1} &= 8 \cdot 5 \cdot 2^{3k-2} + 27 \cdot 3^{3k-1} \\ &= 8(5 \cdot 2^{3k-2} + 3^{3k-1}) + 19 \cdot 3^{3k-1}, \end{aligned}$$

it is evident that $A(k+1)$ is true as well. Indeed, the first summand can be divided by 19 on the assumption that $A(k)$ is true; the second summand can also be divided by 19 because it contains 19 as a factor. Both conditions of the principle of mathematical induction are fulfilled and, consequently, the proposition $A(n)$ is true for all values of n . $\blacktriangle$

Example 2. Prove the formula

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2, \quad n \in \mathbb{N}.$$

$\triangle$ 1. For $n = 1$, both sides of the equality turn into unity and, consequently, the first condition of the principle of mathematical induction is fulfilled.

2. Suppose that the formula is true for $n = k$, i.e.

$$1^3 + 2^3 + 3^3 + \dots + k^3 = \left(\frac{k(k+1)}{2} \right)^2.$$

Let us add $(k+1)^3$ to both sides of the equation and transform the right-hand side. Then we get

$$\begin{aligned} 1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 &= \left(\frac{k(k+1)}{2} \right)^2 + (k+1)^3 = (k+1)^2 \left(\frac{k^2}{4} + k + 1 \right) \\ &= \left(\frac{k+1}{2} \right)^2 (k^2 + 4k + 4) = \left(\frac{(k+1)(k+2)}{2} \right)^2. \end{aligned}$$

Then, from the fact that the formula is true for $n = k$ it follows that it is also true for $n = k + 1$. This assertion is valid for all natural values of k . Thus, the second condition of the principle of mathematical induction is also fulfilled. We have proved the formula. $\blacktriangle$

Example 3. Prove the inequality

$$\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{3n+1} > 1.$$

$\triangle$ 1. For $n=1$, the inequality is true since

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{4} > 1.$$

2. Suppose that the inequality is true for $n=k$, i.e.

$$\frac{1}{k+1} + \frac{1}{k+2} + \dots + \frac{1}{3k+1} > 1.$$

Let us add the sum of the three fractions

$$\frac{1}{3k+2} + \frac{1}{3k+3} + \frac{1}{3k+4}$$

to both sides of the inequality and transfer the first member $\frac{1}{k+1}$, from the left-hand side to the right-hand side. Then we obtain

$$\begin{aligned} \frac{1}{k+2} + \dots + \frac{1}{3k+4} &> 1 + \frac{1}{3k+2} + \frac{1}{3k+3} \\ &\quad + \frac{1}{3k+4} - \frac{1}{k+1}. \end{aligned}$$

The right-hand side of the inequality exceeds unity since

$$\begin{aligned} 1 + \frac{1}{3k+2} + \frac{1}{3k+3} - \frac{1}{3k+4} + \frac{1}{k+1} \\ = 1 + \frac{2}{(3k+2)(3k+3)(3k+4)} > 1. \end{aligned}$$

Consequently, the left-hand side exceeds unity all the more so, i.e.

$$\frac{1}{(k+1)+1} + \dots + \frac{1}{3(k+1)+1} > 1.$$

The last inequality can be obtained from the original inequality for $n = k + 1$.

Thus, having assumed that the inequality is true for $n = k$, we have proved its truth for $n = k + 1$; we have proved the inequality by induction. ▲

Example 4. Assume that $x_1, x_2, \dots, x_n$ are arbitrary positive numbers, with $x_1, x_2, \dots, x_n = 1$. Prove that

$$x_1 + x_2 + \dots + x_n \geq n.$$

△ 1. If $n = 1$, then by the hypothesis, $x_1 = 1$ and, consequently, we can write $x_1 \geq 1$, that is, the statement is true for $n = 1$.

2. Let us assume that the statement is true for $n = k$. Suppose that $x_1, x_2, \dots, x_k, x_{k+1}$ are arbitrary positive numbers and

$$x_1 x_2 \dots x_k x_{k+1} = 1.$$

Two cases can occur here: either all these numbers are equal to unity and then their sum is equal to $k + 1$ and we have proved the inequality, or there is at least one number among these numbers which is not equal to unity and then there must be at least one more number which is not equal to unity. In that case, if one of them is smaller than unity, then the other exceeds unity. Without loss of generality we can assume that $x_k > 1$ and $x_{k+1} < 1$. Let us now consider k numbers

$$x_1, x_2, \dots, x_{k-1}, (x_k x_{k+1}).$$

Their product is equal to unity and, consequently, according to the inductive supposition,

$$x_1 + x_2 + \dots + x_{k-1} + x_k x_{k+1} \geq k.$$

Let us add $x_k + x_{k+1}$ to both sides of the last equation, transfer $x_k x_{k+1}$ to the right-hand side and transform the right-hand side

$$\begin{aligned} x_1 + x_2 + \dots + x_k + x_{k+1} &\geq k - x_k x_{k+1} + x_k + x_{k+1} \\ &= k + 1 + x_k (1 - x_{k+1}) + x_{k+1} - 1 \\ &= k + 1 + x_k (1 - x_{k+1}) - (1 - x_{k+1}) \\ &= k + 1 + (1 - x_{k+1}) (x_k - 1) \geq k + 1. \end{aligned}$$

Thus, the truth of the statement for $n = k$ implies its truth for $n = k + 1$. We have proved the statement. It follows from the proof presented that the equality sign can be

put in the relation being proved if and only if $x_1 = x_2 = \dots$
 $\dots = x_n = 1$. $\blacktriangle$

Example 5. Prove the relation

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n},$$

where $x_1, x_2, \dots, x_n$ are arbitrary positive numbers.

$\triangle$ This important equation of the arithmetic mean and the geometric mean of n numbers is a simple consequence of the relation proved in the preceding example. Indeed, suppose $x_1, x_2, \dots, x_n$ are arbitrary positive numbers. Let us consider n numbers

$$\frac{x_1}{\sqrt[n]{x_1 x_2 \dots x_n}}, \quad \frac{x_2}{\sqrt[n]{x_1 x_2 \dots x_n}}, \quad \dots, \quad \frac{x_n}{\sqrt[n]{x_1 x_2 \dots x_n}}.$$

All these numbers are, evidently, positive and their product is unity. Consequently, in accordance with what was proved in Example 4, their sum is larger than or equal to n , i.e.

$$\frac{x_1}{\sqrt[n]{x_1 x_2 \dots x_n}} + \frac{x_2}{\sqrt[n]{x_1 x_2 \dots x_n}} + \dots + \frac{x_n}{\sqrt[n]{x_1 x_2 \dots x_n}} \geq n.$$

Hence we have

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n},$$

the equality sign obtains if and only if $x_1 = x_2 = \dots$
 $\dots = x_n$. $\blacktriangle$

The inequality between the arithmetic mean and the geometric mean of n numbers often turns out to be useful in proving other inequalities, in seeking the least and greatest values of functions (see 4.6, 4.7).

Let us now discuss some generations of the principle of mathematical induction.

Suppose p is some integer.

The proposition $A(n)$, where n is an integer, is true for all integral values $n \geq p$, provided the following two conditions are fulfilled:

1. The proposition $A(n)$ is true for $n = p$.

2. From the proposition that $A(n)$ is true for $n = k$ (k being an integer, $k \geq p$), it follows that it is true for its successor, $n = k + 1$.

For $p = 1$ we obtain the initial formulation of the principle of mathematical induction.

Example 6. Prove that any sum of money exceeding 7 copecks can be changed only into 3-copeck and 5-copeck coins.

Δ Assume that the sum is equal to n copecks. If $n = 8$, the statement is true. Suppose the statement is also true for $n = k$. Only two cases can occur in changing the sum of k copecks:

- (a) only 3-copeck coins are needed,
- (b) at least one 5-copeck coin is needed.

In case (a) we remove three 3-copeck coins, add two 5-copeck coins and thus change the sum of $k + 1$ copecks. In case (b) we remove one 5-copeck coin, add two 3-copeck coins and thus change the sum of $k + 1$ copecks. $\blacktriangle$

Some problems require the following form of the principle of mathematical induction.

The proposition $A(n)$, where n is an integer, is true for all $n \geq p$ (p being an integer), provided two conditions are fulfilled:

1. The proposition $A(n)$ is true for $n = p$ and $n = p + 1$.
2. From the proposition that $A(n)$ is true for $n = k$ and $n = k - 1$, it follows that it is true for $n = k + 1$ (for any $k > p$).

We shall illustrate the application of this principle by way of the following example.

Example 7. Prove that the function $T_n(x) = \cos(n \cdot \arccos x)$ coincides, on the interval $[-1; 1]$, with a certain polynomial of degree n (n being a nonnegative integer).

1. For $n = 0$ and $n = 1$ we obtain $T_0(x) = 1$ and $T_1(x) = x$ respectively, that is, the statement is true both for $n = 0$ and for $n = 1$.

2. Let us consider $T_{k+1}(x)$ and perform the following transformations:

$$\begin{aligned}
 T_{k+1}(x) &= \cos((k+1) \arccos x) = \cos(k \arccos x) \cdot \cos(\arccos x) \\
 &\quad - \sin(k \arccos x) \times \sin(\arccos x) = T_k(x) T_1(x) \\
 &\quad - \frac{1}{2} (\cos((k-1) \arccos x) - \cos((k+1) \arccos x)) \\
 &= T_k(x) T_1(x) - \frac{1}{2} (T_{k-1}(x) - T_{k+1}(x)).
 \end{aligned}$$

We have obtained the recurrence relation

$$T_{k+1}(x) = 2T_1(x) T_k(x) - T_{k-1}(x)$$

or

$$T_{k+1}(x) = 2xT_k(x) - T_{k-1}(x).$$

Let us make an inductive supposition that the functions $T_{k-1}(x)$ and $T_k(x)$ coincide, on the interval $[-1; 1]$, with the polynomials of degree $k-1$ and of degree k respectively. Then it follows from the recurrence formula obtained that $T_{k+1}(x)$ coincides with the polynomial of degree $k+1$.

In accordance with the principle of mathematical induction, we have proved the statement for all $n \geq 0$. ▲

PROBLEMS OF SECTION I

1. The electric circuit between points M and N has been constructed according to the scheme depicted in Fig. 29. Let us consider the following four sentences:

$A \equiv \{\text{element } k \text{ of the circuit has failed}\},$

$B_i \equiv \{\text{element } l_i \text{ of the circuit has failed}\} \quad (i = 1, 2, 3).$

Is the circuit closed if

(a) the sentence $A + B_1B_2B_3$ is true,

(b) the sentence $\bar{A} (\bar{B}_1 + \bar{B}_2 + \bar{B}_3)$ is true?

2. Prove the formula

$$A \Rightarrow B = \bar{A} + B.$$

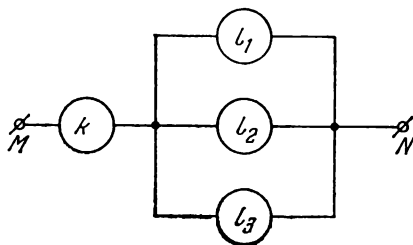


Fig. 29.

3. The question as to who of the three students had studied logic was answered correctly: it is true if the first of them studied logic then so did the second, but it is a false statement that if the third of them studied logic then so did the second.

Who of the students studied logic?

4. Having returned home, Maigret rang his office on quai des Orfèvres.

"Maigret here. Any news?"

"Yes, Chief. The inspectors have reported. Torrence thinks that if François was drunk, then either Etienne is the murderer or François is lying. Justin is of the opinion that either Etienne is the murderer or François was not drunk and the murder occurred after midnight. Inspector Lucas asked me to tell you that if the murder had occurred after midnight then either Etienne is the murderer or François is lying. Then, there was a ring from..."

"That's all. Thanks. That's enough!" The commissar replaced the receiver. He knew that when François was sober he never lied. Now everything was clear to him.

Consider the following sentences:

$A \equiv \{\text{François was drunk}\},$

$B \equiv \{\text{Etienne is the murderer}\},$

$C \equiv \{\text{François is telling a lie}\},$

$D \equiv \{\text{the murder took place after midnight}\}.$

Using logical operations, write down the statements made by inspectors Torrence, Justin and Lucas. Compile the product of these three statements and simplify it. What follows from the evidence of the inspectors? What conclusion has commissar Maigret come to?

5. To prove the equation

$$\sqrt[3]{5\sqrt{2}+7} - \sqrt[3]{5\sqrt{2}-7} = 2$$

the student reasoned as follows: "Cubing both sides of the equation, we get

$$14 - 3(\sqrt[3]{5\sqrt{2}+7})^2\sqrt[3]{5\sqrt{2}-7} + 3\sqrt[3]{5\sqrt{2}+7}(\sqrt[3]{5\sqrt{2}-7})^2 = 8$$

or

$$3\sqrt[3]{5\sqrt{2}+7}\sqrt[3]{5\sqrt{2}-7}(\sqrt[3]{5\sqrt{2}+7} - \sqrt[3]{5\sqrt{2}-7}) = 6;$$

but, since

$$\sqrt[3]{5\sqrt{2}+7} - \sqrt[3]{5\sqrt{2}-7} = 2,$$

it follows that

$$\sqrt[3]{5\sqrt{2}+7}\sqrt[3]{5\sqrt{2}-7} = 1,$$

$$\sqrt[3]{(5\sqrt{2})^2 - 7^2} = 1,$$

$$\sqrt[3]{1} = 1, \quad 1 = 1,$$

and that is what we had to prove."

Did the student prove the equation?

6. A student has to solve the following problem:

"The hypotenuse of a right-angled triangle lies in the plane α , one of the legs makes an angle of 45° with the plane α , and the other makes an angle of 60° . The larger leg is equal to a . Find the hypotenuse."

The student made a drawing (Fig. 30), found

$$|CO| = a \sin 45^\circ = a \sqrt{2}/2$$

from the right-angled triangle AOC , and

$$|BC| = \frac{|CO|}{\sin 60^\circ} = \frac{2|CO|}{\sqrt{3}} = a \sqrt{\frac{2}{3}}$$

from the right-angled triangle BOC , and then determined the hypote-

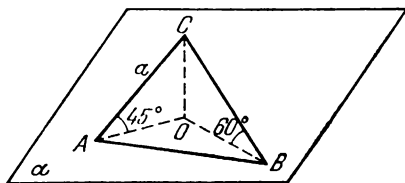


Fig. 30.

nuse by the Pythagorean theorem:

$$|AB| = \sqrt{|AC|^2 + |BC|^2} = a \sqrt{1 + \frac{2}{3}} = a \sqrt{\frac{5}{3}}.$$

Is the result true?

7. Given the inequality $kx + l^2 \leq 0$. At what values of k are the following statements true:

- (a) at any l the inequality has at least one solution,
- (b) there exists l at which the inequality has at least one solution?

8. The following three propositions are given on the set of all natural numbers:

$A(n) \equiv \{\text{the number } n + 48 \text{ is the square of a natural number}\},$

$B(n) \equiv \{\text{the number } n \text{ ends in the digit } 4\},$

$C(n) \equiv \{\text{the number } n - 41 \text{ is the square of a natural number}\}.$

At what values of n are two of the three propositions true and one false?

9. For the theorem "if a quadratic equation does not possess two distinct real roots, then the discriminant of that quadratic equation is nonpositive" formulate the converse theorem, the opposite theorem and the theorem opposite to the converse. Show which of those theorems are true.

10. Prove or disprove the following statements:

(a) For the number $n^2 - 1$ ($n \geq 5$) to be divisible by 24, it is sufficient that n be a prime number.

(b) For the number $n^2 - 1$ ($n \geq 5$) to be divisible by 24, it is necessary that n be a prime number.

11. For what natural values of n is the following sentence true:

$$A(n) \equiv \{2^n > 4n^2 + 1\}?$$

12. There are n straight lines drawn arbitrarily on a plane. Prove that one can use black and white colours to paint the plane so that any two parts having a side in common will be differently coloured.

PROBLEMS OF SECTION II

1. Three shots have been fired at the target. Suppose

$A_k \equiv \{\text{there was a hit at the } k\text{th shot}\}$, $k = 1, 2, 3$.

What do the following sentences mean:

(a) $A_1 + A_2 + A_3$; (b) $A_1 A_2 A_3$; (c) $A_1 \bar{A}_2 \bar{A}_3 + \bar{A}_1 A_2 \bar{A}_3 + \bar{A}_1 \bar{A}_2 A_3$?

2. The case of Brown, Johns and Smith is being heard. One of them performed a criminal deed.

At the inquest each of them made two statements:

Brown. I didn't do it.

Smith did it.

Johns. Smith is innocent.

Brown did it.

Smith. I didn't do it.

Johns didn't do it.

It was established by the Court that one of them had lied twice, another twice spoke the truth and the third lied once and once spoke the truth. Who was the criminal?

3. Six specialists out of the eight applicants, A, B, C, D, E, F, G and H , must be chosen for a polar expedition: a biologist, a hydrologist, a weather forecaster, a radio operator, a mechanic and a doctor. E and G can work as biologists, B and F as hydrologists, F and G as weather forecasters, C and D as radio operators, C and H as mechanics, A and D as doctors. Although some of the applicants have two professions, during the expedition each of them will perform only one duty. Who should be chosen and for what duty if F cannot go without B , D without H and without C , C cannot go together with G , and A cannot go together with B ?

4. Let us consider two definitions of a simple test:

1. The test is considered to be easy if *at least one* student has solved *one* problem.

2. The test is considered to be easy if *at least one* student has solved *all* the problems.

(a) Can the test be easy in the sense of the first definition and difficult (not easy) in the sense of second?

(b) Can the test be easy in the sense of the second definition and difficult in the sense of the first?

5. The senior students of high school 31 boasted of being taller than the senior students of high school 42. The mathematics teacher asked them: "What do you mean by being taller?" The students of school 31 gave the following answers:

1. Any one of us is taller than any one of them.

2. The tallest of us is taller than the tallest of them.

3. For any one of our students there is a student of school 31 who is smaller in height.

4. Every student of school 31 is smaller than at least one of our students.

5. The average height of our students is greater than the average height of their students.

Are there equivalent answers among those cited? If there are, then which are they?

6. The following three propositions are given on the set of all natural numbers:

$A(n) \equiv \{\text{the number } n^2 - 2 \text{ is a multiple of } 7\},$

$B(n) \equiv \{\text{the number } n - 2 \text{ is a multiple of } 7\},$

$C(n) \equiv \{4n^2 - 360n + 8099 < 0\}.$

At what values of n are two of the three given propositions true and one false?

7. Given three propositions defined on the set of all real numbers:

$A(x) \equiv \{x \text{ is an integer}\},$

$B(x) \equiv \{x^2 - 3x \text{ is a negative integer}\},$

$C(x) \equiv \{x + \frac{1}{x} \text{ is a positive integer}\}.$

At what value of x is *one and only one* of the three propositions false?

8. Given the system of equations

$$\begin{cases} x + 2by = a, \\ bx + (1-b)y = c^2 + c, \end{cases}$$

where a, b, c are real numbers.

At what values of a is *there*, for *any* b, c such that the system possesses *at least one solution*?

9. Given two points $A(0; 9)$, $B(3; 6)$ and the system of inequalities

$$\begin{cases} 2x - y + a \leq 0, \\ 6x + 3y + 5a \geq 0. \end{cases}$$

At what values of the parameter a will the following coordinates serve as a solution of the system:

(a) the coordinates of *at least one* point of the line segment AB ,

(b) the coordinates of *every* point of the line segment AB ?

10. For each of the theorems given below formulate the converse, the opposite and the converse of the opposite. Indicate which of the theorems are true.

(a) If a circle can be inscribed in a quadrilateral, then the quadrilateral is a rhombus.

(b) If a parallelogram is a rectangle, then a circle can be circumscribed about it.

(c) If a polygon is a quadrilateral, then the sum of its interior angles is equal to 360° .

11. Which of the following six theorems are converse, opposite and converse of the opposite with respect to one another? Which of these theorems are true?

Theorem 1. If each of the two natural numbers is exactly divisible by 7, then their sum is divisible by 7.

Theorem 2. If neither of the two numbers is divisible by 7, then their sum is not divisible by 7 either.

Theorem 3. If at least one of two numbers is divisible by 7, then their sum is divisible by 7.

Theorem 4. If the sum of two numbers is divisible by 7, then each of the summands is divisible by 7.

Theorem 5. If the sum of two numbers is not divisible by 7, then neither of the summands is divisible by 7.

Theorem 6. If the sum of two numbers is not divisible by 7, then at least one of the summands is not divisible by 7.

12. Given the theorem: In any quadrilateral, which is not a rectangle, the diagonals are not congruent. Formulate the theorems: the converse, the opposite and the opposite to the converse. Which of the four theorems are true?

13. For the theorem "If a function is differentiable at a point, it is continuous at that point" formulate the converse theorem, the opposite theorem, and the theorem opposite to the converse.

Which of the four theorems are true?

14. Replace the dots by the words "it is necessary and sufficient", "it is necessary but not sufficient", "it is sufficient but not necessary" to obtain true statements:

(a) To win a prize in a lottery, ...to have at least one lottery ticket.

(b) For at least two of the given three numbers a , b , c to be equal, ...that the equality

$$(a - b)^2 + (b - c)^2 + (c - a)^2 = 0$$

hold true.

(c) For the sum of two real numbers to be a rational number, ... that each summand should be a rational number.

(d) For the median of a triangle to be equal to half the side which it divides, ... that the triangle be right-angled.

(e) For the function $y = ax^2 + bx + c$ to assume integral values for all integral x , ...that $2a$, $a + b$, c be integers.

15. Prove or disprove the following statements:

(a) For the number $n^2 + 20$ ($n > 3$) to be composite, it is sufficient that n be prime.

(b) For the number $n^2 + 20$ ($n > 3$) to be composite, it is necessary that n be a prime number.

16. Prove that for any natural n :

(a) $n(2n^2 - 3n + 1)$ is a multiple of 6;

(b) $11^{n+1} + 12^{2n-1}$ is exactly divisible by 133;

(c) $n^5 - n$ is a multiple of 5;

(d) $n^7 - n$ is a multiple of 7.

17. Prove that for any natural $n > 1$ the number $2^{2^n} + 1$ ends in the digit 7.

18. Prove by induction the truth of the following equalities for every natural value of n :

(a) $1 \cdot 2 + 2 \cdot 5 + \dots + n(3n - 1) = n^2(n + 1);$

(b) $1^2 + 3^2 + \dots + (2n - 1)^2 = \frac{n(4n^2 - 1)}{3};$

$$(c) 1 \cdot 2^2 + 2 \cdot 3^2 + \dots + (n-1) n^2 = \frac{n(n^2-1)(3n+2)}{12};$$

$$(d) \frac{1}{1 \cdot 5} + \frac{1}{5 \cdot 9} + \dots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1};$$

$$(e) \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{9}\right) \dots \left(1 - \frac{1}{(n+1)^2}\right) = \frac{n+2}{2n+2}.$$

19. Prove that

$$3 + 33 + \dots + 33 \dots 3 = \frac{10^{n+1} - 9^n - 10}{27}, \quad n \in \mathbb{N}$$

(the left-hand side of the equation contains n summands).

20. Prove the equality

$$\sqrt[2]{2 + \sqrt[2]{2 + \dots + \sqrt[2]{2}}} = 2 \cos(\pi/2^{n+1}), \quad n \in \mathbb{N}$$

(the left-hand side contains n roots).

21. At what natural values of n are the following inequalities true:

$$(a) 3^n > 2^n + 7n; \quad (b) 2^n > n^2 + 4n + 5?$$

22. Prove by induction the validity of the following inequalities for all natural $n > 1$:

$$(a) \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \dots \frac{2n-1}{2n} < \frac{1}{\sqrt{3n+1}};$$

$$(b) \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} > \frac{13}{24};$$

$$(c) \sqrt[n]{n} < 1 + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} < 2 \sqrt[n]{n};$$

$$(d) \frac{n}{2} < 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{2^n-1} < n.$$

23. Into how many parts will a sphere be divided by n planes passing through its centre if no three planes pass through one and the same diameter?

24. Suppose $x_1, x_2, \dots, x_n$ are arbitrary nonnegative numbers, with

$$x_1 + x_2 + \dots + x_n \leq 1/2.$$

Prove that

$$(1 - x_1)(1 - x_2) \dots (1 - x_n) \geq 1/2.$$

Chapter 3

Equations and Systems of Equations

3.1. Equations in One and Several Variables

Any proposition of the form

$$f(x) = g(x), \quad (1)$$

where $f(x)$ and $g(x)$ are some functions, is called an *equation in one variable x* (or in one unknown x). The function $f(x)$ is called the *left-hand side* and $g(x)$, the *right-hand side* of equation (1).

The number a is called the *root* (or *solution*) of the given equation in the variable x if its substitution for the variable in both sides of that equation results in a correct numerical equality, that is, if $x = a$, then both sides of the equation are defined and their values coincide.

To *solve* a given equation means to find the set of all roots (solutions) of that equation in the domain $\mathbb{R}$.

Two equations are said to be *equivalent* if they possess the same set of solutions, that is, if every solution of the first equation is also a solution of the second and, conversely, every solution of the second equations is also a solution of the first, or, else, if neither of the equations possesses a solution.

The equations $5x + 1 = 6$ and $2x = 2$, for instance, are equivalent since each of them possesses a single root $x = 1$.

An equation of the form

$$ax = b, \quad (2)$$

where a and b are some given numbers, is called a *linear equation*.

It is evident that if $a = b = 0$, then any number is a solution of equation (2). If $a = 0$ but $b \neq 0$, then equation

(2) has no solutions. Now if $a \neq 0$, then the equation has a unique solution $x = b/a$.

An equation of the form

$$ax^2 + bx + c = 0, \quad (3)$$

where a, b, c are some numbers, with $a \neq 0$, is known as a *quadratic equation*.

The simplest quadratic equation has the form $x^2 + c = 0$. It has no solutions if $c > 0$, possesses one solution $x = 0$ if $c = 0$ and possesses two solutions $x = \sqrt{-c}$ and $x = -\sqrt{-c}$ if $c < 0$.

It is easy to verify that equation (3) is equivalent to the equation

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0.$$

Let us transform the left-hand side of this equation:

$$\left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a^2} = 0.$$

We can see that equation (3) possesses no solutions if $D = b^2 - 4ac < 0$, possesses a unique solution $x = -\frac{b}{2a}$ if $D = 0$, and has two solutions $x = -\frac{b}{2a} \pm \frac{\sqrt{D}}{2a}$ if $D > 0$.

Thus, all solutions of equation (3), provided they exist, can be found by the formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \quad (4)$$

Example 1. Solve the equation

$$\frac{2x+1}{3-x} = \frac{4-x}{x+1}.$$

Δ We multiply both sides of this equation by $(3-x)(x+1)$. Any solution of the given equation is, evidently, a solution of the resulting equation

$$(2x+1)(x+1) = (4-x)(3-x).$$

It is easy to see that it is equivalent to the quadratic equation

$$x^2 + 10x - 11 = 0.$$

Let us find the roots of this equation:

$$x = -5 \pm \sqrt{25 + 11} = -5 \pm 6,$$

i.e. $x = -11$ and $x = 1$.

Consequently, only the numbers -11 and 1 can be solutions of the given equation. Verification confirms that these numbers are solutions of the given equation.

Answer: $\{-11, 1\}$. ▲

Example 2. Solve the equation

$$\frac{3x}{x-1} - \frac{2x}{x+2} = \frac{3x-6}{(x-1)(x+2)}.$$

△ Any solution of this equation is also a solution of the equation

$$3x(x+2) - 2x(x-1) = 3x-6,$$

which results from multiplying the given equation by $(x-1)(x+2)$. This equation is, evidently, equivalent to the following quadratic equation:

$$x^2 + 5x + 6 = 0.$$

We can find its roots by formula (4): $x = -3$ and $x = -2$.

Consequently, only the numbers -3 and -2 can be solutions of this equation. Verification confirms that the number -3 is its root while the number -2 is not since for $x = -2$ the sides of the given equation are not defined.

Answer: $\{-3\}$. ▲

Suppose we are given two equations. Then, if any solution of the first equation is also a solution of the second, the second equation is a *consequence* of the first.

If the equation $f_1(x) = g_1(x)$ is a consequence of the equation $f(x) = g(x)$, we write

$$(f(x) = g(x)) \Rightarrow (f_1(x) = g_1(x)).$$

If the equations $f(x) = g(x)$ and $f_1(x) = g_1(x)$ are equivalent, then, evidently, each of them is a consequence of the other one. In that case, we write

$$(f(x) = g(x)) \Leftrightarrow (f_1(x) = g_1(x)).$$

Let us formulate some statements widely used in solving equations. Their proof is left for the reader as an easy and useful exercise.

1. If the function $\varphi(x)$ is defined for all x for which $f(x)$ and $g(x)$ are defined, then

$$(f(x) = g(x)) \Leftrightarrow (f(x) + \varphi(x) = g(x) + \varphi(x)).$$

In particular,

$$(f(x) = g(x)) \Leftrightarrow (f(x) - g(x) = 0).$$

2. If the function $\varphi(x)$ is defined for all x for which $f(x)$ and $g(x)$ are defined, then

$$(f(x) = g(x)) \Rightarrow (f(x) \varphi(x) = g(x) \varphi(x)).$$

If, in addition, $\varphi(x) \neq 0$ for all x , then

$$(f(x) = g(x)) \Leftrightarrow (f(x) \varphi(x) = g(x) \varphi(x)).$$

In particular, for any number $k \neq 0$

$$(f(x) = g(x)) \Leftrightarrow (kf(x) = kg(x)).$$

3. For any $f(x)$, $g(x)$ and $\varphi(x)$ we have

$$\left(\frac{f(x)}{\varphi(x)} = g(x) \right) \Rightarrow (f(x) = g(x) \varphi(x)).$$

It is evident that any solution x_0 of the equation

$$f(x) = g(x) \varphi(x), \quad (5)$$

such that $\varphi(x_0) \neq 0$, is also a solution of the equation

$$\frac{f(x)}{\varphi(x)} = g(x). \quad (6)$$

Consequently, in the set of all numbers for which $\varphi(x) \neq 0$, equations (5) and (6) possess the same solutions. In that case, we say that under the condition $\varphi(x) \neq 0$ equations (5) and (6) are equivalent and we write

$$\left(\frac{f(x)}{\varphi(x)} = g(x) \right) \Leftrightarrow (\varphi(x) \neq 0, f(x) = g(x) \varphi(x)).$$

4. Every solution of the equation $f(x) g(x) = 0$ is a solution of at least one of the equations $f(x) = 0$ and $g(x) = 0$:

$$(f(x) g(x) = 0) \Rightarrow (f(x) = 0 \text{ or } g(x) = 0).$$

If the function $f(x)$ is defined for all x , where $g(x) = 0$, and the function $g(x)$ is defined for all x , where $f(x) = 0$, then

$$(f(x) g(x) = 0) \Leftrightarrow (f(x) = 0 \text{ or } g(x) = 0).$$

5. For any $f(x)$ and $g(x)$ and any natural n
 $(f(x) = g(x)) \Rightarrow ((f(x))^n = (g(x))^n).$

Example 3. Solve the equation

$$2\sqrt{x+5} = x+2.$$

$\triangle$ We make use of the above statements and the notation introduced. Then we have

$$\begin{aligned} (2\sqrt{x+5} = x+2) &\Rightarrow (4(x+5) = (x+2)^2) \\ &\Leftrightarrow (4x+20 = x^2+4x+4) \Leftrightarrow (x^2 = 16) \\ &\Leftrightarrow (x = 4 \text{ or } x = -4). \end{aligned}$$

Thus only $x = 4$ and $x = -4$ can be solutions of the given equation. Verification shows that only $x = 4$ is a solution of the equation.

Answer: $\{4\}$. $\blacktriangle$

Note that the equations $2\sqrt{x+5} = x+2$ and $4(x+5) = (x+2)^2$ are equivalent under the condition $x+2 \geq 0$. Therefore, we can represent the scheme of solution of the equation as follows:

$$\begin{aligned} (2\sqrt{x+5} = x+2) &\Leftrightarrow (x+2 \geq 0, \\ 4(x+5) = (x+2)^2) &\Leftrightarrow (x \geq -2, x^2 = 16) \Leftrightarrow (x = 4). \end{aligned}$$

Example 4. Solve the equation

$$\sqrt{4+2x-x^2} = x-2.$$

$\triangle$ We have

$$\begin{aligned} (\sqrt{4+2x-x^2} = x-2) &\Leftrightarrow (4+2x-x^2) \\ &= (x-2)^2; x \geq 2) \Leftrightarrow (x \geq 2, 2x^2-6x=0) \\ &\Leftrightarrow (x \geq 2; x=0 \text{ or } x=3) \Leftrightarrow (x=3). \end{aligned}$$

Answer: $\{3\}$. $\blacktriangle$

Any proposition of the form

$$f(x; y) = g(x; y), \quad (7)$$

where $f(x; y)$ and $g(x; y)$ are some functions in the variables x and y , is called an *equation in two variables* x and y or in two unknowns x and y . The function $f(x; y)$ is called the

left-hand side and $g(x, y)$, the *right-hand side* of equation (7).

The *solution* of the given equation in the variables x and y is an ordered pair of numbers $(a; b)$ whose substitution for x and y , respectively, into both sides of the equation results in a true numerical equality.

A set of points of a coordinate plane whose coordinates are solutions of an equation is the *graph* of that equation. Thus, for instance, the graph of the equation $(x - x_0)^2 + (y - y_0)^2 = R^2$, where $R > 0$, is a circle of radius R with centre at the point $(x_0; y_0)$. The graph of the equation $y = ax^2 + bx + c$, where a, b, c are some numbers, with $a \neq 0$, is a parabola.

An equation of the form

$$ax + by + c = 0, \quad (8)$$

where a, b, c are some numbers, is said to be *linear*. The graph of linear equation (8), at least one of whose coefficients, a or b , is nonzero, is a straight line. If $a = b = c = 0$, then any pair of numbers is a solution of (8), and if $a = b = 0$ but $c \neq 0$, then equation (8) has no solutions.

If every solution of equation (7) is also a solution of the equation

$$f_1(x; y) = g_1(x; y), \quad (9)$$

then equation (9) is a *consequence* of equation (7). In that case, the graph of equation (7) is a part of the graph of equation (9) (in a special case they may coincide). If the graphs of the equations coincide, they are said to be *equivalent*.

It is easy to prove that

$$1. (f(x; y) = g(x; y)) \Leftrightarrow (f(x; y) - g(x; y) = 0).$$

$$2. (f(x; y) = g(x; y)) \Leftrightarrow (kf(x; y) = kg(x; y))$$

for any number $k \neq 0$.

$$3. \left(\frac{f(x; y)}{\varphi(x; y)} = g(x; y) \Leftrightarrow (\varphi(x; y) \neq 0, \quad f(x; y) = g(x; y) \varphi(x; y)). \right.$$

$$4. (f(x; y) g(x; y) = 0) \Rightarrow (f(x; y) = 0 \text{ or } g(x; y) = 0).$$

Example 5. Draw the graph of the equation $y^2 = 4x^2$.

$\triangle$ We have

$$\begin{aligned} (y^2 = 4x^2) &\Leftrightarrow (y^2 - 4x^2 = 0) \Leftrightarrow ((y - 2x)(y + 2x) = 0) \\ &\Leftrightarrow (y - 2x = 0 \text{ or } y + 2x = 0) \Leftrightarrow (y = 2x \text{ or } y = -2x). \end{aligned}$$

Thus we see that the graph of the equation $y^2 = 4x^2$ is a pair of intersecting straight lines defined by the equations $y = 2x$ and $y = -2x$ (Fig. 31). ▲

In what follows, besides the equations in one variable and in two variables, we shall also consider equations in three and more variables.

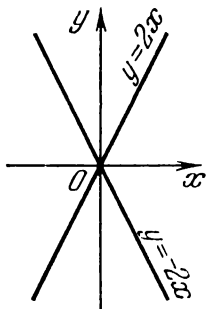


Fig. 31.

In a general case, a *solution* of an equation in n variables is an ordered collection of n numbers whose substitution for the variables into both sides of the equation results in a true numerical equality.

3.2. Systems of Equations

Any finite set of equations is called a *system of equations*.

In a general case, we may consider a system of m equations

in n variables (unknowns), all the three cases, $m = n$, $m < n$ and $m > n$, being possible. A *solution* of a system of equations in n variables is an ordered collection of n numbers which is a solution of each of the equations of the system. To solve a system means to find all its solutions.

Here are some rules governing transformation of systems of equations.

1. If we replace an equation of a system by its equivalent equation and leave the rest of the equations unchanged, then the system resulting from the transformation is equivalent to the original system.

2. Assume that $f = g$ and $\varphi = \psi$ are some two equations of a system. Then, if we replace the equation $f = g$ of the system by the equation $f + \varphi = g + \psi$ (it is called the sum of the equations $f = g$ and $\varphi = \psi$) and leave the rest of the equations unchanged, then the system we shall obtain is equivalent to the original system.

3. Let a system contain an equation $x = \varphi$, where x is a variable and φ is a function independent of x . Then, if we substitute φ for x into all the equations of the system, except for the equation $x = \varphi$, the resulting system will be equivalent to the original system.

This rule underlies the method of solving systems of equations known as the *method of substitution* or the *method of eliminating the unknowns*. It reduces the solution of the given system to the solution of a system of a smaller number of equations in a smaller number of variables (unknowns).

4. If a system contains an equation $f \cdot g = 0$, then it can be separated into two systems in one of which the equation $f \cdot g = 0$ has been replaced by $f = 0$ and in the other by $g = 0$. In this case, every solution of the given system is a solution of one of the systems obtained.

If the functions f and g are defined on one and the same set, then every solution of the systems obtained is a solution of the original system. In a situation of this kind we say that the given system is equivalent to the collection of the systems obtained.

This rule underlies the method of solution of systems known as the *method of factorization*.

Example 1. Solve the system of equations

$$\begin{cases} x^2 + y^2 = 25, \\ x - y = 1. \end{cases}$$

Δ From the second equation we find $x = 1 + y$ and substitute this expression for x into the first equation. As a result we get an equation $(1 + y)^2 + y^2 = 25$ containing the only variable y . The transformation results in a quadratic equation $y^2 + y - 12 = 0$. It has the roots $y_1 = 3$ and $y_2 = -4$. Substituting these values of y into the equation $x = 1 + y$, we find $x_1 = 4$ and $x_2 = -3$. Consequently, the given system possesses the solutions $(4; 3)$ and $(-3; -4)$. Note that in solving this system we have used the method of substitution.

Answer: $\{(4; 3), (-3; -4)\}$. $\blacktriangle$

Example 2. Solve the system of equations

$$\begin{cases} y^2 - 1 = 4x^2 + 4x, \\ 4x^2 + y^2 - 3xy = 1. \end{cases}$$

Δ It follows from the first equation that $y^2 = (2x + 1)^2$, i.e. $y = 2x + 1$ or $y = -2x - 1$. Consequently, the given system is equivalent to the collection of the following two

systems:

$$\begin{cases} y = 2x + 1, \\ 4x^2 + y^2 - 3xy = 1, \end{cases} \quad \begin{cases} y = -2x - 1, \\ 4x^2 + y^2 - 3xy = 1. \end{cases}$$

Let us begin with the first system. Substituting $y = 2x + 1$ into the second equation of the system, we get

$$4x^2 + (2x + 1)^2 - 3x(2x + 1) = 1,$$

$$2x^2 + x + 1 = 1,$$

and, finally, $x(2x + 1) = 0$, i.e. $x = 0$ or $x = -1/2$. Substituting these values of x into the equation $y = 2x + 1$, we obtain $y = 1$ and $y = 0$ respectively. Consequently, the first system has the solutions $(0; 1)$ and $(-1/2; 0)$.

We can solve the second system by analogy. Substituting $y = -2x - 1$ into the second equation, we get $x(2x + 1) = 0$, i.e. $x = 0$ or $x = -1/2$. Consequently, the second system has the solutions $(0; -1)$ and $(-1/2; 0)$.

Answer: $\{(0; 1), (-1/2; 0), (0; -1)\}$. ▲

Example 3. Solve the system of equations

$$\begin{cases} x^2 - y^2 = 5, \\ x^2 - xy + y^2 = 7. \end{cases}$$

△ Multiplying the first equation by 7 and the second by -5 and adding up the results, we get

$$12y^2 - 2x^2 - 5xy = 0.$$

Let us factorize the left-hand side into two factors. For that purpose we solve the equation for y , say, as a quadratic equation:

$$y = \frac{5x \pm \sqrt{25x^2 + 96x^2}}{24} = \frac{5x \pm 11x}{24}.$$

Consequently, $y = \frac{2}{3}x$ or $y = -\frac{1}{4}x$.

The given system is equivalent to the collection of two systems

$$\begin{cases} x^2 - y^2 = 5, \\ y = \frac{2}{3}x, \end{cases} \quad \begin{cases} x^2 - y^2 = 5, \\ y = -\frac{1}{4}x. \end{cases}$$

Substituting $y = \frac{2}{3}x$ into the first equation, we get

$$x^2 - \frac{4}{9}x^2 = 5, \quad x^2 = 9, \quad x = \pm 3.$$

Consequently, the first system has the solutions (3; 2) and (-3; -2). By analogy we obtain solutions of the second system: $(4/\sqrt{3}; -1/\sqrt{3})$ and $(-4/\sqrt{3}; 1/\sqrt{3})$.

Answer: $\{(3; 2), (-3; -2), (4/\sqrt{3}; -1/\sqrt{3}), (-4/\sqrt{3}; 1/\sqrt{3})\}$. ▲

Example 4. Solve the system of equations

$$\begin{cases} x\sqrt{x} - \sqrt{x} = y\sqrt{y} + 8\sqrt{y}, \\ x = y + 5. \end{cases} \quad (1)$$

△ Let us introduce the new variables u and v :

$$\begin{cases} u = \sqrt{x}, \\ v = \sqrt{y}. \end{cases} \quad (2)$$

Then, from the given system with respect to the variables u and v we get the system of equations

$$\begin{cases} u^3 - u = v^3 + 8v, \\ u^2 = v^2 + 5. \end{cases} \quad (3)$$

It is easy to see that the system of equations (1), (2) is equivalent to the system of equations (2), (3). Consequently, to find all the solutions of system (1), we must solve the system of equations (2), (3). In this case, if $(x_1; y_1; u_1; v_1)$ is a solution of that system, then $(x_1; y_1)$ is a solution of system (1).

The system of equations (2), (3) is of a special form: equations (3) do not contain the variables x and y . Therefore, we first solve system (3) and then, substituting the values of u and v obtained into (2), we solve the resulting system of equations for the variables x and y .

Let us solve system (3). For our purpose, we square both sides of the first equation:

$$u^2(u^2 - 1)^2 = v^2(v^2 + 8)^2$$

and make the substitution $u^2 = v^2 + 5$ in the equation obtained. After certain transformations we obtain an equation

$$3v^4 + 8v^2 - 80 = 0.$$

It is quadratic with respect to v^2 . By the formula for the roots of a quadratic equation we find $v^2 = 4$. It follows that any solution of system (3) is a solution of the system of equations $v^2 = 4$, $u^2 = 9$ which has four solutions: $(3; 2)$, $(3; -2)$, $(-3; 2)$, $(-3; -2)$. We make sure by means of verification that system (3) has only two solutions: $(3; 2)$ and $(-3; -2)$.

Now we shall find, from equations (2), the corresponding values of the variables x and y . Substituting into (2) the values $u = 3$, $v = 2$, we get $x = 9$, $y = 4$. At $u = -3$, $v = -2$ we get a system which has no solutions. Consequently, system (1) has the unique solution $x = 9$, $y = 4$.

Answer: $\{(9, 4)\}$. ▲

The solution given here can be represented schematically as follows:

$$\begin{aligned} \{(1); (2)\} &\Leftrightarrow \{(2), (3)\} \\ &\Leftrightarrow \{(2); u = 3, v = 2 \text{ or } u = -3, v = -2\} \\ &\Leftrightarrow \{(2); u = 3, v = 2\} \Leftrightarrow \{x = 9, y = 4, u = 3, v = 2\}. \end{aligned}$$

The method we used in solving system (1) is called the *method of introduction of new variables (unknowns)*. It is widely used in solving systems of equations. Here is one more example of solving a system of equations by the method of introduction of new variables:

Example 5. Solve the system of equations

$$\begin{cases} x + y - xy = 1, \\ x^2 + y^2 - xy = 3. \end{cases}$$

△ Reducing the second equation to the form

$$(x + y)^2 - 3xy = 3,$$

we see that it is convenient to introduce new variables $u = x + y$, $v = xy$. For u , v we get the system of equations

$$\begin{cases} u - v = 1, \\ u^2 - 3v = 3. \end{cases}$$

Finding u from the first equation and substituting the resulting expression into the second equation, we get a quadratic equation

$$v^2 - v - 2 = 0,$$

whose roots are the numbers $v_1 = 2$ and $v_2 = -1$. From the equation $u = v + 1$ we find that $u_1 = 3$ and $u_2 = 0$.

In accordance with what was stated above, the given system is equivalent to the collection of the following two systems:

$$\begin{cases} x + y = 3, \\ xy = 2 \end{cases} \quad \text{and} \quad \begin{cases} x + y = 0, \\ xy = -1. \end{cases}$$

The first system has two solutions: $(1; 2)$ and $(2; 1)$. The second system also has two solutions: $(1; -1)$ and $(-1; 1)$.

Answer: $\{(1; 2), (2; 1), (1; -1), (-1; 1)\}$. ▲

3.3. Systems of Linear Equations

Let us consider a system of two linear equations in two variables:

$$\begin{cases} a_1x + b_1y = c_1, \\ a_2x + b_2y = c_2. \end{cases} \quad (1)$$

If all the numbers $a_1, b_1, c_1, a_2, b_2, c_2$ are equal to zero, then any pair of numbers is a solution of the given system. Now if a_1, b_1, a_2, b_2 are equal to zero but at least one of the numbers c_1 and c_2 is nonzero, then the system has no solutions.

Let us consider the case when at least one of the numbers a_1, a_2, b_1, b_2 is nonzero, say, $a_1 \neq 0$.

Then system (1) is equivalent to the system

$$\begin{cases} x + \frac{b_1}{a_1}y = \frac{c_1}{a_1}, \\ a_2x + b_2y = c_2. \end{cases} \quad (2)$$

Adding the first equation multiplied by $-a_2$ to the second equation, we get

$$\left(b_2 - \frac{b_1}{a_1}a_2\right)y = c_2 - \frac{c_1}{a_1}a_2. \quad (3)$$

Consequently, if $b_2 - \frac{b_1}{a_1} a_2 \neq 0$, i.e. $a_1 b_2 - a_2 b_1 \neq 0$, then

$$y = \frac{a_1 c_2 - a_2 c_1}{a_1 b_2 - a_2 b_1}. \quad (4)$$

Substituting this value of y into the first equation of system (2), we find

$$x = \frac{c_1 b_2 - c_2 b_1}{a_1 b_2 - a_2 b_1}. \quad (5)$$

Thus, if $a_1 b_2 - a_2 b_1 \neq 0$, then system (2), as well as system (1), has a unique solution which can be determined by formulas (4) and (5).

Assume now that $a_1 b_2 - a_2 b_1 = 0$. In that case, if $a_1 c_2 - a_2 c_1 \neq 0$, then equation (3), as well as system (1), has no solutions. Now if $a_1 c_2 - a_2 c_1 = 0$, then any pair of numbers $(x; y)$, where

$$x = \frac{c_1}{a_1} - \frac{b_1}{a_1} y, \quad y \in \mathbb{R},$$

is a solution of system (1).

Thus it follows that if $a_1 b_2 - a_2 b_1 = 0$, then system (1) either has no solutions or possesses an infinite number of solutions.

Let, as before, $a_1 \neq 0$ and $b_2 - \frac{b_1}{a_1} a_2 = 0$. If we designate $a_2/a_1 = k$, then $a_2 = k a_1$, and it follows from the condition $b_2 - \frac{b_1}{a_1} a_2 = 0$ that $b_2 = k b_1$. In that case system (1) has the form

$$\begin{cases} a_1 x + b_1 y = c_1, \\ k a_1 x + k b_1 y = c_2. \end{cases}$$

This system possesses a solution if and only if $c_2 = k c_1$.

Thus we see that if $a_1 \neq 0$ and $a_1 b_2 - a_2 b_1 = 0$, the system possesses a solution if and only if the second equation can be obtained from the first by termwise multiplication by the number $k = a_2/a_1$.

We have assumed somewhat earlier that $a_1 \neq 0$. But if, for example, $b_2 \neq 0$, then changing the places of the equations and the variables in the two equations we arrive at the case we have discussed.

To find solutions of the systems in the case when $a_1b_2 - a_2b_1 \neq 0$, use can be made of formulas (4), (5), or, else, all the transformations may be carried out which were performed in deriving formulas (4), (5).

Example 1. Solve the system of equations

$$\begin{cases} 2x + 3y = 11, \\ 3x + 2y = 9. \end{cases}$$

$\triangle$ We multiply the second equation by $\frac{2}{3}$ and subtract it from the first equation term-by-term. The result is $\frac{5}{3}y = 5$, i.e. $y = 3$. Substituting this value of y into the first equation of the given system, we get $x = 1$.

Answer: $\{(1, 3)\}$. $\blacktriangle$

Example 2. Solve the system of equations

$$\begin{cases} 3x - y = 1, \\ 12x - 4y = 4. \end{cases}$$

$\triangle$ The given system is equivalent to one equation $3x - y = 1$ since the second equation can be obtained from this equation by termwise multiplication by 4. Consequently, it is satisfied by any pair of numbers $x, y = 3x - 1$, and the system has no other solutions.

Answer: $\{(x; 3x - 1), x \in \mathbb{R}\}$. $\blacktriangle$

Example 3. Solve the system of equations

$$\begin{cases} x + y = 3, \\ 2x + 2y = 8. \end{cases}$$

$\triangle$ This system of equations is not satisfied by any pair of numbers. Indeed, if we multiply the first equation by 2 term-by-term and subtract it from the second equation, we get an equation $0 = 2$, which, evidently, has no solutions.

Answer: The system has no solutions. $\blacktriangle$

Example 4. At what values of the parameter a does the system of two equations

$$\begin{cases} (a+1)x + 8y = 4a, \\ ax + (a+3)y = 3a-1 \end{cases} \quad (6)$$

have an infinite number of solutions?

We divide the first equation by 8:

$$\frac{a+1}{8}x + y = \frac{a}{2}. \quad (7)$$

Next we subtract equation (7) multiplied by $a + 3$ from the second equation (6):

$$\left(a - \frac{(a+1)(a+3)}{8}\right)x = 3a - 1 - \frac{a}{2}(a+3). \quad (8)$$

It is evident that for any a the system of equations (7), (8) is equivalent to the given system. Consequently, if in (8) the coefficient in x is nonzero, that is, if

$$a - \frac{(a+1)(a+3)}{8} \neq 0,$$

then system (6) has a unique solution. Therefore, system (6) can have an infinite number of solutions if and only if $a - \frac{(a+1)(a+3)}{8} = 0$, that is, when

$$a^2 - 4a + 3 = 0.$$

This quadratic equation has the roots $a = 3$ and $a = 1$. At $a = 3$, we get from (6) the system

$$\begin{cases} 4x + 8y = 12, \\ 3x + 6y = 8, \end{cases}$$

which, evidently, has no solutions.

At $a = 1$, we get from (6) the system

$$\begin{cases} 2x + 8y = 4, \\ x + 4y = 2, \end{cases}$$

which is equivalent to one equation $x + 4y = 2$. The solution of this equation is any pair of numbers $(2 - 4y; y)$, where $y \in \mathbb{R}$. Consequently, system (6) has an infinite number of solutions only for $a = 1$.

Answer: $a = 1$. ▲

A linear equation in n variables $x_1, x_2, \dots, x_n$ is an equation of the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b, \quad (9)$$

where $a_1, a_2, \dots, a_n, b$ are some numbers. If all the numbers $a_1, \dots, a_n, b$ are equal to zero, then any collection of

n numbers is a solution of this equation. Now if all the numbers $a_1, \dots, a_n$ are zero and $b \neq 0$, then equation (9) has no solutions. We shall assume in what follows that not all the coefficients $a_1, \dots, a_n$ are equal to zero.

When seeking solutions for a system of m linear equations in n variables, it is convenient to use the *Gaussian method*, which is a special case of the method of elimination of the unknowns and consists in reducing a given system to the so-called *triangular form* by means of equivalent transformations.

Example 5. Solve the system of equations

$$\begin{cases} 2x + y + 3z = 13, \\ x + y + z = 6, \\ 3x + y + z = 8. \end{cases}$$

Δ Adding termwise the second equation multiplied by -2 to the first equation, we get

$$-y + z = 1 \text{ or } y - z = -1.$$

Next we add to the third equation of the system the second equation multiplied by -3 and obtain $-2y - 2z = -10$. Finally, we add the equation $y - z = -1$ multiplied by 2 to the equation obtained, and get $-4z = -12$, i.e. $z = 3$.

As a result of some transformations we get the system of equations

$$\begin{cases} x + y + z = 6, \\ y - z = -1, \\ z = 3, \end{cases}$$

which is, evidently, equivalent to the given system.

Systems of this kind are called *triangular*. They are easy to solve. Indeed, from the third, second and first equations we successively find $z = 3$, $y = z - 1 = 2$, $x = 6 - y - z = 1$.

Answer: $\{(1, 2, 3)\}$. $\blacktriangle$

The solution of a system of linear equations by reducing them to a triangular form is known as *Gauss' method*.

Example 6. Solve the system of equations

$$\begin{cases} x_1 + x_2 + x_3 + x_4 = 1, \\ x_1 + x_2 - x_3 = 2, \\ x_2 + x_3 = 0. \end{cases}$$

△ We shall solve this system of three equations in four variables by the Gaussian method.

By subtracting the first equation of the system from the second, we get

$$-2x_3 - x_4 = 1, \quad x_3 + \frac{1}{2}x_4 = -\frac{1}{2}.$$

Let us consider the system of equations

$$\begin{cases} x_1 + x_2 + x_3 + x_4 = 1 \\ x_2 + x_3 = 0, \\ x_3 + \frac{1}{2}x_4 = -\frac{1}{2}. \end{cases}$$

This triangular system is equivalent to the given system. Assuming x_4 to be arbitrary, we successively find

$$x_3 = -\frac{1}{2} - \frac{1}{2}x_4, \\ x_2 = -x_3 = \frac{1}{2} + \frac{1}{2}x_4, \quad x_1 = 1 - x_4.$$

Consequently, any ordered collection of four numbers of the form $(1 - t; \frac{1}{2} + \frac{1}{2}t; -\frac{1}{2} - \frac{1}{2}t; t)$, where $t \in \mathbb{R}$, is a unique solution of the given system of equations.

Answer: $\left\{ \left(1 - t; \frac{1}{2} + \frac{1}{2}t; -\frac{1}{2} - \frac{1}{2}t; t \right), t \in \mathbb{R} \right\}$. ▲

Example 7. Solve the system of equations

$$\begin{cases} x_1 + x_2 - x_3 - x_4 = 1, \\ x_1 + 2x_2 + 3x_3 - x_4 = 2, \\ 3x_1 + 5x_2 + 5x_3 - 3x_4 = 6. \end{cases}$$

△ Subtracting the first equation from the second, we get

$$x_2 + 4x_3 = 1.$$

Next, subtracting the first equation multiplied by 3 from the third equation of the system, we get

$$2x_2 + 8x_3 = 3, \quad x_2 + 4x_3 = 3/2.$$

Finally, subtracting the equation $x_2 + 4x_3 = 1$ from the last equation, we obtain an equation $0 = 1/2$, which, evidently, has no solutions.

Consequently, the given system has no solutions.

Answer: The system is inconsistent. ▲

The examples considered above show that a system of linear equations can have one solution (see Example 5), an infinite number of solutions (see Example 6) and can have no solutions (see Example 7).

3.4. Problems on Setting up Equations

Example 1. The first worker needs 6 minutes less to machine a workpiece than the second worker. How many workpieces will each of them machine in seven hours if during that time the first of them machines eight workpieces more than the second?

△ Let us designate by n the number of workpieces machined by the first worker in seven hours. Then he needs $7/n$ hours to machine one workpiece. The second worker machines $n - 8$ workpieces in seven hours and, consequently, he needs $7/(n - 8)$ hours to machine one workpiece.

Since the first worker spends six minutes less to machine one workpiece, we have

$$\frac{7}{n-8} - \frac{7}{n} = \frac{1}{10}.$$

Let us solve this equation.

Multiplying both sides of the equation by $n(n - 8)$ and collecting like terms, we obtain a quadratic equation

$$n^2 - 8n - 560 = 0.$$

Let us find the roots of this equation:

$$n = 4 \pm \sqrt{16 + 560} = 4 \pm 24.$$

The condition of the problem is satisfied only by positive values of n , hence $n = 28$. Consequently, the first worker machines 28 workpieces and the second machines eight workpieces less, i.e. 20 workpieces.

Answer: 28 workpieces and 20 workpieces. ▲

Example 2. A vessel contained 10 litres of hydrochloric acid. Part of the acid was poured off and the same quantity of water was added to the vessel. Then the same amount of the mixture was again poured off and the same quantity of

water was added. How many litres were poured off each time if, as a result, the vessel contained a 64% solution of hydrochloric acid?

△ Let us designate by x the litres of hydrochloric acid poured off the first time. After the first pouring-off and addition of water, the mixture contained $10 - x$ litres of acid and x litres of water. Consequently, each litre of the mixture contained $\frac{10-x}{10}$ litre of the acid.

After x litres of the mixture were poured off, the vessel contained $10 - x$ litres of the mixture which included

$$(10 - x) \frac{10 - x}{10} = \frac{(10 - x)^2}{10}$$

litres of hydrochloric acid.

After the addition of x litres of water, the vessel contained 10 litres of the mixture of which $\frac{(10-x)^2}{10}$ litres were hydrochloric acid. The percentage of the acid was

$$\frac{(10-x)^2}{10 \cdot 10} \cdot 100\%.$$

On the other hand, by the hypothesis this content is equal to 64%. Therefore, we obtain for x the equation

$$(10 - x)^2 = 64$$

from which it follows that $x = 2$.

Answer: 2 litres. ▲

Example 3. If we divide a two-digit number by the product of its digits we shall have 1 as a quotient and 16 as a remainder. Now if we add the product of its digits to the square of the difference of its digits, we shall obtain the given number. Find the number.

△ Let us designate the first digit in the required two-digit number by a and the second digit by b . Then, the required two-digit number is equal to $10a + b$.

The first condition of the problem yields an equation

$$10a + b = ab + 16, \quad (1)$$

and the second condition, an equation

$$10a + b = (a - b)^2 + ab, \quad (2)$$

As a result, we have obtained the system of equations (1), (2) for a and b . The left-hand sides of (1) and (2) are equal, hence the right-hand sides are equal too, i.e. $ab + 16 = (a - b)^2 + ab$. This equation implies that $(a - b)^2 = 16$, i.e. $a - b = \pm 4$.

Let us consider the case when $a - b = 4$. Substituting $a = b + 4$ into (1), we obtain, after transformations, a quadratic equation

$$b^2 - 7b - 24 = 0.$$

From the formula for the roots of a quadratic equation, we have

$$b = \frac{7 \pm \sqrt{49 + 96}}{2} = \frac{7 \pm \sqrt{145}}{2}.$$

Since b is an integer, the problem has no solution in this case.

Let us now consider the case when $a - b = -4$, i.e. $a = b - 4$. Substituting this expression for a into (1), we get a quadratic equation

$$b^2 - 15b + 56 = 0.$$

Solving this equation, we find that $b_1 = 7$ and $b_2 = 8$. From the formula $a = b - 4$ we get $a_1 = 3$ and $a_2 = 4$.

Answer: 37 and 48. ▲

Example 4. Two sportsmen are running along a closed track in a stadium. The speed of each runner is constant, and to cover the whole track the first sportsman needs five seconds less time than the second. If they start simultaneously from the same place and run in the same direction, then 30 seconds later they will come alongside each other. After what length of time will they meet if they start simultaneously from the same place and run in opposite directions?

△ Suppose v_1 and v_2 are the speeds of the runners. For definiteness, we shall put $v_2 > v_1$. We designate by S the length of the closed track of the stadium. Then, the first sportsman covers the track in S/v_1 seconds and the second in S/v_2 seconds. Since the first sportsman spends 5 seconds more, we arrive at an equation

$$\frac{S}{v_1} - \frac{S}{v_2} = 5. \quad (3)$$

In 30 seconds the sportsmen cover the distances $30v_1$ and $30v_2$ respectively. If they run in the same direction, then by the moment of their meeting the second sportsman will have run one circle more, i.e.

$$S = 30 (v_2 - v_1). \quad (4)$$

We have obtained a system of two equations (3), (4) in three variables v_1 , v_2 , S .

Dividing both sides of (4) by $30S$, we get

$$\frac{v_2}{S} - \frac{v_1}{S} = \frac{1}{30}. \quad (5)$$

Let us introduce the designations $p_1 = v_1/S$ and $p_2 = v_2/S$. Then, from (3), (5) we get for p_1 and p_2 the system

$$\begin{cases} \frac{1}{p_1} - \frac{1}{p_2} = 5, \\ p_2 - p_1 = \frac{1}{30}. \end{cases} \quad (6)$$

From the second equation we find

$$p_2 = p_1 + \frac{1}{30}. \quad (7)$$

Substituting this expression into the first equation (6), we get a quadratic equation

$$150p_1^2 + 5p_1 - 1 = 0.$$

Consequently,

$$p_1 = \frac{-5 + \sqrt{25 + 600}}{300} = \frac{-5 + 25}{300} = \frac{1}{15}.$$

(We have put only the plus sign before the root here since by the hypothesis $p_1 > 0$.) Substituting this value of p_1 into (7), we get $p_2 = 1/10$. Knowing p_1 and p_2 , it is easy to answer the question put in the problem: after what length of time the runners will meet if they start simultaneously from the same place and run in opposite directions. Indeed, this time is equal to

$$\frac{S}{v_1 + v_2} = \frac{1}{\frac{v_1}{S} + \frac{v_2}{S}} = \frac{1}{p_1 + p_2} = \frac{1}{\frac{1}{15} + \frac{1}{10}} = \frac{30}{5} = 6.$$

Consequently, the required time is equal to 6 seconds.

Answer: 6 seconds. ▲

Note that p_1 and p_2 are the speeds of the runners in the case when the length of the closed track is taken to be the unit length.

PROBLEMS OF SECTION I

Solve the equations 1 through 5:

$$1. \frac{3}{x+2} - \frac{2x-1}{x+1} = \frac{2x+1}{x^2+3x+2}.$$

$$2. \sqrt{9-5x} = \sqrt{3-x} + \frac{6}{\sqrt{3-x}}.$$

$$3. 4\sqrt{x+2} = |x+1| + 4. \quad 4. \frac{x-3}{x^2+4x+9} + \frac{x^2+4x+9}{x-3} = -2.$$

$$5. 2x^2+3x-5\sqrt{2x^2+3x+9}+3=0.$$

Solve the systems of linear equations 6 through 8 by Gauss' method:

$$6. \begin{cases} x+2y+3z=8, \\ 3x+y+z=6, \\ 2x+y+2z=6. \end{cases} \quad 7. \begin{cases} x+2y-3z=1, \\ 2x+3y-6z=0, \\ 3x+3y-5z=0, \\ x+y+z=3. \end{cases}$$

$$8. \begin{cases} x_1+x_2+x_3+x_4=1, \\ 2x_1-3x_2+x_3-x_4=2, \\ 4x_1-x_2+3x_3+x_4=4. \end{cases}$$

Solve the systems of equations 9 through 13:

$$9. \begin{cases} x^2+2y^2-2xy=5, \\ x+2y=7. \end{cases} \quad 10. \begin{cases} 2x^2+3y^2-4xy=3, \\ 2x^2-y^2=7. \end{cases}$$

$$11. \begin{cases} 3xy-x^2-y^2=5, \\ 7x^2y^2-x^4-y^4=155. \end{cases} \quad 12. \begin{cases} xy-yz-2x-z=0, \\ xz-z^2+3x-3y=0, \\ x^2-xz+4y+2z=0. \end{cases}$$

$$13. \begin{cases} 27x^3-y^3-13\frac{xy}{z}=0, \\ 3x^2z-4xy+\frac{3}{z}=0, \\ 3xz-yz=1. \end{cases}$$

14. Find all the values of the parameter c for which the system

$$\begin{cases} -4x+cy=3+c \\ (6+c)x+2y=1+c, \end{cases}$$

has no solutions.

15. The points A , B , C are located along the river (B is located between A and C). A boat covered the distance between A and C in 7 hours. Its proper speed (its speed relative to water) was constant at each of the sections AB and BC , being $1\frac{1}{2}$ times less at the section BC than at the section AB . The boat made the journey back, from C to A , in 8 hours, its proper speed being $1\frac{1}{3}$ times greater during the whole return trip than on its way from A to B . Were the boat's proper speed the same on its return trip as on its way from A to B , it would cover the distance from B to A in 6 hours. How long did it take the boat to go from B to C ? (The velocity of the river flow is constant.)

PROBLEMS OF SECTION II

Solve equations 1 through 11:

$$1. 1 + \frac{x-1}{x+2} + \frac{1}{x} = \frac{2+3x}{x(x+2)}. \quad 2. 2 + \frac{2t-1}{t+2} = \frac{4t+3}{2t+1}.$$

$$3. \frac{2y+1}{y-1} + \frac{y+1}{2y+1} = \frac{5y+4}{(y-1)(2y+1)}.$$

$$4. (x+2)^2 + \frac{24}{x^2+4x} = 18. \quad 5. 3x - \sqrt{18x+1} + 1 = 0.$$

$$6. \sqrt{x} - \sqrt{x+3} = 1. \quad 7. \sqrt[3]{x^2} - 2\sqrt[6]{x} + \sqrt{x} = 2.$$

$$8. \sqrt{x-1} + \sqrt{x-2\sqrt{x-1}} = 1. \quad 9. 2x+3 - \sqrt{x^2-2x-3} = 0.$$

$$10. \sqrt{3x^2+1} + \sqrt{x^2+3} = \sqrt{6x^2+10}.$$

$$11. |x^2+3x+2| + 4x+10 = 0.$$

12. At what values of the parameter a do the equations $x^2 + 5x + a = 0$ and $x^2 + 2ax + a^2 - 4a + 25 = 0$ have real roots?

Use Gauss' method to solve equations 13 through 18:

$$13. \begin{cases} 5x + y - 7z = 0, \\ 5x - 2z = 3, \\ 3x - y + 2z = 3, \\ 3x + 2y - 10z = -3. \end{cases} \quad 14. \begin{cases} x + 3y + 2z = 1, \\ 2x + 4y + z = 1, \\ 4x + 6y - z = 1, \\ 3x + 7y + 3z = 2. \end{cases}$$

$$15. \begin{cases} x_1 - x_2 - x_3 + x_4 = 1, \\ x_1 + 2x_2 + x_3 - x_4 = 3, \\ 3x_1 - x_3 + x_4 = 5. \end{cases} \quad 16. \begin{cases} 7x + 2y - 4z = 19, \\ 5x + 3y - 3z = 15, \\ 5x - 3y + 3z = 15. \end{cases}$$

$$17. \begin{cases} x_1 - x_2 - x_3 = -3, \\ 5x_1 - x_2 + x_3 - 2x_4 = 3, \\ 8x_1 - x_2 - x_3 - x_4 = 1, \\ 7x_1 - x_4 = 3. \end{cases} \quad 18. \begin{cases} x_1 + x_2 + x_3 + x_4 = 1, \\ x_1 + x_2 + 2x_3 + x_4 = 3. \end{cases}$$

Solve the systems of equations 19 through 30:

$$19. \begin{cases} x^2 + y^2 = 41, \\ x + y = 9. \end{cases} \quad 20. \begin{cases} x^2 + xy = 28, \\ y^2 + xy = -12. \end{cases}$$

$$21. \begin{cases} x^2 - xy = 6, \\ xy + y^2 = 4. \end{cases} \quad 22. \begin{cases} 2x + y = 3x^2, \\ x + 2y = 3y^2. \end{cases}$$

$$23. \begin{cases} x - xy^3 = 7, \\ xy^2 - xy = 3. \end{cases} \quad 24. \begin{cases} 2x^2 + y^2 + 3xy = 12, \\ (x+y)^2 - \frac{1}{2}y^2 = 7. \end{cases}$$

$$25. \begin{cases} x^2 - xy + y^2 = 7, \\ x^4 + x^2y^2 + y^4 = 133. \end{cases} \quad 26. \begin{cases} 2|x-1| + 3|y+2| = 17, \\ 2x + y = 7. \end{cases}$$

$$27. \begin{cases} \frac{x}{6} + \frac{4}{y} = 2, \\ \frac{18}{x} + \frac{y}{2} = 5. \end{cases} \quad 28. \begin{cases} \sqrt{\frac{x-y}{x+y}} + \sqrt{\frac{x+y}{x-y}} = \frac{10}{3}, \\ xy - 2x - 2y = 2. \end{cases}$$

$$29. \begin{cases} y\sqrt{y} + \sqrt{y} = 5\sqrt{x} - x\sqrt{x}, \\ x = y + 3. \end{cases}$$

$$30. \begin{cases} \sqrt{\frac{2x-1}{y+2}} + \sqrt{\frac{y+2}{2x-1}} = 2, \\ x + y = 12. \end{cases}$$

31. How many real roots does the system of equations

$$\begin{cases} |x| + |y| = 1, \\ x^2 + y^2 = a \end{cases}$$

possess depending on the parameter a ?

32. Determine at what values of the parameter a the system of equations

$$\begin{cases} x^2 + y^2 = 2(1+a), \\ (x+y)^2 = 14 \end{cases}$$

has two solutions.

Solve the systems of equations 33 through 38:

$$33. \begin{cases} xy - xz + 3y + 6z = 0, \\ y^2 - yz + 4x - 4z = 0, \\ zy - z^2 - 2x - y = 0. \end{cases} \quad 34. \begin{cases} xy + 2yz^2 - 4z^3 = 6, \\ y - yz - xz = 0, \\ y^2 - 7xz^2 - 13z^3 = 0. \end{cases}$$

$$35. \begin{cases} y^2 - xy + x^2 = z^2, \\ x^2 - xz + z^2 = y^2, \\ z^3 - y^3 = x^2 + y^2 + z^2. \end{cases} \quad 36. \begin{cases} xy = 5x + 6y - 4z, \\ y^2 = 3x + 5y - z, \\ yz = x + 4y + 2z. \end{cases}$$

$$37. \begin{cases} xy + \frac{6}{xz} = 2, \\ xz - \frac{4}{yz} = 4, \\ yz - \frac{3}{xy} = 1. \end{cases} \quad 38. \begin{cases} y^2 + z^2 = x^2 + 2yz - 2, \\ z^2 + x^2 = y^2 + 2zx - 3, \\ x^2 + y^2 = 5zy - 5zx + 2xy - 6. \end{cases}$$

39. If we divide a two-digit number by the product of its digits, we shall have 3 as a quotient and 9 as a remainder. Now if we subtract the product of its digits from the square of the sum of the digits, forming that number, we shall obtain the given number. Find the number.

40. Given two solutions of common salt of different concentrations. If we mix 100 g of the first solution and 200 g of the second, we shall have a 50% solution. Now if we mix 300 g of the first solution and 200 g of the second, we obtain a 42% solution. Determine the concentrations of the given solutions.

41. Pieces of metal of the same weight were cut off from each of the two pieces of alloy having different copper content and weighing m kg and n kg, respectively. Each of the cut-off pieces was then alloyed with the remainder of the other piece after which procedure the copper content in both alloys became the same. What is the weight of each cut-off piece?

42. Given a reservoir supplied with two pipes. It takes 5 hours less to fill up the reservoir by the first pipe than by the second. If we first turn on the second pipe and eight hours later turn on the first pipe, the reservoir will be full in 18 hours. What is the capacity of the reservoir and what is the hourly flow through each pipe if a total of 20 cu m of water is delivered by the first pipe in five hours and by the second pipe in four hours?

43. The reservoir was filled by means of several pumps of the same capacity which were turned on one after another at equal time intervals. When the last pump was turned on, the reservoir was $1/6$ full. Another time, when the same reservoir was being filled, the capacity of each pump was reduced by 10%, but the time intervals at which the pumps were turned on remained the same. What proportion of the reservoir will the pumps fill with water this time during the first half of their total operating time?

44. Uniform workpieces are being machined on two machine-tools. The capacity of the first machine-tool is 40% greater than that of the second. How many workpieces were machined on each machine-tool per shift if the first of them operated 6 hours during a shift and the second, 7 hours, and together they machined 616 workpieces?

45. Two workers were to produce a batch of identical workpieces. After the first of them had worked seven hours and the second four hours, it turned out that they had fulfilled $5/9$ of the whole volume of work. Having worked four more hours together, they found that they only had $1/18$ of the work quota left to fulfil. How many hours would it take each worker to fulfil the whole work quota if they worked separately?

46. Two teams of collective farmers were to finish harvesting in 12 days. After the teams had worked together for eight days the first

team was given another assignment and, therefore, the second team finished the remaining part of work alone in seven days. How many days would it take each team to do the harvesting if they worked separately?

47. Two workshops purchased some tools for special jobs; the first spent 300 rubles and the second, 270 rubles. The second workshop purchased 5 tools less than the first but paid 3 rubles more for each tool. How many tools were purchased by each workshop?

48. An electronic computer has to solve two problems. The first problem consists of 9 million operations of type *A* and 16 million operations of type *B* and requires 11 minutes and 40 seconds machine time. The second problem contains twice as many operations of type *A* and half as many operations of type *B* and the machine spends 13 minutes and 20 seconds to solve it. How many operations of each type can the machine perform in a second?

49. It takes a Metro passenger 24 seconds to walk down a moving escalator. If the passenger walks down a stationary escalator at the same speed, he will be downstairs in 42 seconds. In how many seconds will he be downstairs if he stands on a moving escalator?

50. Find the speed and the length of the train knowing that it passes a stationary observer in 7 seconds and requires 25 seconds to pass a platform 378 metres long at the same speed.

51. Two cars, travelling along a circular road and in the same direction at constant speeds, come alongside each other every 56 minutes. Moving at the same speeds but in opposite directions the cars meet every 8 minutes. How much time does each car require to cover the whole circular route?

52. Three cyclists left point *A*. The first of them left an hour earlier than the other two, who started out simultaneously. The speed of each cyclist was constant. Some time later, the third cyclist caught up with the first, and two hours later the second cyclist also caught up with the first. Determine the ratio between the speeds of the first and the third cyclist if the ratio between the speeds of the second and the third cyclist is $\frac{2}{3}$.

53. A pedestrian and a cyclist left point *A* for point *B* simultaneously. At point *B* the cyclist turned back and met the pedestrian 20 minutes after they had both started. Without stopping, the cyclist reached point *A*, turned back and again caught up with the pedestrian only 10 minutes after their last meeting. How much time does the pedestrian need to cover the route from *A* to *B*?

54. Starting simultaneously from point *A*, a boat and a bus travel continuously and without making any stops between point *A* and *B*, the first on the river and the second along the road which runs parallel to the river. Their first meeting took place when the bus had covered $\frac{5}{9}$ of the distance between *A* and *B*; their second meeting took place when the bus, having first reached point *B*, covered $\frac{1}{8}$ of the whole distance back from *B* to *A*. The first time, the bus came to point *A* 16 minutes later than the boat. How many hours after starting will the bus and the boat meet at point *A* if the speed of the boat in stationary water and the speed of the bus are constant?

55. Two trains started simultaneously from point *A* to point *B*. First each of them moved with uniform acceleration (the accelerations

of the trains are different, their initial velocities are zero); then, having attained a certain speed, they began to move with uniform speed. Having passed a third of the distance from A to B , one of the trains came alongside the other and from that moment on it moved with uniform speed. One of the trains covered the whole route from A to B 1.2 times as fast as the other. Determine the ratio between the uniform speeds of the trains.

56. Several dump trucks are loaded in turn at point A (the loading time is the same for all the trucks) and carry the load to point B , where they quickly unload and return to point A . The speeds of the trucks are equal, the speed of a loaded truck being $6/7$ that of an empty truck. The driver Petrov was the first to leave point A . On his return trip he met Ivanov, the last driver to leave point A , and arrived back at A 6 minutes after their meeting. Here Petrov began loading at once and after finishing the operation, left for B ; he met Ivanov for the second time, 40 minutes after their first meeting. Not less than 16 (but not more than 19) minutes had passed since their second meeting before Ivanov reached point A . Determine the loading time.

Chapter 4

Algebraic Inequalities

4.1. Functional Inequalities. The Concept of Equivalence of Inequalities

Let us consider two functions, $f(x)$ and $g(x)$, defined on some set X . It is often necessary to know for what values of x the values of the first function are smaller than the respective values of the second function. In other words, it is required to find all the values of the variable such that $f(x) < g(x)$. Problems of this type are known as problems on solving inequalities.

Thus, to solve the inequality

$$f(x) < g(x)$$

means to find all the values of x whose substitution into the inequality results in a true numerical inequality. Such values of x are called *solutions* of the inequality. The collection of all solutions is called a *set of solutions*. To solve an inequality means to find its set of solutions. The inequality $f(x) < g(x)$ is called *strict* and the inequality $f(x) \leq g(x)$ is known as *nonstrict* or *slack*.

In solving inequalities, as well as in proving them (see 4.6 below), of fundamental significance is the concept of equivalence of inequalities.

Two inequalities

$$f_1(x) < g_1(x) \quad \text{and} \quad f_2(x) < g_2(x)$$

are said to be *equivalent on the set M* if every solution of the first inequality belonging to the set M is also a solution of the second inequality and, conversely, every solution of the second inequality belonging to the set M is a solution of the first inequality.

Inequalities belonging to the set M which have no solutions on that set are also considered to be equivalent.

In some cases, a given inequality can be reduced, by successive transformations, to a simpler inequality equivalent to the original one. The following assertions are most often used to establish the equivalence of inequalities.

1. The inequalities

$$f(x) < g(x) \quad \text{and} \quad -f(x) > -g(x)$$

are equivalent on any number set.

2. If $f(x)$ and $g(x)$ assume only positive values on the set M , then the inequalities

$$f(x) < g(x) \quad \text{and} \quad \frac{1}{f(x)} > \frac{1}{g(x)}$$

are equivalent on M .

3. If the functions $f(x)$, $g(x)$, $\varphi(x)$ are defined on the set M , then the inequalities

$$f(x) < g(x) \quad \text{and} \quad f(x) + \varphi(x) < g(x) + \varphi(x)$$

are equivalent on the set M .

This yields the equivalence of the inequalities

$$f(x) < g(x) + \varphi(x) \quad \text{and} \quad f(x) - \varphi(x) < g(x),$$

that is, the rule of transposition of the summands from one side of the inequality to the other.

4. If the functions $f(x)$, $g(x)$, $\varphi(x)$ are defined on the set M and $\varphi(x) > 0$ on that set, then the inequalities

$$f(x) < g(x) \quad \text{and} \quad f(x)\varphi(x) < g(x)\varphi(x)$$

are equivalent on M .

5. If the functions $f(x)$ and $g(x)$ are defined on the set M and assume only positive values, then the inequalities

$$f(x) < g(x) \quad \text{and} \quad f^2(x) < g^2(x)$$

are equivalent on M .

The validity of these assertions easily follows from the corresponding properties of numerical inequalities. Similar assertions hold true for nonstrict inequalities as well.

Note that to prove the nonequivalence of two inequalities on a certain set, it is sufficient to indicate one element of that set which is a solution of one inequality but does not satisfy the other inequality.

Example 1. Are the following inequalities equivalent on the set of all positive numbers

$$(a) \ x^2 \leq x^3 \text{ and } 1 \leq x,$$

$$(b) \ x^3 + \sqrt{x} < x^2 + \sqrt{x} \text{ and } x^3 < x^2,$$

$$(c) \ \sqrt{x} \leq x + 1 \text{ and } x \leq (x + 1)^2?$$

Δ (a) The inequalities are equivalent by virtue of assertion 4 for nonstrict inequalities:

$$f(x) = x^2, \ g(x) = x^3, \ \varphi(x) = 1/x^2.$$

(b) The equivalence of the inequalities follows from assertion 3:

$$f(x) = x^3, \ g(x) = x^2, \ \varphi(x) = \sqrt{x}.$$

(c) The inequalities are equivalent in accordance with assertion 5:

$$f(x) = \sqrt{x}, \ g(x) = x + 1. \ \blacktriangle$$

Example 2. Are the inequalities of Example 1 equivalent on the set of all real numbers?

Δ (a) The inequalities are not equivalent since $x = 0$ satisfies the first inequality and does not satisfy the second.

(b) The inequalities are not equivalent since $x = -1$ satisfies only the second inequality.

(c) The inequalities are not equivalent since $x = -1$ belongs to the set of solutions of the second inequality but does not belong to the set of solutions of the first. $\blacktriangle$

Example 3. Are the inequalities

$$\sqrt{x^3 + x - 2} > x \text{ and } x^3 + x - 2 > x^2$$

equivalent on the set of all real numbers?

Δ If $x^3 + x - 2 \leq 0$, then neither the first nor the second inequality possesses a solution. Now if $x^3 + x - 2 > 0$, then x is also positive and, in accordance with assertion 5, the inequalities are equivalent. Consequently, the given inequalities are equivalent on the whole set of real numbers. $\blacktriangle$

4.2. Rational Inequalities. The Method of Intervals

In Chapter 1 we considered polynomials of the form

$$y = P_n(x) = a_0 + a_1x + \dots + a_nx^n$$

as the simplest numerical functions, and also functions representable as a ratio of two polynomials

$$y = R(x) = \frac{P_n(x)}{Q_m(x)},$$

that is, rational functions.

The number a is called a *zero* of the function $y = P_n(x)$ or a *root* of the polynomial $P_n(x)$ if $P_n(a) = 0$.

The polynomial $P_2(x) = 6 - 5x + x^2$, for instance, has two zeros, $x = 2$ and $x = 3$, since $P_2(2) = 0$ and $P_2(3) = 0$. A polynomial may not possess zeros at all, say, $P_0(x) = 1$ or $P_2(x) = 1 + x^2$. It is known that the number of zeros of a polynomial does not exceed its degree (see 8.7).

We shall call the zeros of the polynomials $P_n(x)$ and $Q_m(x)$ the *critical values* of the variable or the *critical (stationary) points* of the rational function

$$y = \frac{P_n(x)}{Q_m(x)}.$$

For instance, for the function

$$y = \frac{P_3(x)}{Q_2(x)} = \frac{x^3 - 6x^2 - x + 6}{x^2 + 3x + 2} = \frac{(x^2 - 1)(x - 6)}{(x + 1)(x + 2)}$$

the critical values of the variable are

$$x = -2, x = -1, x = 1, x = 6.$$

A *rational inequality* is an inequality which contains only rational functions.

A rational inequality can sometimes be solved by the so-called method of intervals. The method is based on a significant property of a rational function which will be assumed without proof, namely: *a rational function retains sign in the interval between its two neighbouring critical points.*

The *method of intervals* consists in the following. A rational inequality is reduced to a standard form:

$$\frac{P_n(x)}{Q_m(x)} > 0 \text{ or } \frac{P_n(x)}{Q_m(x)} < 0 \text{ (in the case of a strict inequality),}$$

$$\frac{P_n(x)}{Q_m(x)} \geq 0 \text{ or } \frac{P_n(x)}{Q_m(x)} \leq 0 \text{ (in the case of a nonstrict inequality).}$$

Then all the critical points of the rational function are found and marked on the number axis. Then the entire number axis is partitioned by the critical points into a finite number of subintervals on each of which the left-hand side of the inequality retains sign. To determine the sign of the left-hand side on the whole interval, it suffices to determine the sign of $P_n(x)/Q_m(x)$ at some one point of the interval and thus find out whether the interval belongs to the set of solutions of the given inequality.

As to the critical points themselves, they, evidently, do not belong to the set of solutions in the case of the strict inequality $P_n(x)/Q_m(x) > 0$ while in the case of the nonstrict inequality $P_n(x)/Q_m(x) \geq 0$ the zeros of the polynomial $P_n(x)$ belong to the set of solutions, provided they are not the zeros of the polynomial $Q_m(x)$ at the same time.

The method of intervals is not by far always applicable in solving rational inequalities. It is applicable only when the zeros of the polynomials $P_n(x)$ and $Q_m(x)$, that is, the critical values of the variable for the rational function $P_n(x)/Q_m(x)$, are known (or can be obtained). Unfortunately, the problem of seeking zeros of a polynomial cannot always be solved (see 8.7).

Example 1. Solve the inequality

$$\frac{x^3 - 3x^2 - x + 3}{x^2 + 3x + 2} > 0.$$

△ The zeros of the polynomial appearing in the denominator are $x = -1$ and $x = -2$. The zeros of the polynomial appearing in the numerator can be easily found. In fact,

$$\begin{aligned} x^3 - 3x^2 - x + 3 &= x^2(x - 3) - (x - 3) \\ &= (x - 3)(x - 1)(x + 1). \end{aligned}$$

The inequality can now be written as follows:

$$\frac{(x - 3)(x - 1)(x + 1)}{(x + 1)(x + 2)} > 0,$$

The critical points of the rational function are

$$x = -2, x = -1, x = 1, x = 3.$$

These points partition the number axis into 5 subintervals. Let us mark the points on the number axis (Fig. 32). To determine the sign of the function on each subinterval, we can do the following: note that for $x > 3$ all the linear factors in the numerator and the denominator of the rational function are positive and, consequently, on the interval

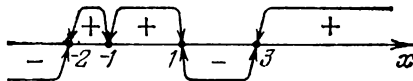


Fig. 32.

$(3; +\infty)$ the function assumes only positive values. We mark the interval $(3; +\infty)$ in Fig. 32 with the plus sign.

When passing from the interval $(3; +\infty)$ to the interval $(1; 3)$, through the point $x = 3$, only one of the linear factors, $x - 3$, changes sign and, consequently, the function becomes negative. So we mark the interval $(1; 3)$ in Fig. 32 with the negative sign.

Proceeding to the left, to the interval $(-1; 1)$, we find that only the factor $x - 1$ changes sign. This means that on passing through the point $x = 1$ the left-hand side of the inequality changes sign. On passing through the point $x = -1$, the sign of the function is, evidently, retained since the factor $x + 1$ appears both in the numerator and in the denominator of the rational function. Finally, the passage to the last interval, $(-\infty; -2)$, is again followed by a change of sign of the function. We mark all the changes of signs in the figure.

The inequality being strict, the critical points themselves are not the solutions. (The arrows in Fig. 32 point out this fact.)

Answer. $(-2; -1) \cup (-1; 1) \cup (3; +\infty)$. ▲

When solving this inequality, it may seem convenient to replace it, from the very beginning, by the simpler inequality

$$\frac{(x-1)(x-3)}{(x+2)} > 0.$$

But such a simplification, performed without any stipulations, will lead to an error. The resulting inequality is not equivalent to the original inequality since its set of solutions includes $x = -1$, the value of the variable which is not a solution of the given inequality.

Example 2. Solve the inequality

$$\frac{(x+3)^2(x^2+x+1)}{(4-x)x} \geq 0.$$

△ The critical points of the rational function are

$$x = -3, x = 0, x = 4.$$

We divide the number axis into four subintervals (Fig. 33), on each of which we easily determine the sign of the func-

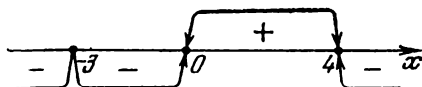


Fig. 33.

tion. When determining the sign, it is only necessary to follow the change of sign of the linear factors in the denominator since the quadratic factors in the numerator, $(x+3)^2$ and x^2+x+1 , are positive on all the four subintervals. Only $x = -3$ out of the three critical points belongs to the set of solutions of the inequality.

Answer: $\{-3\} \cup (0; 4)$. ▲

Example 3. Find the domain of definition of the function

$$\sqrt{\frac{2}{x^2-x+1} - \frac{1}{x+1} - \frac{2x-1}{x^2+1}}.$$

To find the domain of definition of the given function, we must solve the inequality

$$\frac{2}{x^2-x+1} - \frac{1}{x+1} - \frac{2x-1}{x^2+1} \geq 0.$$

We reduce it to the standard form:

$$\frac{2(x+1) - x^2 + x - 1 - 2x + 1}{x^3 + 1} \geq 0, \quad \frac{-x^2 + x + 2}{x^3 + 1} \geq 0.$$

Next we find the critical points $x = -1$ and $x = 2$ and write down the inequality as follows:

$$\frac{-(x+1)(x-2)}{(x+1)(x^2-x+1)} \geq 0.$$

Since $x^2 - x + 1 > 0$ for all the values of the variable, we pass to the equivalent inequality

$$\frac{-(x+1)(x-2)}{x+1} \geq 0.$$

The critical points partition the number axis into three subintervals (Fig. 34). We determine the sign of the left-

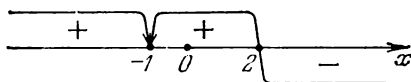


Fig. 34.

hand side of the inequality on each subinterval and investigate the critical points: the point $x = 2$ is a zero of the numerator and, the inequality being nonstrict, it enters into the set of solutions. The point $x = -1$, although it is a zero of the numerator, does not belong to the set of solutions because it turns the denominator into zero.

Answer: $(-\infty; -1) \cup (-1; 2]$. $\blacktriangle$

4.3. Irrational Inequalities

We shall now discuss inequalities containing a variable under the radical sign. Such inequalities are usually solved by reducing them to rational inequalities. Sometimes it is possible to get rid of the radicals by raising both sides of the inequality to a power. Unfortunately, this operation often leads to an inequality non-equivalent to the original one. Therefore, we must be very cautious when solving irrational inequalities. First of all, we must restrict our consideration to only those values of the variable for which both sides of the inequality are meaningful. Let us consider a typical example.

Example 1. Solve the inequality

$$\sqrt{x^2 - 4x + 3} \geq 2 - x,$$

Students often reason as follows: "When solving irrational equations and inequalities, we must first of all get rid of the radical and, therefore, we square both sides of the inequality, that is, write

$$x^2 - 4x + 3 \geq 4 - 4x + x^2.$$

This yields $3 \geq 4$, which is incorrect; hence the inequality in question has no solutions."

How true is the result obtained?

In the given case it is sufficient to look attentively at the inequality in order to see that the result obtained is not only improbable but even incorrect. For $x = 5$, for instance, the left-hand side of the inequality is positive while the right-hand side is negative. Thus it follows that the inequality possesses solutions and, consequently, the reasoning is faulty.

Here is the correct solution of Example 1.

$\triangle$ We should, evidently, consider only those values of x for which $x^2 - 4x + 3 \geq 0$. The zeros of the polynomial $x^2 - 4x + 3$ are $x_1 = 1$, $x_2 = 3$. Consequently, the set of solutions of the inequality $x^2 - 4x + 3 \geq 0$ is $(-\infty; 1] \cup [3; +\infty)$. It is clear that there are no solutions on the interval $(1; 3)$ since the left-hand side of the inequality is senseless for any x from that interval.

Furthermore, $\sqrt{x^2 - 4x + 3} \geq 0$ (the radical is taken in the arithmetic sense) and, since for any $x \geq 3$ the right-hand side of the inequality is less than zero, all $x \geq 3$ are solutions.

Now if $x \leq 1$, then $2 - x > 0$ and, squaring both sides of the inequality, we get the equivalent inequality

$$x^2 - 4x + 3 \geq 4 - 4x + x^2,$$

which has no solutions since the inequality $3 \geq 4$ is false.

Answer: $[3; +\infty)$. $\blacktriangle$

Example 2. Solve the inequality

$$\frac{1 - \sqrt{1 - 4x^2}}{x} > \frac{3}{2}.$$

$\triangle$ The left-hand side of the inequality is meaningful if and only if $|x| \leq 1/2$ and $x \neq 0$,

If $-1/2 \leq x < 0$, then the left-hand side of the inequality is negative and, consequently, there are no solutions on that interval.

Suppose $0 < x \leq 1/2$, then the obvious transformations yield

$$\sqrt{1-4x^2} < 1 - \frac{3}{2}x.$$

Both sides of the inequality are nonnegative and, therefore, squaring both sides, we get the equivalent inequality

$$1 - 4x^2 < 1 - 3x + \frac{9}{4}x^2.$$

Successive simplifications of the inequality yield

$$\frac{25}{4}x^2 - 3x > 0, \quad \frac{25}{4}x - 3 > 0, \quad x > \frac{12}{25}.$$

Taking into account the restriction $0 < x \leq 1/2$, we arrive at the final result $12/25 < x \leq 1/2$.

Answer: $(12/25, 1/2]$. ▲

Example 3. Solve the inequality

$$\sqrt{2x+1} + \sqrt{2x-5} \geq \sqrt{5-2x}.$$

△ Both sides of the inequality are meaningful for those and only those values of x which satisfy the system of the inequalities

$$\begin{cases} 2x+1 \geq 0, \\ 2x-5 \geq 0, \\ 5-2x \geq 0. \end{cases}$$

It is easy to see that the system has only one solution $x = 5/2$. Substituting this value into the inequality, we make sure that $x = 5/2$ is a solution.

Answer: $\{5/2\}$. ▲

4.4. Inequalities with a Modulus

In elementary mathematics we often come across inequalities containing a variable under the sign of the absolute value (modulus).

To solve inequalities of this kind, it is advisable to partition the number axis into subintervals so that we can write

the inequality on each of them without using the sign of the absolute value.

Example 1. Solve the inequality

$$x^2 - |5x + 6| > 0.$$

△ Let us divide the entire number axis into two subintervals

$$(-\infty; -6/5) \quad \text{and} \quad [-6/5 = +\infty)$$

on each of which the inequality can be written without the modulus sign.

The equality $|5x + 6| = -5x - 6$ is true for the subinterval $(-\infty; -6/5)$ and, consequently, the inequality assumes the form

$$x^2 + 5x + 6 > 0 \quad \text{or} \quad (x + 3)(x + 2) > 0,$$

whence we have $x \in (-\infty; -3) \cup (-2; +\infty)$. Taking into account the fact that the variable belongs to the subinterval $(-\infty; -6/5)$, we obtain the solution of the original inequality on that subinterval:

$$(-\infty; -3) \cup (-2; -6/5).$$

There holds the equality $|5x + 6| = 5x + 6$ on the second subinterval, $[-6/5 = +\infty)$, and, consequently, we can write the inequality as follows: $x^2 - 5x - 6 > 0$ or $(x + 1)(x - 6) > 0$, whence we have $x \in (-\infty; -1) \cup (6; +\infty)$. Taking into account that the variable belongs to the subinterval $[-6/5; +\infty)$, we get the set of solutions of the inequality on that subinterval:

$$[-6/5 = -1) \cup (6; +\infty).$$

Answer: $(-\infty; -3) \cup (-2; -1) \cup (6; +\infty)$. ▲

Example 2. Solve the inequality

$$\frac{|2x-1|}{x^2-x-2} > \frac{1}{2}.$$

△ For $x \geq 1/2$ the inequality can be rewritten without the modulus sign:

$$\frac{2x-1}{x^2-x-2} > \frac{1}{2}.$$

This is a rational inequality. We reduce it to the standard form:

$$\frac{x(5-x)}{(x+1)(x-2)} > 0.$$

Applying the method of intervals, we get $x \in (-1; 0) \cup (2; 5)$. The restriction $x \geq 1/2$ makes us retain only the subinterval $(2; 5)$. If $x < 1/2$, the inequality assumes the form

$$\frac{-2x+1}{x^2-x-2} > \frac{1}{2}.$$

We reduce the inequality to the standard form

$$\frac{(1-x)(x+4)}{(x+1)(x-2)} > 0.$$

Applying the method of intervals we easily obtain $x \in (-4; -1) \cup (1; 2)$. The restriction $x < 1/2$ makes us retain only the interval $(-4; -1)$.

Answer: $(-4; -1) \cup (2; 5)$. ▲

Example 3. Solve the inequality

$$|x-1| + |x+1| < 4.$$

△ By the definition of the modulus, we have $|x-1| = -x+1$, $|x+1| = -x-1$ on the interval $(-\infty; -1)$ and, consequently, on that interval the inequality is equivalent to the linear inequality $-2x < 4$ which is valid for $x > -2$. Thus, the set of solutions includes the interval $(-2; -1)$. On the interval $[-1; 1]$ the original inequality is equivalent to the true numerical inequality $2 < 4$. Therefore, all the values of the variable belonging to that interval belong to the set of solutions.

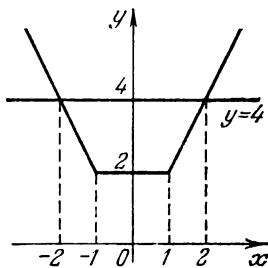


Fig. 35.

On the interval $(1; +\infty)$ we again obtain the linear inequality $2x < 4$ which is true for $x < 2$. Therefore, the interval $(1; 2)$ also belongs to the set of solutions. Collecting all the results obtained we come to the conclusion

that the inequality is satisfied by all the values of the variable from the interval $(-2; 2)$ and only those values.

We can obtain the same result from visual and at the same time strict geometrical considerations. Figure 35 shows the graphs of the functions

$$y = f(x) = |x - 1| + |x + 1| \quad \text{and} \quad y = 4.$$

On the interval $(-2; 2)$ the graph of the function $y = f(x)$ is located below the graph of the function $y = 4$ and this means that the inequality $f(x) < 4$ is valid.

Answer: $(-2; 2)$. ▲

4.5. Inequalities with Parameters

To solve an inequality with one or several parameters is, as a rule, more difficult than to solve an inequality in which there are no parameters.

The inequality

$$\sqrt{a+x} + \sqrt{a-x} > a,$$

for instance, which contains a parameter a , requires more effort for its solution than the inequality

$$\sqrt{1+x} + \sqrt{1-x} > 1.$$

What does it mean to solve the first of these inequalities? This essentially means to solve not one inequality but a whole class, a whole set of inequalities which result upon assigning concrete numerical values to the parameter a . As to the second inequality, it is a special case of the first, since it is obtained from the first inequality at the value $a = 1$.

Thus, to solve an inequality containing parameters means to determine at what values of the parameters the inequality has solutions and to find *all* the solutions for *all* such values of the parameters.

It should be remembered that an inequality with parameters must be considered for *all* the values of the parameters. If *at least one* value of any parameter is neglected, the solution of the problem cannot be considered complete.

Example 1. Solve the linear inequality

$$x - 2 \frac{a-1}{a} \leq \frac{2}{3a}(x+1).$$

$\triangle$ It should be noted at once that for $a = 0$ the inequality has no solutions since both sides of the inequality lose sense.

We transform the inequality as follows:

$$\left(1 - \frac{2}{3a}\right) x \leq \frac{2}{3a} + 2 - \frac{2}{a}, \quad x \left(1 - \frac{2}{3a}\right) \leq 2 \left(1 - \frac{2}{3a}\right).$$

If $1 - \frac{2}{3a} > 0$, then $x \leq 2$. Solving the inequality

$$1 - \frac{2}{3a} > 0,$$

we get $a < 0$ or $a > 2/3$. Thus, if $a < 0$ or $a > 2/3$, then $x \leq 2$.

If

$$1 - \frac{2}{3a} < 0,$$

i.e. $0 < a < 2/3$, then $x \geq 2$. And, finally, if $a = 2/3$, then x is any number.

Answer: If $a < 0$, then $x \in (-\infty; 2]$,
 if $a = 0$, then there are no solutions,
 if $0 < a < 2/3$, then $x \in [2; +\infty)$,
 if $a = 2/3$, then $x \in (+\infty; +\infty)$,
 if $a > 2/3$, then $x \in (-\infty; 2]$. $\blacktriangle$

Example 2. Solve the inequality

$$|x - a| + |x + a| < b, \quad a \neq 0.$$

$\triangle$ A special case of this inequality, $a = 1$, $b = 4$, was considered in Example 3 in 4.4. To solve the given inequality with two parameters a and b , we shall make use of geometrical considerations. Constructed in Fig. 36 are the graphs of the functions

$$y = f(x) = |x - a| + |x + a| \quad \text{and} \quad y = b.$$

It is evident that at $b \leq 2|a|$ the straight line $y = b$ passes not higher than the horizontal segment of the curve $y = |x - a| + |x + a|$ and, consequently, in this case the inequality has no solutions (Fig. 36a). Now if $b > 2|a|$, then the straight line $y = b$ intersects the graph of the

function $y = f(x)$ at two points $(-b/2; b)$ and $(b/2; b)$ (Fig. 36b), and in this case the inequality is valid for $-b/2 < x < b/2$ since at these values of the variable the

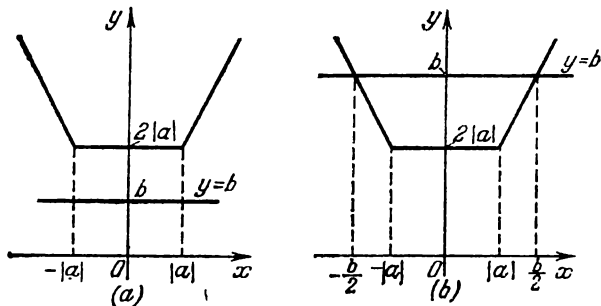


Fig. 36.

curve $y = |x + a| + |x - a|$ lies below the straight line $y = b$.

Answer: If $b \leq 2|a|$, then there are no solutions,

if $b > 2|a|$, then $x \in (-b/2; b/2)$. ▲

Example 3. Solve the inequality

$$\sqrt{a+x} + \sqrt{a-x} > a.$$

△ The left-hand side of the inequality is meaningful if and only if x and a satisfy the following system of inequalities:

$$\begin{cases} a+x \geq 0, \\ a-x \geq 0. \end{cases}$$

At $a < 0$ this system has evidently no solutions. If $a = 0$, the system possesses a unique solution $x = 0$. But at $a = 0$ the value of the variable $x = 0$ does not satisfy the inequality. If $a > 0$, the solutions of the system are all the values of the variable x belonging to the interval $[-a; a]$. Under the conditions $a > 0$ and $|x| \leq a$ we can, without violating equivalence, square the original inequality term-by-term and obtain

$$2a + 2\sqrt{a^2 - x^2} > a^2,$$

whence we get

$$2\sqrt{a^2 - x^2} > a^2 - 2a.$$

We have now to consider three cases:

1°. If $a^2 - 2a < 0$, i.e. $0 < a < 2$, then the inequality is valid for all $|x| \leq a$ since for $|x| \leq a$ the left-hand side of the inequality is nonnegative while the right-hand side is negative.

2°. If $a^2 - 2a = 0$, i.e. $a = 2$, then the inequality has the form $2\sqrt{4 - x^2} > 0$ and is satisfied at $|x| < 2$.

3°. If $a^2 - 2a > 0$, i.e. $a > 2$, then squaring both sides of the inequality we arrive at the equivalent inequality

$$4(a^2 - x^2) > a^4 - 4a^3 + 4a^2,$$

whose simplification results in

$$-4x^2 > a^3(a - 4), \text{ i.e. } x^2 < \frac{a^3(4 - a)}{4}.$$

Now we see that there are no solutions for $a \geq 4$. In the case when $2 < a < 4$, the solutions of the last inequality are all the values of x for which

$$|x| < \frac{a\sqrt{a(4-a)}}{2}.$$

Will all these values of x give solutions of the original inequality? This depends on whether the values of the expression $\frac{a\sqrt{a(4-a)}}{2}$, for $a \in (2; 4)$, exceed a or not.

Recall that we consider only those values of the variable x for which $|x| \leq a$. We shall prove that they will not exceed a , that is,

$$\frac{a\sqrt{a(4-a)}}{2} \leq a \text{ or } \frac{\sqrt{a(4-a)}}{2} \leq 1.$$

Squaring both sides of the inequality, we get

$$\frac{a(4-a)}{4} \leq 1 \text{ or } 4a - a^2 \leq 4,$$

i.e. $a^2 - 4a + 4 \geq 0$ and, consequently, we arrive at the true inequality $(a - 2)^2 \geq 0$. Reasoning in reverse order,

we ascertain the validity of the inequality

$$\frac{a \sqrt{a(4-a)}}{2} \leq a.$$

Answer: If $a \leq 0$, then there are no solutions,

if $0 < a < 2$, then $x \in [-a; a]$,

if $a = 2$, then $x \in (-2; 2)$,

if $2 < a < 4$, then $x \in \left(-\frac{a \sqrt{a(4-a)}}{2}; \frac{a \sqrt{a(4-a)}}{2} \right)$,

if $a \geq 4$, then there are no solutions. ▲

Besides the problems of the type we have considered, in which it is required to solve the inequality for all the values of the parameter, there are problems in which it is necessary to isolate, from all the values of the parameter, only those for which the inequality will possess some preassigned properties; for instance, it will be satisfied for any value of the variable, or will have no solutions at all, or else, will possess only one positive solution, and so on. Let us consider a problem of this type.

Example 4. At what values of the parameter a is the inequality

$$\frac{2-ax-x^2}{1-x+x^2} \leq 3$$

true for all the values of the variable?

△ Since $1 - x + x^2 > 0$ for all x , the given inequality is equivalent to the quadratic inequality

$$2 - ax - x^2 \leq 3 - 3x + 3x^2 \text{ or } 4x^2 + (a - 3)x + 1 \geq 0.$$

The last inequality is valid for all the values of the variable under the condition $(a - 3)^2 - 16 \leq 0$, i.e. $|a - 3| \leq 4$, whence $-4 \leq a - 3 \leq 4$ and, consequently, $-1 \leq a \leq 7$.

Answer: $a \in [-1; 7]$. ▲

4.6. Proving Inequalities

In previous sections we considered problems in which it was required to solve an inequality, that is, to find the set of its solutions. We often come across another kind of problems connected with inequalities in which, besides an

inequality, a certain set of values of the variable is given and it is required to prove that all its elements belong to the set of solutions of the given inequality. Problems of this kind are customarily called the problems concerned with proving inequalities.

If in a problem of this kind the set of values of the variable is not specified, this means that it is required to establish the validity of the inequality for all real values of the variable.

A great variety of methods and techniques are used to solve such problems. In the case when the variable or the parameter appearing in the inequality belongs to the set of natural numbers use can be successfully made of the principle of mathematical induction discussed in Chapter 2.

Let us now consider some other methods of proving inequalities. Sometimes it is possible to prove an inequality by reducing it, by means of equivalent transformations, to an obvious inequality.

Example 1. Prove the validity of the inequality

$$10x^3 - 9x^2 + 9x + \frac{1}{10} > 0$$

for all positive values of the variable.

△ We transform the inequality as follows:

$$x(10x^2 - 9x + 9) + \frac{1}{10} > 0,$$

$$x \left(\left(\sqrt{10}x - \frac{9}{2\sqrt{10}} \right)^2 + 9 - \frac{81}{40} \right) + \frac{1}{10} > 0,$$

$$x \left(\left(\sqrt{10}x - \frac{9}{2\sqrt{10}} \right)^2 + \frac{279}{40} \right) + \frac{1}{10} > 0.$$

The validity of the inequality obtained for all $x > 0$ is obvious. But the last inequality is equivalent to the original one which has thus been proved. ▲

Example 2. Prove the inequality

$$x^2 + y^2 - xy - x - y + 1 \geq 0.$$

△ We transform the inequality as follows:

$$\begin{aligned} 2x^2 + 2y^2 - 2xy - 2x - 2y + 2 &\geq 0, \\ x^2 - 2xy + y^2 + x^2 - 2x + 1 + y^2 - 2y + 1 &\geq 0, \\ (x - y)^2 + (x - 1)^2 + (y - 1)^2 &\geq 0, \end{aligned}$$

and arrive at an obvious inequality. The transformations carried out do not violate the equivalence of the inequalities and, therefore, the given inequality is valid for any values of the variables x and y . ▲

In the example that will follow we cannot reduce the given inequality to an obvious one for all values of the variable at once. However, we can first divide the entire number axis into three subintervals and then easily establish the validity of the inequality for each of the three subintervals.

Example 3. Prove the inequality

$$x^8 - x^5 + x^2 - x + 1 > 0.$$

△ The inequality is true on the interval $(-\infty; 0]$ since the first four terms on the left-hand side are nonnegative. On the interval $(0; 1)$ the validity of the inequality becomes obvious upon the following grouping of the terms:

$$x^8 + x^2(1 - x^3) + (1 - x) > 0.$$

Now if $x \in [1; +\infty)$, then, having rewritten the inequality in the form

$$x^5(x^3 - 1) + x(x - 1) + 1 > 0,$$

we ascertain its validity.

We have proved the inequality for all real values of the variable. ▲

The two examples that will now follow illustrate the application of the method of estimation in proving inequalities.

Example 4. Prove the inequality

$$\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} \geq \frac{1}{2}, \quad n \in \mathbb{N}.$$

△ The left-hand side of the inequality contains n terms each of which is not smaller than the last term. Therefore,

we minorize (estimate from below) the left-hand side of the inequality:

$$\frac{1}{n+1} + \dots + \frac{1}{2n} \geq \frac{1}{2n} + \dots + \frac{1}{2n} = n \cdot \frac{1}{2n} = \frac{1}{2}. \quad \blacktriangle$$

Example 5. Prove the inequality

$$\frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} < 1 - \frac{1}{n}, \quad n \in \mathbb{N}, \quad n > 1.$$

$\triangle$ Each term on the left-hand side of the inequality admits of the following estimation from above:

$$\frac{1}{k^2} < \frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k},$$

and, consequently, the left-hand side of the inequality can be estimated as follows:

$$\begin{aligned} \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} &< \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots \\ &\quad + \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{n}. \end{aligned}$$

And that is what we had to prove. $\blacktriangle$

Let us consider one more widely used method of proving inequalities, which consists in employing certain familiar inequalities. In Chapter 2 we proved a remarkable inequality involving an arithmetic mean and a geometric mean

$$\begin{aligned} \frac{x_1 + x_2 + \dots + x_n}{n} &\geq \sqrt[n]{x_1 x_2 \dots x_n}, \quad x_i > 0 \\ (i = 1, 2, \dots, n), \end{aligned}$$

and emphasized that the inequality is strict if and only if not all x_i are equal to each other. This inequality, which is usually connected with Cauchy, a French mathematician of the 19th century, turns out to be very useful in proving many inequalities.

Example 6. Prove the inequality

$$x^4 + y^4 + z^4 + t^4 \geq 4xyzt.$$

$\triangle$ Let us set

$$x_1 = x^4, \quad x_2 = y^4, \quad x_3 = z^4, \quad x_4 = t^4 \quad \text{and} \quad n = 4$$

in the Cauchy inequality involving an arithmetic mean and a geometric mean. Then we obtain

$$\frac{x^4 + y^4 + z^4 + t^4}{4} \geq \sqrt[4]{x^4 y^4 z^4 t^4} = |xyzt| \geq xyzt.$$

And that is what we had to prove. $\blacktriangle$

Example 7. Prove that the sequence

$$a_n = \left(1 + \frac{1}{n}\right)^n$$

increases monotonically, that is, prove the validity of the inequality

$$a_{n+1} = \left(1 + \frac{1}{n+1}\right)^{n+1} > \left(1 + \frac{1}{n}\right)^n = a_n.$$

$\triangle$ Let us consider $n + 1$ positive numbers

$$x_1 = 1, \quad x_2 = 1 + \frac{1}{n}, \quad x_3 = 1 + \frac{1}{n}, \quad \dots, \quad x_{n+1} = 1 + \frac{1}{n},$$

and apply to them Cauchy's inequality.

We get

$$\frac{1+n\left(1+\frac{1}{n}\right)}{n+1} > \sqrt[n+1]{\left(1+\frac{1}{n}\right)^n},$$

i.e.

$$1 + \frac{1}{n+1} > \sqrt[n+1]{\left(1 + \frac{1}{n}\right)^n},$$

or

$$\left(1 + \frac{1}{n+1}\right)^{n+1} > \left(1 + \frac{1}{n}\right)^n. \quad \blacktriangle$$

Example 8. Prove that the resistance of n conductors connected in series is not less than n^2 times greater than the resistance of the same conductors connected in parallel.

$\triangle$ Let us designate the resistance of the given conductors as $r_1, r_2, \dots, r_n$. Assume that R is the resistance of the conductors connected in series and R_1 is the resistance of the conductors connected in parallel. Then we have

$$R = r_1 + r_2 + \dots + r_n;$$

$$\frac{1}{R_1} = \frac{1}{r_1} + \frac{1}{r_2} + \dots + \frac{1}{r_n}$$

and, consequently,

$$\frac{R}{R_1} = (r_1 + r_2 + \dots + r_n) \left(\frac{1}{r_1} + \frac{1}{r_2} + \dots + \frac{1}{r_n} \right),$$

and in accordance with the theorem on the arithmetic mean we can write

$$\begin{aligned} r_1 + r_2 + \dots + r_n &\geq n \sqrt[n]{r_1 r_2 \dots r_n}, \\ \frac{1}{r_1} + \frac{1}{r_2} + \dots + \frac{1}{r_n} &\geq n \sqrt[n]{\frac{1}{r_1} \frac{1}{r_2} \dots \frac{1}{r_n}}, \end{aligned}$$

whence we get $\frac{R}{R_1} \geq n^2$, and this completes the proof. $\blacktriangle$

4.7. Application of Inequalities to Problems on the Greatest and Least Values

Problems on maximizing and minimizing functions are of especial importance in mathematics, physics and other sciences. A powerful apparatus (differential calculus) has been elaborated in mathematics to solve problems of this kind. Application of differential calculus in seeking the greatest and least values of functions will be considered in 6.8. However, it sometimes turns out to be quicker and simpler to solve such problems by elementary means and the solution proves to be more elegant. If that is the case, the inequalities discussed in 4.6 can be conveniently used.

Two significant statements follow from Cauchy's inequality on the arithmetic mean and geometric mean.

1. If the sum of arbitrary positive numbers $x_1, x_2, \dots, x_n$ is equal to s , then their product $P = x_1 x_2 \dots x_n$ attains the *greatest* value equal to $(s/n)^n$, all the numbers being equal.

2. If the product of arbitrary positive numbers $x_1, x_2, \dots, x_n$ is equal to P , then their sum $s = x_1 + \dots + x_n$ assumes the *least* value, equal to $n \sqrt[n]{P}$, all the numbers being equal.

Example 1. From a round log of radius R saw a rectangular beam so that the amount of waste is at a minimum (Fig. 37).

$\triangle$ The amount of waste is defined by the area of the hatched part of the cross-section of the log and, therefore,

the problem can be reduced to the following: in a circle of radius R inscribe a rectangle of the greatest area. If x and y are the sides of the rectangle and S is its area, then $S = xy$. If the rectangle is inscribed in a circle, then $x^2 + y^2 = 4R^2$. Consequently, $S = x\sqrt{4R^2 - x^2}$. Note that S attains the greatest value in the case when $S^2 = x^2(4R^2 - x^2)$ is of the greatest value.

But the sum of the factors x^2 and $4R^2 - x^2$ is constant and is equal to $4R^2$; therefore, the greatest value of their

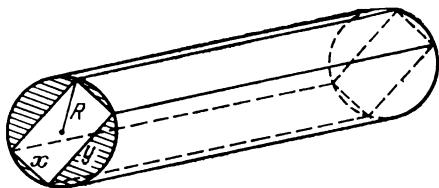


Fig. 37.

product is equal to $(4R^2/2)^2$ and the greatest value of S is equal to $2R^2$, it being attained when $x^2 = 4R^2 - x^2$, that is when $x = \sqrt{2}R$, but then $y = \sqrt{2}R$. It follows that the amount of waste is at a minimum when the cross-section of the beam is a square. ▲

Example 2. Find the least value of the function

$$f(x) = x + \frac{a}{x}, \quad a > 0,$$

on the interval $(0; +\infty)$.

△ We set $x_1 = x$, $x_2 = a/x$. The product $x_1x_2 = a$, i.e. is constant and, hence, the sum $x_1 + x_2$ assumes the least value, equal to $2\sqrt{a}$, for $x = \sqrt{a}$.

Note that this yields a useful inequality

$$\left| x + \frac{a}{x} \right| \geq 2\sqrt{a}, \quad a > 0,$$

the equality being attained at $x = \sqrt{a}$. ▲

Example 3. Find the least value of the rational function

$$f(x) = \frac{x^4 + x^2 + 4}{x}$$

on the interval $(0; +\infty)$.

△ Let us represent $f(x)$ as the sum of six terms

$$f(x) = x^3 + x + \frac{1}{x} + \frac{1}{x} + \frac{1}{x} + \frac{1}{x},$$

whose product is unity. The sum of these terms attains its least value when all the terms are equal, that is, under the condition $x^3 = x = 1/x$, whence it follows that $x = 1$. Consequently, the least value of $f(x)$ on the interval $(0; +\infty)$ is 6. ▲

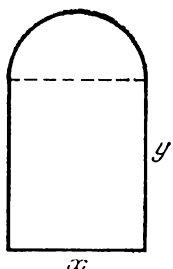


Fig. 38.

It should be pointed out in conclusion that among the problems on the greatest and least values of functions which can be solved by elementary means, without resort to a derivative, students most often have to solve problems requiring an applica-

tion of the following extremum property of a quadratic function. It can be seen from the identity

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a}, \quad a \neq 0$$

that the function $f(x) = ax^2 + bx + c$ has the least (if $a > 0$) and the greatest (if $a < 0$) value equal to $c - b^2/4a$ for $x = -b/2a$.

Example 4. A window is a rectangle complemented by a semicircle (Fig. 38). The perimeter of the window is P . At what ratio of the sides of the rectangle will the window let in the most light?

△ Let us designate the lengths of the sides of the rectangle as x and y ; then the area of the window is $S = xy + \frac{\pi}{2} \left(\frac{x}{2} \right)^2$, but x and y are related as

$$x + 2y + \pi \frac{x}{2} = P.$$

Consequently,

$$S = x \left(\frac{P}{2} - \frac{x}{2} - \frac{\pi}{4} x \right) + \pi \frac{x^2}{8}$$

or

$$S = - \left(\frac{1}{2} + \frac{\pi}{8} \right) x^2 + \frac{P}{2} x.$$

The function S attains the greatest value for $x = \frac{P/2}{1 + \pi/4} = \frac{2P}{4 + \pi}$. It is easy to calculate that $y = \frac{P}{4 + \pi}$.

Thus we see that the height of the rectangular part of the window must be half its width. $\blacktriangle$

PROBLEMS OF SECTION I

1. Are the inequalities $\frac{x}{3-x} > \frac{x-7}{2}$ and $2x > (x-7)(3-x)$ equivalent on the interval $[4; 8]$?

Test inequalities 2-4 for equivalence on the set of all real numbers.

2. $x - 1 \leq \sqrt{x}$ and $(x - 1)^2 \leq x$.

3. $x + 1 \leq \sqrt{x}$ and $(x + 1)^2 \leq x$.

4. $\sqrt{x^3 + x - 2} < x$ and $x^3 + x - 2 < x^2$.

Solve inequalities 5-6.

5. $x > \frac{15}{x+2}$.

6. $\frac{(2-x^2)(x-3)^3}{(x+1)(x^2-3x-4)} \geq 0$.

7. Find the domain of definition of the function

$$f(x) = \sqrt{\frac{2x+1}{x-2} - \frac{x+4}{2x+5}}.$$

Solve inequalities 8 through 12.

8. $\sqrt{x^2 - 40x + 39} \leq x - 1$.

9. $\sqrt{3x-4} + \sqrt{2x-13} > \sqrt{13-2x}$.

10. $\sqrt{45x^2 - 30x + 1} < 7 + 6x - 9x^2$.

11. $\left| x - \frac{4}{x} - 2 \right| \geq 1$.

12. $\frac{2a+1}{ax-3x-2a+6} \geq \frac{x}{x-2}$.

13. Prove the inequality

$$x(1+y) + y(1+z) + z(1+x) \geq 6\sqrt{xyz}$$

for all nonnegative values of the variables x, y, z .

14. Prove the inequality

$$\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{3n+1} < 2, \quad n \in \mathbb{N}.$$

15. Prove the inequality

$$2^n - 1 \geq n2^{(n-1)/2}, \quad n \in \mathbb{N}.$$

16. At what values of a does the system of inequalities

$$\begin{cases} ax - 1 \leq 0, \\ x - 4a \geq 0 \end{cases}$$

possess at least one solution?

17. At what values of a do all the zeros of the function

$$f(x) = (a - 2)x^2 + 2ax + a + 3$$

lie on the interval $(-2; 1)$?

18. Determine at what values of the variable the polynomial

$$P_4(x) = x(x + 1)(x + 2)(x + 3)$$

assumes the least value. What is it equal to?

19. A sportsground 0.9 hectare in area and rectangular in shape, must be bounded by a wooden fence to the north and the south and by a wire fence to the east and the west. One metre of wooden fencing costs five rubles, one metre of wire fencing two rubles. The sum of 1200 rubles is allocated for the construction of the whole fence. Is this sum sufficient?

20. Two straight roads pass through point B , one to point A ,

the other to point C , with $\widehat{ABC} = 120^\circ$. Two cars start out simultaneously, from A to B and from B to C . Find the ratio of their speeds if it is known that the least distance between the cars was at the moment when the car starting from point A had covered one-seventh of its route.

21. A rectangular open box is being made from a square tin sheet with side a , by cutting off congruent squares at the angles and bending the edges. What should be the side length of the cut-off squares for the resulting box to have the greatest capacity?

PROBLEMS OF SECTION II

Test inequalities 1 through 5 for equivalence on the set of all real numbers.

1. $(x + 7)(2x + 1) > (x + 7)^2$ and $2x + 1 > x + 7$.

2. $x^2 + x + \sqrt{x} < 2 + \sqrt{x}$ and $x^2 + x < 2$.

3. $\sqrt{x - 7} \geq x$ and $x - 7 \geq x^2$.

4. $1 - \sqrt{2x^2} > 0$ and $1 - \sqrt{2}x > 0$.

5. $\sqrt{x - 1} \geq x$ and $x \geq x^2 + 5$.

Solve inequalities 6 through 18.

6. $(3x^2 - 13x + 4)(4x^2 + 12x + 9) < 0$.

7. $(2x - 5)(x^2 - 4)(x^3 + 8) \leq 0$.

8. $\frac{x^2-7x+12}{x^2-10x+20} < 0$. 9. $\frac{2x-5}{x^2-6x-7} < \frac{1}{x-3}$.
10. $\frac{x^2-5x+6}{x^2+5x+6} \geq \frac{x+1}{x}$. 11. $\frac{(x^2+3x-4)(x^2-16)}{x(x+1)} < 0$.
12. $\frac{8+4x}{4x+x^2} \leq \frac{2}{x} + \frac{3}{4+x}$. 13. $\frac{2}{x-2} - \frac{2}{x+1} < \frac{3}{(x-2)^2}$.
14. $\frac{x+2}{x^3-x^2} < \frac{2x+1}{x^3+3x^2}$. 15. $\frac{1}{x^2-4} - \frac{1}{(x+2)^2} \geq \frac{1}{x^2} - \frac{1}{2x}$.
16. $\frac{2}{x^2-x+1} - \frac{1}{x+1} \geq \frac{2x-1}{x^2+1}$.
17. $\frac{2}{2-x} + \frac{3}{2+x} \geq \frac{4x}{4-x^2}$.
18. $\frac{2}{x} + \frac{1}{x-1} - \frac{1}{x+1} \leq \frac{2}{x^2} + \frac{2}{(x+1)^2}$.

Find the domain of definition of the functions in problems 19-20.

19. $f(x) = \sqrt{\frac{3-3x}{15-2x-x^2}} - 1$.
20. $f(x) = \frac{\sqrt{3-x}}{\sqrt{x - \frac{x^2+2x-2}{3+2x-x^2}}}$.

Solve inequalities 21 through 40.

21. $\sqrt{2x+5} > x+1$. 22. $\sqrt{x^2-4x} > x-3$.
23. $\sqrt{x^2-x-12} > x-1$. 24. $\sqrt{x^2-3x-4} > x-2$.
25. $\sqrt{x^2-3x-10} < 8-x$. 26. $\sqrt{x^2-x-2} > 2x+3$.
27. $\frac{x^2-3|x-3|}{x^2-5x+6} \geq 1$. 28. $\sqrt{(x(x+3))^2} \geq 2-x^2$.
29. $20 \sqrt{(x^2-x)^2} < 1$. 30. $\sqrt{\left(\frac{x+1}{3-2x}\right)^2} > 1$.
31. $\frac{2x^2}{1-\sqrt{1-x^2}} \leq 3$. 32. $\sqrt{\frac{x^3+8}{x}} > x-2$.
33. $\frac{1-3\sqrt{16-x^2}}{x} \leq 1$. 34. $\sqrt{x+6} > \sqrt{x+1} + \sqrt{2x-4}$.
35. $\sqrt{2-x} \leq \sqrt{x+17} + \sqrt{2x-4}$. 36. $||x|-1| \geq \sqrt{x^2-1}$.
37. $\sqrt{\frac{1}{x^2} - \frac{1}{4}} > \frac{1}{x} - \frac{1}{4}$. 38. $\sqrt{\frac{1}{x^2} - \frac{3}{4}} < \frac{1}{x} - \frac{1}{2}$.

$$39. \sqrt{3x^2+5x+7} - \sqrt{3x^2+5x+2} > 1.$$

Hint. Make a change of variable $3x^2+5x+\frac{9}{2}=t$.

$$40. \sqrt{x^3+233} + \sqrt{x^3-49} - \sqrt{128-x} \leq \frac{56}{x} + 5.$$

Solve the inequalities with parameters in problems 41 through 46.

$$41. \frac{a^2x+1}{2} - \frac{a^2x+3}{3} < \frac{a+9x}{6}.$$

$$42. \frac{ax+1}{3} - \frac{x-4a}{2} \geq \frac{a^2}{6}. \quad 43. \frac{2a+1}{(a-3)(x-2)} \geq \frac{x}{x-2}.$$

$$44. ax > 1/x. \quad 45. x + \sqrt{a-x} > 0, \quad a \geq 0.$$

$$46. |x-a| + |x| + |x+a| \leq b.$$

Hint. It is useful to construct the graphs of the functions

$$f(x) = |x-a| + |x| + |x+a| \text{ and } g(x) = b.$$

47. At what values of the parameter a is the inequality

$$\frac{ax^2+3x+4}{x^2+2x+2} < 5$$

satisfied at all the values of the variable?

48. At what values of the parameter a is the inequality

$$\left| \frac{x^2+ax+1}{x^2+x+1} \right| < 3$$

satisfied for all the values of the variable?

49. At what values of the parameter a is the inequality

$$\frac{x^2+3x+a}{x^2+x+1} < 2$$

valid for all the values of the variable except for one value?

50. At what values of the parameter a is the inequality

$$ax^3 + 4x > 1 - 3a$$

valid for all the positive values of the variable?

51. At what values of a are the roots of the equation

$$(a-3)x^2 - 2ax + 6a = 0$$

positive?

52. At what values of a is one of the roots of the equation

$$(2a+1)x^2 - ax + a - 2 = 0$$

greater than unity and the other smaller than unity?

Prove inequalities 53-55:

$$53. \frac{x^3+3}{\sqrt{x^3+2}} > 2.$$

54. $(x + y + z)^3 \geq 27xyz$, if $x > 0$, $y > 0$, $z > 0$.

55. $(x^3 + x^2 + x + 1)^2 \geq 16x^3$; $x \in [0; +\infty)$.

56. Prove the inequality

$$\frac{xy}{z} + \frac{yz}{x} + \frac{zx}{y} \geq x + y + z$$

for all positive values of the variables x , y , z .

57. Prove the inequality

$$x^2 + y^2 + z^2 \geq 12$$

for all the values of the variables satisfying the condition $x + y + z = 6$.

58. Prove the inequality

$$1 + \sqrt{2} + \sqrt{3} + \dots + \sqrt{n} \leq \frac{n^2x + nx + 2n}{4\sqrt{x}}$$

for all natural values of n and all positive values of x .

59. Find the greatest and least values of the sum $2x^2 + 3y^2$ under the conditions $x + y = 2$, $x \geq 0$, $y \geq 0$.

60. What is the greatest area of a rectangle the sum of whose three sides is equal to one hundred metres?

61. A plane figure consists of a rectangle and an equilateral triangle constructed on one side of the rectangle. How large is the area of that figure if the perimeter is specified?

62. The scales of a salesman got out of order (the arms of the scales turned out to be uneven). The salesman served a customer weighing twice: first time he put a one-kilogram weight on one pan of the scales and some goods on the other, and the second time he changed the places of the weight and the goods. How much did he give the customer: two kilograms, more than two kilograms, or less than two kilograms of goods?

63. A reservoir is shaped like a regular triangle. A ferry-boat goes from point A to point B both located on the bank. A cyclist who has to get from A to B can use the ferry-boat or to ride along the bank. What is the least ratio between the speed of the cyclist and the speed of the boat at which the use of the boat does not give any gain in time, the locations of the points A and B being arbitrary?

64. Two roads intersect at right angles. Two cars move towards the crossing: one uses the first road and travels at a speed of 60 km per hour and the other uses the second road and travels at 80 km per hour. At noon exactly both cars were 10 km from the crossing. At what moment will the distance between the cars be at its least? Where will the cars be at that moment?

65. Two tourists set off from point A to point B . The first tourist travelled the first half of the total time spent on the trip at a speed v_1 and the second half at the speed v_2 ; the second tourist covered the first half of his route at the speed v_1 and the second half at the speed v_2 . Which of them spent less time to cover the distance between A and B ?

66. The distance between points A and B is 100 km. Two cyclists start simultaneously from A to B . The speed of the first cyclist is

10 km/h greater than that of the other. The first cyclist stops on his way to B and rests for 50 minutes but is still the first to arrive at B . Within what speed range must the first cyclist travel.

67. A boat travels along the river from point A to point B and back again. The distance between A and B is a , the velocity of the river flow is v . What must be the speed of the boat for the time spent on the whole trip to be less than t ?

68. At what values of x does the rational function $y = \frac{x^4 + 1}{x^2 + 1}$ assume the least value? What is it equal to?

69. Twenty-four class 2 players take part in a chess tournament. They must get not less than $14\frac{1}{2}$ points to gain first class standard.

What is the greatest number of the chess-players who can reach first class standard? (In chess tournaments the winner gets one point; if the game ends in a draw, each of the participants gets half a point.)

70. A factory must send 1100 articles to a customer. The articles are packed into three kinds of boxes. A box of the first kind can hold 70 articles, a box of the second kind, 40 articles, and a box of the third kind, 25 articles. The cost of the carriage of a box of the first kind is 20 rubles, that of a box of the second kind is 10 rubles, and that of a box of the third kind 7 rubles. What boxes should be used by the factory to minimize the cost of carriage? (Underloading is prohibited.)

Chapter 5

Sequences. Limit of a Sequence. Limit of a Function. The Derivative

5.1. Infinite Sequences. Bounded and Unbounded Sequences

An infinite *number sequence* is a number function defined on the set $\mathbb{N}$ of natural numbers. The argument of such a function is usually designated as n and the function itself by a letter with a subscript: a_n , b_n , x_n and so on. Thus it follows that an infinite sequence is specified if a rule is indicated in accordance with which every natural number n is associated with a definite number a_n .

It is customary to write an infinite sequence a_n , $n \in \mathbb{N}$, as

$$a_1, a_2, \dots, a_n, \dots$$

or, concisely, as (a_n) . The numbers a_1 , a_2 , a_n are called the *elements* or *terms* of the sequence, a_1 being the first (initial) term, a_2 , the second, a_n , the n th term of the sequence.

Let us consider some examples.

(a) We put a number $(n-1)/(n+1)$ into correspondence with every natural n and thus define the sequence

$$0, \frac{1}{3}, \frac{1}{2}, \frac{3}{5}, \dots, \frac{n-1}{n+1}, \dots$$

with the n th term $a_n = (n-1)/(n+1)$.

(b) We associate every natural number n with a number equal to the n th decimal digit of the number $8/33$ in decimal notation. This rule of correspondence defines a sequence in which $a_1 = 2$, $a_2 = 4$, $a_3 = 2$, $a_4 = 4$. It is easy to show that the n th term of this sequence can be written in the form

$$a_n = 3 + (-1)^n.$$

We can use various means to define a sequence. For instance, we can indicate a formula associating the values of the n th term of the sequence with its number for any

natural n (such is the sequence in (a)). The rule of correspondence between the number of a term and the value of that term can be given verbally, as, for instance, in (b). Not infrequently a sequence is defined recursively, by giving several first terms of the sequence and a formula, known as a *recurrence relation*, relating a term of the sequence to its predecessor, like this, for example: $a_1 = 1$, $a_n = 2a_{n-1} + 3$ for $n \geq 2$.

Example 1. Prove that if $a_1 = 1$ and $a_n = 2a_{n-1} + 3$ for $n \geq 2$, then the n th term of the sequence has the form

$$a_n = 2^{n+1} - 3. \quad (1)$$

$\triangle$ We use the method of mathematical induction. Formula (1) is true for $n = 1$. We assume that it is also true for $n = k \geq 1$, i.e. $a_k = 2^{k+1} - 3$. Substituting this expression for the k th term into the recurrence relation $a_{k+1} = 2a_k + 3$, we get $a_{k+1} = 2(2^{k+1} - 3) + 3 = 2^{k+2} - 3$. But this is precisely formula (1) for $n = k + 1$. In accordance with the principle of mathematical induction, formula (1) is valid for any natural n . $\blacktriangle$

An important special case of sequences is that of arithmetic and geometric progressions. They will be treated in 5.4 and 5.5.

A sequence (a_n) is said to be *bounded* if there are two numbers, a and b , such that the inequalities

$$a \leq a_n \leq b \quad (2)$$

are satisfied for all n . In this case we say that the number a is the lower bound of the sequence and the number b is its upper bound.

The sequence $((n-1)/(n+1))$ is bounded since for all n there hold the inequalities $0 \leq (n-1)/(n+1) \leq 1$.

The sequence $((-1)^n)$ is also bounded since for any natural n there hold the inequalities $-1 \leq (-1)^n \leq 1$.

We can show that the given definition is equivalent to the following: the sequence (a_n) is said to be *bounded* if there exists a number M such that for all n there holds the inequality

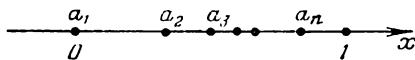
$$|a_n| \leq M.$$

Not every sequence is bounded.

Example 2. Prove that the sequence (n) is not bounded.

△ Let us assume the contrary, that is, suppose that the sequence (n) is bounded. This means that there exists a number M such that for all n there holds the inequality $|n| \leq M$. However, for a natural number $n = [M] + 1$, for instance (here $[M]$ is the *integral part* of the number M), this inequality does not hold. Consequently, our assumption is incorrect, that is, the sequence (n) is not bounded. ▲

A sequence which is not bounded is called *unbounded*. We shall formulate the definition of an unbounded sequence



Fin. 39.

by negating the definition of a bounded sequence: (a_n) is an unbounded sequence if for any number M there is a number n such that $|a_n| > M$.

If we represent the terms of a sequence by points of the coordinate axis, then all the terms of a bounded sequence will lie on some interval. For example, all the terms of the sequence $a_n = (n-1)/(n+1)$, $n \in \mathbb{N}$, lie on the interval $[0; 1]$ (Fig. 39).]

The terms of an unbounded sequence can be found outside any interval.

5.2. Limit of a Sequence.

Theorems on Convergent Sequences

The number a is called a *limit* of the sequence (a_n) if for any positive ε there is a natural number N such that the inequality

$$|a_n - a| < \varepsilon$$

is satisfied for all $n > N$.

The fact that the number a is a limit of the sequence (a_n) is written as

$$\lim_{n \rightarrow \infty} a_n = a \quad \text{or} \quad a_n \rightarrow a \quad \text{as} \quad n \rightarrow \infty.$$

Example 1. Prove that $\lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0$.

△ The inequality $\left| \frac{(-1)^n}{n} - 0 \right| < \varepsilon$, i.e. $\frac{1}{n} < \varepsilon$, is satisfied for all $n > 1/\varepsilon$. Thus, for any $\varepsilon > 0$ we can take, say, $N = [1/\varepsilon] + 1$. If we take into account that $N > 1/\varepsilon$, then for any $n > N$, we shall have $\left| \frac{(-1)^n}{n} - 0 \right| < \varepsilon$. And this precisely means that $(-1)^n/n \rightarrow 0$ as $n \rightarrow \infty$. ▲

Note that the inequality $|a_n - a| < \varepsilon$ is equivalent to the inequality

$$- \varepsilon < a_n - a < \varepsilon \quad \text{or} \quad a - \varepsilon < a_n < a + \varepsilon.$$

This means that the number a_n belongs to the interval $(a - \varepsilon; a + \varepsilon)$. Such an interval is called the ε -neighbourhood of the point a (Fig. 40).

We can paraphrase the definition of a limit as follows, resorting to a more visual geometric interpretation: the

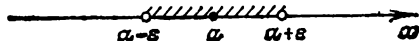


Fig. 40.

number a is called a *limit of the sequence* (a_n) if any ε -neighbourhood of the number a contains all the terms of the sequence, except, maybe, their finite number. In fact, if $a_n \rightarrow a$, as $n \rightarrow \infty$, then for every $\varepsilon > 0$ there is a number N such that all the terms of the sequence with the numbers $n > N$ lie in the ε -neighbourhood of the number a and, hence, only the first N terms of the sequence can lie outside that neighbourhood.

For example, for the sequence $((-1)^n/n)$, the ε -neighbourhood of the point zero, at $\varepsilon = 1/10$, includes all the terms of the sequence except for the first ten terms, at $\varepsilon = 1/100$, all the terms of the sequence except for the first hundred terms, since

$$-\frac{1}{10} < \frac{(-1)^n}{n} < \frac{1}{10} \quad \text{for } n \geq 11,$$

$$-\frac{1}{100} < \frac{(-1)^n}{n} < \frac{1}{100} \quad \text{for } n \geq 101.$$

Example 2. Suppose $\lim_{n \rightarrow \infty} a_n = a$, $a > 0$. Prove that, beginning with some number, all the terms of the sequence (a_n) are positive.

Δ Since $\lim_{n \rightarrow \infty} a_n = a > 0$, there is a natural number N , at $\varepsilon = a/2 > 0$, such that for $n > N$ there holds the inequality $|a_n - a| < a/2$. Let us rewrite this inequality as $-a/2 < a_n - a < a/2$. Then we shall get

$$a_n > a/2 > 0. \quad \blacktriangle$$

The geometrical meaning of the assertion we have proved is obvious: if the number a is positive, then there exists its

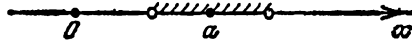


Fig. 41.

neighbourhood, containing no zero, and, according to the definition, this neighbourhood contains all the terms of the sequence, except, maybe, their finite number (Fig. 41).

Example 3. Assume $\lim_{n \rightarrow \infty} a_n = a$ and $a_n \geq 0$. Prove that

$$\lim_{n \rightarrow \infty} \sqrt{a_n} = \sqrt{a}.$$

Δ (a) $a = 0$. By the definition of a limit, at any $\varepsilon > 0$, there is an N , dependent on ε^2 , such that for $n > N$ there holds the inequality $|a_n| < \varepsilon^2$. Since by the hypothesis $a_n \geq 0$, it follows that $0 \leq a_n < \varepsilon^2$. But then, for the same $n > N$ there holds the inequality $|\sqrt{a_n} - 0| = \sqrt{a_n} < \varepsilon$.

(b) $a > 0$. At any $\varepsilon > 0$ there is an N such that for $n > N$ there holds the inequality $|a_n - a| < \varepsilon \sqrt{a}$. Then, for the same $n > N$, there holds the inequality

$$\begin{aligned} |\sqrt{a_n} - \sqrt{a}| &= \left| \frac{(\sqrt{a_n} - \sqrt{a})(\sqrt{a_n} + \sqrt{a})}{\sqrt{a_n} + \sqrt{a}} \right| \\ &= \frac{|a_n - a|}{\sqrt{a_n} + \sqrt{a}} < \frac{|a_n - a|}{\sqrt{a}} < \varepsilon. \end{aligned}$$

In both cases, we have proved that $\lim_{n \rightarrow \infty} \sqrt{a_n} = \sqrt{a}$. $\blacktriangle$

Example 4. Prove that $\lim_{n \rightarrow \infty} q^n = 0$ if $|q| < 1$.

Δ For $q = 0$ the assertion is obvious; now if $q \neq 0$, then $\frac{1}{|q|} > 1$. Let us designate $\alpha = \frac{1}{|q|} - 1$ ($\alpha > 0$), then $|1/q| = 1 + \alpha$ and $|1/q|^n = (1 + \alpha)^n$. Using Bernoulli's inequality*, we get $|1/q|^n = (1 + \alpha)^n > 1 + \alpha n > \alpha n$, which yields

$$|q|^n < 1/\alpha n. \quad (1)$$

We take $N = [1/\alpha\varepsilon] + 1$ for any positive ε . Since $N > 1/\alpha\varepsilon$, equation (1) yields $|q|^n < 1/\alpha n < 1/\alpha N < \varepsilon$ for any $n > N$. This means that $\lim_{n \rightarrow \infty} |q|^n = 0$. $\blacktriangle$

A sequence which has a limit is called *convergent*, that having no limit is called *divergent*.

Example 5. Prove that the sequence with the n th term $a_n = 1 + (-1)^n$ is divergent.

Δ Let us assume the contrary, that is, suppose that there exists a number a which is a limit of that sequence. Then, by the definition of a limit, at $\varepsilon = 1/2$ there is a natural number N such that for any $n > N$ there holds the inequality $|a_n - a| < 1/2$.

It follows that if $n_1 > N$ and $n_2 > N$, then

$$\begin{aligned} |a_{n_1} - a_{n_2}| &= |(a_{n_1} - a) + (a - a_{n_2})| \\ &\leq |a_{n_1} - a| + |a_{n_2} - a| < 1. \end{aligned} \quad (2)$$

However, if we take $n_1 = 2N$ ($n_1 > N$) and $n_2 = 2N + 1$ ($n_2 > N$), then we get for such numbers the expression

$$\begin{aligned} |a_{n_1} - a_{n_2}| &= |(1 + (-1)^{2N}) - (1 + (-1)^{2N+1})| \\ &= |2 - 0| = 2 > 1, \end{aligned}$$

which contradicts (2). Consequently, there is no number a , which is a limit of the sequence $a_n = 1 + (-1)^n$, and the sequence is divergent. $\blacktriangle$

Let us recall the theorems on convergent sequences which are usually included in courses of elementary mathematics.

Theorem 1. *If a sequence converges, then it has only one limit.*

Theorem 2. *If a sequence converges, then it is bounded.*

* See *Algebra and Fundamentals of Analysis*, edited by G. N. Yakovlev, Mir Publishers, 1981, p. 45.

Theorem 3. *If the sequences a_n and b_n converge, then*

$$\lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n;$$

$$\lim_{n \rightarrow \infty} (a_n - b_n) = \lim_{n \rightarrow \infty} a_n - \lim_{n \rightarrow \infty} b_n;$$

$$\lim_{n \rightarrow \infty} (a_n b_n) = \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n.$$

If, in addition, $b_n \neq 0$ for any n and $\lim_{n \rightarrow \infty} b_n \neq 0$, then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}.$$

It follows from Theorem 2 that any unbounded sequence is divergent. The sequences (n) , $((n^2 + 1)/n)$, (2^n) , for example, are divergent since they are unbounded. From Theorem 3 it follows that a constant can be taken outside the limit sign, that is, if the sequence (a_n) converges, then

$$\lim_{n \rightarrow \infty} (ca_n) = c \lim_{n \rightarrow \infty} a_n.$$

Let us consider some examples on calculating limits.

Example 6. Find $\lim_{n \rightarrow \infty} \frac{7n^2 - 3n + 12}{n^2 + 2n + 2}$.

$\triangle$ Applying Theorem 3, we get

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{7n^2 - 3n + 12}{n^2 + 2n + 2} &= \lim_{n \rightarrow \infty} \frac{7 - \frac{3}{n} + \frac{12}{n^2}}{1 + \frac{2}{n} + \frac{2}{n^2}} \\ &= \frac{\lim_{n \rightarrow \infty} \left(7 - \frac{3}{n} + \frac{12}{n^2}\right)}{\lim_{n \rightarrow \infty} \left(1 + \frac{2}{n} + \frac{2}{n^2}\right)} = \frac{7 - 3 \lim_{n \rightarrow \infty} \frac{1}{n} + 12 \left(\lim_{n \rightarrow \infty} \frac{1}{n}\right)^2}{1 + 2 \lim_{n \rightarrow \infty} \frac{1}{n} + 2 \left(\lim_{n \rightarrow \infty} \frac{1}{n}\right)^2} = 7. \quad \blacktriangle \end{aligned}$$

Example 7. Find $\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n)$.

$\triangle$ Since

$$\begin{aligned} \sqrt{n^2 + n} - n &= \frac{(\sqrt{n^2 + n} - n)(\sqrt{n^2 + n} + n)}{\sqrt{n^2 + n} + n} \\ &= \frac{n}{\sqrt{n^2 + n} + n} = \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} \end{aligned}$$

and $\lim_{n \rightarrow \infty} \sqrt{1 + \frac{1}{n}} = 1$ (see Example 3), we have, by Theorem 3

$$\begin{aligned} \lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n) &= \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} \\ &= \frac{1}{\lim_{n \rightarrow \infty} \sqrt{1 + \frac{1}{n}} + 1} = \frac{1}{2}. \quad \blacktriangle \end{aligned}$$

5.3. Monotonic Sequences. Weierstrass' Theorem

The sequence (a_n) is said to be *increasing* if the inequality $a_{n+1} > a_n$ is satisfied for any n . The sequence (a_n) is said to be *decreasing* if the inequality $a_{n+1} < a_n$ is satisfied for any n .

For instance, the sequence (n^2) is increasing since $a_{n+1} = (n+1)^2 > n^2 = a_n$ holds for any natural n . The sequence $(1/n)$ is decreasing since the inequality $1/(n+1) < 1/n$ is valid for any n , i.e. $a_{n+1} < a_n$.

The sequence (a_n) is called *nondecreasing* if $a_{n+1} \geq a_n$ for any n , and *nonincreasing* if $a_{n+1} \leq a_n$ for any n .

For example, a sequence with the n th term $a_n = [(n+1)/2]$ is nondecreasing: 1; 1; 2; 2; 3; 3; . . .

All the sequences of this kind (increasing, decreasing, nondecreasing, nonincreasing) are called *monotonic*.

Example 1. Prove that the sequence whose n th term has the form $a_n = \sqrt{n+1} - \sqrt{n}$ is monotonic.

△ Note that

$$\begin{aligned} a_n = \sqrt{n+1} - \sqrt{n} &= \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n+1} + \sqrt{n}} \\ &= \frac{1}{\sqrt{n+1} + \sqrt{n}}, \end{aligned}$$

therefore, for any n ,

$$\begin{aligned} a_{n+1} - a_n &= \frac{1}{\sqrt{n+2} + \sqrt{n+1}} - \frac{1}{\sqrt{n+1} + \sqrt{n}} \\ &= \frac{-(\sqrt{n+2} - \sqrt{n})}{(\sqrt{n+2} + \sqrt{n+1})(\sqrt{n+1} + \sqrt{n})} < 0. \end{aligned}$$

Thus, $a_{n+1} < a_n$, that is, the sequence is decreasing. ▴

Not every sequence is monotonic. The sequence $((-1)^n/n)$ can serve as an example of a nonmonotonic sequence. Its every term with an even number exceeds both its predecessor and successor

$$-1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, -\frac{1}{5}, \dots, -\frac{1}{2k-1}, \frac{1}{2k}, -\frac{1}{2k+1}, \dots$$

The following theorem is valid for monotonic and bounded sequences.

Weierstrass' Theorem. *If a sequence is monotonic and bounded, then it has a limit.*

This theorem, very significant in mathematical analysis, yields the sufficient conditions for the existence of a limit of the sequence. It follows from Weierstrass' theorem, for instance, that the sequence of the areas of regular n -gons, inscribed in a circle of unit radius, has a limit, since it is an increasing and bounded sequence. We designate the limit of this sequence as π .

Example 2. Prove that the sequence $a_1 = 2$, $a_n = (a_{n-1} + 1)/2$ possesses a limit for $n \geq 2$ and find that limit.

Δ Let us find several first terms of the sequence:

$$a_1 = 2, a_2 = 3/2, a_3 = 5/4, a_4 = 9/8.$$

They all belong to the interval $(1; 2]$. We shall prove that for all n there hold the inequalities

$$1 < a_n \leq 2. \quad (1)$$

Let us resort to mathematical induction. For $n = 1$ inequalities (1) are valid since $a_1 = 2$. Assume that for $n = k \geq 1$ there holds $1 < a_k \leq 2$. Since $a_{k+1} = (a_k + 1)/2$, it follows from $a_k > 1$ that $a_{k+1} > 1$ and from $a_k < 2$ that $a_{k+1} < 2$. Thus we see that $1 < a_{k+1} \leq 2$, i.e. inequalities (1) are satisfied at $n = k + 1$ as well. It follows from the principle of mathematical induction that inequalities (1) are satisfied for any natural n .

Next we have

$$a_{n+1} - a_n = \frac{a_n + 1}{2} - a_n = \frac{1 - a_n}{2},$$

and, taking into account that $a_n > 1$ for any n , we obtain from this equation $a_{n+1} - a_n < 0$. Consequently, $a_{n+1} < a_n$ for all n , the sequence is decreasing. In accordance with the Weierstrass theorem, this sequence has a limit, we shall denote it by a : $a = \lim_{n \rightarrow \infty} a_n$. It is evident that if the sequence (a_n) has a limit, then

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} a_{n+1}.$$

For the sequence in question $a_{n+1} = \frac{a_n}{2} + \frac{1}{2}$ and, therefore,

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \left(\frac{a_n}{2} + \frac{1}{2} \right) = \frac{1}{2} \lim_{n \rightarrow \infty} a_n + \frac{1}{2},$$

that is

$$a = \frac{1}{2} a + \frac{1}{2},$$

whence we find that $a = 1$. Thus we have proved that $\lim_{n \rightarrow \infty} a_n = 1$. $\blacktriangle$

5.4. Arithmetic Progression

The sequence (a_n) , in which $a_1 = a$ and in which, at any n ,

$$a_{n+1} = a_n + d, \quad (1)$$

where a and d are any given numbers, is called an *arithmetic progression*. The number d is called the *difference* of the arithmetic progression.

An arithmetic progression is a monotonic sequence, increasing for $d > 0$ (since $a_{n+1} - a_n = d > 0$), decreasing for $d < 0$ (since $a_{n+1} - a_n = d < 0$), nonincreasing for $d = 0$. For the n th term of an arithmetic progression the following formula holds:

$$a_n = a + d(n - 1). \quad (2)$$

We shall prove it by induction.

$\square$ The formula is true for $n = 1$: $a_1 = a$. Let us assume that the formula is also true for $n = k \geq 1$, i.e. $a_k = a + d(k - 1)$. By the definition of an arithmetic progression $a_{k+1} = a_k + d$. Substituting here the expression for the

k th term, we get

$$a_{k+1} = a + d(k - 1) + d = a + dk,$$

and this is formula (2) for $n = k + 1$. It follows from the principle of mathematical induction that formula (2) is true for any natural n . ■

It follows from formula (2) that if the difference of an arithmetic progression is nonzero, then the arithmetic progression is an unbounded sequence. The proof of this fact is analogous to the proof of an unbounded sequence (n) (see Example 2 in 5.1).

Considered most frequently is an arithmetic progression containing a finite number of terms, which is called a *finite arithmetic progression*.

Properties of an arithmetic progression.

1. *Every term of an arithmetic progression, beginning with the second, is equal to the arithmetic mean of its adjacent terms, that is, for $k \geq 2$ we have*

$$a_k = \frac{a_{k-1} + a_{k+1}}{2}. \quad (3)$$

2. *The sum of the terms of the finite arithmetic progression $a_1, a_2, \dots, a_n$, equidistant from its ends, is equal to the sum of the first and the last term, that is, for $k = 1, 2, \dots, n$*

$$a_k + a_{n-k+1} = a_1 + a_n. \quad (4)$$

3. *The sum of the terms of a finite arithmetic progression is equal to the product of the half-sum of the first and the last term by the number of the terms, that is, if $S_n = a_1 + a_2 + \dots + a_n$, then*

$$S_n = \frac{a_1 + a_n}{2} \cdot n. \quad (5)$$

□ For $k \geq 2$ we have $a_k = a_{k-1} + d$ and $a_k = a_{k+1} - d$. Summing up these equalities term-by-term, we get $2a_k = a_{k-1} + a_{k+1}$, whence follows (3).

In the finite arithmetic progression $a_1, a_2, \dots, a_n$ the terms a_k and a_{n-k+1} are equidistant from the ends. According to formula (2), $a_k = a + d(k - 1)$ and $a_{n-k+1} = a + d(n - k)$. The sum of these terms is $a_k + a_{n-k+1} = 2a + d(n - 1)$ and is equal to the sum of the first and the last term $a_1 + a_n = 2a + d(n - 1)$.

If $S_n = a_1 + a_2 + \dots + a_n$, then $S_n = a_n + a_{n-1} + \dots + a_1$. Summing these equalities termwise and using property 2, we get

$$2S_n = (a_1 + a_n) + (a_2 + a_{n-1}) + \dots + (a_1 + a_n) \\ = n(a_1 + a_n),$$

whence follows formula (5):

$$S_n = \frac{a_1 + a_n}{2} \cdot n. \blacksquare$$

If we substitute expression (2) for the n th term into formula (5), we obtain

$$S_n = \frac{2a + d(n-1)}{2} \cdot n. \quad (6)$$

Example 1. The sum of the third and the seventh term of the arithmetic progression is equal to 6, and their product is equal to 8. Find the sum of the first sixteen terms of the progression.

$\triangle$ Let us designate the first term of the progression as a and its difference as d . Then, by formula (2), we have $a_3 = a + 2d$ and $a_7 = a + 6d$. By the hypothesis, $a_3 + a_7 = 6$, $a_3 a_7 = 8$. Substituting here the expressions for a_3 and a_7 , we get the system

$$\begin{cases} 2a + 8d = 6, \\ (a + 2d)(a + 6d) = 8. \end{cases}$$

Solving this system, we obtain $a = 1$, $d = 1/2$, and $a = 5$, $d = -1/2$. We use formula (6) to calculate the sum of the first 16 terms for each of the progressions

$$S_{16} = \frac{2a + 15d}{2} \cdot 16 = 76 \text{ for } a = 1, d = 1/2,$$

$$S_{16} = 20 \text{ for } a = 5, d = -1/2. \blacktriangle$$

Example 2. The sum of the first n terms of the arithmetic progression, whose difference is nonzero, is equal to half the sum of the next n terms. Find the ratio of the sum of the first $3n$ terms of that progression to the sum of its first n terms.

$\triangle$ If S_k is the sum of the first k terms of the progression, then the hypothesis can be written as follows: $S_n = \frac{1}{2} \times$

$(S_{2n} - S_n)$, whence it follows that $3S_n = S_{2n}$. Using formula (6) for the sum S_k , we can obtain from this

$$2a = d(n + 1). \quad (7)$$

The required ratio S_{3n}/S_n has the form

$$\frac{S_{3n}}{S_n} = 3 \frac{2a + d(3n - 1)}{2a + d(n - 1)}.$$

Substituting into this expression a from (7) and cancelling by $d \neq 0$, we get $S_{3n}/S_n = 6$. $\blacktriangle$

5.5. Geometric Progression

The sequence (a_n) in which $a_1 = a$ and for any n ,

$$a_{n+1} = a_n q, \quad (1)$$

where a and q are any nonzero specified terms, is called a *geometric progression*. The number q is called the *common ratio* (or simply *ratio*) of the geometric progression.

If $a > 0$ and $q > 1$, the geometric progression is an increasing sequence since for any n we have $a_{n+1}/a_n = q > 1$, i.e. $a_{n+1} > a_n$; now if $a > 0$ and $0 < q < 1$, then the geometric progression is decreasing: $a_{n+1} = a_n q < a_n$.

The reverse is true, evidently, for $a < 0$, that is, the geometric progression is decreasing if $q > 1$ and increasing if $0 < q < 1$.

For $q < 0$, the sign of each term of the progression is opposite to that of its successor; the progression is not monotonic.

The formula for the n th term of a geometric progression has the form

$$a_n = a q^{n-1}. \quad (2)$$

We shall prove it by induction.

□ Formula (2) is, evidently, true for $n = 1$. Let us suppose that it is also true for $n = k \geq 1$, i.e. $a_k = a q^{k-1}$. It follows from (1) that $a_{k+1} = a_k q$. Substituting into this equation the expression for the k th term, we get $a_{k+1} = a q^k$, that is, formula (2) is valid for $n = k + 1$ as well. It follows from the principle of mathematical induction that formula (2) is valid for any natural n . $\blacksquare$

A geometric progression containing a finite number of terms is called a *finite geometric progression*.

Assume that (a_n) is a geometric progression with the initial term a and the common ratio q .

Properties of a geometric progression.

1. *The square of each term of a geometric progression, beginning with the second, is equal to the product of the adjacent terms, that is, for $k \geq 2$ we have*

$$a_k^2 = a_{k-1}a_{k+1}. \quad (3)$$

If all the terms of a geometric progression are positive, then this property is formulated as follows: every term of a geometric progression, beginning with the second, is equal to the geometric mean of its adjacent terms, i.e. $a_k = \sqrt{a_{k-1}a_{k+1}}$.

2. *In the finite geometric progression $a_1, a_2, \dots, a_n$ the product of the terms equidistant from its ends is equal to the product of its first and last terms, that is,*

$$a_k a_{n-k+1} = a_1 a_n. \quad (4)$$

3. *If the common ratio q of a geometric progression is not equal to unity and $S_n = a_1 + a_2 + \dots + a_n$, then*

$$S_n = a \frac{1-q^n}{1-q}. \quad (5)$$

The proof of properties 1 and 2 is similar to the proof of the corresponding properties of an arithmetic progression. Let us prove property 3.

□ Assume $S_n = a_1 + a_2 + \dots + a_n$. Using formula (2), we can write

$$S_n = a + aq + \dots + aq^{n-1}, \quad (6)$$

in that case

$$qS_n = aq + aq^2 + \dots + aq^n. \quad (7)$$

Subtracting (7) from (6) termwise, we obtain

$$(1 - q) S_n = a (1 - q^n),$$

which, at $q \neq 1$, yields formula (5). ■

Example 1. The sum of the first four terms of a geometric progression is equal to 30 and the sum of the next four terms is equal to 480. Find the first term of the progression.

△ Assume that a is the first term and q is the ratio of the geometric progression. Formula (5) yields

$$a \frac{1-q^4}{1-q} = 30. \quad (8)$$

The next four terms aq^4, aq^5, aq^6, aq^7 form a geometric progression with the initial term aq^4 and the ratio q . Therefore, we obtain, from formula (5),

$$aq^4 \frac{1-q^4}{1-q} = 480. \quad (9)$$

Formulas (8) and (9) yield $q^4 = 16$, whence it follows that $q = 2$ or $q = -2$. If $q = 2$, then from (8) we find $a = 2$, and if $q = -2$, we get $a = -6$. ▲

We have proved in Example 4 of 5.2 that $\lim_{n \rightarrow \infty} q^n = 0$ if $|q| < 1$. This implies that if the ratio q of the geometric progression (a_n) satisfies the condition $|q| < 1$, then $a_n \rightarrow 0$ for $n \rightarrow \infty$. Indeed, since $a_n = aq^{n-1}$, it follows that

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} aq^{n-1} = a \lim_{n \rightarrow \infty} q^{n-1} = 0.$$

A geometric progression whose common ratio satisfies the condition $|q| < 1$ is said to be *infinitely decreasing*.

If (a_n) is an infinitely decreasing geometric progression, then there exists a limit of the sequence of the sums of its first n terms. From formula (5) it follows that

$$S_n = \frac{a}{1-q} - \frac{a}{1-q} \cdot q^n,$$

and since $q^n \rightarrow 0$ as $n \rightarrow \infty$, we have $\lim_{n \rightarrow \infty} S_n = a/(1-q)$.

This limit is known as the *sum* of an infinitely decreasing geometric progression and is usually denoted by S . Thus we have

$$S = \frac{a}{1-q}. \quad (10)$$

Example 2. Find the sum of the infinitely decreasing geometric progression the sum of whose first three terms is equal to 7 and the product of those terms is equal to 8.

△ Assume that a is the first term and q is the ratio of the geometric progression. By the hypothesis

$$\begin{cases} a(1 + q + q^2) = 7, \\ a^3q^3 = 8. \end{cases}$$

From the second equation of the system we find $aq = 2$ and, substituting $a = 2/q$ into the first equation of the system, we obtain the quadratic equation $2q^2 - 5q + 2 = 0$ whose roots are $q = 1/2$ and $q = 2$. The sequence is infinitely decreasing, $|q| < 1$. Consequently, $q = 1/2$, and in that case $a = 4$. The sum of that progression can be found by formula (10): $S = \frac{a}{1-q} = 8$. ▲

5.6. Limit of a Function. Continuity of a Function

Recall that any interval containing a point a in its interior is called the *neighbourhood* of that point. The symmetric interval $(a - \delta; a + \delta)$ is called the δ -neighbourhood of the point a for any $\delta > 0$.

Let the function $f(x)$ be defined in some neighbourhood of the point a except, perhaps, the point a itself, that is, we have made no assumption as to whether the function is defined at the point a or not.

The number b is a *limit* of the function $f(x)$ as x approaches a (or at the point a) if, for any positive number ε , there is a positive number δ such that

$$|f(x) - b| < \varepsilon$$

for all $x \neq a$ satisfying the inequality $|x - a| < \delta$.

The fact that the number b is a limit of the function $f(x)$ as x tends to a can be written as

$$\lim_{x \rightarrow a} f(x) = b, \text{ or } f(x) \rightarrow b \text{ as } x \rightarrow a.$$

Note that the conditions $|x - a| < \delta$, $x \neq a$, mean that the point x belongs to the δ -neighbourhood of the point a and is distinct from a ; these conditions can be concisely written as

$$0 < |x - a| < \delta.$$

We can illustrate the definition of a limit we have given as follows (Fig. 42). We take a number $\varepsilon > 0$ and mark on the Oy axis the ε -neighbourhood of the point b , that is, the interval $(b - \varepsilon; b + \varepsilon)$, through whose end points we draw straight lines parallel to the Ox axis. We obtain a strip 2ε wide. If for any $\varepsilon > 0$ there is a $\delta > 0$ such that the graph of the function $y = f(x)$ (considered for x from the δ -neighbourhood of the point a and not equal to a) lies entirely in that strip, then the number b is a limit of $f(x)$ as x tends to a .

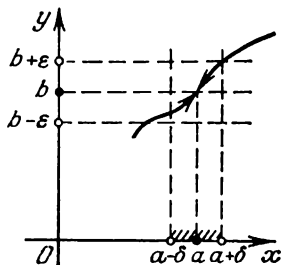


Fig. 42.

Example 1. Prove that $\lim_{x \rightarrow 1} (3x - 1) = 2$.

△ The function $f(x) = 3x - 1$ is defined in any neighbourhood of the point $x = 1$. The inequality $|f(x) - 2| < \varepsilon$, i.e. $|(3x - 1) - 2| = 3|x - 1| < \varepsilon$ is satisfied for all x such that $|x - 1| < \varepsilon/3$. Thus, for any $\varepsilon > 0$ we can take $\delta = \varepsilon/3$, and then, for all x such that $|x - 1| < \delta$, there will hold the inequality $|f(x) - 2| = 3|x - 1| < \varepsilon$. And this means that

$$\lim_{x \rightarrow 1} (3x - 1) = 2. \blacktriangle$$

Example 2. Consider a function $f(x) = (x^2 - 1)/(x - 1)$. It is defined everywhere except for the point $x = 1$. Prove that the function possesses a limit equal to 2 for all x tending to 1.

△ For all $x \neq 1$ we have $(x^2 - 1)/(x - 1) = x + 1$. Consequently, $|f(x) - 2| = |x - 1|$, and the inequality $|f(x) - 2| < \varepsilon$ is satisfied if $|x - 1| < \varepsilon$ and $x \neq 1$. Therefore, for any $\varepsilon > 0$ we can take $\delta = \varepsilon$, and then, for all x satisfying the condition $0 < |x - 1| < \delta$, there will hold the inequality $\left| \frac{x^2 - 1}{x - 1} - 2 \right| = |x - 1| < \varepsilon$. Thus, by the definition, $(x^2 - 1)/(x - 1) \rightarrow 2$ as $x \rightarrow 1$. ▴

The solutions of examples 1 and 2 show, in particular, that δ depends on ε , the dependence being, in a general case, different for different functions.

Example 3. Suppose that $f(x) > 0$ in some neighbourhood of a point a , $\lim_{x \rightarrow a} f(x) = b$ and $b > 0$. Prove that for x tending to a there is a limit of the function $\sqrt[n]{f(x)}$ and $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{b}$.

△ It follows from the definition of a limit that for any $\varepsilon > 0$ there is a positive δ such that $|f(x) - b| < \varepsilon/b$

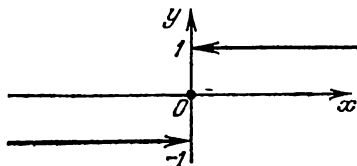


Fig. 43.

for all x satisfying the condition $0 < |x - a| < \delta$. Then, for the same x , it follows from

$$|\sqrt[n]{f(x)} - \sqrt[n]{b}| = \frac{|f(x) - b|}{\sqrt[n]{f(x)} + \sqrt[n]{b}} < \frac{|f(x) - b|}{b}$$

that $|\sqrt[n]{f(x)} - \sqrt[n]{b}| < \varepsilon$. This means that $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{b}$. ▲

Not every function defined in the neighbourhood of the point a has a limit as x tends to a . Consider, for instance, the function

$$\text{sign } x = \begin{cases} 1, & \text{if } x > 0, \\ 0, & \text{if } x = 0, \\ -1, & \text{if } x < 0, \end{cases}$$

whose graph is shown in Fig. 43. Let us prove that the function has no limit at the point $x = 0$. We shall carry out the proof by contradiction.

□ Assume that there is a number b such that

$$\lim_{x \rightarrow 0} \text{sign } x = b.$$

By the definition of a limit, for any positive ε and, consequently, for $\varepsilon = 1/2$, there is a positive δ such that $|\text{sign } x - b| < 1/2$ for all x satisfying the condition $0 < |x| < \delta$.

Thus it follows that if x_1 and x_2 are such that $0 < |x_1| < \delta$ and $0 < |x_2| < \delta$, then $|\text{sign } x_1 - b| < 1/2$ and $|\text{sign } x_2 - b| < 1/2$ and, consequently,

$$|\text{sign } x_1 - \text{sign } x_2| = |\text{sign } x_1 - b + b - \text{sign } x_2| \\ \leq |\text{sign } x_1 - b| + |\text{sign } x_2 - b| < 1.$$

Taking, however, $x_1 = \delta/2$ and $x_2 = -\delta/2$, we shall have $|\text{sign } x_1 - \text{sign } x_2| = |1 - (-1)| = 2 > 1$.

The contradiction obtained shows that the function $\text{sign } x$ has no limit at the point $x = 0$. ■

For functions possessing a limit at a point there holds a theorem analogous to the theorem on convergent sequences.

Theorem. Suppose the functions $f(x)$ and $g(x)$ have limits as x tends to a . Then, as $x \rightarrow a$, there also exist limits of the sum, the difference, and the product of these functions and, in this case,

$$\lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x),$$

$$\lim_{x \rightarrow a} (f(x) - g(x)) = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x),$$

$$\lim_{x \rightarrow a} (f(x) \cdot g(x)) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x).$$

If, in addition, $\lim_{x \rightarrow a} g(x) \neq 0$, then there exists a limit of the quotient $f(x)/g(x)$ and

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}.$$

Example 4. Find $\lim_{x \rightarrow 3} \frac{2x-3}{3+7x}$.

△ We have, by the theorem on limits,

$$\lim_{x \rightarrow 3} \frac{2x-3}{3+7x} = \frac{\lim_{x \rightarrow 3} (2x-3)}{\lim_{x \rightarrow 3} (3+7x)} = \frac{2 \lim_{x \rightarrow 3} x - 3}{3 + 7 \lim_{x \rightarrow 3} x} = \frac{2 \cdot 3 - 3}{3 + 7 \cdot 3} = \frac{1}{8}. \blacktriangle$$

Example 5. Find $\lim_{x \rightarrow 1} \frac{\sqrt{x+3}-2}{x-1}$.

△ For x tending to 1 the limit of the denominator is zero and, therefore, we cannot apply the theorem on the limit

of a quotient. However, transforming the fraction as follows:

$$\frac{\sqrt{x+3}-2}{x-1} = \frac{(\sqrt{x+3}-2)(\sqrt{x+3}+2)}{(x-1)(\sqrt{x+3}+2)} = \frac{x-1}{(x-1)(\sqrt{x+3}+2)}$$

and cancelling by $x-1$ (if $x \rightarrow 1$, then $x \neq 1$), we get

$$\frac{\sqrt{x+3}-2}{x-1} = \frac{1}{\sqrt{x+3}+2}.$$

Since $\lim_{x \rightarrow 1} \sqrt{x+3} = 2$ (see Example 3), we have

$$\lim_{x \rightarrow 1} \frac{\sqrt{x+3}-2}{x-1} = \lim_{x \rightarrow 1} \frac{1}{\sqrt{x+3}+2} = \frac{1}{\lim_{x \rightarrow 1} \sqrt{x+3}+2} = \frac{1}{4}. \blacktriangle$$

For certain functions, when x tends to a , the limit is equal to the value of the function at the point a (see Exam-

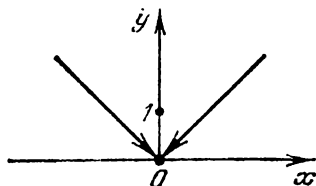


Fig. 44.

ples 1 and 4). This is not always the case, however. For instance, for the function

$$f(x) = \begin{cases} \frac{x^2}{|x|} & \text{for } x \neq 0, \\ 1 & \text{for } x = 0, \end{cases}$$

whose graph is represented in Fig. 44, the limit is zero as $x \rightarrow 0$ and $f(0) = 1$.

If, the limit of the function $f(x)$ at the point a is equal to the value of the function at that point

$$\lim_{x \rightarrow a} f(x) = f(a),$$

then the function $f(x)$ is said to be *continuous* at a .

It follows from the theorem on limits that a linear function is continuous at any point, i.e. for any point x_0 there

holds the equation

$$\lim_{x \rightarrow x_0} (a_0 + a_1x) = a_0 + a_1x_0.$$

The polynomial $P_n(x) = a_0 + a_1x + \dots + a_nx^n$ is also a continuous function at any point:

$$\lim_{x \rightarrow x_0} P_n(x) = P_n(x_0).$$

This follows from the fact that a polynomial is a finite sum of the terms of the form a_kx^k , $k \geq 0$, for each of which

$$\lim_{x \rightarrow x_0} a_kx^k = a_k(x_0)^k$$

(the last fact can be established by induction).

It follows from the continuity of the polynomial and from the theorem on the limit of the quotient that a rational function is continuous at any point belonging to the domain of its definition since

$$\lim_{x \rightarrow x_0} \frac{P_n(x)}{Q_m(x)} = \frac{P_n(x_0)}{Q_m(x_0)}, \quad \text{if } Q_m(x_0) \neq 0.$$

All elementary functions, such as, for instance,

$$x^h, a^h (a > 0), \sin x, \tan x, \log_a x \quad (a > 0, a \neq 1),$$

are continuous at every point in whose neighbourhood these functions are defined.

If a function is continuous at every point of an interval, it is said to be *continuous on that interval*. The graph of a function continuous on an interval can be drawn by an unbroken movement of a pencil over the paper.

Every elementary function is continuous on any interval of its definition.

5.7. The Derivative and Its Geometrical Meaning

Suppose the function $y = f(x)$ is defined on a certain interval $(a; b)$ containing a point x_0 in its interior. For any other point x from the interval $(a; b)$ the difference $x - x_0$ is designated as Δx and is called an *increment of the argument*; the corresponding difference of the values of the function $f(x) - f(x_0)$ is designated as $\Delta f(x_0)$ (or $\Delta y(x_0)$) and is called an *increment of the function*. The equation

$\Delta x = x - x_0$ yields $x = x_0 + \Delta x$ and

$$\Delta f(x_0) = f(x_0 + \Delta x) - f(x_0).$$

If $x \rightarrow x_0$, then, evidently, $\Delta x \rightarrow 0$.

The *derivative* of the function $y = f(x)$ at the point x_0 is the limit of the ratio of the increment of the function to the corresponding increment of the argument, when the latter increment tends to zero. The derivative of the function $y = f(x)$ at the point x_0 is designated as $f'(x_0)$ or $y'(x_0)$. Thus, by the definition,

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{\Delta f(x_0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

For the derivative $f'(x_0)$ to exist, it is necessary that the function $f(x)$ be defined in some neighbourhood of the point x_0 . Let us show that it is also necessary that the function be continuous at the point x_0 . Suppose there exists $f'(x_0)$, then

$$\lim_{x \rightarrow x_0} (f(x) - f(x_0)) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \cdot (x - x_0) = f'(x_0) \cdot 0 = 0.$$

This yields $\lim_{x \rightarrow x_0} f(x) = f(x_0)$, which implies the continuity of the function $f(x)$ at the point x_0 .

The function which has a derivative at the point x_0 is said to be *differentiable at that point*; the operation of finding the derivative is called *differentiation*.

A function differentiable at every point of an interval is said to be *differentiable on that interval*. The derivative of the function $y = f(x)$ differentiable on an interval is itself a function of the argument x . It is designated as $f'(x)$ or $y'(x)$ and is called a *derivative function*.

Example 1. Prove that the function $f(x) = \sqrt{x}$ is differentiable on the interval $(0; +\infty)$ and $(\sqrt{x})' = \frac{1}{2\sqrt{x}}$ for any $x \in (0; +\infty)$.

Δ Assume $x_0 > 0$ and $x > 0$. We have

$$\begin{aligned} f(x) - f(x_0) &= \sqrt{x} - \sqrt{x_0} = \frac{(\sqrt{x} - \sqrt{x_0})(\sqrt{x} + \sqrt{x_0})}{\sqrt{x} + \sqrt{x_0}} \\ &= \frac{x - x_0}{\sqrt{x} + \sqrt{x_0}}, \end{aligned}$$

that is,

$$\Delta f(x_0) = \frac{\Delta x}{\sqrt{x_0 + \Delta x} + \sqrt{x_0}}.$$

From this, knowing that $\lim_{\Delta x \rightarrow 0} \sqrt{x_0 + \Delta x} = \sqrt{x_0}$ (see Example 3 in 5.6), we get

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta x (\sqrt{x_0 + \Delta x} + \sqrt{x_0})} = \frac{1}{2\sqrt{x_0}}.$$

This means that the function $f(x) = \sqrt{x}$ is differentiable at every point of the interval $(0; +\infty)$ and $(\sqrt{x})' = \frac{1}{2\sqrt{x}}$. ▲

Here are the main formulas and rules with whose aid we can, in most cases, carry out differentiation without calculating the derivative as the limit of the ratio $\frac{\Delta f}{\Delta x}$ as $\Delta x \rightarrow 0$.

The derivatives of elementary functions:

- | | |
|---|--|
| 1. $(x^n)' = nx^{n-1}$, $n \in \mathbb{N}$. | 6. $(a^x)' = a^x \ln a$, $a > 0$. |
| 2. $(\sin x)' = \cos x$. | 7. $(\ln x)' = \frac{1}{x}$. |
| 3. $(\cos x)' = -\sin x$. | 8. $(\log_a x)' = \frac{1}{x \ln a}$, |
| 4. $(\tan x)' = \frac{1}{\cos^2 x}$. | $a > 0$, $a \neq 1$. |
| 5. $(e^x)' = e^x$. | |

Rules of differentiation.

Suppose c is a constant, and $u(x)$, $v(x)$ and $u_i(x)$ are functions differentiable on an interval $(a; b)$ on which the following formulas are also valid:

$$\begin{aligned} c' &= 0. \\ (cu)' &= cu'. \\ (u_1 + u_2 + \dots + u_n)' &= u_1' + u_2' + \dots + u_n'. \\ (uv)' &= u'v + uv'. \\ \left(\frac{u}{v}\right)' &= \frac{u'v - uv'}{v^2}, \quad v \neq 0. \end{aligned}$$

Here are some examples of seeking the derivatives.

Example 2. Find the derivative of the function $y = x^2 + \frac{1}{x}$.

$\Delta y' = (x^2)' + (x^{-1})' = 2x - 1 \cdot x^{-2} = 2x - \frac{1}{x^2}$. The formula $(x^{-n})' = -nx^{-n-1}$, $n \in \mathbb{N}$, follows from the rule of differentiation of a quotient if we set $u = 1$, $v = x^n$. $\blacktriangle$

Example 3. Find the derivative of the function $y = x^4 - 2x^3 + 3x - 7$.

$$\Delta y' = (x^4)' - 2(x^3)' + 3(x)' - (7)' = 4x^3 - 6x^2 + 3. \quad \blacktriangle$$

Example 4. Find the derivative of the function $y = x \ln x$.

$$\Delta y' = (x)' \ln x + x (\ln x)' = 1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1. \quad \blacktriangle$$

Example 5. Find the derivative of the function

$$y = \frac{2x-1}{3+x}.$$

$$\begin{aligned} \Delta y' &= \frac{(2x-1)'(3+x) - (2x-1)(3+x)'}{(3+x)^2} \\ &= \frac{2(3+x) - (2x-1) \cdot 1}{(3+x)^2} = \frac{7}{(3+x)^2}. \quad \blacktriangle \end{aligned}$$

The chain rule of differentiation of a composite function.

Assume that $y = F(u)$, $u = u(x)$, and $y(x) = F(u(x))$ is a composite function. If the function $u(x)$ is differentiable at the point x_0 and the function $F(u)$ is differentiable at the point $u_0 = u(x_0)$, then the composite function $y(x) = F(u(x))$ is differentiable at the point x_0 and

$$y'(x_0) = F'(u_0) u'(x_0).$$

Example 6. Find the derivative of the function $y = (\sin x + \cos x)^3$.

Δ We set $u(x) = \sin x + \cos x$, $y = u^3$. In accordance with the chain rule we get

$$\begin{aligned} y'(x) &= (u^3)' u'(x) = 3u^2 u'(x) \\ &= 3(\sin x + \cos x)^2 (\cos x - \sin x) \end{aligned}$$

for any x . $\blacktriangle$

Example 7. Find the derivative of the function $y(x) = \sqrt{x^2 - 2x}$.

△ Setting $u(x) = x^2 - 2x$, we get $y(x) = \sqrt{u(x)}$. Since $(\sqrt{u})' = \frac{1}{2\sqrt{u}}$ (see Example 1), we have

$$y'(x) = (\sqrt{u})' u'(x) = \frac{1}{2\sqrt{u}} u'(x) = \frac{2x-2}{2\sqrt{x^2-2x}} = \frac{x-1}{\sqrt{x^2-2x}}. \quad \blacktriangle$$

The geometrical meaning of the notion of the derivative becomes evident when we consider the graph of the function

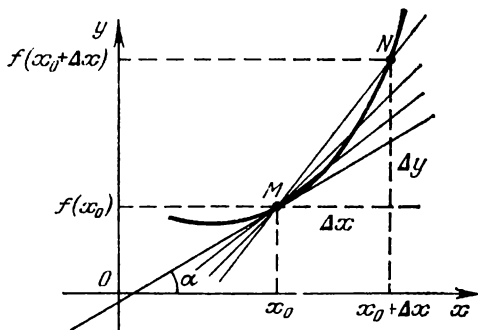


Fig. 45.

$y = f(x)$ defined on some interval $(a; b)$. For the values $x_0 + \Delta x$ and x_0 of the argument belonging to the interval $(a; b)$ the respective ordinates of the graph of the function are equal to $f(x_0 + \Delta x)$ and $f(x_0)$ and their difference is $\Delta y(x_0)$, i.e. $\Delta y(x_0) = f(x_0 + \Delta x) - f(x_0)$. The ratio $\frac{\Delta y(x_0)}{\Delta x}$ is equal to the slope of the line passing through the points $M(x_0; f(x_0))$ and $N(x_0 + \Delta x, f(x_0 + \Delta x))$ of the graph. The line MN is called a *secant* (Fig. 45).

Assume that $\Delta x \rightarrow 0$. Then the point N tends to the point M . If there exists a derivative $f'(x_0)$, that is, the limit of the ratio $\frac{\Delta y(x_0)}{\Delta x}$, then the secant MN tends to the straight line passing through the point M and having the slope $f'(x_0)$. The limiting position the secant MN acquires when N tends to M is a *tangent* to the graph of the function $y = f(x)$ at the point M . The slope of the tangent line is equal to $f'(x_0)$, or, else, $\tan \alpha = f'(x_0)$. The equation of

the tangent, being the equation of the straight line passing through the point $M(x_0, y_0)$ and having the slope $f'(x_0)$, can be written in the form $\frac{y-y_0}{x-x_0} = f'(x_0)$ or (since $y_0 = f(x_0)$) in the form

$$y = f'(x_0)(x - x_0) + f(x_0). \quad (1)$$

Example 8. Find the equation of the tangent to the parabola $y = x - x^2 + 6$ at the point with the abscissa $x_0 = 1$.

$\triangle$ We find the derivative of the function $f(x) = x - x^2 + 6$:

$$f'(x) = 1 - 2x,$$

and, substituting the values $x_0 = 1$, $f(x_0) = 6$ and $f'(x_0) = -1$ into the equation of the tangent (1), we obtain the equation of the tangent $y + x = 7$. $\blacktriangle$

The notion of the derivative is one of the principal concepts in mathematical analysis. For the applications of the derivatives to the investigations of the behaviour of functions see Chapter 6.

5.8. Limit of a Function at Infinity

Assume that the function $f(x)$ is defined outside of some interval $[c; d]$, or, as they say, is defined in the neighbourhood of infinity.

The number b is called a *limit* of the function $f(x)$, as x tends to infinity, if for any positive number ε there is a positive number M such that $|f(x) - b| < \varepsilon$ for all x satisfying the condition $|x| > M$. In this case we write $\lim_{x \rightarrow \infty} f(x) = b$ or $f(x) \rightarrow b$ as $x \rightarrow \infty$.

Example 1. Prove that $\lim_{x \rightarrow \infty} (1/x) = 0$.

$\triangle$ The function $f(x) = 1/x$ is defined for all $x \neq 0$. For any $\varepsilon > 0$ we have $|f(x) - 0| = 1/|x| < \varepsilon$ if $|x| > 1/\varepsilon$. Thus, for any positive ε we can take $M = 1/\varepsilon$, and then, for all x such that $|x| > M$, we have $|\frac{1}{x} - 0| < \varepsilon$. And this means that $1/x \rightarrow 0$ as $x \rightarrow \infty$. $\blacktriangle$

The theorem on limits formulated in 5.6 remains valid if $x \rightarrow a$ is replaced everywhere by $x \rightarrow \infty$.

Example 2. Find

$$\lim_{x \rightarrow \infty} \frac{2x^2 + 3}{x^2 + 5}.$$

△ As was mentioned above, the theorem on limits remains valid when $x \rightarrow \infty$. In particular, if the limits of the numerator and denominator exist and the limit of the denominator is nonzero, then there is a limit of the ratio which is equal to the ratio of the limits, and, therefore, we have

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{2x^2 + 3}{x^2 + 5} &= \lim_{x \rightarrow \infty} \frac{2 + \frac{3}{x^2}}{1 + \frac{5}{x^2}} = \frac{\lim_{x \rightarrow \infty} \left(2 + \frac{3}{x^2}\right)}{\lim_{x \rightarrow \infty} \left(1 + \frac{5}{x^2}\right)} \\ &= \frac{2 + 3 \left(\lim_{x \rightarrow \infty} \frac{1}{x}\right)^2}{1 + 5 \left(\lim_{x \rightarrow \infty} \frac{1}{x}\right)^2} = 2. \quad \blacktriangle \end{aligned}$$

It is sometimes of interest to consider two separate cases of seeking the limits of a function, viz., when x tends to $+\infty$ and when x tends to $-\infty$. We define the limits as follows:

$$(a) \quad \lim_{x \rightarrow \infty} f(x) = b,$$

if for any positive ε there is a number M such that $|f(x) - b| < \varepsilon$ for all x satisfying the condition $x > M$.

$$(b) \quad \lim_{x \rightarrow -\infty} f(x) = b,$$

if for any positive ε there is a number M such that $|f(x) - b| < \varepsilon$ for all x , satisfying the condition $x < M$.

If $\lim_{x \rightarrow \infty} f(x) = b$, then it is easy to see that

$$\lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = b.$$

From $\lim_{x \rightarrow \infty} (1/x) = 0$ it follows, for instance, that $1/x \rightarrow 0$ as $x \rightarrow +\infty$ and $x \rightarrow -\infty$.

Note that the theorem on limits remains valid if $x \rightarrow a$ is everywhere replaced by $x \rightarrow +\infty$ or $x \rightarrow -\infty$.

Example 3. Find the limits of the function $f(x) = \frac{2x}{1 + |x|}$ in the cases when x tends to $+\infty$ and when x

tends to $-\infty$.

△ If $x > 0$, then $\frac{2x}{1+|x|} = \frac{2x}{1+x}$ and

$$\lim_{x \rightarrow +\infty} \frac{2x}{1+|x|} = \lim_{x \rightarrow +\infty} \frac{2x}{1+x} = \lim_{x \rightarrow +\infty} \frac{2}{\frac{1}{x}+1} = 2.$$

Now if $x < 0$, then $\frac{2x}{1+|x|} = \frac{2x}{1-x}$ and, therefore,

$$\lim_{x \rightarrow -\infty} \frac{2x}{1+|x|} = \lim_{x \rightarrow -\infty} \frac{2x}{1-x} = \lim_{x \rightarrow -\infty} \frac{2}{\frac{1}{x}-1} = -2. \quad \blacktriangle$$

5.9. One-Sided Limits. Infinite Limits

The function $\text{sign } x$ considered in 5.6 (see Fig. 43) has no limit at the point $x = 0$. However, if we restrict our consideration to positive x , then the limit, as x tends to zero, exists and is equal to $+1$. Now, if we consider only negative x , then the limit, as x tends to zero, also exists and is equal to -1 . It is customary to say that when x tends to zero the function $\text{sign } x$ possesses *one-sided* limits: the *right-hand limit* equal to $+1$ and the *left-hand limit* equal to -1 .

We shall assume that any interval $(a; a + c)$, for $c > 0$, is the *right-hand neighbourhood* of a point a . Suppose that the function $f(x)$ is defined in some right-hand neighbourhood of a . The number b will be the *right-hand limit* (or the *limit on the right*) of the function $f(x)$ at the point a if for any positive number ε there is a positive number δ such that $|f(x) - b| < \varepsilon$ for all x , satisfying the condition $a < x < a + \delta$, that is, belonging to the right-hand δ -neighbourhood of the point a . In that case we write

$$\lim_{x \rightarrow a+0} f(x) = b.$$

The definition of the limit on the left is quite analogous. The *left-hand neighbourhood* of a point a is any interval $(a - c; a)$ for $c > 0$. Suppose the function $f(x)$ is defined on some left-hand neighbourhood of the point a .

The number b is called the *left-hand limit* (or the *limit on the left*) of the function $f(x)$ at the point a if for any positive

number ε there is a positive number δ such that $|f(x) - b| < \varepsilon$ for all x , satisfying the condition $a - \delta < x < a$, that is, belonging to the left-hand neighbourhood of the point a . If the left-hand limit at the point a is equal to b , then we write

$$\lim_{x \rightarrow a-0} f(x) = b.$$

In a special case, when the right-hand and left-hand limits are considered at the point zero, they are written in the form

$$\lim_{x \rightarrow +0} f(x) \quad \text{and} \quad \lim_{x \rightarrow -0} f(x).$$

Thus, for the function $\text{sign } x$ we have

$$\lim_{x \rightarrow +0} \text{sign } x = 1, \quad \lim_{x \rightarrow -0} \text{sign } x = -1.$$

Example 1. Prove that

$$\lim_{x \rightarrow +0} \frac{\sqrt{x} - x}{\sqrt{x} + x} = 1.$$

Δ The function $f(x) = \frac{\sqrt{x} - x}{\sqrt{x} + x}$ is defined for $x > 0$ and

$$\frac{\sqrt{x} - x}{\sqrt{x} + x} = \frac{\sqrt{x}(1 - \sqrt{x})}{\sqrt{x}(1 + \sqrt{x})} = \frac{1 - \sqrt{x}}{1 + \sqrt{x}}.$$

Therefore

$$|f(x) - 1| = \left| \frac{1 - \sqrt{x}}{1 + \sqrt{x}} - 1 \right| = \frac{2\sqrt{x}}{1 + \sqrt{x}} < 2\sqrt{x}.$$

Let us take $\delta = \varepsilon^2/4$ for any $\varepsilon > 0$. Then, if $0 < x < \delta$, then $|f(x) - 1| < 2\sqrt{x} < 2\sqrt{\delta} = \varepsilon$. And this means that $\lim_{x \rightarrow +0} \frac{\sqrt{x} - x}{\sqrt{x} + x} = 1$. $\blacktriangle$

The statement of the theorem on limits of functions formulated in 5.6 remains valid if $x \rightarrow a$ is everywhere replaced by $x \rightarrow a + 0$ (by $x \rightarrow a - 0$ respectively).

Infinite limits. Assume that the function $f(x)$ is defined in some right-hand neighbourhood of a point a and the func-

tion $\frac{1}{f(x)}$ possesses a right-hand limit equal to zero:

$$\lim_{x \rightarrow a+0} \frac{1}{f(x)} = 0.$$

For example, the function $f(x) = 1/\sqrt{x-1}$ is defined for $x > 1$ and $\lim_{x \rightarrow 1+0} (1/f(x)) = \lim_{x \rightarrow 1+0} \sqrt{x-1} = 0$.

If $f(x) > 0$ in some right-hand neighbourhood of the point a and $\lim_{x \rightarrow a+0} (1/f(x)) = 0$, then we say that the function $f(x)$ *tends to plus infinity* when x tends to a from the right and we write

$$\lim_{x \rightarrow a+0} f(x) = +\infty$$

(read: the right-hand limit of the function $f(x)$ at the point a is equal to plus infinity).

If $f(x) < 0$ in some right-hand neighbourhood of a point a and $\lim_{x \rightarrow a+0} (1/f(x)) = 0$, then we say that the function $f(x)$ *tends to minus infinity* when x tends to a from the right and we write

$$\lim_{x \rightarrow a+0} f(x) = -\infty$$

(read: the right-hand limit of the function $f(x)$ at the point a is equal to minus infinity).

Thus we have $\lim_{x \rightarrow 1+0} (1/\sqrt{x-1}) = +\infty$ since $1/\sqrt{x-1} > 0$ for $x > 1$ and $\lim_{x \rightarrow 1+0} \sqrt{x-1} = 0$.

The definition of the *left-hand infinite limits* at the point a for the function $f(x)$ specified in some left-hand neighbourhood of the point a is quite analogous:

$$\lim_{x \rightarrow a-0} f(x) = +\infty,$$

if $f(x) > 0$ in some left-hand neighbourhood of the point a and $\lim_{x \rightarrow a-0} (1/f(x)) = 0$;

$$\lim_{x \rightarrow a-0} f(x) = -\infty,$$

if $f(x) < 0$ in some left-hand neighbourhood of the point a and $\lim_{x \rightarrow a-0} (1/f(x)) = 0$.

Example 2. Find the right-hand and the left-hand limit of the function $f(x) = \frac{x+1}{x-1}$ at the point $x=1$.

$\triangle$ The function $f(x) = \frac{x+1}{x-1}$ is defined in the neighbourhood of the point $x=1$ but not at that point itself, and

$$\lim_{x \rightarrow 1} \frac{1}{f(x)} = \lim_{x \rightarrow 1} \frac{x-1}{x+1} = 0.$$

Furthermore, the function $f(x) = \frac{x+1}{x-1}$ is positive on the interval $(1; 3/2)$ and negative on the interval $(1/2; 1)$, therefore, by the definition of right-hand and left-hand infinite limits we have

$$\lim_{x \rightarrow 1+0} f(x) = +\infty \quad \text{and} \quad \lim_{x \rightarrow 1-0} f(x) = -\infty \quad (\text{Fig 46}). \quad \blacktriangle$$

If for x tending to a from the right and from the left the function $f(x)$ possesses an infinite limit, then to make

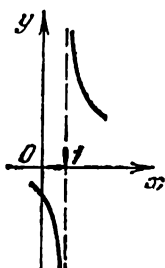


Fig. 46.

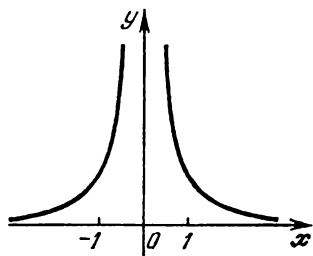


Fig. 47.

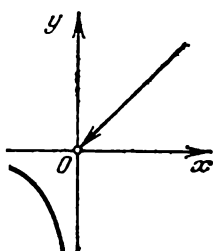


Fig. 48.

the graphical representation of that function more picturesque it is good practice to draw a straight line $x = a$.

For a function defined in the right-hand and left-hand neighbourhood of the point a , the right-hand and left-hand limits may be both infinite and of unlike signs (Fig. 46), or both infinite and of the same sign, as, for instance, for the function $f(x) = 1/x^2$ at the point $x = 0$ (Fig. 47).

It may also happen that one of the one-sided limits is finite and the other infinite (Fig. 48).

Infinite limits can be also defined when x tends to $+\infty$ or $-\infty$; for instance, $\lim_{x \rightarrow -\infty} f(x) = +\infty$ if $f(x)$ is defined

and positive for all $x < M$ for a certain M and $\lim_{x \rightarrow -\infty} (1/f(x)) = 0$.

Example 3.

$$\lim_{x \rightarrow +\infty} \frac{1-2x-x^2}{x+2} = -\infty, \quad \lim_{x \rightarrow -\infty} \frac{1-2x-x^2}{x+2} = +\infty.$$

The concept of an infinite limit is widely used in investigating the behaviour of functions and constructing their graphs (see Ch. 6).

PROBLEMS OF SECTION I

1. Proceeding from the definition of a limit, prove that $\lim_{n \rightarrow \infty} \frac{n^2 - 1}{n^2 + 1} = 1$ and fill in the following table showing the relationship between N and ε

ε	1/10	1/100	1/1000
N			

2. Prove that the sequence $(n^{(-1)^n})$ is unbounded.
3. Prove that the sequence $\left(\frac{n + (-1)^n n}{n + 2}\right)$ is divergent.
4. Prove that $\lim_{n \rightarrow \infty} (1/n^\alpha) = 0$ for any $\alpha > 0$.
5. Assume that the sequences (a_n) and (b_n) are such that $|a_n| \leq |b_n|$ for any n and $\lim_{n \rightarrow \infty} b_n = 0$. Prove that $\lim_{n \rightarrow \infty} a_n = 0$.
6. Prove that $\lim_{n \rightarrow \infty} (\sqrt[n]{n+1} - \sqrt[n]{n}) = 0$.
7. Find the formula for the n th term of the sequence defined recursively: $a_1 = 2$, $a_{n+1} = \frac{1}{3}a_n + \frac{1}{3}$, $n \in \mathbb{N}$. Find the limit of that sequence.

Find the limits of the following sequences:

8. $\lim_{n \rightarrow \infty} \frac{1+a+a^2+\dots+a^n}{1+b+b^2+\dots+b^n}$ for $|a| < 1$ and $|b| < 1$
9. $\lim_{n \rightarrow \infty} \frac{n \sqrt{1+3+5+\dots+(2n-1)}}{2n^2+n+1}$.
10. Prove that $\lim_{n \rightarrow \infty} \sqrt[n]{a} = 1$ for any $a > 0$.
11. Suppose that the sequences (a_n) , (b_n) , and (x_n) are such that $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = a$ and the inequalities $a_n \leq x_n \leq b_n$ hold true for any n . Prove that $\lim_{n \rightarrow \infty} x_n = a$.

12. Find $\lim_{n \rightarrow \infty} \sqrt[n]{3^n + 5^n}$.

13. Prove that the recursively defined sequence $a_1 = \sqrt{2}$, $a_{n+1} = \sqrt{2 + a_n}$, $n \in \mathbb{N}$, satisfies the Weierstrass theorem. Find the limit of that sequence.

14. Find the number of terms of the arithmetic progression in which: the ratio of the sum of the first 13 terms to the sum of the last 13 terms is $1/2$; and the ratio of the sum of all the terms, except for the first three terms, to the sum of all the terms, except for the last three terms is $4/3$.

15. The sum of the terms of the arithmetic progression and its first term are positive. If we increase the difference of this progression by 3, without changing the first term, the sum of its terms will be doubled. Now if we increase the first term of the original progression four times, without changing its difference, the sum of its terms will also be doubled. Find the difference of the original progression.

16. An infinitely decreasing geometric progression, whose sum is equal to $16/3$, contains a term equal to $1/6$. The ratio of the sum of all the terms preceding that term to the sum of all the terms succeeding it is 30. Determine the ordinal number of that term of the progression.

17. Find the value of the product $0.2(7) \cdot 0.1(63)$ in the form of a common fraction.

18. Proceeding from the definition of a limit of a function at a point, prove that $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$.

Find the limits of the following functions:

$$19. \lim_{x \rightarrow 2} \frac{x+2}{\sqrt{x+3}-1} \quad 20. \lim_{x \rightarrow 1} \frac{x^3-1}{x-1}.$$

$$21. \lim_{x \rightarrow -1} \frac{2x^2+x-1}{x^2-6x-7} \quad 22. \lim_{x \rightarrow -\frac{\pi}{4}} \frac{\cos 2x}{\sin x + \cos x}.$$

23. Find the limits on the right and on the left at the point $x = 0$ of the function

$$f(x) = \frac{x - |x|}{2x}.$$

24. Find the limits on the right and on the left at the point $x = 1$ of the function

$$f(x) = \frac{x^2 - 1}{x^2 - 2|x - 1| - 1}.$$

$$25. \text{ Find } \lim_{x \rightarrow \infty} \frac{(x-1)^2(7x+2)^2}{(2x+1)^4}.$$

$$26. \text{ Find } \lim_{x \rightarrow +\infty} (\sqrt{x + \sqrt{x}} - \sqrt{x}).$$

27. Prove that as $x \rightarrow +\infty$ the function $f(x) = \sin x$ does not possess either a finite or an infinite limit.

28. Find the derivative of the function $y = \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^{50}$.

29. Find the equation of the tangent to the graph of the function $y = \sqrt{4x - 3 - x^2}$ at the point with the abscissa $x = 3/2$.

30. Find the equation of the common tangent to the parabolas $y = x^2 - 2x + 5$ and $y = x^2 + 2x - 11$.

PROBLEMS OF SECTION II

1. Prove that the following sequences are bounded:

(a) $\left(\frac{2n^2 - 1}{2 + n^2} \right)$; (b) $\left(\frac{1 - n}{\sqrt{n^2 + 1}} \right)$; (c) $\left(\frac{n + (-1)^n}{3n - 1} \right)$.

2. Proceeding from the definition of a limit, prove that

(a) $\lim_{n \rightarrow \infty} \frac{2n}{n+1} = 2$; (b) $\lim_{n \rightarrow \infty} \frac{1 + (-1)^n}{n} = 0$;

(c) $\lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0$; (d) $\lim_{n \rightarrow \infty} \frac{n - n^2}{2n^2 + 3} = -\frac{1}{2}$.

For each of the sequences fill in the table showing the relationship between N and ε :

ε	1/10	1/100	1/1000
N			

3. Prove that the following sequences are unbounded:

(a) $a_n = (-1)^n \cdot n$, $n \in \mathbb{N}$; (b) $a_n = n^2 - n$, $n \in \mathbb{N}$;

(c) $a_n = \frac{1 - n}{\sqrt{n}}$, $n \in \mathbb{N}$; (d) $a_n = n + (-1)^n \cdot n$, $n \in \mathbb{N}$.

4. Prove that the following sequences are divergent:

(a) $a_n = \sin \frac{\pi n}{2}$, $n \in \mathbb{N}$; (b) $a_n = \frac{1}{n} + (-1)^n$, $n \in \mathbb{N}$;

(c) $a_n = \frac{1 - n}{\sqrt{n}}$, $n \in \mathbb{N}$; (d) $a_n = \left[1 + \frac{(-1)^n}{n} \right]$, $n \in \mathbb{N}$ (here $[]$

designates the integral part of the number).

Find the limits of the sequences considered in problems 5-14:

5. $\lim_{n \rightarrow \infty} \frac{10^5 n}{10^5 + n}$. 6. $\lim_{n \rightarrow \infty} \frac{6n^3 - 2n + 1}{n^3 - 2n}$.

$$7. \lim_{n \rightarrow \infty} \frac{\sqrt[n]{n+1}}{\sqrt{n+1}}. \quad 8. \lim_{n \rightarrow \infty} \frac{1-n+2n^2}{5n^2+n}$$

$$9. \lim_{n \rightarrow \infty} \frac{1+2+\dots+n}{n^2}. \quad 10. \lim_{n \rightarrow \infty} \frac{\sqrt[3]{n^3+n}}{n+2}.$$

$$11. \lim_{n \rightarrow \infty} \sqrt[n]{1+2^n}. \quad 12. \lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{(n-1) \cdot n} \right).$$

$$13. \lim_{n \rightarrow \infty} \frac{3^n+2^n}{3^n-2^n}. \quad 14. \lim_{n \rightarrow \infty} \frac{2^{1/n}-1}{2^{1/n}+1}.$$

15. Prove that the following sequences are monotonic:

(a) $\left(\frac{n+1}{2n-1} \right);$ (b) $\left(\frac{1-n}{\sqrt{n}} \right);$

(c) $(2^n - 1);$

(d) Beginning with what n is the sequence $(n \cdot q^n)$ monotonic for $0 < q < 1$?

16. Find at what ratios of a, b, c and d the sequence $\left(\frac{an+b}{cn+d} \right)$ is

(a) increasing, (b) decreasing.

17. Prove that $\lim_{n \rightarrow \infty} n \cdot q^n = 0$ for $|q| < 1$.

18. Prove that each of the following sequences has a limit and find it:

(a) $a_1 = 1/2, a_n = 1/(2 - a_{n-1})$ for $n \geq 2$;

(b) $a_1 = 1/2, a_n = 2/(3 - a_{n-1})$ for $n \geq 2$.

19. Find the formula of the n th term for each of the following sequences and find its limit:

(a) $a_1 = 3, a_{n+1} = \frac{a_n+1}{2}$ for $n > 1$;

(b) $a_1 = 1, a_2 = \frac{5}{2}, a_n = \frac{3}{2} a_{n-1} - \frac{1}{2} a_{n-2}$ for $n \geq 3$.

20. Write the following expressions as an ordinary fraction in its lowest terms:

(a) $\frac{0.2(5)}{0.12(7)};$ (b) $0.3(8) \cdot 0.5(45);$ (c) $\frac{0.2(27)}{0.(63)}.$

21. Find the number of terms of an arithmetic progression in which the ratio of the sum of the first 23 terms to the sum of the last 23 terms is $2/5$, and the ratio of the sum of all the terms except for the first seven to the sum of all the terms without the last seven terms is $10/7$.

22. Find the number of terms of a geometric progression in which the ratio of the sum of the first 11 terms to the sum of the last 11 terms is $1/8$, and the ratio of the sum of all the terms without the first nine to the sum of all the terms without the last nine terms is 2.

23. Three numbers $x; y; z$, put in the indicated order, form a geometric progression, and three numbers, $x; 2y; 3z$, in the indicated order, form an arithmetic progression. Find the common ratio of the geometric progression differing from 1.

24. The second, first and third terms of the arithmetic progression, whose difference is nonzero, form, in the indicated order, a geometric progression. Find its common ratio.

25. The difference of the arithmetic progression is nonzero. The numbers equal to the products of the first term of the progression by the second, of the second term by the third and of the third by the first, form in the indicated order a geometric progression. Find its common ratio.

26. Find four integral numbers $a; b; c; d$, the first three of which, $a; b; c$, in the indicated order, form a geometric progression and the last three, $b; c; d$, in the indicated order, form an arithmetic progression, if the sum of the end numbers is 21 and the sum of the middle numbers is 18.

27. In the arithmetic progression whose difference is nonzero the sum of the first $3n$ terms is equal to the sum of the next n terms. Find the ratio of the sum of the first $2n$ terms to the sum of the next $2n$ terms.

28. Find the first term of the geometric progression in which the ratio of the sum of the first five terms to the sum of their inverse values is 49, and the sum of the first and the third term is 35.

29. Find the common ratio of an infinitely decreasing geometric progression in which the ratio of the sum of the cubes of all the terms to the sum of the squares of all the terms is 3, and the ratio of the sum of all the terms to the sum of the squares of all the terms is $3/7$.

30. Three numbers form a geometric progression. If we reduce the second term by 4, the resulting three numbers, in the same order, again form a geometric progression. If we reduce the third term of the new progression by 9, we obtain an arithmetic progression. Find the numbers.

31. Three numbers, whose sum is 78, form a geometric progression. They can also be taken as the first, the third and the ninth term of an arithmetic progression. Find the numbers.

32. Find the sum of the infinitely decreasing geometric progression whose second term is equal to 6 and whose sum is equal to $1/8$ of the sum of the squares of its terms.

33. Find the sum of the infinitely decreasing geometric progression in which the sum of the squares of the first n terms is equal to the sum of the first $2n$ terms, and the sum of the cubes of the first n terms is equal to $1/3$ of the sum of the first $3n$ terms.

34. Find the sum of the infinitely decreasing geometric progression the sum of whose first and fifth terms is 34 and the product of the first term by the ninth is 4.

35. Find the sum of the infinitely decreasing geometric progression whose all terms are positive, if the sum of the first three terms of that progression is 39 and the sum of their inverse values is $13/27$.

36. Find the sum S of the infinitely decreasing geometric progression if that sum is greater by 6 than the sum of its first three terms and the sum of the first three terms is greater by 1 than the sum of the next three terms.

37. Two arithmetic progressions have the same number of terms. The ratio of the last term of the first progression to the first term of the second progression is equal to the ratio of the last term of the second progression to the first term of the first progression and is equal to 4. The ratio of the sum of the terms of the first progression to the sum of the terms of the second progression is equal to 2. Determine the ratio of the differences of the progressions.

38. The sum and the difference of the terms of the arithmetic progression are positive. If we increase the difference by 2, without changing its first term, the sum of its terms will increase three times. Now if we increase the difference of the original progression four times, without changing the first term, the sum of the terms of the progression will increase five times. Determine the difference of the original progression.

39. Two infinitely decreasing geometric progressions are such that the first term of the first progression is the common ratio of the second and the common ratio of the first progression is the first term of the second. The sum of the sums of the two progressions is equal to 2. Find the first term of the first progression if its common ratio is equal to $1/3$.

40. Proceeding from the definition of a limit of a function at a point, prove that

$$(a) \lim_{x \rightarrow 2} (x^2 - 1) = 3; \quad (b) \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = 6; \quad (c) \lim_{x \rightarrow 0} \sqrt[3]{x} \cos \frac{1}{x} = 0.$$

Find the limits of the functions presented in problems 41 through 51.

$$41. \lim_{x \rightarrow 0} \frac{\sqrt{1+x}+3}{2-x}. \quad 42. \lim_{x \rightarrow 1} \frac{x^2+4x-5}{x^2-1}. \quad 43. \lim_{x \rightarrow 2} \frac{\sqrt{x-1}-1}{x-2}.$$

$$44. \lim_{x \rightarrow 1} \left(\frac{1}{1-x} - \frac{3}{1-x^3} \right). \quad 45. \lim_{x \rightarrow 0} \frac{\sqrt{1+x}-1}{x}.$$

$$46. \lim_{x \rightarrow 1} \frac{(x-1)\sqrt{2-x}}{x^2-1}. \quad 47. \lim_{x \rightarrow 1} \frac{x^3-2x+1}{x^2-2x+1}.$$

$$48. \lim_{x \rightarrow -3} \frac{x+3}{\sqrt{x+4}-1}. \quad 49. \lim_{x \rightarrow \infty} \frac{x^2-1}{2x^2+1}.$$

$$50. \lim_{x \rightarrow \infty} \left(\frac{x^3}{x^2+1} - x \right). \quad 51. \lim_{x \rightarrow \infty} x(\sqrt{x^2+1}-x).$$

52. Find the limits on the right and on the left at the point $x = 0$ of the function

$$f(x) = \frac{x^2 + |x| + x}{x^2 - |x| + 3x}.$$

53. Find the limits on the right and on the left at the point $x = 0$ of the function

$$f(x) = \frac{x^2 - |x|}{x^2 + x}.$$

54. Find the limits on the right and on the left at the point $x = 1$ of the function

$$f(x) = \frac{x-1+|x-1|}{x^2-1}.$$

55. For the function $f(x) = \frac{2(x-x^3)+|x-x^3|}{2(x-x^3)-|x-x^3|}$ find (a) the right-hand and left-hand limits at the point $x = 0$, (b) the right-hand and left-hand limits at the point $x = 1$.

56. Prove that $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$.

57. Find the right-hand and left-hand limits of the function $f(x) = 2^{1/x}$ at the point $x = 0$.

58. Find the right-hand and left-hand limits of the function $f(x) = \frac{1}{x-[x]}$ at the point $x = 1$.

Find the derivative of the function $y(x)$ appearing in problems 59 through 64.

59. $y = \frac{x-1}{\sqrt{x+1}}$. 60. $y = (x-2) \cos x - \sin x$.

61. $x = \frac{3x-\sqrt{x}}{x+1}$. 62. $y = \frac{\sin x - \cos x}{\sin x + \cos x}$.

63. $y = x \ln(x^2-1)$. 64. $y = 2^{x \cdot (1-x)}$.

65. Simplify the expression for $f(x)$ and find $f'(x)$:

$$f(x) = \left(\frac{\sqrt{x-2}}{\sqrt{x+2} + \sqrt{x-2}} + \frac{x-2}{\sqrt{x^2-4-x+2}} \right)^{-2}.$$

66. Find the point x at which the derivative of the function $f(x) = e^{2x-1} + 2e^{1-2x} + 7x - 5$ vanishes.

67. Solve the inequality $f'(x) < g'(x)$ if $f(x) = 2/x$ and $g(x) = x - x^2$.

68. Find the equation of the tangent to the parabola $y = 3x^2 - x + 1$ at the point with the abscissa $x = 1/2$.

69. Find the angle the tangent to the parabola $y = x^2 - 5x + 7$ at the point $M(2; 1)$ makes with the abscissa axis.

70. Find the equation of the common tangent to the parabolas $y = x^2 + 4x + 8$ and $y = x^2 + 8x + 4$.

71. Find the equation of the straight line which passes through the point $(1/2; 2)$, touches the parabola $y = 2 - \frac{x^2}{2}$, and intersects

the graph of the function $y = \sqrt{4-x^2}$: (a) at one point, (b) at two points.

Chapter 6

Investigating the Behaviour of Functions and Constructing Their Graphs

6.1. Even and Odd Functions

The function $f(x)$ defined on the set X is said to be *even* if, for any $x \in X$, the following conditions are fulfilled:

$$-x \in X \text{ and } f(-x) = f(x).$$

It is evident that the functions $f(x) = 1$ and $f(x) = x^2$, defined on the entire number axis, are even. However, the function $f(x) = x^2$, $x \in [1; 2]$, defined only on the interval $[1; 2]$, is not even since the point 1.5 belongs to the domain of definition of the function but the point -1.5 does not.

The graph of an even function on the coordinate plane is symmetric about the axis of ordinates.

□ Indeed, let the function $f(x)$, $x \in X$ be even and suppose that the point M with the coordinates x_0 , y_0 belongs to the graph of that function, i.e. $x_0 \in X$ and $y_0 = f(x_0)$. Then, by the definition of an even function, we have

$$-x_0 \in X, f(-x_0) = f(x_0) = y_0,$$

and this means that the point $M'(-x_0; y_0)$, which is symmetric with respect to the point $M(x_0; y_0)$ about the Oy axis, belongs to the graph of the given function. ■

The converse statement is also true: if the graph of the function $f(x)$ is symmetric about the axis of ordinates, then the function $f(x)$ is even.

If two even functions possess the same domain of definition, then the sum, the difference and the product of those functions are even functions.

The function $f(x)$, defined on the set X , is said to be *odd* if, for any $x \in X$, the following conditions are fulfilled:

$$-x \in X \text{ and } f(-x) = -f(x).$$

For example, the functions $f(x) = x^3$ and $f(x) = \sin x$, defined on the entire number axis, are odd.

It is easy to show that the graph of an odd function on the coordinate plane Oxy is symmetric about the origin O . The converse is also true: if the graph of the function $f(x)$ on the coordinate plane Oxy is symmetric about the origin O , then the function $f(x)$ is odd.

If two odd functions $f(x)$ and $g(x)$ have the same domain of definition X , then the functions $f(x) + g(x)$ and $f(x) - g(x)$ are also odd while the function $f(x) \cdot g(x)$ is even.

Note that the domains of definition of even and odd functions are symmetric about the point O on the coordinate line Ox .

For instance, the function $\ln x$ is neither even nor odd since it is defined only for $x > 0$. The function $f(x) = 1/(x-1)$ is also neither even nor odd since the point -1 belongs to the domain of definition of the function while the point 1 does not. Although the function $f(x) = x^2 + 3x$ is defined throughout the number axis, it is neither even nor odd since, for instance, $f(1) = 4$ and $f(-1) = -2$, i.e. $f(1) \neq f(-1)$ and $f(1) \neq -f(-1)$.

The examples cited above show that there are functions which are neither even nor odd. The following assertion can be proved for any function $f(x)$.

If the function $f(x)$ is defined on the set X , which is symmetric about the point O on the coordinate line Ox , then it can be represented as the sum of two functions, even and odd, and such a representation is unique.

□ Indeed, assume that there exist an even function $f_1(x)$ and an odd function $f_2(x)$ such that

$$f(x) = f_1(x) + f_2(x)$$

for any $x \in X$. Then we have $f(-x) = f_1(-x) + f_2(-x) = f_1(x) - f_2(x)$ for every $x \in X$. To define the functions $f_1(x)$ and $f_2(x)$, we obtain the following system of equations:

$$f(x) = f_1(x) + f_2(x),$$

$$f(-x) = f_1(x) - f_2(x),$$

which yields

$$f_1(x) = \frac{f(x) + f(-x)}{2}, \quad f_2(x) = \frac{f(x) - f(-x)}{2}. \quad (1)$$

Consequently, the required functions can be unambiguously found from the given function by formulas (1). The same formulas imply that the desired representation always exists. ■

Example 1. Represent the function

$$f(x) = 3x^5 - 4x^4 + x^2 + 1$$

as the sum of an even and an odd function.

Δ The given function is the sum of the odd function $f_2(x) = 3x^5$ and the even function $f_1(x) = -4x^4 + x^2 + 1$. The same functions can, evidently, be obtained by formulas (1) as well.

Note that the function $f_1(x)$ contains only those summands in which x appears in an even degree and the function $f_2(x)$ contains the summands in which x is of an odd degree. ▲

Example 2. Suppose the function $f(x) = 2x + 1$ is given on the interval $[0; 1]$. Can we extend the definition of the function to all the points of the number axis so that the resulting function be (a) even, (b) odd?

Δ (a) The definition of the function $f(x)$ can be extended to the entire number axis so that the resulting function is even. One of such functions is $F(x) = 1 + 2|x|$, $x \in \mathbb{R}$ (Fig. 49).

It is evident that such a function cannot be defined unambiguously. In fact, this property is inherent in the function $F_1(x)$ as well, which is equal to $1 + 2|x|$ on the interval $[-1; 1]$ and is zero for all x , $|x| > 1$.

(b) Note that if an odd function is defined for $x = 0$, then $f(0) = f(-0) = -f(0)$ and, consequently, $f(0) = 0$.

Since for $x = 0$ the given function assumes the value 1, it is impossible to extend its definition to the entire number axis so that it should be odd. ▲

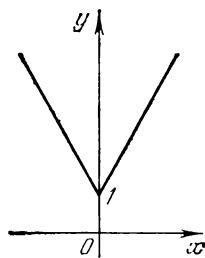


Fig. 49.

6.2. Periodic Functions

Many important processes in nature and engineering are periodic. To study such processes, it is necessary to consider the behaviour of periodic functions.

The function $f(x)$, defined on the set X , is said to be *periodic* if there is a number $T > 0$ such that

$$x + T \in X, \quad x - T \in X$$

and

$$f(x + T) = f(x - T) = f(x)$$

for any $x \in X$. The number T is called a *period* of the function $f(x)$, $x \in X$.

It is evident that if the number T is a period of some function, then any number nT , where $n \in \mathbb{N}$, is also a period of that function.

We shall now show that if the number $T_0 > 0$ is the least period of some function, then any period of that function has the form nT_0 , $n \in \mathbb{N}$.

□ Suppose T_0 is the least period and T is some period of the function $f(x)$, $x \in X$. We shall find the number $n \in \mathbb{N}$ such that $T = nT_0 + \alpha T_0$, where $0 \leq \alpha < 1$. Then, for any $x \in X$, we have

$$f(x) = f(x + T) = f(x + nT_0 + \alpha T_0) = f(x + \alpha T_0).$$

T_0 being the least period, it follows that $\alpha = 0$, i.e. $T = nT_0$. ■

Let us consider the function $\sin x$. It follows from its definition that it is periodic and that the number 2π is its period. We shall show that any positive number $T < 2\pi$ is not its period. Indeed, $\sin(\pi/2) = 1$, but $\sin(\frac{\pi}{2} + T) < 1$.

Consequently, 2π is the least period of the function $\sin x$.

Not every periodic function possesses the least period. The function $f(x) = 2$, $x \in \mathbb{R}$, for instance, is periodic, its period is any positive number. This function, evidently, does not have the least period.

It follows from the definition itself that if the periodic function $f(x)$ is not defined at the point x_0 , then it is not defined at all the points of the form $x_0 \pm nT$ either, where $n \in \mathbb{N}$ and T is a period of the function $f(x)$. Therefore, any function defined at all the points of the number axis, except for one point, is not periodic.

From the definition itself it also follows that if the function is not defined, say, for all $x \leq a$ or $x \geq a$, then it is not periodic. Thus, for instance, the function $f(x) = \sin \sqrt{x}$, defined only for $x \geq 0$, is not periodic.

The function $f(x) = x^4$ is defined throughout the number line, but, nevertheless, it is not periodic. Indeed, if the number $T > 0$ is a period of that function, then $f(0) = f(T)$, i.e. $0 = T^4$, which is impossible.

Example 1. Prove that the function $\cos^2 x$ is periodic and find its least period.

Δ Since the function $\cos x$ is periodic with the period 2π , the function $\cos^2 x$ is also periodic and the number 2π is its period. Let us find its least period.

It is easy to verify that the number π is a period of the function $\cos^2 x$. In fact, for any $x \in \mathbb{R}$ we have

$$\cos^2(x + \pi) = (-\cos x)^2 = \cos^2 x.$$

And since $\cos^2 0 = \cos^2 \pi = 1$ and $\cos^2 x < 1$ for any $x \in (0; \pi)$, it follows that the number π is the least period of the given function. $\blacktriangle$

It is evident that if two periodic functions, $f_1(x)$ and $f_2(x)$, $x \in X$, have the same period T , then their sum, difference, and product are also periodic functions and the number T is their period.

If the periodic functions $f_1(x)$ and $f_2(x)$, $x \in X$, have commensurable periods T_1 and T_2 , then they possess a common period.

Indeed, T_1 and T_2 being commensurable, there are natural numbers n and m such that $mT_1 = nT_2 = T$. Consequently, the number T is the common period of the functions $f_1(x)$ and $f_2(x)$. In particular, in that case the functions $f_1(x) + f_2(x)$, $f_1(x) - f_2(x)$, $f_1(x) \cdot f_2(x)$ are periodic with period T .

Example 2. Prove that the function

$$f(x) = \cos 3x + \cos 4x$$

is periodic and find its least period.

Δ The function $\cos 3x$ has a period $T_1 = 2\pi/3$ and the function $\cos 4x$ has a period $T_2 = 2\pi/4$. The periods T_1 and T_2 are commensurable: $3T_1 = 4T_2 = 2\pi$. This means

that the number 2π is a period of the given function. Let us show that this period is the least.

The greatest value of the function $f(x)$ is 2 and it is attained at the points where $\cos 3x = 1$ and $\cos 4x = 1$. On the interval $[0; 2\pi]$ the function $\cos 3x$ assumes the value 1 at the points 0, $2\pi/3$, $4\pi/3$, 2π , and the function $\cos 4x$ assumes the value 1 at the points 0, $\pi/2$, π , $3\pi/2$, 2π . Consequently, on the interval $[0; 2\pi]$ the function $f(x) = \cos 3x + \cos 4x$ assumes the value 2 only at the points 0 and 2π . Hence it follows that the number 2π is the least period. ▲

Now if the periods of the functions $f_1(x)$ and $f_2(x)$, $x \in X$, are incommensurable, then the function $f_1(x) + f_2(x)$ is not periodic.

6.3. Asymptotes

When investigating a function and constructing its graph, it is important to elucidate its behaviour in the neighbourhood of the points of discontinuity and of the points where the function is not defined, and also

find out how it behaves as $x \rightarrow +\infty$ and $x \rightarrow -\infty$.

Let us consider the function

$$f(x) = \frac{x^2 + 2x}{x + 1}. \quad (1)$$

It is defined for all $x \in \mathbb{R}$, except for $x = -1$. We shall see how function (1) behaves as $x \rightarrow +\infty$. Since $f(x) =$

$x + 1 - \frac{1}{x+1}$, the difference $f(x) - (x + 1)$ tends to zero as $x \rightarrow +\infty$. Note that the absolute value of the difference $f(x) - (x + 1)$ is equal to the distance from the point $M(x; f(x))$ to the point $M'(x; x + 1)$ of the line l defined by the equation $y = x + 1$. And since $|MM'| = \sqrt{2}|MN|$ (Fig. 50), where N is the projection of the point M on l , it follows that the distance from M to the line l tends to zero as $x \rightarrow +\infty$. In such a case the straight line l is called an *asymptote* to the graph of the function $y = f(x)$

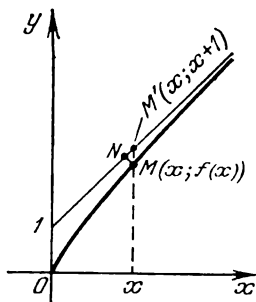


Fig. 50.

as $x \rightarrow +\infty$. It can be shown by analogy that in the given case the distance from the point M of the graph of the function to the line l tends to zero as $x \rightarrow -\infty$. The straight line l is called an *asymptote* to the graph of the function $y = f(x)$ as $x \rightarrow -\infty$.

Here are requisite definitions.

Let the function $f(x)$ be defined on the infinite interval $(a; +\infty)$. The straight line $y = kx + b$ is called an *asymptote* to the graph of the function $y = f(x)$ as $x \rightarrow +\infty$, if

$$\lim_{x \rightarrow +\infty} (f(x) - (kx + b)) = 0.$$

Similarly, the straight line $y = kx + b$ is called an *asymptote* to the graph of the function $y = f(x)$, $x \in (-\infty; b)$ as $x \rightarrow -\infty$, if

$$\lim_{x \rightarrow -\infty} (f(x) - (kx + b)) = 0.$$

Let us now see how function (1) behaves as $x \rightarrow -1$.

Since, as $x \rightarrow -1$, the limit of the numerator is equal to -1 and the limit of the denominator to 0 , with $x + 1 < 0$ for $x < -1$ and $x + 1 > 0$ for $x > -1$, it follows that

$$\lim_{x \rightarrow -1-0} f(x) = +\infty \quad \text{and} \quad \lim_{x \rightarrow -1+0} f(x) = -\infty.$$

Hence it follows that the graph of the given function in the neighbourhood of the point $x = -1$ has the form given in Fig. 51. As can be seen, the point $M(x; f(x))$ of the graph of the function approaches the straight line $x = -1$ as $x \rightarrow -1$, the distance from the point M to the straight line $x = -1$, equal to $x + 1$, tends to zero as $x \rightarrow -1$.

The straight line $x = -1$ is known as a *vertical asymptote* to the graph of the function $y = f(x)$ as $x \rightarrow -1$ from the left and from the right.

Here is a requisite definition.

Let the function $f(x)$ be defined on some interval $(a; b)$. If $f(x) \rightarrow \infty$ as $x \rightarrow a + 0$, then the straight line $x = a$ is a *vertical asymptote* to the graph of the function as $x \rightarrow a$

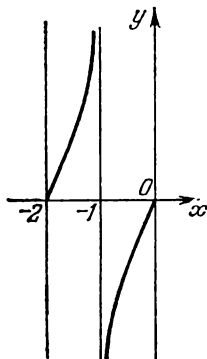


Fig. 51.

from the right. Now if $f(x) \rightarrow \infty$ as $x \rightarrow b - 0$, then the straight line $x = b$ is a *vertical asymptote* to the graph of the function $y = f(x)$ as $x \rightarrow b$ from the left.

Suppose the straight line $y = kx + b$ is an asymptote to the graph of the function as $x \rightarrow +\infty$. Let us consider the function $\alpha(x) = f(x) - kx - b$. It follows from the definition of an asymptote that

$$\lim_{x \rightarrow +\infty} \alpha(x) = 0.$$

Since $b = f(x) - kx - \alpha(x)$, we have

$$b = \lim_{x \rightarrow +\infty} (f(x) - kx), \quad (2)$$

and since

$$k = \frac{f(x)}{x} - \frac{b}{x} - \frac{\alpha(x)}{x},$$

we have

$$k = \lim_{x \rightarrow +\infty} \frac{f(x)}{x}. \quad (3)$$

It is evident that if the limits (2) and (3) exist, then the straight line $y = kx + b$ is the asymptote to the graph of the function corresponding to $x \rightarrow +\infty$.

It can be shown by analogy that if the function $y = f(x)$ is defined on the interval $(-\infty; a)$ and if there exist the limits

$$\lim_{x \rightarrow -\infty} \frac{f(x)}{x} = k, \quad \lim_{x \rightarrow -\infty} (f(x) - kx) = b,$$

then the straight line $y = kx + b$ is an asymptote to the graph of the function $y = f(x)$ as $x \rightarrow -\infty$.

The straight line $y = kx + b$ is known as an *inclined asymptote* if $k \neq 0$. If $k = 0$, then the line $y = b$ is called a *horizontal asymptote*.

Example 1. Find the asymptotes to the graph of the function

$$f(x) = \frac{x^2}{|x| + 1}.$$

Δ The function $f(x)$ can be represented in the form

$$f(x) = \frac{|x|^2}{|x| + 1} = \frac{|x^2| + |x| - |x| - 1 + 1}{|x| + 1} = |x| - 1 + \frac{1}{|x| + 1}.$$

Since $\lim_{x \rightarrow \infty} \frac{1}{|x|+1} = 0$, the straight line $y = x - 1$ is an asymptote to the graph of the function $y = f(x)$ as $x \rightarrow +\infty$, and the line $y = -x - 1$ is an asymptote to the graph of the function $y = f(x)$ as $x \rightarrow -\infty$. Since $\frac{1}{|x|+1} > 0$ for any x , the graph of the function $y = f(x)$ lies above the asymptotes.

The graph of the function $y = f(x)$ has no vertical asymptotes. $\blacktriangle$

Example 2. Find the asymptotes to the graph of the function $f(x) = \sqrt{x^2 - 1}$.

Δ The function $y = f(x)$ is defined for $|x| \geq 1$. There are no vertical asymptotes to the graph of the function $f(x)$.

By (3) we have

$$k = \lim_{x \rightarrow +\infty} \frac{\sqrt{x^2 - 1}}{x} = \lim_{x \rightarrow +\infty} \frac{x \sqrt{1 - \frac{1}{x^2}}}{x} = \lim_{x \rightarrow +\infty} \sqrt{1 - \frac{1}{x^2}} = 1.$$

By (2) we have

$$\begin{aligned} b &= \lim_{x \rightarrow +\infty} [\sqrt{x^2 - 1} - x] = \lim_{x \rightarrow +\infty} \frac{(\sqrt{x^2 - 1} - x)(\sqrt{x^2 - 1} + x)}{\sqrt{x^2 - 1} + x} \\ &= \lim_{x \rightarrow +\infty} \frac{x^2 - 1 - x^2}{\sqrt{x^2 - 1} + x} = \lim_{x \rightarrow +\infty} \frac{-1}{\sqrt{x^2 - 1} + x} = 0. \end{aligned}$$

Thus we see that the graph of the function $y = f(x)$ has an asymptote $y = x$ as $x \rightarrow +\infty$.

Let us find the asymptote to the graph of the function $y = f(x)$ corresponding to $x \rightarrow -\infty$.

$$\begin{aligned} k &= \lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 - 1}}{x} = \lim_{x \rightarrow -\infty} \frac{|x| \sqrt{1 - \frac{1}{x^2}}}{x} \\ &= \lim_{x \rightarrow -\infty} \frac{-x \sqrt{1 - \frac{1}{x^2}}}{x} = \lim_{x \rightarrow -\infty} \left(-\sqrt{1 - \frac{1}{x^2}} \right) = -1, \end{aligned}$$

$$\begin{aligned} b &= \lim_{x \rightarrow -\infty} (\sqrt{x^2 - 1} + x) = \lim_{x \rightarrow -\infty} \frac{(\sqrt{x^2 - 1} + x)(\sqrt{x^2 - 1} - x)}{\sqrt{x^2 - 1} - x} \\ &= \lim_{x \rightarrow -\infty} \frac{x^2 - 1 - x^2}{\sqrt{x^2 - 1} - x} = \lim_{x \rightarrow -\infty} \frac{-1}{\sqrt{x^2 - 1} - x} = 0. \end{aligned}$$

Consequently, the graph of the function $y = f(x)$ has an asymptote $y = -x$ as $x \rightarrow -\infty$. ▲

It should be noted that the graph of the function $f(x) = \sqrt{x^2 - 1}$ possesses different asymptotes corresponding to $x \rightarrow +\infty$ and $x \rightarrow -\infty$. The graph of the function (1) possesses one and the same asymptote as $x \rightarrow +\infty$ and $x \rightarrow -\infty$.

Example 3. Find the asymptotes to the graph of the function $f(x) = x + \frac{1}{x}$.

△ Since $\lim_{x \rightarrow 0} f(x) = \infty$, the straight line $x = 0$ is a vertical asymptote to the graph of the function $y = f(x)$ as $x \rightarrow 0$. Since $\lim_{x \rightarrow \infty} (1/x) = 0$, the straight line $y = x$ is an inclined asymptote to the graph of the function $y = f(x)$ as $x \rightarrow +\infty$ and $x \rightarrow -\infty$. ▲

Example 4. Find the asymptotes to the graph of the function $f(x) = \frac{\sin x}{x}$, $x \in [\pi; +\infty)$.

△ The function $y = f(x)$ is the product of the bounded function $\sin x$ by the function $1/x$, which tends to zero as $x \rightarrow +\infty$. Therefore,

$$\lim_{x \rightarrow +\infty} f(x) = 0,$$

that is, the straight line $y = 0$ is a horizontal asymptote to the graph of the function $y = f(x)$ as $x \rightarrow +\infty$.

This example is of especial interest since the graph of the function $f(x)$ intersects the asymptote $y = 0$ infinitely many times. ▲

6.4. Transforming the Graphs of Functions

It will be shown in this section how we can obtain, from the graph of the function $y = f(x)$, the graphs of the functions of the form $y = Af(ax + b) + B$, where A , B , a , b are some real numbers.

1. The graph of the function $y = f(x) + b$ can be obtained from that of the function $y = f(x)$ by means of the translation $\tau(0; b)$.

If $b > 0$, then the graph is displaced parallel to the axis

of ordinates by the distance b upwards, and if $b < 0$, then by the distance $|b|$ downwards.

Figure 52 shows the graphs of the functions $y = f(x)$, $y = f(x) + 1$ and $y = f(x) - 2$.

2. The graph of the function $y = f(x + a)$ can be obtained from that of the function $y = f(x)$ by means of the translation $\mathbf{r}(-a; 0)$.

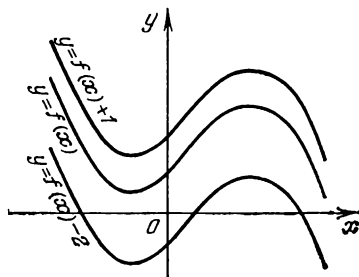


Fig. 52.

If $a > 0$, then the graph is displaced parallel to the abscissa axis to the left, by the distance a , and if $a < 0$, then to the right, by the distance $|a|$.

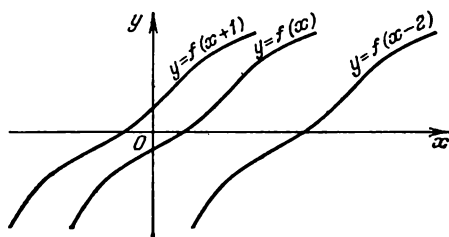


Fig. 53.

Figure 53 illustrates the graphs of the functions $y = f(x)$, $y = f(x + 1)$ and $y = f(x - 2)$.

3. The graph of the function $y = Af(x)$, where $A > 0$, can be obtained from the graph of the function $y = f(x)$ by contraction towards the Ox axis or expansion from the Ox axis in the ratio $1:A$, that is, if $0 < A < 1$, then by $1/A$ -fold contraction and if $A > 1$, then by A -fold expansion.

Figure 54 shows the graphs of the functions $y = f(x)$, $y = 2f(x)$ and $y = \frac{1}{2}f(x)$.

4. The graph of the function $y = -f(x)$ can be obtained from the graph of the function $y = f(x)$ by means of mirror reflection about the abscissa axis.

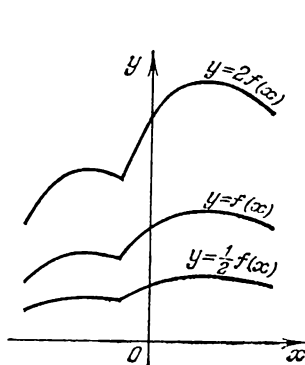


Fig. 54.

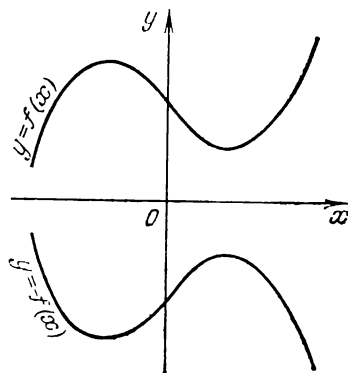


Fig. 55.

Figure 55 illustrates the graphs of the functions $y = f(x)$ and $y = -f(x)$.

5. The graph of the function $y = f(kx)$, where $k > 0$, can be obtained from the graph of the function $y = f(x)$ by

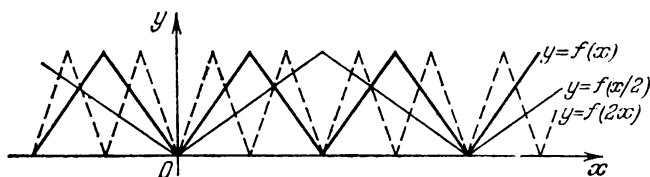


Fig. 56.

contraction towards the axis of ordinates or expansion from the axis of ordinates in the ratio $1 : \frac{1}{k}$, that is, if $0 < k < 1$, then by $1/k$ -fold expansion and if $k > 1$, then by k -fold contraction.

Figure 56 shows the graphs of the functions $y = f(x)$, $y = f(2x)$ and $y = f(x/2)$.

6. The graph of the function $y = f(-x)$ can be obtained from the graph of the function $y = f(x)$ by a mirror reflection about the axis of ordinates.

Figure 57 illustrates the graphs of the functions $y = f(x)$ and $y = f(-x)$.

7. Let us now see how the graph of the function $y = Af(ax + b) + B$, where $A > 0$ and $a > 0$, can be

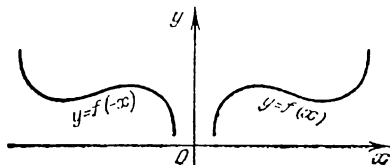


Fig. 57.

obtained from the graph of the function $y = f(x)$. Since

$$y = Af\left(a\left(x + \frac{b}{a}\right)\right) + B,$$

we can obtain the graph of the function $y = f(ax)$ from the graph of the function $y = f(x)$ by means of contraction to the axis of ordinates in the ratio $1 : \frac{1}{a}$. By contraction towards the abscissa axis in the ratio $1 : A$ we can obtain the graph of the function $y = Af(ax)$ from the graph of the function $y = f(ax)$ and then, from the latter graph, we can obtain the graph of the function $y = Af(ax + b) + B$ by means of the translation $\mathbf{r}(-b/a; B)$.

6.5. Elementary Functions and Their Graphs

1. A *linear function* is a function of the form $y = kx + b$, where k and b are some real numbers. It is defined throughout the number axis $\mathbb{R}$. If $k \neq 0$, then its range is the set $\mathbb{R}$, now if $k = 0$, then the range of the function is the single number b .

The function $y = kx + b$ is monotonic: for $k > 0$ it increases on $\mathbb{R}$, for $k < 0$ it decreases, and for $k = 0$ it is constant.

The graph of the linear function $y = kx + b$ is a straight line which passes through the point $(0; b)$ with the slope k .

For $k \neq 0$ this straight line intersects the abscissa axis at the point $x = -b/k$, and for $k = 0$ it is parallel to the abscissa axis.

Shown in Fig. 58 are the graphs of the straight lines $y = x$, $y = 2$, $y = -2x + 1$.

2. A *quadratic function* is a function of the form

$$y = ax^2 + bx + c,$$

where a , b and c are some real numbers, with $a \neq 0$. It is defined throughout the number line $\mathbb{R}$ and assumes real values.

The simplest quadratic function is the function of the form $y = ax^2$, where $a > 0$. This function is even, non-

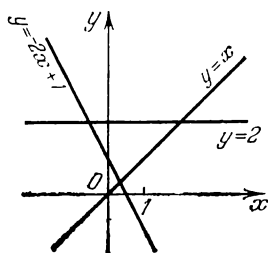


Fig. 58.

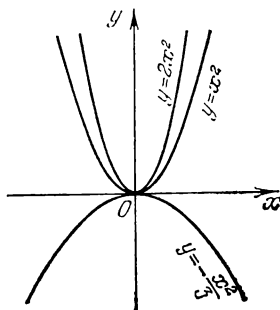


Fig. 59.

negative and can assume any value from the set $[0; +\infty)$. It decreases on the interval $(-\infty; 0)$ and increases on the interval $[0; +\infty)$. The point $x = 0$ is the point of minimum. For any $a > 0$ the graph of the function $y = ax^2$ can be obtained from that of the function $y = x^2$ by means of contraction towards the abscissa axis in the ratio $1 : a$.

The graph of the function $y = ax^2$, where $a < 0$, can be obtained from that of the function $y = |a| x^2$ by means of a mirror reflection about the abscissa axis. For $a < 0$ the function $y = ax^2$ increases on the interval $(-\infty; 0)$ and decreases on the interval $(0; +\infty)$. The point $x = 0$ is its point of maximum.

Shown in Fig. 59 are the graphs of the functions $y = x^2$, $y = 2x^2$, $y = -x^2/3$.

For any $a \neq 0$ the function $y = ax^2$ is even and, therefore, the Oy axis is the symmetry axis of the graph of that function. The graph of the function $y = ax^2$ is called a *parabola*, the point where the parabola cuts its axis of symmetry is called the *vertex* of the parabola. It is evident that the vertex of the parabola defined by the equation $y = ax^2$ is at the point $O(0; 0)$. For $a > 0$ the parabola is concave up and for $a < 0$ it is concave down.

We shall show now that the graph of any quadratic function is a parabola with vertex at a certain point $M_0(x_0; y_0)$ and the axis of symmetry parallel to the axis of ordinates. Indeed, since

$$\begin{aligned} ax^2 + bx + c &= a \left(x^2 + \frac{b}{a}x \right) + c \\ &= a \left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} \right) - \frac{b^2}{4a} + c = a \left(x + \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a}, \end{aligned}$$

the graph of the function $y = ax^2 + bx + c$ can be obtained from the graph of the function $y = ax^2$ by means of the translation $\mathbf{r} \left(-\frac{b}{2a}; \frac{4ac - b^2}{4a} \right)$. Thus, the vertex of the parabola defined by the equation $y = ax^2 + bx + c$ is at the point with the coordinates

$$x_0 = -b/2a, y_0 = (4ac - b^2)/4a.$$

The symmetry axis is the straight line $x = -b/2a$, the parabola being concave up if $a > 0$ and concave down if $a < 0$.

Example 1. Construct the graph of the function $y = -2x^2 + 4x + 1$.

Δ Since

$$\begin{aligned} y &= -2x^2 + 4x + 1 = -2(x^2 - 2x) + 1 \\ &= -2(x - 1)^2 + 3, \end{aligned}$$

its graph can be obtained from that of the function $y = -2x^2$ by means of the translation $\mathbf{r}(1; 3)$. Thus, the graph of the given function is a parabola concave down with vertex at the point $M_0(1; 3)$ (Fig. 60). $\blacktriangle$

3. A *linear-fractional* (homographic) function is defined by the formula

$$y = \frac{ax + b}{cx + d}, \quad (1)$$

where a , b , c , and d are some real numbers. It is evident that if $c = d = 0$, formula (1) does not define any function. Now if $c = 0$, $d \neq 0$, we have a linear function

$$y = \frac{a}{d}x + \frac{b}{d}.$$

We can assume, therefore, that the linear-fractional function is given by formula (1), where $c \neq 0$, and is defined for all $x \in \mathbb{R}$, except $x = -d/c$. In that case, we can transform the right-hand side of formula (1) as follows:

$$\frac{ax+b}{cx+d} = \frac{a\left(x + \frac{d}{c}\right) + b - \frac{ad}{c}}{c\left(x + \frac{d}{c}\right)} = \frac{a}{c} + \frac{bc-ad}{c^2} \frac{1}{x + \frac{d}{c}}. \quad (2)$$

It can be seen from this formula that the function defined by (1), where $bc - ad = 0$, assumes the constant value a/c at all points of its definition. The only distinction between this function and the constant function is that it is not defined at the point $-d/c$.

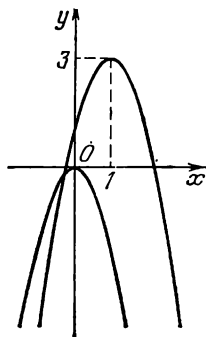


Fig. 60.

Our discussion of a linear-fractional function naturally leads to the following definition: a *linear-fractional function* is a function of the form (1), where $c \neq 0$ and $bc - ad \neq 0$.

It follows from formula (2) that the graph of the linear-fractional function (1) can be obtained from the graph of the function

$$y = \frac{k}{x}, \text{ where } k = \frac{bc-ad}{c^2}$$

by the parallel displacement $\mathbf{r}(-d/c; a/c)$.

As is known, the graphs of the functions of the form $y = k/x$, where $k \neq 0$, are called *hyperbolas*. It follows from (2) that the graph of any linear-fractional function is a hyperbola.

The straight line $y = 0$ is a horizontal asymptote to the hyperbola $y = k/x$ as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$, and

the straight line $x = 0$ is its vertical asymptote. It follows from (2) that the straight line $y = a/c$ is a horizontal asymptote to the graph of the linear-fractional function (1) as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$, and the line $x = -d/c$ is its vertical asymptote.

4. An *exponential function* is a function of the form $y = a^x$, where $a > 0$, $a \neq 1$. The exponential function is

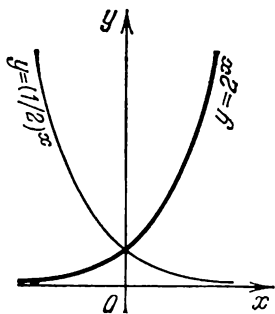


Fig. 61.

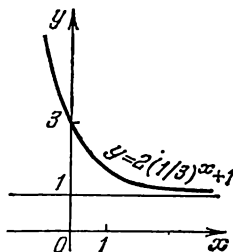


Fig. 62.

defined throughout the number line and maps the interval $(-\infty; +\infty)$ onto the interval $(0; +\infty)$ one-to-one.

For $a > 1$ the function $y = a^x$ increases and for $0 < a < 1$ it decreases.

The straight line $y = 0$ is a horizontal asymptote to the graph of the function $y = a^x$ as $x \rightarrow -\infty$ if $a > 1$ and as $x \rightarrow +\infty$ if $0 < a < 1$ respectively.

Figure 61 shows the graphs of the functions $y = 2^x$, $y = (1/2)^x$.

Example 2. Construct the graph of the function

$$y = 2(1/3)^x + 1.$$

Δ The graph of the given function results from the parallel displacement $r(0; 1)$ of the graph of the function $y = 2(1/3)^x$, which, in its turn, is obtained from the graph of the exponential function $y = (1/3)^x$ by a two-fold expansion from the abscissa axis (Fig. 62). The straight line $y = 1$ is a horizontal asymptote to the graph of the function $y = 2(1/3)^x + 1$ as $x \rightarrow +\infty$. ▲

5. A *logarithmic function* is a function of the form

$$y = \log_a x, \quad (3)$$

where $a > 0$, $a \neq 1$. It is defined only for positive values of x and maps one-to-one the interval $(0; +\infty)$ onto the interval $(-\infty; +\infty)$. It follows from the definition of the logarithm of a number to the base a that the logarithmic function $\log_a x$ is an inverse of the exponential function a^x .

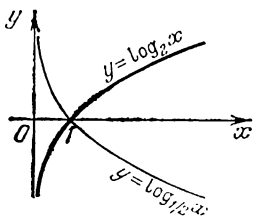


Fig. 63.

If $a > 1$, then the logarithmic function (3) is increasing, and if $0 < a < 1$, it is decreasing.

The graph of the logarithmic function cuts the abscissa axis at the point $x = 1$ and does not intersect the axis of ordinates. It should be noted that

the graph of the logarithmic function (3) can be obtained from the graph of the function $y = a^x$ by means of a mirror reflection about the straight line $y = x$.

Shown in Fig. 63 are the graphs of the functions $y = \log_2 x$ and $\log_{1/2} x$.

Example 3. Construct the graph of the function

$$y = \log_{1/3} |27x - 54|. \quad (4)$$

$\triangle$ Since

$$y = \log_{1/3} |27x - 54| = -3 + \log_{1/3} |x - 2|,$$

the graph of the given function can be obtained from the graph of the function

$$y = \log_{1/3} |x| \quad (5)$$

by the translation $\mathbf{r}(2; -3)$. This function is defined for all $x \neq 0$ and is even. Consequently, a part of its graph, corresponding to $x < 0$, can be obtained from the graph of the function $y = \log_{1/3} x$, where $x > 0$, by means of a mirror reflection about the axis of ordinates. Figure 64 shows the graph of the function $y = \log_{1/3} |x|$, and Fig. 65, the graph of the function $y = \log_{1/3} |27x - 54|$. $\blacktriangle$

6. A *power function* with the exponent p is a function defined by the formula

$$y = x^p, \quad (6)$$

where p is some real number.

Some remarks are due concerning the natural domain of definition of the function specified by formula (6).

It follows from the definition of a power that if p is a natural number, then formula (6) defines the function through-

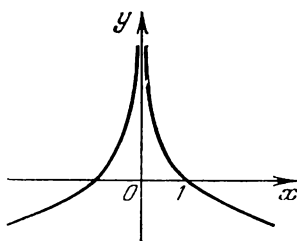


Fig. 64.

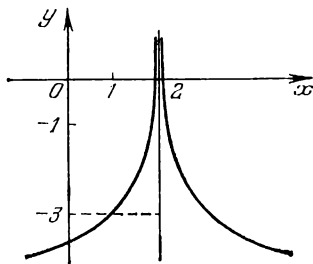


Fig. 65.

out the number line. In particular, for $p = 1$ we have a linear function and for $p = 2$, a quadratic function. If p is a negative integer, then function (6) is defined for all real values of x , except for $x = 0$. In particular, if $p = -1$, then function (6) is a simplest linear-fractional function.

If p is not an integer, then the degree of x^p is generally defined only for $x > 0$. For example, the function $y = x^{1/2}$ is defined on the infinite interval $[0; +\infty)$, and the function $y = x^{-1/2}$ is defined on the infinite interval $(0; +\infty)$.

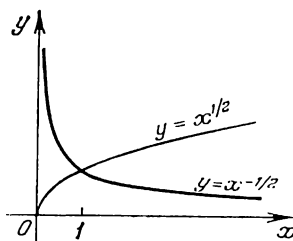


Fig. 66.

The graphs of these functions are illustrated in Fig. 66.

Thus we see that the natural domain of definition of the function specified by formula (6) essentially depends on the exponent p .

It is to be noted, however, that in practical applications we often understand the domain of definition of a power function as the natural domain of definition of the corresponding formula. For instance, it is natural to assume that the function $y = x^{1/3}$, which is an inverse of the function $y = x^3$, is defined on the entire number line.

7. The basic trigonometric functions are the *sine*, *cosine*, *tangent* and *cotangent* functions:

$$\sin x, \cos x, \tan x = \frac{\sin x}{\cos x}, \cot x = \frac{\cos x}{\sin x}.$$

The sine function is defined on the set $\mathbb{R}$ of all real numbers, it is odd and periodic, with the smallest period 2π .

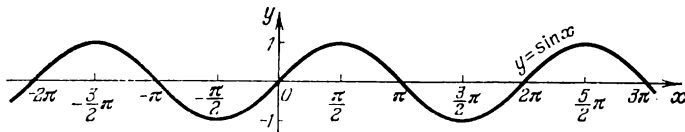


Fig. 67.

The graph of the sine function is a *sinusoid* (Fig. 67). The function $\sin x$ increases from -1 to $+1$ on the interval $[-\pi/2; \pi/2]$, decreases from $+1$ to -1 on the interval $[\pi/2; 3\pi/2]$, and so on. The function $\sin x$ being periodic, with period 2π , its graph, on any interval $[\alpha + 2\pi n; \alpha + 2\pi(n+1)]$, 2π long, can be obtained from its graph on the interval $[\alpha; \alpha + 2\pi]$ by the translation $\mathbf{r}(2\pi n; 0)$.

The cosine function is defined on the set $\mathbb{R}$ of all real numbers, it is even and periodic, with the smallest period 2π . On the interval $[0; \pi]$ it decreases from $+1$ to -1 and on the interval $[\pi; 2\pi]$ it increases from -1 to $+1$. Since $\cos x = \sin(x + \frac{\pi}{2})$ for any $x \in \mathbb{R}$, the graph of the function $y = \cos x$ is a sinusoid, which is obtained from the sinusoid $y = \sin x$, shown in Fig. 67, by the translation $\mathbf{r}(-\pi/2; 0)$. The graph of the function $y = \cos x$ is shown in Fig. 68.

The function

$$\tan x = \frac{\sin x}{\cos x}$$

is defined on the set of all real numbers, except for the numbers $\frac{\pi}{2} + \pi n$, where $n \in \mathbb{Z}$, that is, except the points at which $\cos x$ vanishes. The function $\tan x$ is odd and periodic, with the smallest period π . On the interval $(-\pi/2; \pi/2)$ it increases from $-\infty$ to $+\infty$. The straight lines $x = -\pi/2$, $x = \pi/2$ are vertical asymptotes of the graph of the func-

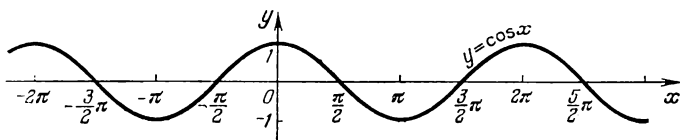


Fig. 68.

tion $y = \tan x$. The function $\tan x$ being periodic, with period π , its graph on any interval of the form $(-\frac{\pi}{2} +$

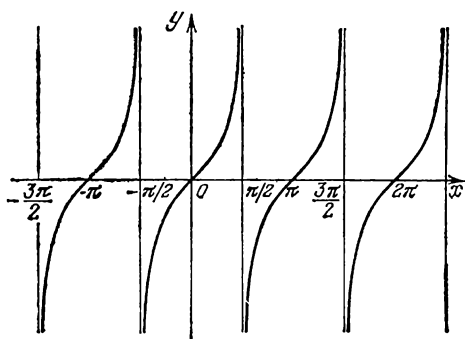


Fig. 69.

$n\pi; \frac{\pi}{2} + n\pi)$ can be obtained from its graph on $(-\pi/2; \pi/2)$ by the translation $\tau(n\pi; 0)$. The graph of the function $y = \tan x$ is given in Fig. 69.

The function

$$\cot x = \frac{\cos x}{\sin x}$$

is defined on the set of all real numbers, except for the numbers of the form $n\pi$, where $n \in \mathbb{Z}$. The function $\cot x$ is odd and

periodic, with the smallest period π . On the interval $[0; \pi]$ it decreases from $+\infty$ to $-\infty$. The straight lines $x = 0$, $x = \pi$ are vertical asymptotes to the graph of the function

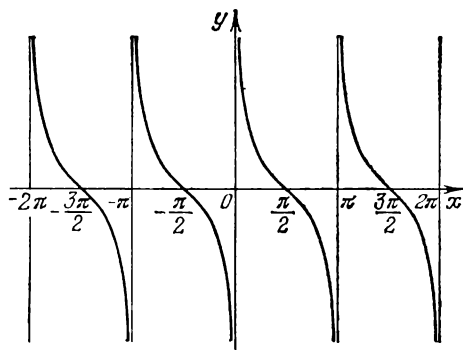


Fig. 70

$y = \cot x$. The function $\cot x$ being periodic, with period π , its graph on any interval $(n\pi; (n+1)\pi)$ can be obtained from its graph on $(0; \pi)$ by means of the translation $\tau(n\pi; 0)$.

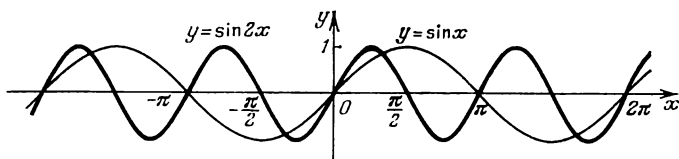


Fig. 71.

The graph of the function $y = \cot x$ is shown in Fig. 70.

Example 4. Construct the graph of the function $y = 2 \sin \left(2x - \frac{\pi}{3} \right)$.

$\triangle$ Since

$$y = 2 \sin 2 \left(x - \frac{\pi}{6} \right),$$

the required graph can be obtained from that of the function $y = \sin x$ by successive applications of the following operations: (1) by two-fold contraction towards the axis of ordinates we can obtain from the sine curve $y = \sin x$ the graph of the function $y = \sin 2x$, which is a sinusoid with period

it (Fig. 71), (2) by the translation $\tau(0; \pi/6)$ we can obtain from the sine curve $y = \sin 2x$ the graph of the function

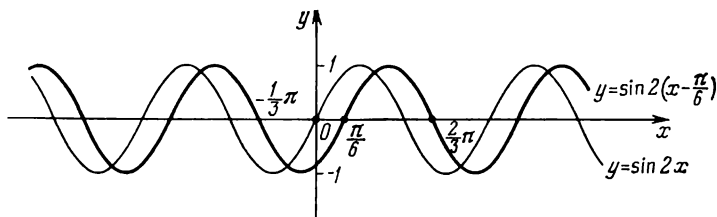


Fig. 72.

$y = \sin 2\left(x - \frac{\pi}{6}\right)$ (Fig. 72), (3) by two-fold expansion in the direction from the abscissa axis we can get the required

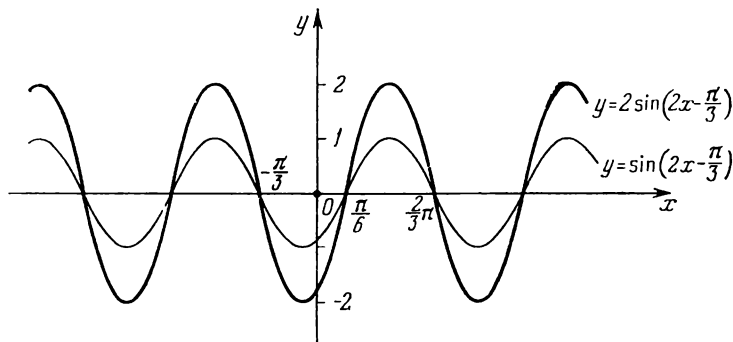


Fig. 73.

graph (Fig. 73) from the sine curve $y = \sin 2\left(x - \frac{\pi}{6}\right)$.

It is also a sinusoid. ▲

In general, the graph of the function $y = A \sin \omega(x - \alpha)$, where $A > 0$ and $\omega > 0$, is called a *sinusoid with period* $2\pi/\omega$, *amplitude* A and *initial phase* α .

6.6. Constructing the Graphs of Functions

When constructing the graphs of functions it is expedient to follow the scheme given below:

(1) to find the domain of definition of the function (if it is not given);

- (2) to determine whether the function is even or odd;
- (3) to determine whether the function is periodic;
- (4) to find the asymptotes;
- (5) to find the intervals of increase and decrease of the function;
- (6) to find the points at which the graph of the function cuts the axes of coordinates;
- (7) to find the extrema of the function.

It is good practice to plot the data obtained immediately on the drawing.

Example 1. Construct the graph of the function

$$f(x) = \frac{x^2 - 1}{x^2 + 1}. \quad (1)$$

$\triangle$ The function $f(x)$ is defined on the entire number axis and is even since $f(x) = f(-x)$ for any $x \in \mathbb{R}$.

The function $f(x)$ is not periodic.

Since

$$f(x) = 1 - \frac{2}{x^2 + 1} \rightarrow 1 \text{ as } x \rightarrow +\infty \text{ and as } x \rightarrow -\infty,$$

the straight line $y = 1$ is a horizontal asymptote to the graph of the function as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$.

Note that $f(x) < 1$ for any $x \in \mathbb{R}$ and, consequently, the graph of the function $f(x)$ lies below the line $y = 1$. Since the function $\frac{2}{x^2 + 1}$ decreases on the interval $(0; +\infty)$,

the function $f(x) = 1 - \frac{2}{x^2 + 1}$ increases on that interval.

On the interval $(-\infty; 0)$ the function $\frac{2}{x^2 + 1}$ increases and,

therefore, the function $f(x) = 1 - \frac{2}{x^2 + 1}$ decreases on the interval $(-\infty; 0)$.

The graph of the function $f(x)$ cuts the abscissa axis at the points $x = 1$ and $x = -1$ and the axis of ordinates at the point $y = -1$.

The function $\frac{2}{x^2 + 1}$ assumes the greatest value at $x = 0$ and, consequently, at $x = 0$, the function $f(x)$ assumes the least value equal to -1 .

The graph of the function (1) is shown in Fig. 74. $\blacktriangle$

Example 2. Construct the graph of the function

$$f(x) = x + \frac{1}{x}.$$

△ The function $f(x)$ is defined for all $x \neq 0$ and is odd, since $f(x) = -f(-x)$ for any $x \neq 0$. The function $f(x)$ is not periodic. Indeed, any number $T > 0$ belongs to its

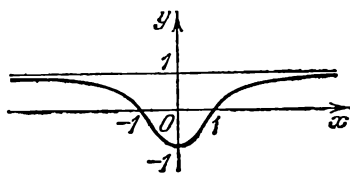


Fig. 74.

domain of definition but the number $T - T = 0$ does not. Consequently, not any number $T > 0$ can be a period of the function $f(x)$.

Let us construct the graph of the function $f(x)$ for $x > 0$. Since

$$\lim_{x \rightarrow +0} f(x) = \lim_{x \rightarrow +0} \left(x + \frac{1}{x} \right) = +\infty,$$

$$\lim_{x \rightarrow +\infty} (f(x) - x) = \lim_{x \rightarrow +\infty} \frac{1}{x} = 0,$$

the straight line $x = 0$ is a vertical asymptote and the straight line $y = x$ is an inclined asymptote to the graph of the function $f(x)$ as $x \rightarrow +\infty$.

On the interval $(0; 1]$ the function $f(x)$ decreases and on $[1; +\infty)$ it increases. In fact, let $0 < x_1 < x_2$. Then

$$f(x_2) - f(x_1) = (x_2 - x_1) - \frac{x_2 - x_1}{x_1 x_2} = (x_2 - x_1) \left(1 - \frac{1}{x_1 x_2} \right).$$

Hence it follows that if $0 < x_1 < x_2 \leq 1$, then $1 - \frac{1}{x_1 x_2} < 0$, and, consequently, $f(x_2) < f(x_1)$. Now if $1 \leq x_1 < x_2$, then $1 - \frac{1}{x_1 x_2} > 0$ and $f(x_2) > f(x_1)$. Since on $(0; 1)$ the

function $f(x)$ decreases and on $[1; +\infty)$ it increases, the point $x = 1$ is a point of minimum of the function: $f(x) > f(1) = 2$ for any $x > 0$, $x \neq 1$. The graph of the function $f(x)$ is shown in Fig. 75. ▲

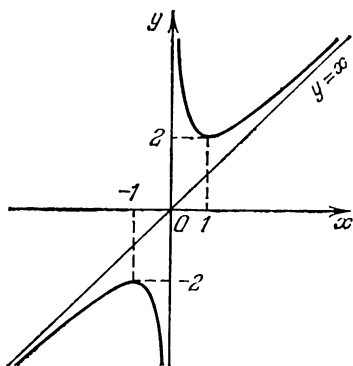


Fig. 75.

Example 3. Construct the graph of the function

$$f(x) = \frac{1}{1 + e^{1/x}}. \quad (2)$$

The function (2) is defined for all $x \in \mathbb{R}$, except, $x = 0$. If $x > 0$, then $e^{1/x} > 1$ and, consequently, $f(x) < 1/2$. On the interval $(0; +\infty)$ the function $e^{1/x}$ decreases and,

therefore, the function $f(x)$ increases. In this case

$$\lim_{x \rightarrow +0} \frac{1}{1 + e^{1/x}} = 0, \quad \lim_{x \rightarrow +\infty} \frac{1}{1 + e^{1/x}} = \frac{1}{2}.$$

It follows that the straight line $y = 1/2$ is a horizontal asymptote to the graph of the function (2) as $x \rightarrow +\infty$, the graph

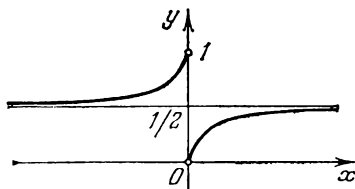


Fig. 76.

of the function (2) lying below that straight line for $x > 0$. If $x < 0$, then $e^{1/x} < 1$ and, consequently, $f(x) > 1/2$. The function $e^{1/x}$ decreases on the interval $(-\infty; 0)$ and, therefore, the function $f(x)$ increases. In this case

$$\lim_{x \rightarrow -0} \frac{1}{1 + e^{1/x}} = 1, \quad \lim_{x \rightarrow -\infty} \frac{1}{1 + e^{1/x}} = \frac{1}{2}.$$

Consequently, the line $x = 1/2$ is a horizontal asymptote to the graph of the function (2) as $x \rightarrow -\infty$, the graph of the function (2) lying above the asymptote for $x < 0$ (see Fig. 76).

6.7. Application of the Derivative to the Investigation of the Behaviour of Functions and Constructing Their Graphs

The derivatives are widely used in investigating the behaviour of functions and constructing their graphs. They aid in finding the intervals of increase and decrease of the given function and its points of extrema.

The following assertions are true:

If the function $f(x)$ possesses a positive derivative $f'(x)$ at all points of some interval, then it increases on that interval, and if it possesses a negative derivative, then $f(x)$ decreases.

Example 1. Find the intervals of increase and decrease of the function

$$f(x) = 2x^3 - 9x^2 + 12x. \quad (1)$$

△ The given function is defined and possesses a derivative on the entire number line. Since

$$f'(x) = 6x^2 - 18x + 12 = 6(x-1)(x-2),$$

it follows that $f'(x) > 0$ on the intervals $(-\infty; 1)$ and $(2; +\infty)$, and $f'(x) < 0$ on $(1; 2)$. Consequently, function (1) increases on $(-\infty; 1)$ and $(2; +\infty)$ and decreases on $(1; 2)$.

Answer: $(-\infty; 1)$, $(2; +\infty)$ are the intervals of increase and $(1; 2)$ is the interval of decrease of the function (1). ▲

The intervals on which the function increases or decreases are called the *intervals of its monotonicity*.

Example 2. Find the intervals of increase and decrease of the function

$$f(x) = \frac{x}{\ln x}. \quad (2)$$

△ The function (2) is defined for all positive $x \neq 1$, that is, on the intervals $(0; 1)$ and $(1; +\infty)$. Since

$$f'(x) = \frac{\ln x - 1}{(\ln x)^2},$$

it follows that $f'(x) > 0$ if $x > e$, and $f'(x) < 0$ if $x < e$. Consequently, on the intervals $(0; 1)$ and $(1; e)$ the function (2) decreases and on the interval $(e; +\infty)$ it increases. It follows from the continuity of the function (2) at the point

$x = e$ that $(1; e]$ is the interval of decrease and $[e; +\infty)$ is the interval of increase of the function (2).

Answer: $(0; 1)$, $(1; e]$ are the intervals of decrease, $[e; +\infty)$ is the interval of increase of the function (2). ▲

The point x_0 is called a *point of minimum* of the function $f(x)$ if there is $\delta > 0$ such that this function is defined in the δ -neighbourhood $(x_0 - \delta; x_0 + \delta)$ of the point x_0 and $f(x) > f(x_0)$ for all $x \neq x_0$ from that δ -neighbourhood. Now if $f(x) < f(x_0)$ for all $x \neq x_0$ from $(x_0 - \delta; x_0 + \delta)$,

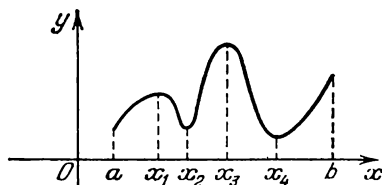


Fig. 77.

then the point x_0 is called a *point of maximum* of the function $f(x)$. The points of minimum and maximum of a function are called its *points of extremum* and the values of the function at those points are called the *extrema* of the given function.

For the function whose graph is shown in Fig. 77 the points x_1 and x_3 are points of maximum and the points x_2 and x_4 are points of minimum. The points a and b are not points of extremum of that function since they do not have neighbourhoods lying entirely in the domain of definition of the function.

For the function $f(x) = |x|$, defined throughout the number line, $f(0) = 0$ and $f(x) > 0$ for any $x \neq 0$ and, therefore, the point $x = 0$ is a point of minimum of the function $f(x) = |x|$.

For the function $f(x) = -(x - 1)^2 + 3$ the point $x = 1$ is a point of maximum since $f(x) < f(1) = 3$ for any $x \neq 1$.

The function $f(x) = \sin(1/x)$, $x > 0$, possesses an infinity of points of extremum. Indeed, every point x satisfying the condition $\frac{1}{x} = \frac{\pi}{2} + 2\pi n$, where $n = 0, 1, 2, \dots$, is its point of maximum, and every point x satisfying the

condition $\frac{1}{x} = -\frac{\pi}{2} + 2\pi n$, is its point of minimum. The extrema of this function are equal to 1 at all points of maximum, and to -1 at all points of minimum.

The following Fermat theorem is of especial significance in seeking the points of extremum of a given function.

If the point x_0 is a point of extremum of the differentiable function $f(x)$, then $f'(x_0) = 0$.

It follows from Fermat's theorem that the points of extremum of the given function should be sought among the points at which the derivative of the function is equal to zero or does not exist.

The points at which the derivative of the function vanishes or does not exist are called *critical* or *stationary points* of the function.

The points of extremum of a function are its critical points. Not every critical point, however, is a point of extremum. The point $x = 0$, for instance, is critical for the function $f(x) = x^3$ since $f'(0) = 0$ but is, evidently, not a point of extremum.

Example 3. Find the points of extremum and the intervals of increase and decrease of the function $f(x) = xe^{-x}$.

△ The given function is defined and possesses a derivative for all $x \in \mathbb{R}$:

$$f'(x) = e^{-x} - xe^{-x} = e^{-x}(1 - x).$$

The equation $f'(x) = 0$ yields the critical points of the function $f(x)$. The given function has one critical point $x = 1$. Since $f'(x) > 0$ for $x < 1$ and $f'(x) < 0$ for $x > 1$, the function $f(x)$ increases on the interval $(-\infty; 1)$ and decreases on the interval $(1; +\infty)$. Consequently, the point $x = 1$ is a point of maximum of the function $f(x)$.

Answer: $f(x)$ increases on $(-\infty; 1)$, decreases on $(1; +\infty)$; $x = 1$ is its point of maximum. ▲

It is expedient to use the following sufficient conditions for the points of maximum and minimum when investigating the points of extremum of a function.

The point x_0 is a point of maximum of the function $f(x)$ if the point x_0 possesses a neighbourhood such that $f(x)$ is continuous in it,

$$f'(x) > 0 \quad \text{for} \quad x < x_0$$

and

$$f'(x) < 0 \text{ for } x > x_0.$$

Now if $f'(x) < 0$ for $x < x_0$ and $f'(x) > 0$ for $x > x_0$, then x_0 is a point of minimum.

It was shown in the examples discussed above how to apply the derivative when seeking the intervals of increase and decrease and the points of maximum and minimum of a given function.

Having carried out all the investigations, we can, evidently, draw a graph of the function more precisely.

Example 4. Construct the graph of the function

$$y = \frac{(x-1)^2}{x+1}. \quad (3)$$

$\triangle$ The function (3) is defined for all $x \in \mathbb{R}$, except $x = -1$. It is neither even nor odd, nor is it periodic. Its

graph cuts the coordinate axes at the points $(0; 1)$ and $(1; 0)$. Since $y(x) \rightarrow -\infty$ as $x \rightarrow 1 - 0$ and $y(x) \rightarrow +\infty$ as $x \rightarrow 1 + 0$, the straight line $x = -1$ is a vertical asymptote and there are no other vertical asymptotes. To find the inclined asymptotes we should note that

$$\frac{(x-1)^2}{x+1} = x - 3 + \frac{4}{x+1}.$$

Consequently, the line $y = x - 3$ is an asymptote as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$. Let us find the derivative

$$y' = \frac{2(x-1)(x+1) - (x-1)^2}{(x+1)^2} = \frac{(x-1)(x+3)}{(x+1)^2}.$$

The points $x = -3$, $x = 1$ are critical for the function (3) since at these points the derivative vanishes.

It is easy to see that $y'(x) > 0$ for $x < -3$, $y'(x) < 0$ for $x \in (-3; 1)$ and for $x \in (-1; 1)$, $y'(x) > 0$ for $x > 1$. Consequently, the function (3) increases from $-\infty$ to -8 on the interval $(-\infty; -3]$, decreases from -8 to $-\infty$ on

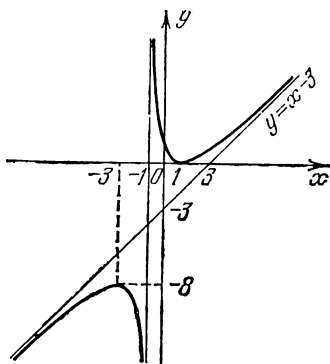


Fig. 78.

$[-3; -1]$, decreases from $+\infty$ to 0 on $(-1; 1]$ and increases from 0 to $+\infty$ on the interval $[1; +\infty)$. Hence it follows that the function (3) possesses a maximum at $x = -3$ and a minimum at $x = 1$. We compile a table:

x	$(-\infty; -3)$	-3	$(-3; -1)$	$(-1; 1)$	1	$(1; +\infty)$
y	$(-\infty; -8)$	-8	$(-8; -\infty)$	$(+\infty; 0)$	0	$(0; +\infty)$
y'	$+$	0	$-$	$-$	0	$+$
	$\nearrow$	max	$\searrow$	$\searrow$	min	$\nearrow$

The graph of the function (3) is shown in Fig. 78. ▲

6.8. The Greatest and the Least Value of a Function

Suppose the function $f(x)$ is defined and continuous on the closed interval $[a; b]$ and assume that $x_1, \dots, x_k$ are the critical points of the function $f(x)$, that is, the points at which the function has no derivative and the points at which the derivative is zero. Then, on each of the intervals $(a; x_1), \dots, (x_k; b)$ the derivative $f'(x)$ retains sign and, consequently, on each of these intervals the function $f(x)$ either increases or decreases. Therefore, the greatest of the numbers $f(a), f(x_1), \dots, f(x_k), f(b)$ is the greatest value of the function $f(x)$ on the interval $[a; b]$ and the smallest of these numbers is the least value of the function on $[a; b]$.

Note that the greatest and the least value of the function $f(x)$, $x \in [a; b]$ can be found among the numbers $f(a), f(x_1), \dots, f(x_k), f(b)$, where $x_1, \dots, x_k$ are the critical points of the function $f(x)$ on $(a; b)$.

Example 1. Find the greatest and the least value of the function

$$f(x) = x^3 - 6x^2 + 9x + 1: \quad (1)$$

(a) on the interval $[0; 2]$; (b) on the interval $[-2; -1]$.

△ Function (1) is defined and possesses a derivative on the whole set $\mathbb{R}$ of real numbers. By the differentiation

rules we find

$$f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3).$$

Thus, function (1) possesses two critical points, $x = 1$ and $x = 3$. In the given case $f(x)$ increases from $-\infty$ to $f(1) = 5$ on the interval $(-\infty; 1]$, decreases from $f(1) = 5$ to $f(3) = 1$ on $[1; 3]$ and again increases from $f(3) = 1$ to $+\infty$ on $[3; +\infty)$.

On the interval $(0; 2)$ function (1) has one critical point $x = 1$. Consequently,

$$\min_{x \in [0; 2]} f(x) = \min \{f(0); f(1); f(2)\} = \min \{1; 5; 3\} = 1,$$

$$\max_{x \in [0; 2]} f(x) = \max \{f(0); f(1); f(2)\} = 5.$$

On the interval $(-2; -1)$, function (1) has no critical points and is increasing. It follows that on the set $[-2; -1)$ the least value of function (1) is equal to $f(-2) = -49$. However, when $x \in [-2; -1)$, there is no greatest value among the values of the function $f(x)$ since the point $x = -1$ does not belong to that set.

Answer: (a) The least value of the function is equal to 1, the greatest value is equal to 5; (b) the least value of the function is equal to -49 , and there is no greatest value. $\blacktriangle$

Example 2. Find the greatest value of the function

$$f(x) = 3x - \tan x, \quad x \in [0; \pi/2).$$

Δ The given function is continuous and possesses a derivative

$$f'(x) = 3 - \frac{1}{\cos^2 x}, \quad x \in (0; \pi/2).$$

On the interval $(0; \pi/2)$ it vanishes at the point $x_0 = \arccos(1/\sqrt{3})$, $f'(x)$ being positive for $x < x_0$ and negative for $x > x_0$. Consequently, the point $x_0 = \arccos(1/\sqrt{3})$ is a point of maximum of the given function:

$$\begin{aligned} f\left(\arccos \frac{1}{\sqrt{3}}\right) &= 3 \arccos \frac{1}{\sqrt{3}} - \tan\left(\arccos \frac{1}{\sqrt{3}}\right) \\ &= 3 \arccos \frac{1}{\sqrt{3}} - \sqrt{2}. \end{aligned}$$

The function $f(x)$ increases on the interval $[0; \arccos(1/\sqrt{3})]$ and decreases on the interval $[\arccos(1/\sqrt{3}); \pi/2)$. Consequently, the greatest value of the function $f(x) = 3x - \tan x$, $x \in [0; \pi/2)$, is equal to $3\arccos(1/\sqrt{3}) - \sqrt{2}$. $\blacktriangle$

Example 3. Find the greatest volume of the cylinder inscribed in the given cone.

$\triangle$ Suppose we are given a cone with the height H and the radius of the base R (Fig. 79). Let us denote by h the height of the cylinder and by r , the radius of the base of the cylinder inscribed in the given cone. We also introduce the notation $BM = x$. Then, $h = PB = x \cdot \tan \widehat{KML} = x \frac{H}{R}$ and $r = R - x$. The volume of the cylinder V is $\pi r^2 h$. In our case

$$V(x) = \pi (R - x)^2 \frac{xH}{R}.$$

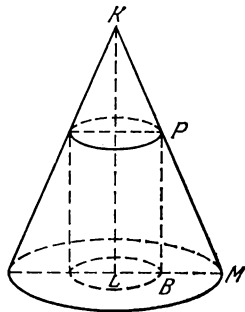


Fig. 79.

Let us determine at what value of x the volume of the cylinder assumes the greatest value. We find the derivative $V'(x)$:

$$\begin{aligned} V'(x) &= \frac{\pi H}{R} (R - x)^2 + \frac{\pi H}{R} x \cdot 2 \cdot (R - x) (-1) \\ &= \frac{\pi H}{R} (R - x) (R - x - 2x) = \frac{\pi H}{R} (R - x) (R - 3x). \end{aligned}$$

We have $V'(x) = 0$ for $x = R/3$, $V'(x) > 0$ for $x < R/3$ and $V'(x) < 0$ for $x > R/3$. Consequently, at the point $x = R/3$ the function $V(x)$ possesses a maximum. Since x can vary from zero to R , with $V(0) = V(R) = 0$, the number

$$V(R/3) = \frac{4}{27} \pi H R^2$$

is the greatest value of the volume of the inscribed cylinders. $\blacktriangle$

PROBLEMS OF SECTION I

1. Determine which of the functions given below are even, which are odd, and which of them are neither even nor odd:

$$f_1(x) = \sqrt{x^2 - 5x + 6}, \quad f_2(x) = \sqrt{3x - 2}, \quad f_3(x) = x^3 + 3 \sin x,$$

$$f_4(x) = x^4 - 5 \cos x, \quad f_5(x) = \frac{e^{-x} - 1}{e^x + 1}, \quad f_6(x) = \ln(x + \sqrt{x^2 + 1}).$$

2. Test the following functions for periodicity and determine the least period, provided it exists:

$$(a) f(x) = \begin{cases} 1, & \text{if } x \text{ is rational,} \\ 0, & \text{if } x \text{ is irrational;} \end{cases}$$

$$(b) f(x) = \cos^3 x \sin x - \sin^3 x \cos x.$$

3. Find all the asymptotes to the graphs of the following functions:

$$(a) f(x) = \frac{x^2 + 4}{x^2 - 9}; \quad (b) f(x) = \frac{x^2 - 3}{x - 1}; \quad (c) f(x) = \sqrt{x^2 + 6x + 8}.$$

4. Construct the graphs of the following functions (without applying the derivative):

$$(a) f_1(x) = x \left(|x - 1| + \frac{x}{3} - 1 \right); \quad (b) f_2(x) = \log(\sin x);$$

$$(c) f_3(x) = 2^{\cos x}.$$

5. Applying the derivative, construct the graphs of the following functions:

$$(a) f(x) = \frac{(2-x)^3}{(x-3)^2};$$

$$(b) f(x) = x + e^{-x}; \quad (c) f(x) = \frac{6 \sin x}{2 + \cos x}.$$

6. Find the intervals of monotonicity and the points of extrema of the function

$$f(x) = 2 \ln(x - 2) - x^2 + 4x + 1.$$

7. Find the greatest and the least value of the function $f(x) = \frac{1}{3}x^3 - \frac{5}{2}x^2 + 6x$ on the interval $[1; 4]$.

8. Circumscribe a cone of the least volume about the given sphere.

PROBLEMS OF SECTION II

1. Determine which of the functions given below are even and which are odd in the domain of their definition, and which of them are neither even nor odd:

$$f_1(x) = 1 + x^2 - x^4, \quad f_3(x) = \sqrt{1+x},$$

$$f_2(x) = x + x^3 - 3x^5, \quad f_4(x) = \sqrt{1-x^2},$$

$$f_5(x) = \sqrt{x^2 - 3x - 4}, \quad f_7(x) = 3^x + 3^{-x},$$

$$f_6(x) = \ln \frac{1-x}{1+x}, \quad f_8(x) = x \sin 3x.$$

2. Represent the following functions as the sum of the even and odd functions in the domain of their definition:

$$f_1(x) = \sin 5x + 2^x \tan x, \quad f_2(x) = x \ln |x| + \frac{x-1}{x+1},$$

$$f_3(x) = \frac{x^3 + x^2 + x}{x^4 - 1}, \quad f_4(x) = \frac{3^x + x^2}{\sin x}.$$

3. Prove that the derivative of an even function is odd and the derivative of an odd function is even.

4. Determine whether the following assertions are true: (a) if $f'(x)$ is even, then $f(x)$ is an odd function, (b) if $f'(x)$ is odd, then $f(x)$ is an even function.

5. Are there functions, defined everywhere, which are simultaneously (a) even and increasing on $\mathbb{R}$, (b) odd and decreasing on $\mathbb{R}$, (c) odd and positive on $\mathbb{R}$?

6. Suppose we are given the function $f(x) = e^x$, $x > 0$. Extend the definition of the function $f(x)$ to the region $x \leq 0$ so that the resulting function be (a) even, (b) odd, on the set $(-\infty; +\infty)$.

7. Test the following functions for periodicity and find the smallest period, provided it exists:

$$f_1(x) = x^2 + 1, \quad f_2(x) = 2 \tan \frac{x}{2} + 4 \tan \frac{x}{3},$$

$$f_3(x) = \frac{\sin x}{[x/\pi] + 1}, \quad f_4(x) = 3 \cos 4x + 5 \sin 4x.$$

8. Can the sum of two nonperiodic functions, defined everywhere, be a periodic function?

9. Suppose we are given the differentiable periodic function $f(x)$, $x \in \mathbb{R}$. Prove that $f'(x)$ is also a periodic function on $\mathbb{R}$.

10. Find all the asymptotes to the graphs of the following functions:

$$(1) f_1(x) = \frac{1}{x^2 - 1}; \quad (2) f_2(x) = \frac{2^x}{3x - 2};$$

$$(3) f_3(x) = \frac{\sin x}{x^2 - x}; \quad (4) f_4(x) = \frac{1}{\cos x};$$

$$(5) f_5(x) = \frac{1}{e^x - 1}; \quad (6) f_6(x) = \frac{\{x\}}{[x]}, \text{ where } \{x\} = x - [x].$$

11. Construct the graphs of the following functions (without resort to the derivative):

$$(1) f_1(x) = e^{1/\sin x}; \quad (2) f_2(x) = \cos(\ln x);$$

$$(3) f_3(x) = \arcsin(\sin x); \quad (4) f_4(x) = \frac{\sin x}{[x/\pi] + 0.5};$$

(5) $f_5(x) = r(x)$, where $r(x)$ is the distance from x to the nearest integer.

12. Applying the derivative, construct the graphs of the following functions:

$$(1) f_1(x) = \frac{(x+3)^2}{(x+1)^2}; \quad (2) f_2(x) = \frac{x^3}{3-x^2};$$

$$(3) f_3(x) = (1+x)|x|^{2/3}; \quad (4) f_4(x) = \frac{\sqrt{x^2-4x}}{2-x};$$

$$(5) f_5(x) = \sin^4 x + \cos^4 x; \quad (6) f_6(x) = \ln x / \sqrt{x}.$$

13. Find all the values of a at which the function

$$f(x) = \frac{a^2-1}{3} x^3 + (a-1)x^2 + 2x + 1$$

increases on $\mathbb{R}$.

14. Suppose x_1 and x_2 are a point of maximum and a point of minimum of the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$ respectively. At what a is the equality $x_1^2 = x_2$ true?

15. Find the greatest and the least value of the function

$$y = x^2(2x-3) - 12(3x-2)$$

on the interval $[-3; 6]$.

16. Find the greatest and the least value of the function $f(x) = \log_{1/3}(x^2+x-2)$ on the interval $[3; 6]$.

17. Find the extrema of the function $f(x) = (x-3)e^{x+1}$ on the interval $(-2; 4)$ and its greatest and least values on the interval $[-2; 4]$.

18. Find the greatest and the least value of the function $f(x) = \cos^2 x + \cos x + 3$.

19. Find the radius of the base of the cylinder having the smallest full surface for the given volume V .

20. Find the cosine of the angle at the vertex of an isosceles triangle having the greatest area for the given constant length l of the median drawn to its lateral side.

Chapter 7

Vectors

7.1. Some Necessary Definitions and Designations

Any point A lying on a straight line partitions that line into two rays emanating from A . A ray originating at the point A and containing a point B is designated $[AB)$ (Fig. 80).

Two rays lying on the same straight line are said to have the *same direction* if their intersection is a ray, and have

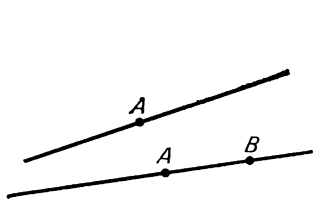


Fig. 80.

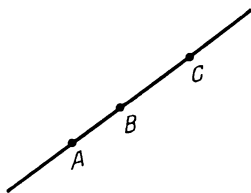


Fig. 81.

opposite directions if their intersection is not a ray. For example, the rays $[AB)$ and $[BC)$ in Fig. 81 are of the same direction and the rays $[AB)$ and $[CB)$ are of opposite directions.

If two rays lie on parallel non-coinciding straight lines, then they lie in a certain plane. The straight line passing through their origin divides the plane into two half-planes. If the rays lie in the same half-plane, they are said to have the *same direction* (Fig. 82); if they lie in different half-planes, they are *oppositely directed* (Fig. 83). The fact that the rays $[AB)$ and $[CD)$ are of the same direction is designated as

$$[AB) \uparrow\uparrow [CD).$$

If the directions of the rays $[AB)$ and $[CD)$ are opposite, then we write

$$[AB) \nparallel [CD).$$

It is evident that when the direction of each of two rays is similar to that of a third ray, then they are similarly directed.

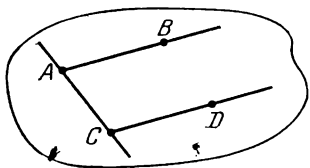


Fig. 82.

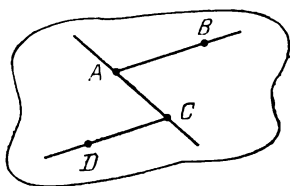


Fig. 83.

A *parallel displacement* (translation) of space (plane) defined by an ordered pair (A, B) of non-coinciding points is such a transformation of space (plane) under which every point M is mapped onto a point N such that the ray $[MN)$ is of the same direction as the ray $[AB)$ and the distance $|MN|$ is equal to the distance $|AB|$. In other words, all the points of space (plane) are displaced the distance $|AB|$ in the direction of the ray $[AB)$ (Fig. 84).

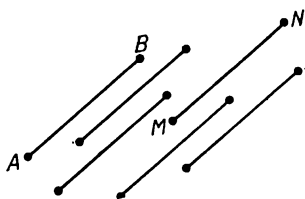


Fig. 84.

An identity transformation which maps every point onto itself is called a translation, or parallel displacement, by a distance zero.

Properties of a translation.

1. A translation is a parallel displacement, that is mapping preserving distances.

2. A translation maps a straight line onto a straight line parallel to it and a ray onto a ray of the same direction.

A translation of space maps a plane onto a plane parallel to it.

3. A combination of two translations (that is, two translations performed successively) is a translation.

4. A mapping converse to a translation is a translation.

An angle between rays. Suppose two rays l_1 and l_2 are neither similarly directed nor oppositely directed (Fig. 85). An arbitrary point O is the origin of one ray $[OA)$ directed similarly to the ray l_1 and of one ray $[OB)$ directed similarly to the ray l_2 . The rays $[OA)$ and $[OB)$ bound the convex angle

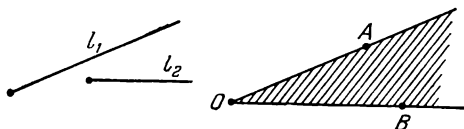


Fig. 85.

AOB which is less than 180° . Now if the rays l_1 and l_2 are oppositely directed, then the rays $[OA)$ and $[OB)$, being of the same direction as l_1 and l_2 and having the common origin, bound two convex straight angles (Fig. 86).

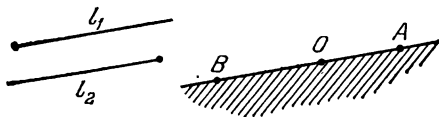


Fig. 86.

The *angle between the rays l_1 and l_2* is the magnitude of the convex angle between any two rays having the common origin and being of the same direction as the given rays.

For example, the angle between the rays l_1 and l_2 in Fig. 85 is the magnitude of the angle AOB , and the angle between the rays l_1 and l_2 in Fig. 86 is equal to 180° .

The following theorem states that the angle between the rays is independent of the choice of the point which is the origin of the rays whose directions are similar to those of the given rays.

If the respective sides of two convex angles are similarly directed, then the angles are congruent.

The angle between the similarly directed rays is assumed to be 0° . Thus, the angle between the rays can assume values from 0° to 180° .

7.2. Vectors, Their Designation and Representation. Collinear and Coplanar Vectors

A *vector* is a parallel displacement. The letters **a**, **b**, **x**, etc. printed in boldface type are used to denote them. Vectors are considered on a plane (parallel displacement of a plane) and in space (parallel displacement of space). In both cases a vector is defined by an ordered pair of points, that is, by specifying a point and its image.

A vector **a**, defined by an ordered pair (A, B) of non-coinciding points, is represented as a *directed line segment* (an arrow) with *origin* at A and *terminus* at B (Fig. 87). The

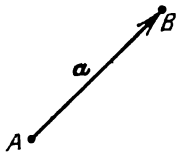


Fig. 87.

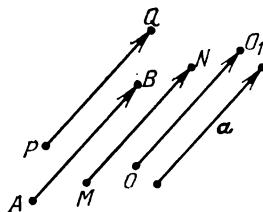


Fig. 88.

vector **a** can be also designated as $\overrightarrow{AB}$ and written as $\mathbf{a} = \overrightarrow{AB}$.

By definition, the vector $\mathbf{a} = \overrightarrow{AB}$ maps every point M onto a point N such that the ray $[MN)$ is directed similarly to the ray $[AB)$ and $|MN| = |AB|$. The direction of the ray $[AB)$ is called the *direction* of the vector $\mathbf{a} = \overrightarrow{AB}$. The distance $|AB|$ is called the *length* or *modulus* of the vector

$\mathbf{a} = \overrightarrow{AB}$ (the notation is $|\overrightarrow{AB}|$ or $|\mathbf{a}|$).

If $[MN) \parallel [AB)$ and $|MN| = |AB|$, then the vector $\mathbf{a} = \overrightarrow{AB}$ can also be designated as $\overrightarrow{MN}$ and represented by a directed line segment with origin at the point M and terminus at the point N . When representing a vector by a directed segment, any point can, evidently, be taken as its origin, that is, a vector has an infinite number of representations. In Fig. 88, for instance, the vector $\mathbf{a} = \overrightarrow{AB}$ is represented by directed line segments PQ , AB , MN , and OO_1 .

The construction of the directed segment MN such that $\vec{MN} = \mathbf{a}$ is called *laying off* the vector $\mathbf{a}$ from the point M .

If $\vec{AB}$ and $\vec{CD}$ are the designations of one and the same vector, then we write $\vec{AB} = \vec{CD}$ and say that the vectors $\vec{AB}$ and $\vec{CD}$ are *equal*. It should be emphasized that the equality $\vec{AB} = \vec{CD}$ means

$$|AB| = |CD| \text{ and } [AB) \uparrow\uparrow [CD).$$

A translation by the distance zero is called a *zero (null) vector* and is designated as $\mathbf{0}$. A zero vector maps every point

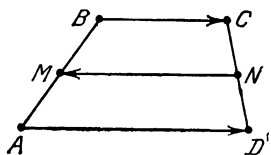


Fig. 89.

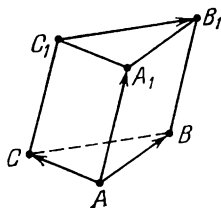


Fig. 90.

onto itself and, therefore, is also designated as $\vec{AA}$, $\vec{BB}$ and the like. The length of a zero vector is zero. A zero vector has no definite direction.

Two nonzero vectors $\vec{AB}$ and $\vec{CD}$ are said to have the *same direction* (designated as $\vec{AB} \uparrow\uparrow \vec{CD}$) if the rays $[AB)$ and $[CD)$ are of the same direction, and to have *opposite directions* (designated as $\vec{AB} \updownarrow \vec{CD}$) if the rays $[AB)$ and $[CD)$ are of opposite directions.

Two nonzero vectors are called *collinear* if they are of the same or opposite directions. It is clear that collinear vectors are represented by line segments of parallel directions.

Figure 89 shows a trapezoid $ABCD$, MN being its midline. The vectors $\vec{AD}$, $\vec{BC}$, and $\vec{NM}$ are collinear: $\vec{AD} \uparrow\uparrow \vec{BC}$ and $\vec{AD} \updownarrow \vec{NM}$.

A zero vector is considered to be collinear with any vector.

We say that the nonzero vector $\vec{AB}$ is parallel to a plane if the straight line AB is parallel to that plane.

Nonzero vectors are said to be *complanar* if they are parallel to the same plane. Any two vectors are always coplanar while three vectors may not be coplanar. Figure 90 shows a triangular prism $ABCA_1B_1C_1$. The vectors $\vec{AC}$, $\vec{AB}$, and $\vec{C_1B_1}$ are coplanar, while the vectors $\vec{AC}$, $\vec{AB}$, and $\vec{AA_1}$ are not.

7.3. The Sum of Vectors. An Opposite Vector. The Difference of Vectors

The composition of two vectors $\mathbf{a}$ and $\mathbf{b}$ is a vector (property 3 of parallel displacement, see 7.1), which is called the *sum* of the vectors $\mathbf{a}$ and $\mathbf{b}$ and is designated as $\mathbf{a} + \mathbf{b}$.

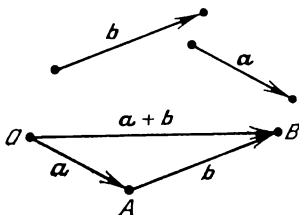


Fig. 91.

The sum of the nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ can be found by the *rule of triangle* (Fig. 91):

a vector $\vec{OA} = \mathbf{a}$ is laid off from an arbitrary point O , and a vector $\vec{AB} = \mathbf{b}$ is laid off from the point A , which is the tip of the directed segment

$\vec{OA}$; then it follows that $\vec{OB} = \mathbf{a} + \mathbf{b}$. The rule of triangle governing addition of vectors can be easily memorized in letter notation:

$$\begin{array}{ccc} \vec{OA} & + & \vec{AB} = \vec{OB} \\ \downarrow & & \downarrow \\ \text{O} & \text{A} & \text{B} \end{array}$$

(the arrows below indicate the sequence of the letters in the notation of the vector of the sum).

Note that this rule is also applicable in the case when the vectors $\mathbf{a}$ and $\mathbf{b}$ are collinear (Fig. 92).

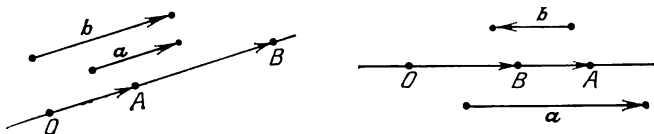


Fig. 92.

Laws governing addition of vectors.

1. $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ (*commutative law*).
2. $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$ (*associative law*).
3. $\mathbf{a} + \mathbf{0} = \mathbf{a}$.

The commutative law yields the *parallelogram law* of vector addition stating that the sum of two noncollinear vectors is a vector represented by a diagonal of the parallelogram constructed on directed line segments representing the given vectors and having the common origin (Fig. 93).

The sum of three vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ is defined as the sum of the vector $\mathbf{a} + \mathbf{b}$ and the vector $\mathbf{c}$ (Fig. 94).

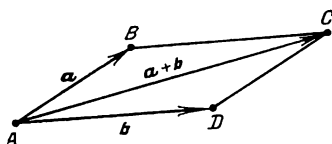


Fig. 93.

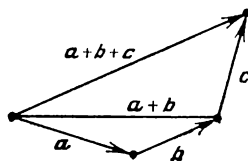


Fig. 94.

By virtue of the associative law we have $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$ and, therefore, the sum of three vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ is written without brackets: $\mathbf{a} + \mathbf{b} + \mathbf{c}$. A similar definition can be applied to the sum of any number of vectors, say, $\mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d} = (\mathbf{a} + \mathbf{b} + \mathbf{c}) + \mathbf{d}$. Figure 95 shows how to find the sum of the vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, and $\mathbf{d}$ by the *rule of a polygon*.

If three vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are not coplanar, their sum can be found by the *parallelepiped law*: the vector $\mathbf{a} + \mathbf{b} + \mathbf{c}$ is represented by a diagonal of the parallelepiped constructed on the directed segments depicting the vectors $\mathbf{a}$, $\mathbf{b}$, and

(the arrows below show the sequence of letters in the notation of the vector of difference).

Note that if we can construct a parallelogram on the vectors $\mathbf{a}$ and $\mathbf{b}$, laid off from the common origin O (Fig. 98),

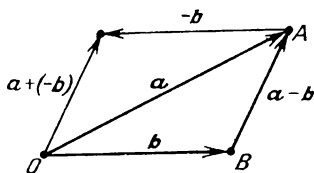


Fig. 97.

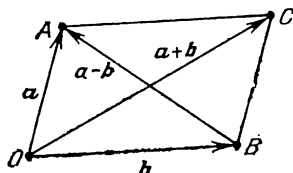


Fig. 98.

then the length of the diagonal, having the same origin O , is equal to the length of the vector $\mathbf{a} + \mathbf{b}$ and the length of the other diagonal is equal to the length of the vector $\mathbf{a} - \mathbf{b}$:

$$|OC| = |\mathbf{a} + \mathbf{b}|, \quad |BA| = |\mathbf{a} - \mathbf{b}|.$$

If $\mathbf{c} = \mathbf{a} - \mathbf{b}$, then $\mathbf{a} = \mathbf{c} + \mathbf{b}$. Indeed,

$$\begin{aligned} \mathbf{c} + \mathbf{b} &= (\mathbf{a} - \mathbf{b}) + \mathbf{b} = (\mathbf{a} + (-\mathbf{b})) + \mathbf{b} \\ &= \mathbf{a} + ((-\mathbf{b}) + \mathbf{b}) = \mathbf{a} + \mathbf{0} = \mathbf{a}. \end{aligned}$$

It shows that we can transfer the summands in vector equalities from one side of the equality to the other, changing the signs of these summands to the opposite.

Example 1. The diagonals of the parallelogram $ABCD$ intersect at the point O (Fig. 99). Find the sum of the vectors $\vec{OA} + \vec{OB} + \vec{OC} + \vec{OD}$.

Δ When they intersect, the diagonals of the parallelogram are divided in half: $|OA| = |OC|$. The directions of the rays $[OA]$ and $[OC]$ are opposite. This means that the vectors $\vec{OA}$ and $\vec{OC}$ are opposite and their sum is equal to the zero vector: $\vec{OA} + \vec{OC} = \mathbf{0}$. It can be established by analogy that $\vec{OB}$ and $\vec{OD}$ are opposite vectors. Thus we have $\vec{OA} + \vec{OB} + \vec{OC} + \vec{OD} = (\vec{OA} + \vec{OC}) + (\vec{OB} + \vec{OD}) = \mathbf{0}$. $\blacktriangle$

Example 2. Prove that the inequality $|\mathbf{a} + \mathbf{b}| \leq |\mathbf{a}| + |\mathbf{b}|$ is true for any vectors $\mathbf{a}$ and $\mathbf{b}$.

△ If one of the vectors $\mathbf{a}$ and $\mathbf{b}$ is zero, the inequality, obviously, holds true. Suppose $\mathbf{a}$ and $\mathbf{b}$ are nonzero vectors, $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{AB} = \mathbf{b}$ (Fig. 91). Then, by the rule of triangle, $\overrightarrow{OB} = \mathbf{a} + \mathbf{b}$. By the property of distance, for any three

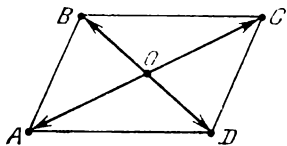


Fig. 99.

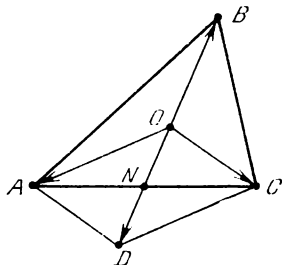


Fig. 100.

points O , A , and B there holds the inequality $|\overrightarrow{OB}| \leq |\overrightarrow{OA}| + |\overrightarrow{AB}|$, therefore $|\mathbf{a} + \mathbf{b}| \leq |\mathbf{a}| + |\mathbf{b}|$.

It should be noted that for distinct points O , A , and B the equality $|\overrightarrow{OB}| = |\overrightarrow{OA}| + |\overrightarrow{AB}|$ holds if and only if the points O , A , and B belong to one straight line, the point A lying between O and B . Consequently, for the non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ the equality $|\mathbf{a} + \mathbf{b}| = |\mathbf{a}| + |\mathbf{b}|$ holds if and only if they are of the same direction. ▲

Example 3. The medians of the triangle ABC intersect at the point O (Fig. 100). Find the sum of the vectors

$$\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}.$$

△ We construct the parallelogram $AOCD$. By the parallelogram rule, $\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{OC}$. When they intersect, the diagonals of the parallelogram are divided in half. Consequently, $|\overrightarrow{OD}| = 2|\overrightarrow{ON}|$, the point N is the midpoint of the side AC , and BN is the median. When they intersect, the medians of the triangle are divided in the ratio 2:1, measuring from the vertex, and, therefore, $|\overrightarrow{OB}| = 2 \times |\overrightarrow{ON}|$. It follows that $|\overrightarrow{OD}| = |\overrightarrow{OB}|$, and, since the

rays $[OD)$ and $[OB)$ are of opposite directions, we have $\vec{OD} = -\vec{OB}$.

Thus we have

$$\begin{aligned}\vec{OA} + \vec{OB} + \vec{OC} &= (\vec{OA} + \vec{OC}) + \vec{OB} \\ &= \vec{OD} + \vec{OB} = -\vec{OB} + \vec{OB} = \mathbf{0}. \quad \blacktriangle\end{aligned}$$

7.4. Multiplication of a Vector by a Scalar. Criterion of Collinearity

The *product* of the nonzero vector $\mathbf{a}$ by the scalar $x \neq 0$ is a vector whose length is equal to $|x| \cdot |\mathbf{a}|$ and whose direction is the same as that of $\mathbf{a}$ at $x > 0$ and opposite at $x < 0$. The product of the vector $\mathbf{a}$ by the scalar x is designated as $x \cdot \mathbf{a}$.

By definition, the product of a zero vector by any scalar and the product of any vector by zero are considered to be equal to a zero vector:

$$x \cdot \mathbf{0} = \mathbf{0}, \quad 0 \cdot \mathbf{a} = \mathbf{0}.$$

Laws of multiplication of vectors by scalars.

1. $x(y\mathbf{a}) = (xy)\mathbf{a}$ (associative law).
2. $xa + ya = (x + y)\mathbf{a}$,
3. $xa + xb = x(\mathbf{a} + \mathbf{b})$,
4. $0 \cdot \mathbf{a} = x \cdot \mathbf{0} = \mathbf{0}$.

The definition of the product of a vector by a scalar implies the following *criterion of collinearity*.

The vector $\mathbf{b}$ is collinear with the nonzero vector $\mathbf{a}$ if and only if there exists a number x such that $\mathbf{b} = x\mathbf{a}$.

Example 1. By what number x must the nonzero vector $\mathbf{a}$ be multiplied for the length of the vector $\mathbf{m} = x\mathbf{a}$ to be equal to unity and (a) for the vector $\mathbf{m}$ to point in the same direction as the vector $\mathbf{a}$; (b) for the vector $\mathbf{m}$ to point in the direction opposite to that of the vector $\mathbf{a}$?

Δ (a) $\mathbf{m} \uparrow \uparrow \mathbf{a}$, when $x > 0$; since $|x| = x$ and $|\mathbf{m}| = x|\mathbf{a}|$ for $x > 0$, the condition $|\mathbf{m}| = 1$ implies $x = 1/|\mathbf{a}|$.

(b) $\mathbf{m} \uparrow \downarrow \mathbf{a}$, when $x < 0$; since $|x| = -x$ for $x < 0$, it follows that $|\mathbf{m}| = -x|\mathbf{a}|$. From the condition $|\mathbf{m}| = 1$, we find $x = -1/|\mathbf{a}|$. $\blacktriangle$

Note that the vector whose length is equal to unity is called a *unit vector*.

It follows from the criterion of collinearity that for the noncollinear vectors $\mathbf{a}$ and $\mathbf{b}$ the equality $x\mathbf{a} + y\mathbf{b} = \mathbf{0}$ is satisfied if and only if $x = y = 0$. In fact, let us suppose, for instance, that $x \neq 0$, then from $x\mathbf{a} + y\mathbf{b} = \mathbf{0}$ we get $\mathbf{a} = -\frac{y}{x}\mathbf{b}$, but this contradicts the fact that the vectors $\mathbf{a}$ and $\mathbf{b}$ are not collinear. Consequently, $x = 0$. We can establish by analogy that $y = 0$.

Example 2. The vectors $\mathbf{a}$ and $\mathbf{b}$ are not collinear. Find at what x the vectors $\mathbf{c} = (x-2)\mathbf{a} + \mathbf{b}$ and $\mathbf{d} = (2x+1)\mathbf{a} - \mathbf{b}$ are collinear.

$\triangle$ The vector $\mathbf{c}$ is nonzero since the coefficient in $\mathbf{b}$ is different from zero and, consequently, there is a number y such that $\mathbf{d} = y\mathbf{c}$, that is,

$$(2x+1)\mathbf{a} - \mathbf{b} = y(x-2)\mathbf{a} + y\mathbf{b}.$$

As mentioned above, we can transfer the summands in a vector equality from one side to the other, changing signs of those summands to the opposite. Therefore, we have

$$(yx - 2y - 2x - 1)\mathbf{a} + (y + 1)\mathbf{b} = \mathbf{0}.$$

The vectors $\mathbf{a}$ and $\mathbf{b}$ are not collinear and, therefore,

$$\begin{aligned} yx - 2y - 2x - 1 &= 0, \\ y + 1 &= 0. \end{aligned}$$

Solving this system, we find $y = -1$ and $x = 1/3$. For $x = 1/3$, we have $\mathbf{c} = -\frac{5}{3}\mathbf{a} + \mathbf{b}$, $\mathbf{d} = \frac{5}{3}\mathbf{a} - \mathbf{b}$. It is easy to see that the vectors are opposite: $\mathbf{d} = -\mathbf{c}$. $\blacktriangle$

7.5. The Condition of Coplanarity of Vectors. Representing a Vector in Terms of Three Noncoplanar Vectors

Assume that the vectors $\mathbf{a}$ and $\mathbf{b}$ are noncollinear. Let us lay them off from one point: $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$ (Fig. 101). By definition, any nonzero vector $\mathbf{c}$, coplanar with the vectors $\mathbf{a}$ and $\mathbf{b}$, is parallel to the plane OAB .

If we construct the vector $\overrightarrow{OC} = \mathbf{c}$, then the point C lies in the plane OAB and we say, therefore, that any three coplanar vectors can be transferred to one plane.

Theorem. *If the vectors $\mathbf{a}$ and $\mathbf{b}$ are noncollinear, then any vector $\mathbf{c}$, coplanar with them, can be uniquely represented in the form $\mathbf{c} = x\mathbf{a} + y\mathbf{b}$.*

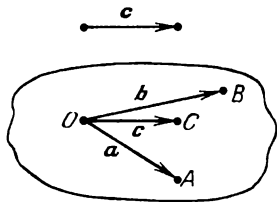


Fig. 101.

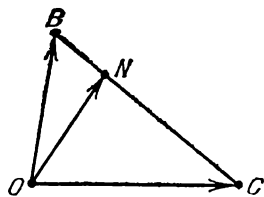


Fig. 102.

This representation is the *resolution* of the vector $\mathbf{c}$ in a plane into components with respect to two noncollinear vectors $\mathbf{a}$ and $\mathbf{b}$.

Example 1. Given the parallelogram $ABCD$ (see Fig. 93). Represent the vectors $\overrightarrow{AB}$ and $\overrightarrow{AD}$ in terms of the vectors $\overrightarrow{AC}$ and $\overrightarrow{BD}$.

△ Recall that $\overrightarrow{AC} = \overrightarrow{AD} + \overrightarrow{AB}$ and $\overrightarrow{BD} = \overrightarrow{AD} - \overrightarrow{AB}$. This means that $2\overrightarrow{AD} = \overrightarrow{AC} + \overrightarrow{BD}$ and $2\overrightarrow{AB} = \overrightarrow{AC} - \overrightarrow{BD}$. Thus we have

$$\overrightarrow{AD} = \frac{1}{2}(\overrightarrow{AC} + \overrightarrow{BD}) \text{ and } \overrightarrow{AB} = \frac{1}{2}(\overrightarrow{AC} - \overrightarrow{BD}). \blacktriangle$$

Example 2. A point N is located on the side BC of the triangle OBC so that $|BN| : |BC| = n$ (Fig. 102). Represent the vector $\overrightarrow{ON}$ in terms of the vectors $\overrightarrow{OB}$ and $\overrightarrow{OC}$.

△ The vectors $\overrightarrow{BN}$ and $\overrightarrow{BC}$ are collinear and point in the same direction. Consequently, $\overrightarrow{BN} = x\overrightarrow{BC}$ and $x \geq 0$. Since $|BN| = n|BC|$, it follows that $x = n$ and $\overrightarrow{BN} =$

$n\vec{BC}$. Since $\vec{BC} = \vec{OC} - \vec{OB}$ and $\vec{ON} = \vec{OB} + \vec{BN}$, we have

$$\vec{ON} = \vec{OB} + n(\vec{OC} - \vec{OB}) = n\vec{OC} + (1 - n)\vec{OB}.$$

Note that for $n = 1/2$, the point N is the midpoint of the side BC and ON is the median of the triangle. In this case

$$\vec{ON} = \frac{1}{2}(\vec{OC} + \vec{OB}). \blacktriangle$$

It is easy to see that the converse statement is also true: if some vector $\mathbf{c}$ is represented in terms of two noncollinear

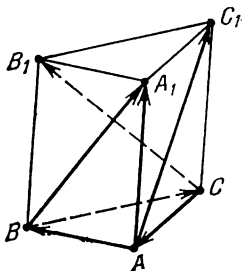


Fig. 103.

vectors $\mathbf{a}$ and $\mathbf{b}$, then the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are coplanar. Indeed, assume $\mathbf{c} = x\mathbf{a} + y\mathbf{b}$. If the scalars x and y are nonzero, then the vectors $x\mathbf{a}$ and $y\mathbf{b}$ are noncollinear either. Let us lay off these vectors from some point O : $x\mathbf{a} = \vec{OA}$ and $y\mathbf{b} = \vec{OB}$.

Then $\mathbf{c} = \vec{OC} = \vec{OA} + \vec{OB}$ is depicted by the diagonal of the parallelogram constructed on the

vectors $\vec{OA}$ and $\vec{OB}$. Consequently, the points O , A , B , and C lie in one plane and the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are coplanar. Now if one of the scalars, say, x is equal to zero, then $\mathbf{c} = y\mathbf{b}$ and, consequently, the vector $\mathbf{c}$ is collinear with the vector $\mathbf{b}$ and is, therefore, coplanar with the vectors $\mathbf{a}$ and $\mathbf{b}$.

Thus, the following *criterion of coplanarity* of three vectors holds true.

If the vectors $\mathbf{a}$ and $\mathbf{b}$ are not collinear, then the vector $\mathbf{c}$ is coplanar with $\mathbf{a}$ and $\mathbf{b}$ if and only if the resolution $\mathbf{c} = x\mathbf{a} + y\mathbf{b}$ holds true.

By definition, a zero vector is considered to be coplanar with any two vectors.

Theorem. *If the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are not coplanar, then any vector $\mathbf{d}$ can be uniquely represented in the form $\mathbf{d} = x\mathbf{a} + y\mathbf{b} + z\mathbf{c}$.*

This representation is the *resolution* of the vector $\mathbf{d}$ into components with respect to three noncoplanar vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$.

Example 3. Given the triangular prism $ABC A_1B_1C_1$ (Fig. 103). Represent the vector $\overrightarrow{AA_1}$ in terms of the vectors $\overrightarrow{AC_1}$, $\overrightarrow{BA_1}$ and $\overrightarrow{CB_1}$.

Δ In accordance with the rule of triangle we have

$$\overrightarrow{AA_1} = \overrightarrow{AB} + \overrightarrow{BA_1}, \quad \overrightarrow{BB_1} = \overrightarrow{BC} + \overrightarrow{CB_1}, \quad \overrightarrow{CC_1} = \overrightarrow{CA} + \overrightarrow{AC_1}.$$

Adding up the left-hand and right-hand sides of these vector equalities, we get

$$\begin{aligned} \overrightarrow{AA_1} + \overrightarrow{BB_1} + \overrightarrow{CC_1} &= (\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA}) \\ &\quad + \overrightarrow{BA_1} + \overrightarrow{CB_1} + \overrightarrow{AC_1}. \end{aligned}$$

Since $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \overrightarrow{AA} = \mathbf{0}$ and $\overrightarrow{AA_1} = \overrightarrow{BB_1} = \overrightarrow{CC_1}$, it follows that $3\overrightarrow{AA_1} = \overrightarrow{BA_1} + \overrightarrow{CB_1} + \overrightarrow{AC_1}$ and, consequently,

$$\overrightarrow{AA_1} = \frac{1}{3}(\overrightarrow{BA_1} + \overrightarrow{CB_1} + \overrightarrow{AC_1}). \quad \blacktriangle$$

7.6. An Angle Between Vectors. Scalar Product of Vectors

The *angle between the vectors* $\mathbf{a} = \overrightarrow{AB}$ and $\mathbf{b} = \overrightarrow{CD}$ is understood to be the angle between the rays $[AB)$ and $[CD)$.

Thus it follows that if we lay off two vectors $\mathbf{a} = \overrightarrow{OM}$ and $\mathbf{b} = \overrightarrow{ON}$ from one point (Fig. 104), the magnitude of the angle MON is, in accordance with the definition, the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$. This angle is denoted by $(\mathbf{a}, \mathbf{b})$.

Like the angle between rays, the angle between vectors can assume values from 0° to 180° . If the vectors are of the same direction, the angle between them is equal to 0° and the angle between the vectors opposite in directions is equal to 180° .

If the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$ is equal to 90° (Fig. 105), then the vectors $\mathbf{a}$ and $\mathbf{b}$ are said to be *perpendicular* or *orthogonal*. The perpendicularity of the vectors $\mathbf{a}$ and $\mathbf{b}$ is designated as $\mathbf{a} \perp \mathbf{b}$.

Example 1. Assume that $\mathbf{a}$ and $\mathbf{b}$ are nonzero noncollinear vectors. Prove that the vector $\mathbf{c} = \frac{\mathbf{a}}{|\mathbf{a}|} + \frac{\mathbf{b}}{|\mathbf{b}|}$ makes equal angles with the vectors $\mathbf{a}$ and $\mathbf{b}$.

$\triangle$ The vectors $\mathbf{a}_1 = \mathbf{a}/|\mathbf{a}|$ and $\mathbf{b}_1 = \mathbf{b}/|\mathbf{b}|$ are unit vectors: $|\mathbf{a}_1| = |\mathbf{b}_1| = 1$. Let us lay off the vectors $\mathbf{a}_1$

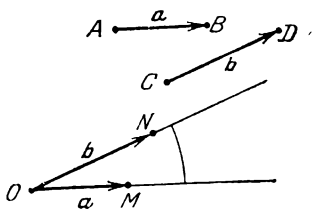


Fig. 104.

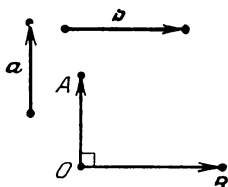


Fig. 105.

and $\mathbf{b}_1$ from one point, $\vec{OA} = \mathbf{a}_1$ and $\vec{OB} = \mathbf{b}_1$, and construct a parallelogram $OACB$ (Fig. 106). Then, $\vec{OC} = \mathbf{a}_1 + \mathbf{b}_1 =$

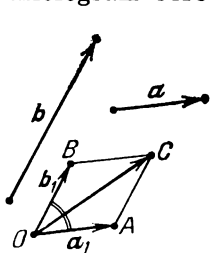


Fig. 106.

$\frac{\mathbf{a}}{|\mathbf{a}|} + \frac{\mathbf{b}}{|\mathbf{b}|} = \mathbf{c}$. In the parallelogram $OACB$ the lengths of the sides OA and OB are equal and, consequently, $OACB$ is a rhombus and its diagonal OC is the bisector of the angle AOB . Hence we conclude that

the vector $\mathbf{c} = \vec{OC}$ makes equal angles with the vectors $\mathbf{a}_1 = \vec{OA}$ and $\mathbf{b}_1 = \vec{OB}$ and with the

vectors $\mathbf{a}$ and $\mathbf{b}$ collinear with them, and is directed along the bisector of the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$. $\blacktriangle$

The *scalar product* of the nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ is the number equal to the product of the lengths of those vectors by the cosine of the angle between them. The scalar product

of the vectors $\mathbf{a}$ and $\mathbf{b}$ is designated $\mathbf{a} \cdot \mathbf{b}$. Thus we have

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \cdot |\mathbf{b}| \cos (\widehat{\mathbf{a}, \mathbf{b}}).$$

If one of the vectors is zero, then, in accordance with the definition, the scalar product is equal to zero: $\mathbf{a} \cdot \mathbf{0} = \mathbf{0} \cdot \mathbf{b} = 0$.

It should be emphasized that a scalar product is a number (when vectors are being considered, the numbers are often called scalars).

The scalar product of a vector by itself is equal to the square of its length: $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$.

The scalar product of vectors is positive if the angle between them is acute and is negative if the angle between the vectors is obtuse.

If the angle between the vectors is 90° , then the cosine of that angle is zero and the scalar product of the vectors

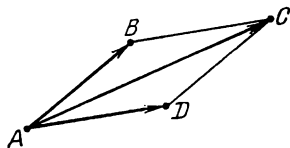


Fig. 107.

is also zero. The converse statement is also true: if the scalar product of nonzero vectors is zero, then the vectors are perpendicular. Consequently, *two nonzero vectors are orthogonal if and only if their scalar product is equal to zero.*

Properties of a scalar product.

1. $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$.
2. $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$.
3. $(x\mathbf{a}) \cdot \mathbf{b} = x(\mathbf{a} \cdot \mathbf{b})$.
4. $(\mathbf{a} + \mathbf{c}) \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{b} + \mathbf{c} \cdot \mathbf{b}$.

For example, using these properties, we get

$$\begin{aligned} (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b}) &= \mathbf{a} \cdot (\mathbf{a} - \mathbf{b}) + \mathbf{b} \cdot (\mathbf{a} - \mathbf{b}) \\ &= \mathbf{a} \cdot \mathbf{a} - \mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{a} - \mathbf{b} \cdot \mathbf{b} = |\mathbf{a}|^2 - |\mathbf{b}|^2. \end{aligned}$$

Example 2. Find the length of the diagonal AC of the rhombus $ABCD$ (Fig. 107), in which the lengths of the sides are equal to unity and the angle BAD is 30° ,

△ By the parallelogram rule, $\vec{AC} = \vec{AB} + \vec{AD}$. It follows from the properties of a scalar product that

$$|\vec{AC}|^2 = (\vec{AB} + \vec{AD})^2 = |\vec{AB}|^2 + 2\vec{AB} \cdot \vec{AD} + |\vec{AD}|^2. \quad (1)$$

Since $|\vec{AB}| = |\vec{AD}| = 1$ and $\widehat{(\vec{AB}, \vec{AD})} = 30^\circ$, we have $\vec{AB} \cdot \vec{AD} = \sqrt{3}/2$. Taking this into account, we get from (1) $|\vec{AC}|^2 = 2 + \sqrt{3}$, whence we find $|\vec{AC}| = \sqrt{2 + \sqrt{3}}$. ▲

If the vectors **a** and **b** are nonzero, then the cosine of the angle between them can be found by the formula

$$\cos \widehat{(\mathbf{a}, \mathbf{b})} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| \cdot |\mathbf{b}|}. \quad (2)$$

Example 3. The lengths of the nonzero vectors **a** and **b** are equal. Find the angle between the vectors if the vectors $\mathbf{p} = \mathbf{a} + 2\mathbf{b}$ and $\mathbf{q} = 5\mathbf{a} - 4\mathbf{b}$ are known to be perpendicular.

△ The vectors **p** and **q** being perpendicular, their scalar product is equal to zero: $\mathbf{p} \cdot \mathbf{q} = (\mathbf{a} + 2\mathbf{b}) \cdot (5\mathbf{a} - 4\mathbf{b}) = 0$. Using the properties of a scalar product, we get $(\mathbf{a} + 2\mathbf{b}) \times (5\mathbf{a} - 4\mathbf{b}) = 5|\mathbf{a}|^2 + 6\mathbf{a} \cdot \mathbf{b} - 8|\mathbf{b}|^2$. Thus we have

$$5|\mathbf{a}|^2 + 6|\mathbf{a}||\mathbf{b}|\cos \widehat{(\mathbf{a}, \mathbf{b})} - 8|\mathbf{b}|^2 = 0,$$

whence, for $|\mathbf{a}| = |\mathbf{b}|$, we get

$$6|\mathbf{a}|^2 \cos \widehat{(\mathbf{a}, \mathbf{b})} - 3|\mathbf{a}|^2 = 0.$$

Since $|\mathbf{a}| \neq 0$, we can cancel by $|\mathbf{a}|^2$ and get $\cos \widehat{(\mathbf{a}, \mathbf{b})} = 1/2$. Consequently, the angle between the vectors **a** and **b** is 60° . ▲

7.7. The Basis. The Coordinates of a Vector. Operations on Vectors Defined by Their Coordinates

Three mutually perpendicular (orthogonal) unit vectors, taken in a certain order, form a *Cartesian (Euclidean) basis* in space (Fig. 108). It is customary to denote the vectors of a Cartesian basis by the letters **i**, **j**, **k**.

Every vector can be uniquely represented in terms of any three noncoplanar vectors (see 7.4) and, therefore, every vector $\mathbf{a}$ can be uniquely represented as

$$\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}.$$

The numbers x, y, z are called the *coordinates* of the vector $\mathbf{a}$ in the basis $(\mathbf{i}, \mathbf{j}, \mathbf{k})$. If $\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, we write $\mathbf{a} = (x; y; z)$. For example, $\mathbf{i} = (1; 0; 0)$, $\mathbf{j} = (0; 1; 0)$, $\mathbf{k} = (0; 0; 1)$, $\mathbf{0} = (0; 0; 0)$.

Assume $\mathbf{a} = (x_1; y_1; z_1)$ and $\mathbf{b} = (x_2; y_2; z_2)$. The uniqueness of resolution implies that the vector equality $\mathbf{a} = \mathbf{b}$ is equivalent to the system of three scalar equalities:

$$x_1 = x_2, \quad y_1 = y_2, \quad z_1 = z_2.$$

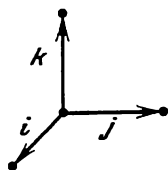


Fig. 108.

The following rules of operations on vectors, defined by their coordinates, are valid.

1. When adding the vectors $\mathbf{a} = (x_1; y_1; z_1)$ and $\mathbf{b} = (x_2; y_2; z_2)$, their respective coordinates are added up:

$$\mathbf{a} + \mathbf{b} = (x_1 + x_2; y_1 + y_2; z_1 + z_2).$$

2. When multiplying the vector $\mathbf{a} = (x; y; z)$ by a scalar, all the coordinates of $\mathbf{a}$ are multiplied by that scalar:

$$m\mathbf{a} = (mx; my; mz).$$

3. The scalar product of the vectors $\mathbf{a} = (x_1; y_1; z_1)$ and $\mathbf{b} = (x_2; y_2; z_2)$ is equal to the sum of the products of the respective coordinates:

$$\mathbf{a} \cdot \mathbf{b} = x_1x_2 + y_1y_2 + z_1z_2.$$

4. The length of the vector $\mathbf{a} = (x; y; z)$ is equal to the square root of the sum of the squares of its coordinates:

$$|\mathbf{a}| = \sqrt{x^2 + y^2 + z^2}.$$

Example 1. Find the coordinates and the length of the vector $2\mathbf{a} - 3\mathbf{b}$ if $\mathbf{a} = (0; 3; 2)$ and $\mathbf{b} = (-2; 3; 2)$.

△ In accordance with the rules of operations on vectors we find $2\mathbf{a} = (0; 6; 4)$, $3\mathbf{b} = (-6; 9; 6)$ and, finally, $2\mathbf{a} - 3\mathbf{b} = (6; -3; -2)$. Now we can find the length of the vector: $|2\mathbf{a} - 3\mathbf{b}| = \sqrt{6^2 + 3^2 + 2^2} = 7$. ▲

Example 2. At what values of m and n are the vectors $\mathbf{a} = (1; m; -2)$ and $\mathbf{b} = (-2; 3; n)$ collinear?

$\triangle$ The vector $\mathbf{b}$ is collinear with the vector $\mathbf{a} \neq 0$ if and only if there is a number p such that $\mathbf{b} = p\mathbf{a}$. For the given vectors $\mathbf{a}$ and $\mathbf{b}$ this vector equality is equivalent to the system $-2 = p$, $3 = mp$, $n = -2p$, from which we find $p = -2$, $m = -3/2$, $n = 4$.

Thus it follows that $\mathbf{a} = (1; -3/2; -2)$; $\mathbf{b} = (-2; 3; 4)$. $\blacktriangle$

Example 3. At what value of m are the vectors $\mathbf{a} = (1; 3; -2)$ and $\mathbf{b} = (-1; m; 4)$ orthogonal.

$\triangle$ Nonzero vectors are orthogonal if and only if their scalar product is zero. Since $\mathbf{a} \cdot \mathbf{b} = -9 + 3m$, we have $\mathbf{a} \cdot \mathbf{b} = 0$ for $m = 3$. $\blacktriangle$

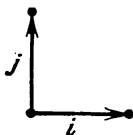


Fig. 109.

Example 4. Find the coordinates of the unit vector $\mathbf{b}$ pointing in the same direction as the vector $\mathbf{a} = (2; -3; 6)$.

$\triangle$ We have $\mathbf{b} \uparrow \mathbf{a}$ if $\mathbf{b} = m\mathbf{a}$ and $m > 0$. Taking into account that $|\mathbf{b}| = |m| |\mathbf{a}| = m |\mathbf{a}|$ for the condition $|\mathbf{b}| = 1$, that $m = 1/|\mathbf{a}|$. We calculate the length of the vector $\mathbf{a}$: $|\mathbf{a}| = 7$. Thus we have $m = 1/7$ and $\mathbf{b} = (2/7; -3/7; 6/7)$. $\blacktriangle$

Example 5. Find the cosines of the angles the vector $\mathbf{a} = (3; 0; -4)$ makes with the base vectors.

$\triangle$ Let us calculate the scalar products of the vector $\mathbf{a}$ with each of the base vectors. Since $\mathbf{i} = (1; 0; 0)$, $\mathbf{j} = (0; 1; 0)$, and $\mathbf{k} = (0; 0; 1)$, it follows that $\mathbf{a} \cdot \mathbf{i} = 3$; $\mathbf{a} \cdot \mathbf{j} = 0$; $\mathbf{a} \cdot \mathbf{k} = -4$. The lengths of the base vectors are equal to unity. We calculate the length of the vector $\mathbf{a}$: $|\mathbf{a}| = 5$. Now, from formula (2) in 7.6, we find

$$\cos \widehat{(\mathbf{a}, \mathbf{i})} = \frac{\mathbf{a} \cdot \mathbf{i}}{|\mathbf{a}| |\mathbf{i}|} = \frac{3}{5}, \quad \cos \widehat{(\mathbf{a}, \mathbf{j})} = \frac{\mathbf{a} \cdot \mathbf{j}}{|\mathbf{a}| |\mathbf{j}|} = 0,$$

$$\cos \widehat{(\mathbf{a}, \mathbf{k})} = \frac{\mathbf{a} \cdot \mathbf{k}}{|\mathbf{a}| |\mathbf{k}|} = -\frac{4}{5}. \quad \blacktriangle$$

Vectors in a plane. Two orthogonal unit vectors taken in a definite order constitute a *rectangular basis* in a plane (Fig. 109). A basis in a plane is designated as $(\mathbf{i}, \mathbf{j})$.

Any vector $\mathbf{a}$ in a plane in which a basis $(\mathbf{i}, \mathbf{j})$ is chosen, can be represented, in a unique manner, in terms of the base vectors: $\mathbf{a} = x\mathbf{i} + y\mathbf{j}$. We call the scalars x and y the *coordinates* of the vector $\mathbf{a}$, and write $\mathbf{a} = (x; y)$.

Suppose $\mathbf{a} = (x_1; y_1)$, $\mathbf{b} = (x_2; y_2)$, then

$$1. \mathbf{a} + \mathbf{b} = (x_1 + x_2; y_1 + y_2).$$

$$2. m\mathbf{a} = (mx_1; my_1).$$

$$3. \mathbf{a} \cdot \mathbf{b} = x_1x_2 + y_1y_2.$$

$$4. |\mathbf{a}| = \sqrt{x_1^2 + y_1^2}.$$

The vector equality $\mathbf{a} = \mathbf{b}$ is equivalent to the system of two scalar equalities: $x_1 = x_2$, $y_1 = y_2$.

Example 6. Determine the angle between the vectors $\mathbf{c} = 4\mathbf{a} + \mathbf{b}$ and $\mathbf{d} = -\frac{1}{4}\mathbf{a} + \frac{7}{4}\mathbf{b}$, if $\mathbf{a} = (-1; 1)$ and $\mathbf{b} = (1; 3)$.

△ We find the coordinates of the vectors $\mathbf{c}$ and $\mathbf{d}$:

$$\mathbf{c} = (-4 + 1; 4 + 3) = (-3; 7),$$

$$\mathbf{d} = \left(\frac{1}{4} + \frac{7}{4}; -\frac{1}{4} + \frac{21}{4}\right) = (2; 5).$$

We calculate the lengths of the vectors $\mathbf{c}$ and $\mathbf{d}$ and their scalar product: $c = \sqrt{58}$, $d = \sqrt{29}$, $\mathbf{c} \cdot \mathbf{d} = 29$. By formula (2), Sec. 7.6, we find

$$\cos(\widehat{\mathbf{c}, \mathbf{d}}) = \frac{\mathbf{c} \cdot \mathbf{d}}{|\mathbf{c}| |\mathbf{d}|} = \frac{29}{\sqrt{58} \sqrt{29}} = \frac{1}{\sqrt{2}},$$

consequently, the angle between the vectors $\mathbf{c}$ and $\mathbf{d}$ is 45° . ▲

7.8. Cartesian System of Coordinates. Equation of a Plane

If we are given a point O and a Cartesian basis $\mathbf{i}, \mathbf{j}, \mathbf{k}$, then we say that a *Cartesian system of coordinates* is given in space (Fig. 110). The point O is called the *origin of coordinates* or simply *origin*. The origin and each of the basis vectors define a *coordinate axis*: the vector $\overrightarrow{OA} = \mathbf{i}$ defines the *abscissa axis* (designated as Ox), the vector $\overrightarrow{OB} = \mathbf{j}$ defines the *axis of ordinates* (Oy), the vector $\overrightarrow{OC} = \mathbf{k}$ defines

the *applicate axis* (Oz). Planes passing through every two coordinate axes are called *coordinates planes*.

For every point M , the coordinates of the vector $\vec{OM}$ in the basis ($\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$) are called the *coordinates of the point M* in the rectangular (Cartesian) system of coordinates $Oxyz$. Thus, if

$$\vec{OM} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k},$$

then the numbers x , y , z are the coordinates of the point M ; they are: x , the *abscissa*, y , the *ordinate*, z , the *applicate* of the point M . The coordinates of the point M are written in parentheses next to the letter: $M(x; y; z)$.

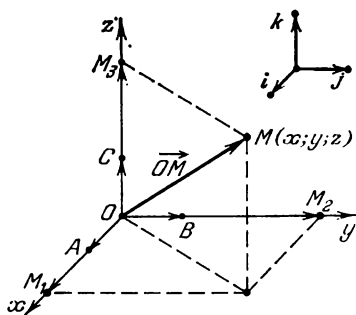


Fig. 110.

If $\vec{OM} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, then the numbers x , y , and z are the coordinates of the points M_1 , M_2 , and M_3 on the axes of abscissa, ordinate and applicate, respectively, such that $\vec{OM}_1 = x\mathbf{i}$, $\vec{OM}_2 = y\mathbf{j}$, $\vec{OM}_3 = z\mathbf{k}$ (see Fig. 110).

To construct the point $M(x; y; z)$, it is necessary to construct the vectors $\vec{OM}_1 = x\mathbf{i}$, $\vec{OM}_2 = y\mathbf{j}$, and $\vec{OM}_3 = z\mathbf{k}$ and the vector $\vec{OM} = \vec{OM}_1 + \vec{OM}_2 + \vec{OM}_3$.

Assume that $\mathbf{a} = \vec{AB}$ and the points A and B are defined by their coordinates: $A(x_1; y_1; z_1)$, $B(x_2; y_2; z_2)$. Since $\vec{AB} = \vec{OB} - \vec{OA}$, we have $\vec{AB} = (x_2 - x_1; y_2 - y_1; z_2 - z_1)$, that is, to find the coordinates of a vector, it is necessary to subtract the coordinates of its origin from the respective coordinates of its terminus.

The distance between the points A and B is equal to the length of the vector $\vec{AB}$, therefore,

$$|\vec{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

Example 1. In a triangle with vertices at the points $A(3; 2; -3)$, $B(5; 1; -1)$, and $C(1; -2; 1)$ determine the angle at the vertex A .

$\triangle$ The angle at the vertex A is equal to the angle between the vectors $\vec{AC}$ and $\vec{AB}$ (Fig. 111). The cosine of the angle φ between the vectors $\vec{AC}$ and $\vec{AB}$ can be found by formula (2), Sec. 6. We determine the coordinates of the vectors $\vec{AC}$

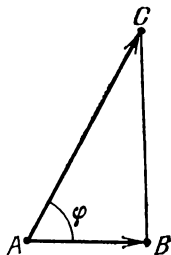


Fig. 111.

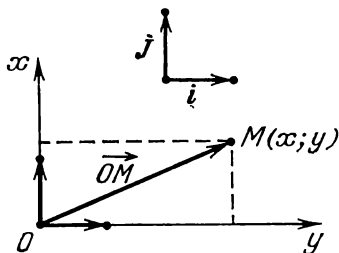


Fig. 112.

and $\vec{AB}$: $\vec{AC} = (-2; -4; 4)$, $\vec{AB} = (2; -1; 2)$. We now find the lengths of these vectors and their scalar product

$$|\vec{AC}| = \sqrt{2^2 + 4^2 + 4^2} = 6; \quad |\vec{AB}| = \sqrt{2^2 + 1^2 + 2^2} = 3;$$

$$\vec{AC} \cdot \vec{AB} = 8.$$

Then $\cos \varphi = 8/(6 \cdot 3) = 4/9$ and, consequently, $\varphi = \arccos(4/9)$. $\blacktriangle$

The rectangular system of coordinates in a plane can be determined by analogy, that is, by defining a point O and a rectangular basis (i, j). The axes of coordinates are called, respectively, the *axis of abscissas* and the *axis of ordinates*. The *coordinates of the point M* are the coordinates $(x; y)$ of the vector $\vec{OM}$, which fact is written as $M(x; y)$ (Fig. 112).

If the points are defined by their coordinates, $A(x_1; y_1)$ and $B(x_2; y_2)$, then

$$\vec{AB} = (x_2 - x_1; y_2 - y_1)$$

and

$$|\overrightarrow{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Example 2. Given the point $A(-1; 1)$ and the vector $\mathbf{a} = (3; 2)$. Find the coordinates of the point B such that $\overrightarrow{AB} = \mathbf{a}$.

Δ If we assume that $(x; y)$ are the coordinates of the point B , then $\overrightarrow{AB} = (x + 1; y - 1)$, and if $\mathbf{a} = \overrightarrow{AB}$, then $x + 1 = 3$ and $y - 1 = 2$. We find $x = 2$, $y = 3$. Then B has the coordinates $(2; 3)$. $\blacktriangle$

Example 3. Given a triangle with vertices at the points $A(1; 1)$, $B(-4; 3)$, and $C(2; 2)$. Find the length of the median AN .

Δ Suppose O is the origin, then $\overrightarrow{OB} = (-4; 3)$ and $\overrightarrow{OC} = (2; 2)$. If N is the midpoint of the side BC , then (see Example 2 in 7.5) $\overrightarrow{ON} = (\overrightarrow{OB} + \overrightarrow{OC})/2$, whence it follows that the coordinates of the mid-point of the segment BC are equal to the half-sum of the respective coordinates of the points B and C . We find $\overrightarrow{ON} = ((-4 + 2)/2; (3 + 2)/2)$. Then $N(-1; 5/2)$ and $|\overrightarrow{AN}| = \sqrt{(-1 - 1)^2 + \left(\frac{5}{2} - 1\right)^2} = \frac{5}{2}$. $\blacktriangle$

Equation of a plane. In the rectangular system of coordinates, any equation

$$ax + by + cz + d = 0, \quad (1)$$

in which at least one of the coefficients a , b , c is nonzero, defines a plane. The converse is also true: every plane can be given by an equation of the form (1).

Example 4. Set up an equation of the plane passing through the points $A(0; 1; 5)$, $B(3; 0; 0)$ and $C(-1; 1; 6)$.

Δ Suppose the equation of the plane is $ax + by + cz + d = 0$. The coordinates of the points A , B , and C satisfy this equation and, consequently,

$$\begin{aligned} b + 5c + d &= 0, \\ 3a + d &= 0, \\ -a + b + 6c + d &= 0. \end{aligned}$$

From the second equation we find $d = -3a$. Substituting this expression into the first and third equations, we get the system

$$b + 5c = 3a,$$

$$b + 6c = 4a.$$

Subtracting the second equation from the first term-by-term, we get $c = a$, and then we find $b = -2a$. The equation of the plane has the form $ax - 2ay + az - 3a = 0$ or (since a , b , and c are simultaneously nonzero)

$$x - 2y + z - 3 = 0. \quad \blacktriangle$$

The nonzero vector $\mathbf{n} = \overrightarrow{AB}$ is said to be perpendicular to a plane if the straight line (AB) is perpendicular to that plane.

Suppose the vector $\mathbf{n} = (a; b; c)$ is perpendicular to the plane passing through the point $M(x_0; y_0; z_0)$. Then for any point $M_1(x; y; z)$ of that plane the vectors $\mathbf{n}$ and $\overrightarrow{MM_1}$ are perpendicular and, consequently, $\mathbf{n} \cdot \overrightarrow{MM_1} = 0$. Writing this scalar product in the coordinates, we obtain the following equation of the plane:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0, \quad (2)$$

which we can reduce to the form (1) by setting $d = -ax_0 - by_0 - cz_0$. Thus, the equation $ax + by + cz + d = 0$ defines the plane perpendicular to the vector $\mathbf{n} = (a; b; c)$.

Example 5. Find the equation of the plane passing through the origin at right angles to the vector $\mathbf{n} = (-2; 1; 3)$.

$\triangle$ Substituting the coordinates of the vector $\mathbf{n}$ and those of the point $M(0; 0; 0)$ into equation (2), we get the equation of the plane: $-2x + y + 3z = 0$. $\blacktriangle$

Let two planes be defined by the equations

$$a_1x + b_1y + c_1z + d_1 = 0, \quad (3)$$

$$a_2x + b_2y + c_2z + d_2 = 0. \quad (4)$$

These two planes are parallel if and only if the vectors $\mathbf{n}_1 = (a_1; b_1; c_1)$ and $\mathbf{n}_2 = (a_2; b_2; c_2)$, perpendicular to them, are collinear, that is, there is a number $m \neq 0$ such that $a_1 = ma_2$; $b_1 = mb_2$; $c_1 = mc_2$. The planes defined by equa-

tions (3) and (4) are perpendicular if and only if the vectors $\mathbf{n}_1 = (a_1; b_1; c_1)$ and $\mathbf{n}_2 = (a_2; b_2; c_2)$ are orthogonal, that is, if $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$.

When they intersect, two planes form four dihedral angles, pairwise equal in magnitude. The *angle α between the planes* is the smallest of these dihedral angles. It is equal to the angle φ between the vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, if $0 \leq \varphi \leq 90^\circ$ and to $180^\circ - \varphi$ if $90^\circ \leq \varphi \leq 180^\circ$ (Fig. 113). In both

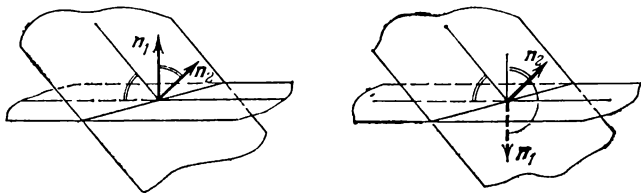


Fig. 113.

cases $\cos \alpha = |\cos \varphi|$. Thus, the cosine of the angle φ between the planes defined by equations (3) and (4) can be found by the formula

$$\cos \alpha = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1| |\mathbf{n}_2|}. \quad (5)$$

Example 6. Find the angle between the planes

$$-3y + z + 2 = 0 \quad \text{and} \quad 2y + z - 5 = 0.$$

△ The vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, perpendicular to those planes, have the coordinates $\mathbf{n}_1 = (0; -3; 1)$ and $\mathbf{n}_2 = (0; 2; 1)$. Let us find the lengths of these vectors and their scalar product: $|\mathbf{n}_1| = \sqrt{10}$, $|\mathbf{n}_2| = \sqrt{5}$; $\mathbf{n}_1 \cdot \mathbf{n}_2 = -5$. Then we get by formula (5): $\cos \alpha = |-5| / (\sqrt{10} \sqrt{5}) = 1/\sqrt{2}$, whence $\alpha = 45^\circ$. ▲

PROBLEMS OF SECTION I

1. In the trapezoid $ABCD$ the ratio of the length of the base BC to the length of the base AD is n . The diagonals of the trapezoid intersect at the point O . Represent the vector $\vec{AO}$ in terms of the vectors $\vec{AB}$ and $\vec{AD}$.

2. Given three nonzero vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ every two of which are noncollinear. Find their sum if the vector $\mathbf{a} + \mathbf{b}$ is collinear with $\mathbf{c}$, and the vector $\mathbf{b} + \mathbf{c}$ is collinear with the vector $\mathbf{a}$.

3. The medians of the faces OAB and OAC of the tetrahedron $OABC$ intersect at the points M and N respectively. Prove that the vectors $\vec{MN}$ and $\vec{BC}$ are collinear and find the ratio $|\vec{MN}| : |\vec{BC}|$.

4. The points M , N , P , and Q lie, respectively, on the sides AB , BC , CD , and DA of the parallelogram $ABCD$, with $|\vec{AM}| : |\vec{AB}| = |\vec{BN}| : |\vec{BC}| = |\vec{CP}| : |\vec{CD}| = |\vec{DQ}| : |\vec{DA}|$. Prove that $MNPQ$ is a parallelogram.

5. Prove that any two nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ are collinear if and only if there are numbers x and y , which are simultaneously nonzero, such that $x\mathbf{a} + y\mathbf{b} = \mathbf{0}$.

6. The two nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ are such that $|\mathbf{a} + \mathbf{b}| = |\mathbf{a} - \mathbf{b}|$. Prove that the vectors $\mathbf{a}$ and $\mathbf{b}$ are orthogonal.

7. Knowing that $|\mathbf{a}| = 11$, $|\mathbf{b}| = 23$, and $|\mathbf{a} - \mathbf{b}| = 30$, find $|\mathbf{a} + \mathbf{b}|$.

8. The vector $\mathbf{a} + 3\mathbf{b}$ is orthogonal to the vector $7\mathbf{a} - 5\mathbf{b}$ and the vector $\mathbf{a} - 4\mathbf{b}$ is orthogonal to the vector $7\mathbf{a} - 2\mathbf{b}$. Find the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$.

9. The unit vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ satisfy the condition $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$. Calculate $\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}$.

10. Find the number m such that the vectors $\mathbf{a} = \mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $\mathbf{b} = 3\mathbf{i} + \mathbf{j}$, and $\mathbf{c} = m\mathbf{i} + 2\mathbf{k}$ are coplanar.

11. Find the coordinates of the vector $\mathbf{b}$, collinear with the vector $\mathbf{a} = (1; 1; -1/2)$, making an acute angle with the base vector $\mathbf{k}$, and such that $|\mathbf{b}| = 3$.

12. Find the coordinates of the unit vector $\mathbf{p}$, orthogonal to the vectors $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 3\mathbf{j} - \mathbf{k}$ and making an obtuse angle with the basis vector $\mathbf{j}$.

13. Find the coordinates of the vector $\mathbf{b}$, coplanar with the vectors $\mathbf{i}$ and $\mathbf{j}$, orthogonal to the vector $\mathbf{a} = 4\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$ and such that $|\mathbf{a}| = |\mathbf{b}|$.

14. Find the coordinates of the point belonging to the axis of ordinates and equidistant from the points $A(2; -1; 1)$ and $B(0; 1; 3)$.

15. The vectors $\mathbf{a} = (-4, 0, 3)$ and $\mathbf{b} = (14, 2, -5)$ are laid off from one point. Find the vector $\mathbf{d}$, which, being laid off from the same point, divides the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$ in half and whose length is equal to $\sqrt{6}$.

16. Given $A(3; 2)$, $B(5; 1)$ and $D(1; -2)$. Find the length of the diagonal AC of the parallelogram $ABCD$.

17. Given $B(2; -19; 16)$, $C(-4; 29; -20)$, and $M(1; -1; 1)$. Find the distance from the point M to the midpoint of the line segment BC and the distance from the point M to the point N belonging to the segment BC and such that $|\vec{BN}| : |\vec{BC}| = 1/3$.

18. Set up an equation of the plane passing through the point $A(1; -3; 2)$ at right angles to the plane $4x - 2y - z + 7 = 0$.

19. The point $A(-1; -1; 2)$ is the foot of the perpendicular dropped from the origin on a plane. Set up an equation of that plane.

20. Find the distance from the point $A(-2; 3; -4)$ to the plane $2x + 2y - z + 3 = 0$.

21. Find the angle between the plane passing through the points $A(0; 0; 0)$, $B(1; 1; 1)$ and $C(3; 2; 1)$ and the plane passing through the points $A(0; 0; 0)$, $B(1; 1; 1)$ and $D(3; 1; 2)$.

PROBLEMS OF SECTION II

1. $ABCDEF$ is a regular hexagon. Represent the vectors $\vec{BC}$ and $\vec{BD}$ in terms of the vectors $\vec{AB}$ and $\vec{AF}$.

2. $OABC$ is a tetrahedron, AM is the median of the face ABC . Represent the vector $\vec{AM}$ in terms of the vectors $\vec{OA}$, $\vec{OB}$, and $\vec{OC}$.

3. $ABCD$ is a parallelogram, M is the midpoint of the side CD . Represent the vectors $\vec{BD}$ and $\vec{AM}$ in terms of the vectors $\vec{BM}$ and $\vec{MC}$.

4. Given the parallelepiped $ABCD A_1 B_1 C_1 D_1$ (Fig. 114). Represent the vectors $\vec{AA_1}$, $\vec{AC}$, and $\vec{DB}$ in terms of the vectors $\vec{DA_1}$, $\vec{DC_1}$, and $\vec{DB_1}$.

5. In the triangle ABC , M and N are the midpoints of the sides AB and AC . Represent the vectors $\vec{AB}$, $\vec{AC}$, and $\vec{MN}$ in terms of the vectors $\vec{BN}$ and $\vec{CM}$.

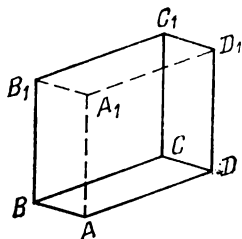


Fig. 114.

6. In the tetrahedron $OABC$, M and N are the midpoints of the edges OB and OC . Represent the vectors $\vec{AM}$, $\vec{BN}$ and $\vec{MN}$ in terms of the vectors $\vec{OA}$, $\vec{OB}$, and $\vec{OC}$.

7. In the triangular prism $ABCA_1 B_1 C_1$ the diagonals of the face $BB_1 C_1 C$ intersect at the point M . Represent the vectors $\vec{AM}$ and $\vec{A_1 M}$ in terms of the vectors $\vec{BA}$, $\vec{BB_1}$, and $\vec{BC}$.

8. Can the length of the vector $\mathbf{a} - \mathbf{b}$ be (a) less than, (b) equal to, or (c) larger than the sum of the lengths of the vectors $\mathbf{a}$ and $\mathbf{b}$?

9. Prove that any two nonzero vectors $\mathbf{a}$ and $\mathbf{b}$, of similar directions, satisfy the condition $\mathbf{a}/|\mathbf{a}| = \mathbf{b}/|\mathbf{b}|$.

10. Find the number x if the length of the vector $\mathbf{b} = x\mathbf{a}$ is equal to $3|\mathbf{a}|$, and the vector $\mathbf{b}$ is (a) of similar direction as the vector $\mathbf{a}$, and (b) of the direction opposite to that of the vector $\mathbf{a}$.

11. The vectors $\mathbf{a}$ and $\mathbf{b}$ are not collinear. Find the numbers x and y if the vectors $x\mathbf{a} + y\mathbf{b}$ and $(y+1)\mathbf{a} + (2-x)\mathbf{b}$ are equal.

12. The vectors $\mathbf{a}$ and $\mathbf{b}$ are not collinear. Find the numbers x and y if the vectors $(2-x)\mathbf{a} + \mathbf{b}$ and $y\mathbf{a} + (x-3)\mathbf{b}$ are equal.

13. The vectors $\mathbf{a}$ and $\mathbf{b}$ are not collinear. Find the number x if the vectors $(x-1)\mathbf{a} + 2\mathbf{b}$ and $3\mathbf{a} + x\mathbf{b}$ are collinear.

14. The vectors $\mathbf{a}$ and $\mathbf{b}$ are not collinear. Find the number x if the vectors $3\mathbf{a} + x\mathbf{b}$ and $(1-x)\mathbf{a} - \frac{2}{3}\mathbf{b}$ are of the same direction.

15. The vectors $\mathbf{a}$ and $\mathbf{b}$ form the angle 120° . Find x from the conditions that $|\mathbf{b}| = 2|\mathbf{a}|$ and the vector $\mathbf{a} + x\mathbf{b}$ is orthogonal to the vector $\mathbf{a} - \mathbf{b}$.

16. Determine at what x and y the vector $\mathbf{a} = -2\mathbf{i} + 3\mathbf{j} + y\mathbf{k}$ is collinear with the vector $\mathbf{b} = x\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}$.

17. Find the unit vector which is of the same direction as the vector $\mathbf{a} = -6\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$.

18. Determine the lengths of the vectors $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$ if $\mathbf{a} = (3; -5; 8)$ and $\mathbf{b} = (-1; 1; -4)$.

19. Given: $|\mathbf{a}| = 13$, $|\mathbf{b}| = 19$ and $|\mathbf{a} - \mathbf{b}| = 22$. Find $|\mathbf{a} + \mathbf{b}|$.

20. Given: $|\mathbf{a}| = 6$, $|\mathbf{a} + \mathbf{b}| = 11$, $|\mathbf{a} - \mathbf{b}| = 7$. Find $|\mathbf{b}|$.

21. Determine the lengths of the vectors $2\mathbf{a} - \mathbf{b}$ and $3\mathbf{b} - \mathbf{a}$ if $\mathbf{a} = -2\mathbf{i} + \mathbf{j}$ and $\mathbf{b} = -\mathbf{i} - 2\mathbf{j}$.

22. Find the angle between the diagonals of the parallelogram constructed on the vectors $\mathbf{a} = (1; 2)$ and $\mathbf{b} = (2; -1)$.

23. Find the angle between the vectors $\mathbf{a} = (4; 0; 3)$ and $\mathbf{b} = (2; -2; 1)$.

24. The angle between the vectors $\mathbf{a}$ and $\mathbf{b}$ is 120° , $|\mathbf{a}| = 3$ and $|\mathbf{b}| = 2$. Find: (a) $\mathbf{a} \cdot \mathbf{b}$, (b) $(\mathbf{a} + \mathbf{b})^2$, (c) $(\mathbf{a} - \mathbf{b})^2$, and (d) $(\mathbf{a} + 2\mathbf{b}) \cdot (2\mathbf{a} - \mathbf{b})$.

25. The vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ satisfy the condition $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$. Calculate $\mathbf{ab} + \mathbf{bc} + \mathbf{ca}$ if $|\mathbf{a}| = 1$, $|\mathbf{b}| = 3$, and $|\mathbf{c}| = 4$.

26. Find the coordinates of the vector $\mathbf{b}$, collinear with the vector $\mathbf{a} = (2; -3)$ and satisfying the condition $\mathbf{a} \cdot \mathbf{b} = -26$.

27. Calculate the scalar product of the vectors $4\mathbf{a} - \mathbf{b}$ and $2\mathbf{a} + 3\mathbf{b}$ if $|\mathbf{a}| = 2$ and $|\mathbf{b}| = 3$ and the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$ is 120° .

28. In a plane find the coordinates of the unit vector $\mathbf{c}$ orthogonal to the vector $\overrightarrow{AB}$ if $A(1; -1)$ and $B(3; 0)$.

29. Find $\mathbf{a} \cdot \mathbf{b}$ if $\mathbf{a} = 4\mathbf{i} + 7\mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} - 5\mathbf{j} + \mathbf{k}$.

30. Find $(6\mathbf{e}_1 + \mathbf{e}_2) \cdot (\mathbf{e}_1 - 2\mathbf{e}_2)$ if the angle between the unit vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ is 60° .

31. Find the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$ if $|\mathbf{a}| = 2|\mathbf{b}|$ and the vector $2\mathbf{a} + \mathbf{b}$ is orthogonal to the vector $\mathbf{a} - 3\mathbf{b}$.

32. Find the coordinates of the vector $\mathbf{b}$, collinear with the vector $\mathbf{a} = (-1; 1; -2)$, if $\mathbf{a} \cdot \mathbf{b} = 12$.

33. Find the coordinates of the vector $\mathbf{b}$, collinear with the vector $\mathbf{a} = (-1; 2)$ if $|\mathbf{b}| = \sqrt{10}$.

34. Find at what m and n the vector $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j} + m\mathbf{k}$ is collinear with the vector $\mathbf{b} = n\mathbf{i} + \mathbf{j} - 2\mathbf{k}$.

35. Find at what m the vector $\mathbf{a} = (m; 7; -2)$ is orthogonal to the vector $\mathbf{b} = (-3; m; +2)$.

36. Find the coordinates of the vector $\mathbf{b}$, orthogonal to the vector $\mathbf{a} = (-2; 1)$ if $|\mathbf{b}| = \sqrt{5}$.

37. Find the cosines of the angles made by the vector $\mathbf{a} = (2; -1; -2)$ with the basis vectors.

38. Let $\mathbf{e}_1$ and $\mathbf{e}_2$ be noncollinear unit vectors. Calculate $(2\mathbf{e}_1 - 5\mathbf{e}_2) \cdot (3\mathbf{e}_1 + \mathbf{e}_2)$ if $|\mathbf{e}_1 + \mathbf{e}_2| = \sqrt{3}$.

39. Find the angle between the vectors $\overrightarrow{AB}$ and $\mathbf{a} = (1; -3; 1)$ if $A(-5; 7; -8)$ and $B(-7; 8; -9)$.

40. Find the coordinates of the unit vector of the direction opposite to that of the vector $\overrightarrow{AB}$, if $A(7; 4; -2)$ and $B(1; 2; 1)$.

41. Find the angle between the vectors $\mathbf{a} + \mathbf{b}$ and $2\mathbf{a} - \mathbf{c}$ if $\mathbf{a} = -\mathbf{i} + \mathbf{j} - \mathbf{k}$, $\mathbf{b} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, and $\mathbf{c} = -2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$.

42. Find the angle between the vectors $\vec{AB}$ and $\vec{CD}$ if $A(-5; 1)$, $B(-1; 4)$, $C(1; -4)$, and $D(2; 3)$.

43. Find the coordinates of the vectors $\vec{AB}$ and $2\vec{BA}$ if $A(3; -1; 2)$ and $B(-1; 2; 1)$.

44. Find the coordinates of the points M_1 and M_2 symmetric with respect to the point $M(1; -2; 5)$ about the abscissa axis and about the plane Oxy , respectively.

45. Given: $A(-1; -2; 4)$, $B(3; 2; -2)$, and $C(3; -2; 1)$. Find the angle at the vertex C of the triangle ABC .

46. Find the length of the median AM of the triangle ABC if $A(2; 3/2; -4)$, $B(3; -4; 2)$, and $C(1; 3; -7)$.

47. Find the distance from the point $M(-2; 0; 1)$ to the midpoint of the line segment AB is $A(2; -1; 0)$ and $B(-2; 3; 2)$.

48. Given: $A(-1; 3; -7)$, $B(2; -1; 5)$, and $C(0; 1; -5)$. Find $|AB|$ and $(2\vec{AB} - \vec{CB}) \cdot (2\vec{BC} + \vec{BA})$.

49. Given: $A(1; -2; 2)$, $B(1; 4; 0)$, $C(-4; 1; 1)$, and $D(-5; -5; 3)$. Find the angle between the vectors $\vec{AC}$ and $\vec{BD}$.

50. Calculate the lengths of the diagonals of the parallelogram constructed on the vectors $5\mathbf{a} + 2\mathbf{b}$ and $\mathbf{a} - 3\mathbf{b}$ if $|\mathbf{a}| = 2\sqrt{2}$,

$|\mathbf{b}| = 3$, and $\widehat{(\mathbf{a}, \mathbf{b})} = 45^\circ$.

51. Find the angle between the vectors $\mathbf{p} + \mathbf{q}$ and $\mathbf{p} - \mathbf{q}$ if $\mathbf{p} = 8\mathbf{i} + 4\mathbf{j}$ and $\mathbf{q} = 4\mathbf{i} + \mathbf{j}$.

52. The vector $\mathbf{a}$ whose first coordinate is twice as large as the second, makes an angle of 135° with the basis vector $\mathbf{k}$. Find its coordinates if $|\mathbf{a}| = 5\sqrt{2}$.

53. The vector $\mathbf{a}$, collinear with the vector $\mathbf{b} = (12; -16; 15)$, makes an acute angle with the basis vector $\mathbf{k}$. Knowing that $|\mathbf{a}| = 100$, find the coordinates of the vector $\mathbf{a}$.

54. Find the coordinates of the vector $\mathbf{a}$ which is orthogonal to the basis vector $\mathbf{j}$ and to the vector $\mathbf{b} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ if $|\mathbf{a}| = \sqrt{13}$.

55. Find the coordinates of the unit vector $\mathbf{a}$ orthogonal to the vectors $\mathbf{i} + \mathbf{j}$ and $\mathbf{j} + \mathbf{k}$.

56. Find the coordinates of the vector $\mathbf{a}$, orthogonal to the vectors $\mathbf{i} - \mathbf{j}$ and $\mathbf{j} - \mathbf{k}$ if $|\mathbf{a}| = \sqrt{3}$.

57. Find the coordinates of the vector $\mathbf{a}$, collinear with the vector $\mathbf{b} = (6; 8; -7.5)$ and making an obtuse angle with the basis vector $\mathbf{j}$, if $|\mathbf{a}| = 50$.

58. Find m and n if the vector $\mathbf{a} = (3; m; -1)$ is orthogonal to the vector $\mathbf{b} = (2; 1; n)$ and $|\mathbf{a}| = |\mathbf{b}|$.

59. Find the coordinates of the vector $\mathbf{a}$ orthogonal to the vectors $\mathbf{i}$ and $\mathbf{b} = 3\mathbf{i} + \mathbf{j} - \mathbf{k}$ if $|\mathbf{a}| = \sqrt{2}$.

60. Find the coordinates of the unit vector $\mathbf{a}$ orthogonal to the vector $\mathbf{b} = (-1; 2; 2)$ and making equal angles with the vectors $\mathbf{i}$ and $\mathbf{j}$.

61. Find the equation of the plane passing through the point $A(1; 3; 1)$ at right angles to the vector $\mathbf{n} = (-1; 2; -5)$.

62. Set up an equation of the plane passing through the point $A(1; -3; 2)$ parallel to the plane $2x + y + z - 1 = 0$.

63. Which of the following pairs of planes are (a) parallel or (b) perpendicular?

(1) $2x + 3y - z + 6 = 0$, $x - y - z - 7 = 0$;

(2) $2x - 3y + 5z - 1 = 0$, $2x - 3y + 5z + 3 = 0$.

64. At what m is the plane $2x + my - 3z - 1 = 0$ perpendicular to the plane $5x + y + 3z + 1 = 0$?

65. Calculate the distance from the origin to the plane $x - 2y + 2z - 6 = 0$.

66. Find the angle between the plane Oxy and the plane $\sqrt{2}x + y - 3z + 17 = 0$.

67. Find the angle between the planes $x - y\sqrt{2} + z - 1 = 0$ and $x + y\sqrt{2} - z + 3 = 0$.

Chapter 8

Complex Numbers

In this chapter we shall introduce and discuss complex numbers which are a generalization of real numbers. The last section is dedicated to algebraic equations with complex coefficients.

Complex numbers and functions of a complex variable play an especially significant role in many divisions of mathematics and physics.

8.1. Definition of a Complex Number

Let us consider a set whose elements are all ordered pairs of real numbers. A pair of numbers is said to be ordered if it is indicated which of the numbers in the pair is the first and which is the second. An element of the set under consideration will be designated as follows: $(a; b)$, the first number of the pair occupying the first place and the second, the second place. The elements $(a; b)$ and $(b; a)$ are taken to be different if $a \neq b$. The elements of our set, for instance, are the pairs $(0; 0)$, $(1; 0)$, $(0; 1)$, $(0; \sqrt{2})$, $(\pi; 1)$, $(-1; 10^6)$. The elements $(a_1; b_1)$ and $(a_2; b_2)$ are considered to be *equal* if and only if

$$a_1 = a_2, \quad b_1 = b_2.$$

Let us now introduce the algebraic operations of addition of two pairs and multiplication of two pairs on the set of all ordered pairs of real numbers. We shall do it as follows.

The elements

$$(a_1 + a_2; b_1 + b_2)$$

will be termed the *sum* of the elements $(a_1; b_1)$ and $(a_2; b_2)$

The pair

$$(a_1a_2 - b_1b_2; \quad a_1b_2 + b_1a_2)$$

will be termed the *product* of the pairs $(a_1; b_1)$ and $(a_2; b_2)$.

We shall use the sign “+” to denote the operation of addition of two elements of our set, and shall write the elements side-by-side, omitting the dot designating multiplication, when putting down a product of the factors.

Thus, the operations we have introduced are specified by the equations

$$(a_1; b_1) + (a_2; b_2) = (a_1 + a_2; b_1 + b_2), \quad (1)$$

$$(a_1; b_1) (a_2; b_2) = (a_1a_2 - b_1b_2; a_1b_2 + b_1a_2). \quad (2)$$

Thus, for instance, the sum and the product of the pairs $(3; -7)$ and $(2; 9)$ can be calculated as follows:

$$(3; -7) + (2; 9) = (3 + 2; -7 + 9) = (5; 2),$$

$$(3; -7) (2; 9) = (3 \cdot 2 - (-7) \cdot 9; 3 \cdot 9 + (-7) \cdot 2) = (69; 13).$$

Now we can give the definition of a complex number.

Complex numbers are ordered pairs of real numbers for which the operations of addition and multiplication are defined by formulas (1) and (2).

Complex numbers are often denoted by a single letter, the letters z and w being customarily used for that purpose, sometimes with indices, as, for example, z_1, z_2, w_0 . The equation $z = (a; b)$ implies, for instance, that the complex number $(a; b)$ is denoted by the single letter z .

8.2. Properties of the Operations of Addition and Multiplication

The operations of addition and multiplication introduced in 8.1 possess the following properties:

1. $z_1 + z_2 = z_2 + z_1$ (commutative law for addition).
2. $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$ (associative law for addition).
3. For any complex numbers z_1 and z_2 there is a complex number z such that $z_1 + z = z_2$. This number is known as the *difference* of the numbers z_2 and z_1 and is designated $z_2 - z_1$.
4. $z_1z_2 = z_2z_1$ (commutative law for multiplication).
5. $(z_1z_2)z_3 = z_1(z_2z_3)$ (associative law for multiplication).
6. For any complex numbers $z_1 \neq (0; 0)$ and z_2 there is

a number z such that $z_1 z = z_2$. This number is known as the *quotient* of the complex numbers z_2 and z_1 and is designated z_2/z_1 . The division by the complex number $(0; 0)$, called a *zero*, is impossible.

7. $z_1 (z_2 + z_3) = z_1 z_2 + z_1 z_3$ (distributive law).

All the enumerated properties of the operations of addition and multiplication follow from formulas (1) and (2) of 8.1, defining these operations. We shall prove properties 3 and 6 and leave the verification of the rest of the properties for the reader.

□ *Proof of property 3.* Suppose $z_1 = (a_1; b_1)$, $z_2 = (a_2; b_2)$ and $z = (x; y)$. Then the equality $z_1 + z = z_2$ assumes the form

$$(a_1; b_1) + (x; y) = (a_2; b_2).$$

Adding up the complex numbers on the left-hand side of the equality, we obtain

$$(a_1 + x; b_1 + y) = (a_2; b_2).$$

It follows from the definition of the equality of ordered pairs of real numbers that x and y satisfy the system of two equations

$$a_1 + x = a_2, \quad b_1 + y = b_2.$$

This system has a unique solution $x = a_2 - a_1$, $y = b_2 - b_1$. Consequently, the difference $z_2 - z_1$ always exists, with $z = z_2 - z_1 = (a_2; b_2) - (a_1; b_1) = (a_2 - a_1; b_2 - b_1)$. (1)

This formula yields the rule of subtraction of complex numbers. ■

□ *Proof of property 6.* Suppose $z_1 = (a_1; b_1)$, $z_2 = (a_2; b_2)$ and $z = (x; y)$. Then the equality $z_1 z = z_2$ assumes the form

$$(a_1; b_1) (x; y) = (a_2; b_2).$$

Multiplying the complex numbers on the left-hand side of the equation, we obtain

$$(a_1 x - b_1 y; a_2 y + b_1 x) = (a_2; b_2).$$

It follows from the definition of the equality of ordered pairs that x and y satisfy the system of two linear equations

$$\begin{cases} a_1 x - b_1 y = a_2, \\ b_1 x + a_1 y = b_2. \end{cases}$$

Multiplying termwise the first equation by a_1 and the second equation by b_1 and then adding the resulting equations term-by-term, we obtain

$$(a_1^2 + b_1^2) x = a_1 a_2 + b_1 b_2.$$

By analogy we get

$$(a_1^2 + b_1^2) y = a_1 b_2 - a_2 b_1.$$

Thus we see that our system is equivalent to the following system:

$$\begin{cases} (a_1^2 + b_1^2) x = a_1 a_2 + b_1 b_2, \\ (a_1^2 + b_1^2) y = a_1 b_2 - a_2 b_1. \end{cases}$$

Since $z_1 = (a_1; b_1) \neq (0; 0)$ and, hence, $a_1^2 + b_1^2 \neq 0$, the system possesses a unique solution

$$x = \frac{a_1 a_2 + b_1 b_2}{a_1^2 + b_1^2}, \quad y = \frac{a_1 b_2 - a_2 b_1}{a_1^2 + b_1^2}.$$

Thus, under the condition that the divisor is nonzero, the quotient of two complex numbers always exists, with

$$z = \frac{z_2}{z_1} = \frac{(a_2; b_2)}{(a_1; b_1)} = \left(\frac{a_1 a_2 + b_1 b_2}{a_1^2 + b_1^2}; \frac{a_1 b_2 - a_2 b_1}{a_1^2 + b_1^2} \right). \quad (2)$$

Formula (2) yields the rule of division of complex numbers. ■

The operations of addition and multiplication introduced above permit us to consider complex numbers as a generalization of real numbers and treat real numbers as a special case of complex numbers. Indeed, let us take not all complex numbers, but only those of the form $(a; 0)$. We can immediately see from the formulas of addition, multiplication, subtraction, and division that the addition, multiplication, subtraction and division ($a \neq 0$) of such numbers always result in numbers of the same form. Besides, it is evident that the rules of operating with complex numbers of the form $(a; 0)$ completely coincide with the respective rules of operating with real numbers. Because of this, the complex number $(a; 0)$ is identified with the *real* number a and it is taken that $(a; 0) = a$. For example, $(0; 0) = 0$, $(2; 0) = 2$, $(-7; 0) = -7$.

In this case, the set of real numbers becomes a subset of the set of complex numbers.

Complex numbers of the form $(0; b)$ are called *imaginary*. The number $(0; 1)$ is called an *imaginary unit number* and the letter i is used to designate it, i.e. $(0; 1) = i$. Using the multiplication formula, it is easy to verify that the arbitrary imaginary number $(0; b)$ can be represented as the product of the numbers $(b; 0)$ and $(0; 1)$, i.e. $(0; b) = (b; 0)(0; 1)$. But since $(b; 0) = b$ and $(0; 1) = i$, the imaginary number $(0; b)$ is written as bi . For instance, $(0; 2) = 2i$, $(0; -1) = -i$.

8.3. The Algebraic Form of Complex Numbers.

Rules of Operating with Complex Numbers Written in the Algebraic Form

Every complex number $z = (a; b)$ can be represented as follows:

$$z = (a; b) = (a; 0) + (0; b) = (a; 0) + (b; 0)(0; 1).$$

Taking into account that $(a; 0) = a$, $(b; 0) = b$, $(0; 1) = i$, we obtain

$$z = (a; b) = a + bi.$$

The representation of the complex number $z = (a; b)$ in the form $a + bi$ is called the *algebraic form* of a complex number.

The real number a is called the *real part* of the complex number $a + bi$, and the real number b is called the *imaginary part* of the complex number $a + bi$.

The discussion carried out in the preceding sections implies that the following operations can be performed on complex numbers written in the algebraic form.

Two complex numbers $z_1 = a_1 + b_1i$ and $z_2 = a_2 + b_2i$ are *equal* if and only if $a_1 = a_2$ and $b_1 = b_2$, that is, if the real parts and the imaginary parts of these numbers are equal to each other. Pay attention to the fact that *one* equality $z_1 = z_2$ of complex numbers is equivalent to *two* equalities $a_1 = a_2$ and $b_1 = b_2$ of real numbers. It should also be noted that the notions of "larger than" and "less than" are not defined for complex numbers. The notations $i > 0$, $1 + i < 2$ and the like are *meaningless*.

In the new designations, the summation formula is written as

$$(a_1 + b_1i) + (a_2 + b_2i) = a_1 + a_2 + (b_1 + b_2)i. \quad (1)$$

It yields the *rule of summation* of complex numbers written in algebraic form.

Multiplication of complex numbers given in algebraic form is performed as follows:

$$(a_1 + b_1 i)(a_2 + b_2 i) = a_1 a_2 - b_1 b_2 + (a_1 b_2 + a_2 b_1) i. \quad (2)$$

This formula is nothing other than formula (2) of 8.1 written in the new designations.

Putting $a_1 = a_2 = 0$ and $b_1 = b_2 = 1$ in (2), we get the important relation

$$ii = -1,$$

or, using the concise notation i^2 for the product ii , i.e. $ii = i^2$, we have

$$i^2 = -1. \quad (3)$$

We want to point out that formula (2) need not necessarily be memorized since it results from formal multiplication of the binomials $a_1 + b_1 i$ and $a_2 + b_2 i$ by the ordinary rule of multiplication of binomials and the subsequent replacement of i^2 by -1 in accordance with formula (3).

Example 1. Find the sum and the product of the complex numbers $z_1 = -3 + 2i$ and $z_2 = 13 - i$.

$$\begin{aligned} \Delta \quad z_1 + z_2 &= (-3 + 2i) + (13 - i) = 10 + i, \\ z_1 z_2 &= (-3 + 2i)(13 - i) = -39 + 3i + 26i + 2 \\ &= -37 + 29i. \quad \blacktriangle \end{aligned}$$

Example 2. Find the sum and the product of the complex numbers $z_1 = a + bi$ and $z_2 = a - bi$.

$$\begin{aligned} \Delta \quad z_1 + z_2 &= (a + bi) + (a - bi) = 2a, \\ z_1 z_2 &= (a + bi)(a - bi) = a^2 - abi + abi - (bi)^2 \\ &= a^2 + b^2. \quad \blacktriangle \end{aligned}$$

The complex numbers $a + bi$ and $a - bi$, that is, the numbers which differ only in the sign of the imaginary part are called *conjugate*. The number conjugate to the number z is designated as $\bar{z}$. Example 2 shows that the sum $z + \bar{z}$ of conjugate numbers is always a real number, and the product $z\bar{z}$ is a real number and is, moreover, nonnegative.

Before passing to the converse operations, let us note one more special case of formula (2). Setting $a_1 = \alpha$, $b_1 = 0$,

$a_2 = a$, and $b_2 = b$, we get the rule of multiplication of a real number by a complex number:

$$\alpha (a + bi) = \alpha a + \alpha bi.$$

The converse operations, those of subtraction and division of complex numbers, written in algebraic form, are performed by the following formulas:

$$(a_2 + b_2 i) - (a_1 + b_1 i) = a_2 - a_1 + (b_2 - b_1) i, \quad (4)$$

$$\frac{a_2 + b_2 i}{a_1 + b_1 i} = \frac{a_1 a_2 + b_1 b_2}{a_1^2 + b_1^2} + \frac{a_1 b_2 - a_2 b_1}{a_1^2 + b_1^2} i. \quad (5)$$

Formulas (4) and (5) are nothing other than formulas (1) and (2) of 8.2 in different notation. Setting $a_2 = a$, $b_2 = b$, $a_1 = \alpha$, and $b_1 = 0$ in formula (5), we get the rule of division of the complex number $a + bi$ by the real number α :

$$\frac{a + bi}{\alpha} = \frac{a}{\alpha} + \frac{b}{\alpha} i.$$

Example 3. Given the complex numbers $z_1 = -3 + 2i$ and $z_2 = 13 - i$. Find the difference $z_2 - z_1$ and the quotient z_2/z_1 .

△ Formula (4) yields

$$z_2 - z_1 = (13 - i) - (-3 + 2i) = 16 - 3i.$$

Using formula (5), we get the quotient

$$\begin{aligned} \frac{z_2}{z_1} &= \frac{13 - i}{-3 + 2i} = \frac{(-3)13 + 2(-1)}{(-3)^2 + 2^2} + \frac{(-3)(-1) - 13 \cdot 2}{(-3)^2 + 2^2} \\ &= -\frac{41}{13} - \frac{23}{13} i. \quad \blacktriangle \end{aligned}$$

Formula (5) is rather unwieldy and cannot be easily memorized. Therefore, it is better to avoid it. It is much easier to multiply the numerator and the denominator of the fraction z_2/z_1 by $\bar{z}_1$, that is, by the number conjugate to the denominator. Then, seeking the quotient z_2/z_1 reduces to multiplying the number z_2 by the number $\bar{z}_1$ and dividing the resulting product by the positive number $z_1 \bar{z}_1$ (see Example 2).

To put it otherwise, to divide a complex number, it is a good practice to use the formula

$$\frac{z_2}{z_1} = \frac{z_2 \bar{z}_1}{z_1 \bar{z}_1}.$$

For the numbers considered in Example 3, we shall have

$$\begin{aligned}\frac{13-i}{-3+2i} &= \frac{(13-i)(-3-2i)}{(-3+2i)(-3-2i)} \\ &= \frac{-39-26i+3i-2}{9+4} = -\frac{41}{13} - \frac{23}{13}i.\end{aligned}$$

Example 4. Write the complex number

$$z = \frac{3+i}{(1+i)(1-2i)}$$

in algebraic form.

$$\begin{aligned}\triangle z &= \frac{3+i}{(1+i)(1-2i)} = \frac{3+i}{1-2i+i+2} = \frac{3+i}{3-i} \\ &= \frac{(3+i)(3+i)}{(3-i)(3+i)} = \frac{9+6i-1}{9+1} = \frac{4}{5} + \frac{3}{5}i. \quad \blacktriangle\end{aligned}$$

8.4. Geometric Interpretation of Complex Numbers. The Modulus and the Arguments of a Complex Number

It is well known that a one-to-one correspondence can be established between the set of all ordered pairs of real numbers and the set of all points of a plane. For that purpose it is sufficient to choose a coordinate system in the plane and associate every ordered pair of numbers $(a; b)$ with a point $M(a; b)$, that is, a point with abscissa $x = a$ and ordinate $y = b$ (Fig. 115). We have defined complex numbers as ordered pairs of real numbers for which the operations of addition and multiplication are specified. Consequently, every complex number $(a; b) = a + bi$ can be put in correspondence with a point $M(a; b)$, and, conversely, every point $M(a; b)$ of the plane can be put in correspondence with a complex number $(a; b) = a + bi$. The correspondence thus established is, evidently, one-to-one. It makes it possible to consider complex numbers as points of a coordinate plane. That plane is called a *complex plane*. The abscissa axis is called the *real axis* (or the *axis of reals*), containing the points that correspond to the numbers $(a; 0) = a$, and the axis of ordinates is called the *imaginary axis* (*axis of imaginaries*), to which belong the points corresponding to the imaginary numbers $(0; b) = bi$.

It is often convenient to interpret the complex number $(a; b) = a + bi$ as the vector $\vec{OM}$ (Fig. 115). It is evident that every vector of the plane beginning at the point $O(0; 0)$ and terminating at the point $M(a; b)$ can be associated with the complex number $(a; b) = a + bi$ and vice versa. The point $O(0; 0)$ is associated with the zero vector.

The correspondence we have established between the set of complex numbers on one hand and the set of points or

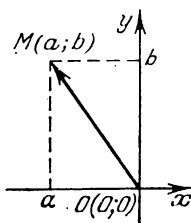


Fig. 115.

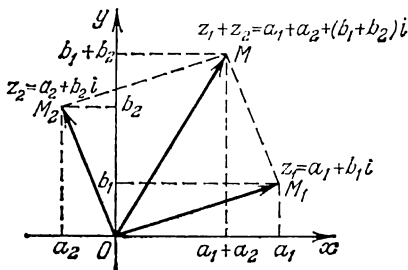


Fig. 116.

vectors of the plane, on the other hand, enables us to consider complex numbers as points or vectors and speak of, say, the vector $a + bi$ or the point $a + bi$.

Representation of complex numbers as vectors admits of a simple geometrical interpretation of operations on complex numbers. Let us now discuss the summation of complex numbers.

When the numbers $z_1 = a_1 + b_1i$ and $z_2 = a_2 + b_2i$ are added, their real and imaginary parts are added up (Fig. 116).

When adding up the vectors $\vec{OM}_1$ and $\vec{OM}_2$ corresponding to them, their coordinates are added. Therefore, with the correspondence we have established between complex numbers and vectors, the sum $z_1 + z_2$ of the numbers z_1 and z_2 will be associated with the vector $\vec{OM}$ equal to the sum of the vectors $\vec{OM}_1$ and $\vec{OM}_2$. Thus, a sum of complex numbers can be interpreted in terms of geometry as a vector equal to the sum of the vectors corresponding to the complex numbers being added up.

Definition. The *modulus* of the complex number $(a; b) = a + ib$ is the length of the vector corresponding to that number.

The modulus of the number z is designated as $|z|$. By the Pythagorean theorem (see Fig. 115), the following significant formula can be easily obtained for the modulus of the complex number $z = a + ib$:

$$|z| = \sqrt{a^2 + b^2}, \quad (1)$$

expressing the modulus of the complex number in terms of its real and imaginary parts.

Conjugate complex numbers have equal moduli. In fact,

$$|z| = |a + bi| = \sqrt{a^2 + b^2}, \quad |\bar{z}| = |a - bi| = \sqrt{a^2 + b^2}.$$

Note that the modulus of the real number $z = a + i0$ coincides with its absolute value:

$$|z| = |a + i0| = \sqrt{a^2} = |a|.$$

Example 1. Find the moduli of the complex numbers

$$z_1 = 2 - i, \quad z_2 = 2\sqrt{6} + 5i, \quad z_3 = i.$$

△ We get by formula (1)

$$|z_1| = \sqrt{4 + 1} = \sqrt{5}, \quad |z_2| = \sqrt{(2\sqrt{6})^2 + 5^2} = 7.$$

Formula (1) need not be used to calculate the modulus of z_3 . The length of the vector $z_3 = i$ is, evidently, unity and therefore, $|z_3| = 1$. ▲

The complex numbers z possessing the same modulus $|z| = r$ evidently correspond to the points of the complex plane located on the circle of radius r with centre at the origin (Fig. 117).

If $|z| \neq 0$, then there are infinitely many complex numbers with the given modulus. A modulus equal to zero has only one complex number, namely, $z = 0$.

It is evident, from the point of view of geometry, that the complex number $z \neq 0$ is defined if, in addition to

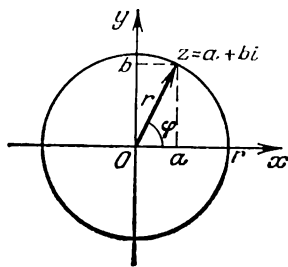


Fig. 117.

the modulus of the number z , we also indicate the direction of the vector z by specifying, for instance, the magnitude of the angle φ (Fig. 117).

Definition. The *argument* of the complex number $z \neq 0$ is the magnitude of the angle between the positive direction of the real axis and the vector z , the angle being considered positive if we measure counterclockwise and negative if we measure clockwise.

Specification of the modulus and the argument results in a unique definition of a complex number. For the number

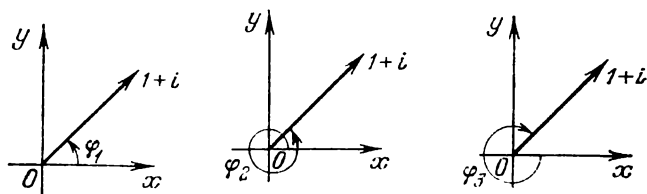


Fig. 118.

$z = 0$ the argument is not defined, and in this and only this case the number is specified exclusively by its modulus.

As distinct from the modulus, the argument of a complex number is not defined uniquely. For example, the arguments of the number $z = 1 + i$ are the following angles: $\varphi_1 = \frac{\pi}{4}$, $\varphi_2 = \frac{\pi}{4} + 2\pi = \frac{9}{4}\pi$, and $\varphi_3 = \frac{\pi}{4} - 2\pi = -\frac{7\pi}{4}$ (Fig. 118); and, in general, each of the angles $\varphi_k = \frac{\pi}{4} + 2\pi k$, where k is an arbitrary integer.

Any two arguments of a complex number differ by a number which is a multiple of 2π . For instance, the difference $\varphi_2 - \varphi_3$ of the arguments $\varphi_2 = 9\pi/4$ and $\varphi_3 = -7\pi/4$ of the number $z = 1 + i$ is equal to 4π . To designate the set of all arguments of the number $z = a + bi$, use can be made of the notation $\arg z$ or $\arg(a + bi)$. If we speak of one of the arguments, we usually designate it as φ .

Example 2. Find the arguments of the complex numbers $z_1 = -i$, $z_2 = 1$, and $z_3 = -1 + i$.

△ Having constructed the vectors z_1 , z_2 , and z_3 (Fig. 119), we find one of the arguments for every number: $\varphi_1 = -\pi/2$,

$\varphi_2 = 0$, and $\varphi_3 = 3\pi/4$. Consequently,

$$\arg z_1 = -\frac{\pi}{2} + 2\pi k, \arg z_2 = 2\pi k, \arg z_3 = \frac{3\pi}{4} + 2\pi k,$$

where k is an arbitrary integer. ▲

The real and the imaginary part of the complex number $z = a + bi$ are expressed in terms of its modulus $|z| = r$ and argument φ as follows

$$a = r \cos \varphi, \quad b = r \sin \varphi. \quad (2)$$

This relationship can be easily established by studying Fig. 117.

Thus, the arguments φ of a complex number can be easily found from the system of equations

$$\cos \varphi = \frac{a}{\sqrt{a^2 + b^2}}, \quad \sin \varphi = \frac{b}{\sqrt{a^2 + b^2}}. \quad (3)$$

Example 3. Find the arguments of the complex number $z = -1 - \sqrt{3}i$.

△ In the given case, $a = -1$ and $b = -\sqrt{3}$. System (3) has the form

$$\cos \varphi = -1/2, \quad \sin \varphi = -\sqrt{3}/2.$$

Solving it, we find $\varphi_k = \frac{4\pi}{3} + 2\pi k, k \in \mathbb{Z}$. Consequently,

$$\arg z = \frac{4\pi}{3} + 2\pi k, k \in \mathbb{Z}. \quad \blacktriangle$$

There is another technique of seeking the arguments of a complex number. It can be seen from formulas (3) that each of the arguments satisfies the equation

$$\tan \varphi = b/a.$$

This equation is not equivalent to system (3), it has more solutions, but the selection of the required solutions (the arguments of the complex number) does not present any difficulties since it is always clear from the algebraic notation of the complex number in what quadrant of the complex plane it is located.

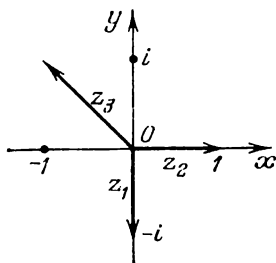


Fig. 119.

Example 4. Find the arguments of the complex number $z = -\sqrt{3} + i$.

Δ Each of the arguments φ of this number satisfies the equation

$$\tan \varphi = -1/\sqrt{3}.$$

It follows from the equation that $\varphi_k = -\frac{\pi}{6} + \pi k$, $k \in \mathbb{Z}$.

Since the number $z = -\sqrt{3} + i$ lies in the second quad-

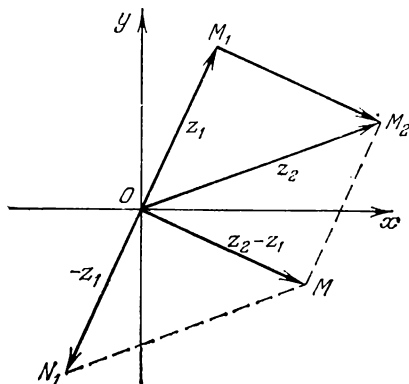


Fig. 120.

rant of the complex plane, its arguments are the numbers φ_k for odd values of k . Consequently,

$$\arg(-\sqrt{3} + i) = -\frac{\pi}{6} + \pi(2n + 1) = \frac{5\pi}{6} + 2\pi n, \quad n \in \mathbb{Z}. \quad \blacktriangle$$

To conclude this section, we shall discuss the geometrical meaning of the modulus of the difference between two complex numbers. Let us construct the vector $\mathbf{z}_2 - \mathbf{z}_1$ as the sum of the vectors $\mathbf{z}_2$ and $(-\mathbf{z}_1)$ (Fig. 120). By the definition of the modulus, the number $|\mathbf{z}_2 - \mathbf{z}_1|$ is the length of the vector $\mathbf{z}_2 - \mathbf{z}_1$, that is, the length of the vector $\overrightarrow{OM}$. The congruence of the triangles OMN_1 and M_1M_2O yields $|\overrightarrow{OM}| = |\overrightarrow{M_1M_2}|$. Thus, the length of the vector $\mathbf{z}_2 - \mathbf{z}_1$ is equal to the distance between the points z_1 and z_2 . We can now say that the *modulus of the difference of two complex numbers*

is the distance between the points of the complex plane corresponding to those numbers.

This important geometrical interpretation of the modulus of the difference between two complex numbers makes it possible to use simple geometrical facts in solving certain problems.

Example 5. What sets of points of the complex plane are defined by the conditions

- (a) $|z - i| = 1$, (b) $|2 + z| < |2 - z|$, and
(c) $2 \leq |z - 1 + 2i| < 3$?

Δ (a) The condition $|z - i| = 1$ is satisfied by those and only those points of the complex plane which are

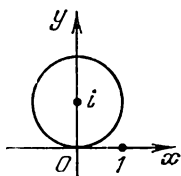


Fig. 121.

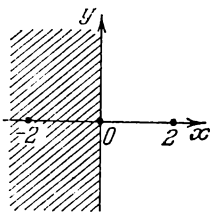


Fig. 122.

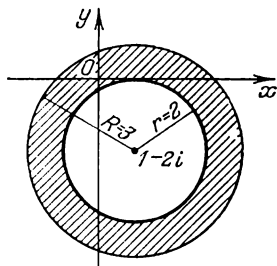


Fig. 123.

removed from the point i by the distance equal to unity. Such points lie on the circle of unit radius with centre at the point i (Fig. 121).

(b) Using the geometrical interpretation of the modulus of the difference between two complex numbers, we can give a different formulation of the problem: what is the set of points of the complex plane which are located closer to the point $z = -2$ than to the point $z = 2$? It is clear that this property is possessed by all the points of the plane which lie to the left of the imaginary axis and only by those points. In Fig. 122 the required set of points is hatched.

(c) The complex numbers z satisfying the inequalities

$$2 \leq |z - (1 - 2i)| < 3,$$

are removed from the point $1 - 2i$ by the distance greater than or equal to 2, but less than 3. Such points lie in the interior and on the inner boundary of the ring formed by two concentric circles with centre at the point $1 - 2i$ and the radii $r = 2$ and $R = 3$. In Fig. 123 the required set is shown hatched. ▲

8.5. Trigonometric Form of a Complex Number. Multiplication and Division of Complex Numbers Written in Trigonometric Form

Let us return to formulas (2) of the preceding section:

$$a = r \cos \varphi, \quad b = r \sin \varphi.$$

These formulas relate the real and the imaginary part of the complex number $z = a + bi$ with its modulus r and argument φ .

Every complex number $z = a + bi$, different from zero, can, consequently, be written in the form

$$z = r (\cos \varphi + i \sin \varphi). \quad (1)$$

The notation of the complex number

$$r (\cos \varphi + i \sin \varphi),$$

where r is the modulus of the number and φ is one (any) of its arguments, is called the *trigonometric notation*.

To pass from the algebraic notation to trigonometric, it is sufficient to find the modulus of the complex number and one of its arguments.

Example 1. Write the numbers $z_1 = -1 - i$, $z_2 = -2$, and $z_3 = i$ in trigonometric form.

△ Since $|z_1| = \sqrt{2}$ and $\varphi_1 = -3\pi/4$, we have

$$z_1 = \sqrt{2} (\cos (-3\pi/4) + i \sin (-3\pi/4)).$$

The modulus of z_2 is equal to 2 and one of the arguments of z_2 is the angle $\varphi_2 = \pi$, hence

$$z_2 = 2 (\cos \pi + i \sin \pi).$$

Taking into account that $|z_3| = 1$, and $\varphi_3 = \pi/2$ is one of the arguments of z_3 , we obtain the trigonometric form

for z_3 :

$$z_3 = \cos (\pi/2) + i \sin (\pi/2). \quad \blacktriangle$$

Example 2. Write the numbers

$$z_1 = 2 \cos (7\pi/4) - 2i \sin (\pi/4) \quad \text{and}$$

$$z_2 = -\cos (\pi/17) + i \sin (\pi/17)$$

in trigonometric form.

$\triangle$ To write the numbers z_1 and z_2 in trigonometric form, we need not find their moduli and arguments (although it is easy to do). We make use of the fact that

$$\cos (7\pi/4) = \cos (-\pi/4) \quad \text{and} \quad -\sin (\pi/4) = \sin (-\pi/4),$$

and immediately obtain the trigonometric form for the first number:

$$z_1 = 2 (\cos (-\pi/4) + i \sin (-\pi/4)).$$

By analogy, taking into account that

$$-\cos \frac{\pi}{17} = \cos \left(\pi - \frac{\pi}{17} \right) = \cos \frac{16\pi}{17},$$

$$\sin \frac{\pi}{17} = \sin \left(\pi - \frac{\pi}{17} \right) = \sin \frac{16\pi}{17},$$

we obtain

$$z_2 = \cos (16\pi/17) + i \sin (16\pi/17). \quad \blacktriangle$$

The trigonometric form of complex numbers proves to be very convenient in multiplication and division of numbers. Before considering these operations, we make note of the fact that two complex numbers

$$z_1 = r_1 (\cos \varphi_1 + i \sin \varphi_1) \quad \text{and} \quad z_2 = r_2 (\cos \varphi_2 + i \sin \varphi_2)$$

are equal if and only if

$$r_1 = r_2, \quad \varphi_1 = \varphi_2 + 2\pi k, \quad k \in \mathbb{Z},$$

that is, if the moduli of the numbers are equal and the arguments differ by $2\pi k$, where k is some integer.

Assume now that

$$z_1 = r_1 (\cos \varphi_1 + i \sin \varphi_1) \quad \text{and} \quad z_2 = r_2 (\cos \varphi_2 + i \sin \varphi_2)$$

are two numbers written in trigonometric form.

Let us represent in trigonometric form their product

$$z_1 z_2 = r_1 r_2 (\cos \varphi_1 \cos \varphi_2 - \sin \varphi_1 \sin \varphi_2 + i \sin \varphi_1 \cos \varphi_2 + i \cos \varphi_1 \sin \varphi_2),$$

$$z_1 z_2 = r_1 r_2 (\cos (\varphi_1 + \varphi_2) + i \sin (\varphi_1 + \varphi_2)). \quad (2)$$

Consequently,

$$|z_1 z_2| = r_1 r_2, \arg(z_1 z_2) = \varphi_1 + \varphi_2 + 2\pi k, k \in \mathbb{Z}.$$

Thus, the following assertion is true: *the modulus of the product of two complex numbers is equal to the product of the moduli of those numbers, the sum of the arguments of the factors being the argument of the product.*

Example 3. Find the product of the numbers

$$z_1 = \sqrt{2} \left(\cos \frac{11\pi}{4} + i \sin \frac{11\pi}{4} \right) \quad \text{and}$$

$$z_2 = \sqrt{8} \left(\cos \frac{3\pi}{8} + i \sin \frac{3\pi}{8} \right).$$

△ Since $|z_1| = \sqrt{2}$, $|z_2| = \sqrt{8}$, we have $|z_1 z_2| = \sqrt{2} \sqrt{8} = 4$. The argument of the product $z_1 z_2$ of the given numbers is the sum $\varphi_1 + \varphi_2 = \frac{11\pi}{4} + \frac{3\pi}{8} = \frac{25}{8}\pi$. Consequently,

$$z_1 z_2 = 4 (\cos (25\pi/8) + i \sin (25\pi/8)),$$

or

$$z_1 z_2 = 4 (\cos (9\pi/8) + i \sin (9\pi/8)). \quad \blacktriangle$$

Let us now take the division of numbers. We write the quotient of two complex numbers

$$z_1 = r_1 (\cos \varphi_1 + i \sin \varphi_1) \quad \text{and} \quad z_2 = r_2 (\cos \varphi_2 + i \sin \varphi_1)$$

in trigonometric form. Multiplying the numerator and the denominator of the quotient z_1/z_2 by $\cos \varphi_2 - i \sin \varphi_2$, we get

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{r_1 (\cos \varphi_1 + i \sin \varphi_1) (\cos \varphi_2 - i \sin \varphi_2)}{r_2 (\cos \varphi_2 + i \sin \varphi_2) (\cos \varphi_2 - i \sin \varphi_2)} \\ &= \frac{r_1 (\cos \varphi_1 \cos \varphi_2 + \sin \varphi_1 \sin \varphi_2 + i \sin \varphi_1 \cos \varphi_2 - i \cos \varphi_1 \sin \varphi_2)}{r_2 (\cos^2 \varphi_2 + \sin^2 \varphi_2)}, \end{aligned}$$

or

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} (\cos (\varphi_1 - \varphi_2) + i \sin (\varphi_1 - \varphi_2)). \quad (3)$$

Consequently,

$$|z_1/z_2| = r_1/r_2, \quad \arg(z_1/z_2) = \varphi_1 - \varphi_2 + 2\pi k, \quad k \in \mathbb{Z}.$$

Thus we can say that *the modulus of the quotient of two complex numbers is equal to the quotient of the moduli of those numbers, the difference between the arguments of the dividend and the divisor being the argument of the quotient.*

Example 4. Write the number $z = \frac{i-1}{\cos(\pi/3) + i \sin(\pi/3)}$ in trigonometric form.

$\triangle$ We introduce the notation $z_1 = i - 1$ and $z_2 = \cos(\pi/3) + i \sin(\pi/3)$. The number z_2 is written in trigonometric form. It is evident that $|z_2| = 1$ and $\varphi_2 = \pi/3$. Let us find the modulus and the argument of the number z_1 : $|z_1| = \sqrt{2}$ and $\varphi_1 = 3\pi/4$. By formula (3) we obtain

$$z = \frac{z_1}{z_2} = \sqrt{2} \left(\cos \left(\frac{3\pi}{4} - \frac{\pi}{3} \right) + i \sin \left(\frac{3\pi}{4} - \frac{\pi}{3} \right) \right),$$

that is,

$$z = \sqrt{2} (\cos(5\pi/12) + i \sin(5\pi/12)). \quad \blacktriangle$$

8.6. Raising to a Power and Extracting a Root

The formula for the product of two complex numbers can be generalized to the case of n factors. By mathematical induction, we can easily obtain the following result: the modulus of the product of n complex numbers is equal to the product of the moduli of all the factors, the sum of the arguments of all the factors being the argument of their product.

Hence, as a special case, we obtain the formula

$$(r(\cos \varphi + i \sin \varphi))^n = r^n (\cos n\varphi + i \sin n\varphi), \quad (1)$$

specifying the rule of raising of the complex number $r(\cos \varphi + i \sin \varphi)$ to an integral positive power.

When raising a complex number to a power with a natural exponent, its modulus is raised to the power with the same exponent and the argument is multiplied by the exponent.

Example 1. Write the number $z = (i - \sqrt{3})^{13}$ in algebraic form.

△ First we write the given number in trigonometric form and then pass to the algebraic form. We find the modulus and one of the arguments of the number $i - \sqrt{3}$:

$$r = |i - \sqrt{3}| = 2, \quad \varphi = 5\pi/6.$$

We represent the number $i - \sqrt{3}$ in trigonometric form:

$$i - \sqrt{3} = 2 (\cos (5\pi/6) + i \sin (5\pi/6)).$$

Now, applying formula (1), we get

$$\begin{aligned} (i - \sqrt{3})^{13} &= (2 (\cos (5\pi/6) + i \sin (5\pi/6)))^{13} \\ &= 2^{13} (\cos (65\pi/6) + i \sin (65\pi/6)) \\ &= 2^{13} (\cos (5\pi/6) + i \sin (5\pi/6)). \end{aligned}$$

That is the trigonometric form of the given number. Let us write this number in algebraic form:

$$\begin{aligned} (i - \sqrt{3})^{13} &= 2^{13} (\cos (5\pi/6) + i \sin (5\pi/6)) \\ &= 2^{13} \left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i \right) = -2^{12}\sqrt{3} + 2^{12}i. \quad \blacktriangle \end{aligned}$$

Let us now find the root of the given degree of a complex number. The number z is called the *root of degree n* of the number w (the notation is $\sqrt[n]{w}$), if $z^n = w$.

For instance, the numbers $z_1 = i$ and $z_2 = -i$ are the roots of degree 2 (square roots) of the number $w = -1$ since $i^2 = -1$ and $(-i)^2 = -1$. It follows from the definition that every solution of the equation $z^n = w$ is a root of degree n of the number w . To put it otherwise, to extract the root of degree n of the number w , it is sufficient to solve the equation $z^n = w$.

If $w = 0$, then for any n the equation $z^n = w$ possesses one and only one solution $z = 0$. Now if $w \neq 0$, then $z \neq 0$ as well and, consequently, both z and w can be represented in trigonometric form:

$$z = r (\cos \varphi + i \sin \varphi), \quad w = \rho (\cos \alpha + i \sin \alpha).$$

The equation $z^n = w$ assumes the form

$$r^n (\cos n\varphi + i \sin n\varphi) = \rho (\cos \alpha + i \sin \alpha).$$

Two complex numbers are equal if and only if their moduli are equal and the arguments differ by $2\pi k$, k being a certain

integer. Hence,

$$r^n = \rho \quad \text{and} \quad n\varphi = \alpha + 2\pi k$$

or

$$r = \sqrt[n]{\rho} \quad \text{and} \quad \varphi = \frac{\alpha}{n} + \frac{2\pi}{n} k, \quad k \in \mathbb{Z}.$$

Thus, all the solutions of the equation $z^n = w$ can be written as follows:

$$z_k = \sqrt[n]{\rho} \left(\cos \left(\frac{\alpha}{n} + \frac{2\pi}{n} k \right) + i \sin \left(\frac{\alpha}{n} + \frac{2\pi}{n} k \right) \right), \quad k \in \mathbb{Z}.$$

It is easy to see that all the numbers z_k , obtained for $k = 0, 1, 2, \dots, n-1$, are different. If we take the values $k \geq n$, then we cannot obtain any other complex numbers distinct from $z_0, z_1, z_2, \dots, z_{n-1}$.

For instance, for $k = n$ we get

$$\begin{aligned} z_n &= \sqrt[n]{\rho} \left(\cos \left(\frac{\alpha}{n} + 2\pi \right) + i \sin \left(\frac{\alpha}{n} + 2\pi \right) \right) \\ &= \sqrt[n]{\rho} (\cos(\alpha/n) + i \sin(\alpha/n)) = z_0. \end{aligned}$$

Thus it follows that if $w \neq 0$, then there are exactly n roots of degree n of the number w ; they are all obtained from the formula

$$\begin{aligned} z_k &= \sqrt[n]{\rho} \left(\cos \left(\frac{\alpha}{n} + \frac{2\pi}{n} k \right) + i \sin \left(\frac{\alpha}{n} + \frac{2\pi}{n} k \right) \right), \quad (2) \\ k &= 0, 1, 2, \dots, n-1. \end{aligned}$$

It can be seen from formula (2) that all the roots of degree n of the number w have one and the same modulus but distinct arguments differing from each other by $\frac{2\pi}{n} k$, where k is some integer.

Hence it follows that complex numbers which are the roots of degree n of the complex number w correspond to the points of the complex plane lying at the vertices of a regular n -gon inscribed in a circle of radius $\sqrt[n]{\rho}$ with centre at the point $z = 0$.

One more remark is due concerning the designation $\sqrt[n]{w}$. We usually understand $\sqrt[n]{w}$ as the set of all roots of degree n of w . For instance, $\sqrt{-1}$ is understood as a set consisting

of two numbers, i and $-i$. Sometimes $\sqrt[n]{w}$ is understood as a root of degree n of w . In such cases, it must be indicated what value of the root is meant.

Example 2. Find all the values of $\sqrt[6]{-64}$.

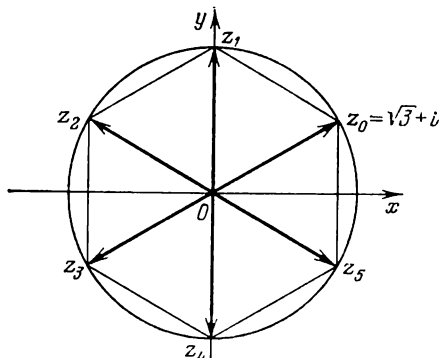


Fig. 124.

△ Let us write the number $w = -64$ in trigonometric form:

$$-64 = 64 (\cos \pi + i \sin \pi).$$

Applying formula (2), we get

$$z_k = \sqrt[6]{64} \left(\cos \left(\frac{\pi}{6} + \frac{2\pi}{6} k \right) + i \sin \left(\frac{\pi}{6} + \frac{2\pi}{6} k \right) \right),$$

$$k = 0, 1, 2, 3, 4, 5.$$

Consequently,

$$z_0 = 2 (\cos (\pi/6) + i \sin (\pi/6)) = \sqrt{3} + i,$$

$$z_1 = 2 (\cos (\pi/2) + i \sin (\pi/2)) = 2i,$$

$$z_2 = 2 (\cos (5\pi/6) + i \sin (5\pi/6)) = -\sqrt{3} + i,$$

$$z_3 = 2 (\cos (7\pi/6) + i \sin (7\pi/6)) = -\sqrt{3} - i,$$

$$z_4 = 2 (\cos (3\pi/2) + i \sin (3\pi/2)) = -2i,$$

$$z_5 = 2 (\cos (11\pi/6) + i \sin (11\pi/6)) = \sqrt{3} - i.$$

The points corresponding to the numbers z_k lie at the vertices of a regular hexagon inscribed in a circle of radius 2 with centre at the point $z = 0$ (Fig. 124). ▲

8.7. Algebraic Equations

Let us discuss algebraic equations of degree n with complex coefficients, that is, equations of the form

$$a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0 = 0, \quad a_n \neq 0, \quad n \in \mathbb{N}. \quad (1)$$

The number z_0 is called a *solution* or a *root* of the equation if the substitution of z_0 for z in the equation results in a true numerical equality. Consequently, if z_0 is a root of equation (1), then we have

$$a_n z_0^n + a_{n-1} z_0^{n-1} + \dots + a_1 z_0 + a_0 = 0.$$

For instance, the equations

$$\begin{aligned} z + 1 - i &= 0, \\ z^2 - 1 &= 0, \\ z^5 + z^3 + iz^2 + i &= 0, \\ iz^7 + 32(1 - i)z &= 0 \end{aligned}$$

are algebraic equations of the 1st, the 2nd, the 5th, and the 7th degree, respectively. The root of the first equation is the number $z_0 = -1 + i$, the second equation has two roots, $z_0 = 1$ and $z_1 = -1$. The number $z_0 = i$ is the root of the third equation since $i^5 + i^3 + i \cdot i^2 + i = 0$. It is evident that $z_0 = 0$ is the root of the fourth equation. It follows from the preceding section that besides the root $z_0 = 0$ the fourth equation possesses six more roots.

To solve an equation in the set of complex numbers means to find all the roots of the equation.

The general form of an algebraic equation of the first degree is

$$a_1 z + a_0 = 0, \quad a_1 \neq 0.$$

Such an equation evidently possesses one and only one solution $z_0 = -a_0/a_1$. An equation of the second degree (quadratic equation) is generally written as follows:

$$a_2 z^2 + a_1 z + a_0 = 0, \quad a_2 \neq 0.$$

To solve this equation, let us perform an identity transformation of its left-hand side:

$$\begin{aligned} a_2 \left(z^2 + \frac{a_1}{a_2} z + \frac{a_0}{a_2} \right) &= 0, \\ a_2 \left(\left(z + \frac{a_1}{2a_2} \right)^2 + \frac{a_0}{a_2} - \frac{a_1^2}{4a_2^2} \right) &= 0, \\ a_2 \left(\left(z + \frac{a_1}{2a_2} \right)^2 - \frac{a_1^2 - 4a_0a_2}{4a_2^2} \right) &= 0 \end{aligned}$$

and find the roots

$$\begin{aligned} \left(z + \frac{a_1}{2a_2} \right)^2 &= \frac{a_1^2 - 4a_0a_2}{4a_2^2}, \quad z = -\frac{a_1}{2a_2} + \frac{\sqrt{a_1^2 - 4a_0a_2}}{2a_2}, \\ z &= \frac{-a_1 \pm \sqrt{D}}{2a_2}. \end{aligned} \quad (2)$$

In the last formula we have introduced the notation $D = a_1^2 - 4a_0a_2$ (the discriminant of the quadratic equation is designated by the letter D), $\sqrt{D}$ being understood as all the values of the root.

Formula (2) for the roots of a quadratic equation has the same form as in the case when the coefficients of the equation are real numbers and the solutions are sought in the set of real numbers. But inasmuch as in the set of complex numbers the operation of extracting a square root is meaningful for any complex number, the restriction $D > 0$ becomes superfluous. Moreover, the restriction loses sense since the discriminant may prove to be not a real number, and the notions of "greater than", "less than" are not defined for such numbers.

Thus, in the set of complex numbers the equation

$$a_2 z^2 + a_1 z + a_0 = 0$$

(a_2, a_1, a_0 being complex numbers, $a_2 \neq 0$) is always solvable.

If $D = a_1^2 - 4a_0a_2 = 0$, then the equation has one root, now if $D \neq 0$, the equation has two roots. In all cases, the roots of a quadratic equation can be found by formula (2).

Example 1. Solve the equation $z^2 + 3z + 3 = 0$.

Δ By formula (2) we find

$$z = \frac{-3 + \sqrt{9-12}}{2} = \frac{-3 + \sqrt{-3}}{2}.$$

Taking into consideration that $\sqrt{-3} = \pm i\sqrt{3}$, we get

$$z_1 = -\frac{3}{2} + \frac{\sqrt{3}}{2}i, \quad z_2 = -\frac{3}{2} - \frac{\sqrt{3}}{2}i. \quad \blacktriangle$$

Example 2. Solve the equation

$$z^2 - 8z - 3iz + 13 + 13i = 0.$$

Δ By the formula for the roots of a quadratic equation we have

$$z = \frac{8+3i + \sqrt{(8+3i)^2 - 4(13+13i)}}{2} = \frac{8+3i + \sqrt{3-4i}}{2}.$$

To find all the values of $\sqrt{3-4i}$ we can use formula (2) of 8.6, but another technique is much simpler. Let us put

$$\sqrt{3-4i} = x + yi;$$

then $3-4i = x^2 + 2xyi - y^2$ and, consequently, x and y satisfy the system of equations

$$\begin{cases} x^2 - y^2 = 3, \\ xy = -2, \end{cases}$$

x and y being real numbers. The system has two real solutions, $x = 2$, $y = -1$ and $x = -2$, $y = 1$. Therefore,

$$\sqrt{3-4i} = \begin{cases} 2-i, \\ -2+i \end{cases}$$

and

$$z_1 = \frac{8+3i+2-i}{2} = 5+i, \quad z_2 = \frac{8+3i-2+i}{2} = 3+2i. \quad \blacktriangle$$

Let us now discuss algebraic equations of higher degrees. It is much more difficult, as a rule, to solve equation (1) for $n > 2$. The great German mathematician Carl Gauss proved, in 1799, the following theorem:

Every algebraic equation possesses at least one root in the set of complex numbers.

This theorem is customarily called the *fundamental theorem of algebra*. It has been named after Gauss. Its proof is beyond the scope of elementary mathematics and is not, therefore, presented here (see A. D. Myškis, *Introductory Mathematics for Engineers*. Lectures in higher mathematics, Mir Publishers, Moscow, 1978).

Proceeding from Gauss' theorem, we can prove that the left-hand side of equation (1) always admits of a representation in the form of a product:

$$a_n (z - z_1)^{\alpha_1} (z - z_2)^{\alpha_2} \dots (z - z_k)^{\alpha_k},$$

where $z_1, z_2, \dots, z_k$ are some distinct complex numbers, and $\alpha_1, \alpha_2, \dots, \alpha_k$ are natural numbers, with

$$\alpha_1 + \alpha_2 + \dots + \alpha_k = n.$$

Hence it follows that the numbers $z_1, z_2, \dots, z_k$ and only those numbers are the roots of equation (1). In this case we say that z_1 is a root of multiplicity α_1 , z_2 is a root of multiplicity α_2 and so on. If we agree to count the root of the equation as many times as is its multiplicity, then we can formulate the following theorem.

Every algebraic equation of degree n possesses in the set of complex numbers exactly n roots.

Both Gauss' theorem and the theorem we have just formulated are typical theorems of existence. They present a comprehensive solution of the problem on the existence of roots of an arbitrary algebraic equation but, unfortunately, do not say how to find these roots. While the root of the first-degree equation $a_1 z + a_0 = 0$ is determined by the formula $z = -a_0/a_1$, and while the roots of the second-degree equation $a_2 z^2 + a_1 z + a_0 = 0$ can always be easily found by formula (2), the analogous formulas for third- and fourth-degree equations are so cumbersome that they are often avoided, and for equations of the degree higher than the fourth such formulas do not exist at all.* The absence of a general method for solving algebraic equations does not prevent us, of course, from finding all its roots in special cases, depending on the specific nature of the equation.

* Note that diverse methods have been elaborated in higher mathematics making it possible to find *approximate* values of the roots of algebraic equations with the required accuracy.

Formula (2), 8.6, for instance, enables us to find all the roots of the equation $a_n z^n + a_0 = 0$, that is, of a binomial equation of degree n .

To solve equations with integral coefficients, the following theorem often proves to be of use.

Integral roots of any algebraic equation with integral coefficients are the divisors of the constant term.

□ The proof of this theorem is quite easy. Suppose $z = k$ is an integral root of the equation

$$a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0 = 0$$

with integral coefficients. Then

$$a_n k^n + a_{n-1} k^{n-1} + \dots + a_1 k + a_0 = 0$$

and, consequently,

$$a_0 = -k (a_n k^{n-1} + \dots + a_1).$$

Under the assumptions made, the number $a_n k^{n-1} + \dots + a_1$ is, evidently, integral; hence k is the divisor of the number a_0 . ■

Example 3. Solve the equation

$$z^3 - 6z - 9 = 0.$$

△ Considering the divisors of the constant term, we make sure that $z = 3$ alone is an integral root of the equation. We divide the left-hand side of the equation by $z - 3$:

$$\begin{array}{r} z^2 + 3z + 3 \\ z - 3 \overline{) z^3 - 6z - 9} \\ \underline{z^3 - 3z^2} \\ 3z^2 - 6z \\ \underline{3z^2 - 9z} \\ 3z - 9 \\ \underline{3z - 9} \\ 0 \end{array}$$

and, solving the quadratic equation

$$z^2 + 3z + 3 = 0,$$

we obtain the rest of the roots. Thus we have

$$z_1 = 3, \quad z_2 = -\frac{3}{2} + i\frac{\sqrt{3}}{2}, \quad z_3 = -\frac{3}{2} - i\frac{\sqrt{3}}{2}. \quad \blacktriangle$$

Example 4. Find the integral roots of the equation

$$2z^3 - 5z^2 - 2z - 2 = 0.$$

$\triangle$ Only ± 1 and ± 2 can be integral roots of the equation. Substitution in the equation shows that neither of these four numbers satisfies it. The given equation does not possess integral roots. $\blacktriangle$

Example 5. Solve the equation

$$z^5 - 2z^4 - 13z^3 + 26z^2 + 36z - 72 = 0.$$

$\triangle$ Testing the divisors of the constant term, we find that $z = \pm 2$ are the roots of the equation. Dividing the left-hand side of the equation by $z^2 - 4$, we arrive at an equation

$$z^3 - 2z^2 - 9z + 18 = 0,$$

whose root is $z = 2$. Dividing the left-hand side of this equation by $z - 2$, we get $z^2 - 9$. Thus we see that the left-hand side of the equation can be factored:

$$(z + 3)(z + 2)(z - 2)(z - 3).$$

Hence, the equation possesses three simple roots $z = -3$, $z = -2$, and $z = 3$ and one double root $z = 2$. $\blacktriangle$

PROBLEMS OF SECTION I

1. Represent the complex number

$$z = \frac{(1+2i)^2 - (1-i)^3}{(3+2i)^3 - (2+i)^2}$$

in algebraic form.

2. Find the moduli and the arguments of the following complex numbers:

$$(a) \quad z = \frac{(1+i)^{13}}{(1-i)^7}, \quad (b) \quad z = \sin(6\pi/5) + i(1 + \cos(6\pi/5)).$$

3. Represent z in algebraic and trigonometric forms:

$$(a) \quad z = (\sqrt{3} - i)^{100}, \quad (b) \quad z = 1 + \cos(10\pi/9) + i \sin(10\pi/9).$$

4. Represent the number $z = (\tan 1 - i)^4$ in trigonometric form.

5. Solve the equation $z^2 + |z|^2 = 0$.

6. What set of points of a complex plane is given by the condition
 (a) $|z + 1| = |z - i|$, (b) $|z|^2 + 3z + 3\bar{z} = 0$, (c) $\sin |z| > 0$?
 7. The points $z_1 = 6 + 8i$ and $z_2 = 4 - 3i$ are given on a complex plane. Find the complex numbers corresponding to the points belonging to the bisector of the angle formed by the vectors z_1 and z_2 .
 8. Among the complex numbers z satisfying the condition

$$|z + 3 - \sqrt{3}i| = \sqrt{3}$$

find the number having the least positive argument.

9. Solve the equation $z^4 + 8 + 8\sqrt{3}i = 0$.

10. Write

$$z = \frac{\sqrt{5+12i} + \sqrt{5-12i}}{\sqrt{5+12i} - \sqrt{5-12i}}$$

in algebraic form under the condition that the real part of the complex numbers $\sqrt{5+12i}$, $\sqrt{5-12i}$ is negative.

11. Prove that the remainder of the division of the polynomial $P_n(z)$ by $z - z_0$ is equal to $P_n(z_0)$ (*Bézout's theorem*).

PROBLEMS OF SECTION II

1. Find $z_1 + z_2$, $z_1 z_2$, $z_1 - z_2$, z_1/z_2 , if
 (a) $z_1 = 2 + 5i$, $z_2 = 1 - 7i$, (b) $z_1 = \sqrt{2} - \sqrt{3}i$, $z_2 = \sqrt{2} + \sqrt{3}i$.
 2. Write z in algebraic form:
 (a) $z = \frac{-41+63i}{50} - \frac{6i+1}{1-7i}$, (b) $z = \left(\frac{-1+\sqrt{3}i}{2i} \right)^2$,
 (c) $z = \frac{13+12i}{6i-8} + \frac{(2i+1)^2}{i+2}$, (d) $z = (2+i)^6$.
 3. Find the complex number z satisfying the condition
 $(i - z)(1 + 2i) + (1 - iz)(3 - 4i) = 1 + 7i$,
 and write it in algebraic and trigonometric forms.
 4. Solve the system of equations

$$\begin{cases} z_1 + 2z_2 = 1 + i, \\ 3z_1 + iz_2 = 2 - 3i. \end{cases}$$

 5. At what real values of x and y are the complex numbers
 $z_1 = 9y^2 - 4 - 10xi^5$ and $z_2 = 8y^2 + 20i^{11}$
 conjugate?

6. Prove the equalities

$$\overline{z_1 + z_2 + \dots + z_n} = \overline{z_1} + \overline{z_2} + \dots + \overline{z_n},$$

$$\overline{z_1 z_2 \dots z_n} = \overline{z_1} \overline{z_2} \dots \overline{z_n}.$$

7. Solve the equations

(a) $z^2 + \bar{z} = 0$, (b) $z^2 + |z| = 0$,

(c) $|z| - iz = 1 - 2i$, (d) $z^2 = \bar{z}^3$,

(e) $z^2 + z|z| + |z^2| = 0$.

8. Solve the system of equations

$$|z + 1 - i| = |3 - z + 2i| = |z + i|.$$

9. Prove that the system of equations

$$\begin{cases} |z + 1 - i| = \sqrt{2}, \\ |z| = 3 \end{cases}$$

has no solutions.

10. The points z_1 , z_2 , and z_3 , being the vertices of a triangle, are given on a complex plane. Find the point of intersection of its medians.

11. Find the set of points of a complex plane which satisfy the condition

(a) $|z + 1 - i| = |z - 1 + i|$, (b) $(1 - i)\bar{z} = (1 + i)z$,

(c) $|z + 1 + 2i| \leq 0$, (d) $\arg z = (2n + 1)\pi$, $n \in \mathbb{Z}$,

(e) $|z + 1| \geq |z + i|$, (f) $|z + 2 - i| = \sqrt{3}$,

(g) $\log |z + i| \leq 1$.

(h) $\frac{\pi}{4}(8n + 1) < \arg(z + i) < \frac{\pi}{2}(4n + 1)$, $n \in \mathbb{Z}$.

(i) $|z - 2|^2 + |z + 2|^2 = 26$.

12. Among the complex numbers z satisfying the condition

(a) $|z + 1 - i| \leq 1$, (b) $|z - 5i| \leq 3$,

find the number having the least positive argument.

13. Write z in trigonometric form:

(a) $z = -\sqrt{3} + i$, (b) $z = -1$,

(c) $z = -\cos \frac{\pi}{12} - i \sin \frac{\pi}{12}$, (d) $z = \frac{1 + i\sqrt{3}}{2i(\cos 60^\circ + i \sin 60^\circ)}$,

(e) $z = \frac{i - 1}{i \left(1 - \cos \frac{2\pi}{5}\right) + \sin \frac{2\pi}{5}}$.

14. Represent z in algebraic form:

(a) $z = \frac{(i - \sqrt{3}) \left(\cos \frac{\pi}{12} - i \sin \frac{\pi}{12}\right)}{1 - i}$.

(b) $z = \left(-\frac{\sqrt{3}}{2} + \frac{i}{2}\right)^{13}$. (c) $z = \left(\frac{1 + i}{1 - i}\right)^{2n}$, $n \in \mathbb{N}$.

(d) $z = \frac{\left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3}\right) (1 + \sqrt{3}i)^7}{i^6}$.

15. Find all the values of $\sqrt[n]{w}$ if

(a) $w = +1$, $n = 3$, (b) $w = -1$, $n = 4$.

(c) $w = -4 + \sqrt{48}i$, $n = 3$, (d) $w = 1 + i$, $n = 8$.

16. Assume that A_k ($k = 1, 2, \dots, n$) are the vertices of a regular n -gon inscribed in a circle of radius $R = 1$. Find:

(a) $|A_1A_2|^2 + |A_1A_3|^2 + \dots + |A_1A_n|^2$.

(b) $|A_1A_2| \cdot |A_1A_3| \cdot \dots \cdot |A_1A_n|$.

17. Solve the equations

(a) $z^3 - 2iz - 5 = 0$, (b) $z^3 - 20z + 92 + 6i = 0$,

(c) $z^5 - 1 - i\sqrt{3} = 0$, (d) $z^5 + z^4 + z^3 + z^2 + z + 1 = 0$,

(e) $z^4 - 4z^3 + 7z^2 - 16z + 12 = 0$,

(f) $z^5 + 2z^4 + 4z^3 + 8z^2 + 16z + 32 = 0$.

18. Prove that if the equation

$$a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0 = 0$$

with real coefficients has the root z_0 , then the number $\bar{z}_0$ is also a root of that equation.

19. Ascertain that the number $1 + i$ is a root of the equation

$$3z^4 - 5z^3 + 3z^2 + 4z - 2 = 0,$$

and find the rest of the roots.

20. Find the common roots of the equations

$$z^3 + 2z^2 + 2z + 1 = 0,$$

$$z^{1982} + z^{100} + 1 = 0.$$

21. Dividing $P_n(z)$ by $z - i$, we obtain the remainder i , and dividing it by $z + i$ we get the remainder $1 + i$. Find the remainder upon the division of $P_n(z)$ by $z^2 + 1$.

22. Can the point $z = 0$ belong to some polygon whose vertices are at the points

$$z_k = 1 + z + z^2 + \dots + z^{k-1}, \quad |z| < 1?$$

Solutions to Problems of Section I

CHAPTER 1

1. We have

$$A = \{1; 3; 5; 15\}, \quad B = \{2; 3; 5; 7\}, \quad C = \{2; 4; 6; 8\}.$$

Proceeding from the definitions of the union and intersection of sets, we obtain $A \cup B = \{1; 2; 3; 5; 7; 15\}$, $B \cap C = \{2\}$, $A \cup C = \{1; 2; 3; 4; 5; 6; 8; 15\}$, therefore, $(A \cup C) \cap B = \{2; 3; 5\}$. The sets A and C do not possess any elements in common; consequently, $A \cap C = \emptyset$ and $A \cap B \cap C = \emptyset$.

2. We easily find that $A \cup C = [-1, 2)$, $A \cap B = [-1; 0)$, $A \cup B \cup C = (-\infty; 2)$, $(A \cup B) \cap C = [0; 1]$, $B \cap C = \emptyset$.

3. If the equality

$$A \cap X = A \cup Y \tag{1}$$

holds true for any subset A of the set E , then it is also valid in the case when A coincides with the set E and when A is an empty set, since, by definition, $E \subset E$ and $\emptyset \subset E$ for any set E . At $A = E$, formula (1) yields $E \cap X = E \cup Y$. X and Y being subsets of the set E , it follows that $E \cap X = X$ and $E \cup Y = E$. Thus we get $X = E$.

If $A = \emptyset$, formula (1) yields $\emptyset \cap X = \emptyset \cup Y$. Note that $\emptyset \cap X = \emptyset$ and $\emptyset \cup Y = Y$. Consequently, $Y = \emptyset$. Thus we have $X = E$ and $Y = \emptyset$.

4. It is easy to see that if the point $(x; y)$ belongs to the given set, then the points $(-x; -y)$, $(-x; y)$, and $(x; -y)$ also belong to the set, that is, the set is symmetric about the coordinate axes and the origin. We shall, therefore, consider the points lying in the first quadrant: $x \geq 0$, $y \geq 0$. The coordinates of the points belonging to that set satisfy the equation $x + y = 1$, and, since $x \geq 0$ and $y \geq 0$, this is a line segment with the end points $(0; 1)$ and $(1, 0)$.

From this segment we get the whole set (Fig. 125) first by means of the symmetry transformation about the Ox axis and then by the symmetry transformation of the resulting set about the Oy axis.

5. Let us rewrite the condition $|x + y| \leq 1$ in the form $-1 \leq x + y \leq 1$. The set of points, whose coordinates satisfy the condition $x + y \leq 1$, is a half-plane whose boundary is the straight line $x + y = 1$, and it is the half-plane which contains the point $(0; 0)$. The set of points defined by the inequality $x + y \geq -1$ is also a half-plane which contains the point $(0; 0)$ and whose boundary is the straight line $x + y = -1$. The set of points for whose coordinates

both inequalities hold true is the intersection of these half-planes, that is, a strip whose boundaries are parallel lines $x + y = 1$ and $x + y = -1$.

It can be established by analogy that the condition $|x - y| \leq 1$ defines a strip whose boundaries are the parallel lines $x - y = 1$ and $x - y = -1$. If the coordinates of the point M simultaneously satisfy the conditions $|x + y| \leq 1$ and $|x - y| \leq 1$, then the point M

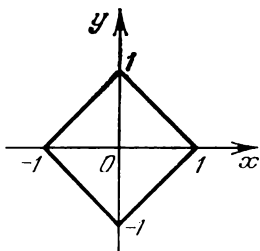


Fig. 125.

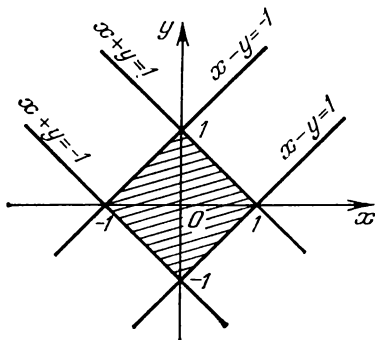


Fig. 126.

belongs to the intersection of the sets defined by each of these conditions, that is, the required set is the intersection of the indicated strips (Fig. 126) which is a square with vertices at the points $(0; 1)$, $(1; 0)$, $(-1; 0)$, $(0; -1)$.

6. Since $y = -x^2 + 2x$ is a parabola which cuts the abscissa axis at the points $x = 0$ and $x = 2$, which has a straight line $x = 1$ as

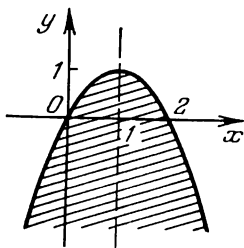


Fig. 127.

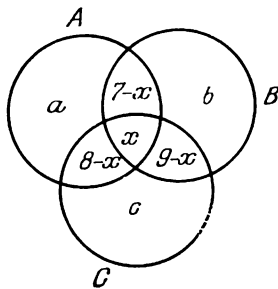


Fig. 128.

its axis of symmetry, and whose vertex has the coordinates $(1; 1)$, it follows that the set of points, whose coordinate satisfy the condition $y \leq -x^2 + 2x$, is bounded by this parabola, the points of the set lying on the parabola and below it (Fig. 127).

7. We shall denote the set of tourists of the given group speaking German by A and those speaking French by B . Then the tourists

speaking both languages form the set $A \cap B$. By the hypothesis, $m(A) = 70$, $m(B) = 45$, $m(A \cap B) = 23$. By formula (1) from 1.1 we find that $m(A \cup B) = 92$. The set $A \cup B$ is the set of tourists speaking at least one of the two languages. It follows that eight tourists of the group speak neither German nor French.

8. It is much easier to solve the problem with the aid of the visual representation of the sets (Fig. 128). We introduce the following notation: A is the set of students who solved the problem in algebra, B in geometry, and C in trigonometry.

If x is the number of students who solved all the three problems, then $(7 - x)$ is the number of students who solved the problems in algebra and geometry and did not solve the problem in trigonometry; $(8 - x)$ are those who solved the problems in algebra and trigonometry but did not solve the problem in geometry, and $(9 - x)$ are those who solved the problems in geometry and trigonometry but did not solve the problem in algebra.

If a , b , c is the number of students who solved only one problem, in algebra, geometry or trigonometry respectively, then we have

$$m(A) = 20 = 15 + a - x,$$

$$m(B) = 18 = 16 + b - x,$$

$$m(C) = 18 = 17 + c - x,$$

whence we find that $a = 5 + x$, $b = 2 + x$, $c = 1 + x$.

Altogether 40 students took part in the mathematics Olympiad all in all. Three of them did not solve a single problem; consequently, 37 students solved at least one problem. They form the set $A \cup B \cup C$. Thus we have $m(A \cup B \cup C) = 37$. And since $m(A \cup B \cup C) = a + b + c + 24 - 2x$, we can substitute here the expressions for a , b , and c and find that $x = 5$.

Thus it follows that five students solved all the three problems. The number of students who solved two problems each is equal to $24 - 3x = 9$.

9. Assume that C is the set of roots of the equation $f(x)g(x) = 0$. If $a \in C$, then $f(a)g(a) = 0$ and either $a \in A$ or $a \in B$, i.e. $a \in A \cup B$. The converse is also true: $a \in A \cup B$ yields $a \in C$. Indeed, if $a \in A \cup B$, then either $a \in A$ or $a \in B$. If, for example, $a \in A$, then $f(a) = 0$. The function $g(x)$ is a polynomial, its domain of definition is the entire number axis. Consequently, the function g is defined at the point a and $f(a)g(a) = 0$. The case $a \in B$ can be treated similarly. It follows from the afore-said that $C = A \cup B$.

If one of the functions, say, $g(x)$, is not a polynomial but a rational function, then the assertion $C = A \cup B$ is, in general, no longer true. For instance, if $g(x) = \frac{x^2 + 3}{x^2 - 1}$, $f(x) = x - 1$, then $x = 1 \in A$ ($f(1) = 0$); for $x = 1$, however, the function $g(x)$ is not defined. Consequently, for $x = 1$ the function $f(x)g(x)$ is not defined and 1 cannot belong to the set of its zeros.

Let us now consider the equation $f^2(x) + g^2(x) = 0$ and designate the set of its zeros as D . If $a \in D$, then $f(a) = 0$, $g(a) = 0$, i.e. $a \in A \cap B$. Conversely, if $a \in A \cap B$, then $a \in A$ and $f(a) = 0$,

$a \in B$ and $g(x) = 0$, whence it follows that $a \in D$. Thus we have $D = A \cap B$.

10. The function $f(x) = \sqrt{3+x} + \sqrt{3-x}$ is defined if $3+x \geq 0$ and $3-x \geq 0$, that is, if it is defined on the intersection of the sets specified by the inequalities $x \geq -3$ and $x \leq 3$. Thus we have $D(f) = [-3; 3]$.

11. The function is defined for all the values of x except for $x = \pm 1$. Its domain of definition is

$$(-\infty; -1) \cup (-1; +1) \cup (1; +\infty).$$

12. The function is defined for all the values of the argument for which $2^x - 3^x \geq 0$. Let us rewrite this inequality in the form $(3/2)^x \leq 1$, whence, taking the logarithms to the base $3/2$ (note that $3/2 > 1$), we get $x \leq 0$. The domain of definition of the function is $(-\infty; 0]$.

13. The logarithm is defined for any positive number to any positive base not equal to unity. Thus, the function is defined for all the values of x for which $x^2 - 1 > 0$ and $3+x > 0$, $3+x \neq 1$. Solving the inequalities simultaneously, we find that

$$x \in (-3; -1) \cup (1; +\infty).$$

Bearing in mind that $x \neq -2$, we obtain the following domain of definition of the function:

$$(-3; -2) \cup (-2; -1) \cup (1; +\infty).$$

14. The function is defined, provided the conditions $x+5 \geq 0$, $9-x > 0$, $9-x \neq 1$ are satisfied, i.e. $x \geq -5$, $x < 9$, $x \neq 8$. Consequently, $D(f) = [-5; 8) \cup (8; 9)$.

15. The function $\arccos t$ is defined on the interval $[-1; 1]$, and vanishes for $t = 1$. Thus it follows that the function $f(x)$ is defined providing the conditions $-1 \leq 2x-1 < 1$ and $2x > 0$, $2x \neq 1$ are fulfilled. The first inequality yields $0 \leq x < 1$, whence, taking into account that $x \neq 1/2$ and $x \neq 0$, we get the following domain of the function:

$$(0; 1/2) \cup (1/2; 1).$$

16. The function is defined for the values of the argument for which the expression under the logarithm sign is positive, i.e. $(1.25)^{1-x^2} - (0.4096)^{1+x} > 0$. It should be noted $1.25 = 5/4$, $0.4096 = (5/4)^{-4}$ and therefore, we can rewrite the inequality in the form $(5/4)^{-(x^2-1)} > (5/4)^{-(4+4x)}$, whence it follows that x satisfies the inequality $x^2 - 1 < 4 + 4x$ or $x^2 - 4x - 5 < 0$. The roots of the equation $x^2 - 4x - 5 = 0$ are -1 and 5 , and, therefore, $x^2 - 4x - 5 < 0$ for $-1 < x < 5$. The domain of the function is $(-1; 5)$.

17. We represent the quadratic function $\varphi(x) = x^2 + 2x + 2$ in the form $\varphi(x) = (x+1)^2 + 1$. It is evident that for $x = -1$ it assumes the least value equal to 1. The range of the function $\varphi(x)$ is the interval $[1; +\infty)$. Hence it follows that the range of the function $\sqrt{\varphi(x)}$ is again the interval $[1; +\infty)$.

18. Note that the function is odd. Let us consider $x \geq 0$. It follows from the inequality $2ab \leq a^2 + b^2$ that $2x \leq 1 + x^2$ for all x , the equality being attained if and only if $x = 1$ ($a = b$). Thus, for $x \geq 0$

we have

$$0 \leq \frac{2x}{1+x^2} \leq 1.$$

The function being odd, its range is the interval $[-1; +1]$.

19. On the interval $(-\infty; 0)$ the function $y = x^2 + 1$ decreases monotonically. Indeed, if $x_1 < x_2 \leq 0$, then

$$y(x_1) - y(x_2) = x_1^2 - x_2^2 = (x_1 - x_2)(x_1 + x_2) > 0$$

(since both parentheses are negative). Consequently, there exists an inverse function. We solve the equation $y = x^2 + 1$ for x , taking into account that x is negative, and get $x = -\sqrt{y-1}$. Thus we have $f^{-1}(x) = -\sqrt{x-1}$.

20. Since $y(x_1) - y(x_2) = \frac{1}{x_1} - \frac{1}{x_2}$, we have $y(x_1) = y(x_2)$ only when $x_1 = x_2$. The function establishes a one-to-one correspondence between the domain of definition and the range and, therefore, has an inverse. Solving the equation $y = 1 + \frac{1}{x}$ for x , we get $x = \frac{1}{y-1}$, whence it follows that $f^{-1}(x) = 1/(x-1)$.

21. Since $y = \frac{x(x+3)}{x(x-5)}$, the function $y = \frac{x+3}{x-5}$ for $x \neq 0$, and it is not defined for $x = 0$. Furthermore,

$$y(x_1) - y(x_2) = \frac{8(x_2 - x_1)}{(x_1 - 5)(x_2 - 5)},$$

therefore, $y(x_1) = y(x_2)$ only when $x_1 = x_2$. Thus we see that this function establishes a one-to-one correspondence between the domain of definition and the range and, consequently, possesses an inverse.

Solving the equation $y = (x+3)/(x-5)$ for x , we get $x = (3+5y)/(y-1)$. Recalling that $x \neq 0$, we exclude the value $y = -3/5$ and thus obtain

$$f^{-1}(x) = \frac{3+5x}{x-1}, \quad x \neq -\frac{3}{5}.$$

22. For an inverse function to exist it is necessary and sufficient that different values of the function should correspond to different values of the argument. Suppose $x_1 \neq x_2$. Since $y(x_1) - y(x_2) = a(x_1 - x_2)$, the inverse function exists only when $a \neq 0$. For $a \neq 0$ the equation $y = ax + b$ can be solved for x : $x = (y-b)/a$. Therefore, an inverse function exists when $a \neq 0$ and has the form

$$y = \frac{x}{a} - \frac{b}{a}.$$

The function $y = ax + b$ coincides with the function $y = \frac{x}{a} - \frac{b}{a}$ of the equality

$$ax + b = \frac{x}{a} - \frac{b}{a}$$

is valid for any x . Setting $x = 0$, we get $b = -b/a$, and setting $x = 1$ we get $a = 1/a$ under the condition $b = -b/a$. Solving the system

$$\begin{cases} b = -b/a, \\ a = 1/a, \end{cases}$$

we find that $a = 1$, $b = 0$ or $a = -1$, b being arbitrary. In the first case the function $y = \frac{x}{a} - \frac{b}{a}$ assumes the form $y = x$, in the second, $y = -x + b$. The necessary condition we have obtained is at the same time sufficient. In fact, the functions $y = x$ and $y = -x + b$ evidently coincide with their inverse functions. Note that the straight line $y = x$ coincides with the bisector of the first and third quadrants and any straight line $y = -x + b$ is perpendicular to that bisector, and when the line is symmetric about the bisector it is mapped onto itself.

CHAPTER 2

1. (a) The circuit is open since either the element k has failed or all the three elements l_i have failed simultaneously, or, else, both accidents occurred. (b) The current passes along the circuit since the element k and at least one of the elements l_i operate.

2. Let us compile the truth table for the propositions of the form $\bar{A} + B$:

A	B	$\bar{A}$	$\bar{A} + B$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

The truth table for $\bar{A} + B$ coincides with that for the implication $A \Rightarrow B$. We have proved the formula.

3. Let us denote by A_1 , A_2 , and A_3 the propositions stating, respectively, that the first, the second, and the third student studied logic. The hypothesis implies the truth of the sentence

$$P = (A_1 \Rightarrow A_2) (\bar{A}_3 \Rightarrow A_2).$$

Let us eliminate the implication using the formula

$$A \Rightarrow B = \bar{A} + B$$

proved in the preceding problem. Then we get

$$P = (\bar{A}_1 + A_2) (\bar{A}_3 + A_2).$$

To simplify the expression still further, we shall make use of formulas 7 and 9 of 2.1:

$$\overline{A + B} = \overline{A} \overline{B} \text{ and } \overline{\overline{A}} = A.$$

We obtain

$$P = (\overline{A_1} + A_2) A_3 A_2,$$

whence we get

$$P = \overline{A_1} A_3 \overline{A_2} + A_3 A_2 \overline{A_2} = \overline{A_1} A_3 \overline{A_2} + L = \overline{A_1} A_3 \overline{A_2},$$

that is, logic was studied by the third student, and the first and the second student did not study it.

4. The inspectors of Commissar Maigret established the truth of the sentences

$$A \Rightarrow (B + C), \quad B + \overline{A} D, \quad D \Rightarrow (B + C).$$

The product

$$P = (A \Rightarrow (B + C)) (B + \overline{A} D) (D \Rightarrow (B + C))$$

of these three sentences is true. Eliminating the implication with the aid of the formula proved in problem 2, we get

$$P = (\overline{A} + B + C) (B + \overline{A} D) (\overline{D} + B + C).$$

Let us simplify the product of the first and the third parenthesis in accordance with the law 6 given in 2.1. Then we shall have

$$P = (\overline{A} \overline{D} + B + C) (B + \overline{A} D).$$

Applying this law once again, we obtain

$$P = B + (\overline{A} \overline{D} + C) \overline{A} D$$

or

$$P = B + C \overline{A} D.$$

Thus, it can only follow from the evidence of the inspectors that either Etienne is the murderer, or the following three events occurred simultaneously: François lied, François was not drunk, and the murder took place after midnight. But Commissar Maigret knew that when François was sober, he never lied, i.e. $\overline{A} C = L$ and, consequently $P = B + C \overline{A} D = B + L = B$, that is B is true. Thus it follows that Etienne is the murderer.

5. The equality is not proved. In the course of "proving" it, the student twice used the relation

$$\sqrt[3]{5 \sqrt{2} + 7} - \sqrt[3]{5 \sqrt{2} - 7} = 2.$$

and it was precisely its truth that he had to prove.

To prove the equality we can do the following. We designate the left-hand side by a , that is, set

$$\sqrt[3]{5 \sqrt{2} + 7} - \sqrt[3]{5 \sqrt{2} - 7} = a.$$

Cubing both sides of the equation, we get

$$14 - 3 \left(\sqrt[3]{5\sqrt{2}+7} \right)^2 \sqrt[3]{5\sqrt{2}-7} + 3 \sqrt[3]{5\sqrt{2}+7} \left(\sqrt[3]{5\sqrt{2}-7} \right)^2 = a^3,$$

or

$$3 \sqrt[3]{5\sqrt{2}+7} \sqrt[3]{5\sqrt{2}-7} \left(\sqrt[3]{5\sqrt{2}+7} - \sqrt[3]{5\sqrt{2}-7} \right) = 14 - a^3.$$

But since

$$\sqrt[3]{5\sqrt{2}+7} - \sqrt[3]{5\sqrt{2}-7} = a,$$

it follows that

$$3 \sqrt[3]{50-49} a = 14 - a^3,$$

that is, the left-hand side of the equation is equal to one of the roots of the equation

$$a^3 + 3a - 14 = 0.$$

Since $a = 2$ is the only real root of that equation, we have proved the equality.

6. The reasoning and the computations carried out only prove the following: if the right-hand triangle mentioned in the hypothesis exists, then its hypotenuse is equal to $a\sqrt{5/3}$. But the existence of such a triangle cannot be derived from any source. It is easy to make sure that actually such a triangle does not exist, otherwise the plane angle ACB of the trihedral angle $CAOB$ would be greater than the sum of the two remaining plane angles ACO and OCB .

7. Statement (a) is true for all $k \neq 0$.

If $k = 0$, then, say for $l = 1$ the inequality has no solutions.

If $k > 0$, then, whatever the value of l , the inequality has the solution $x \in (-\infty; -l^2/k]$.

If $k_l < 0$, then for any l the inequality has the solution $x \in [-l^2/k; +\infty)$.

Statement (b) is true for all the values of k . In fact, for any k there is an l , namely, $l = 0$, such that the inequality has the solution $x = 0$.

8. If the number n ends with the digit 4, then the number $n + 48$ ends with the digit 2 and, consequently, cannot be the square of a natural number since the square of a natural number must end with one of the following digits: 0, 1, 4, 5, 6, 9. Thus, for any n the proposition $A(n) B(n)$ is false.

The proposition $C(n) B(n)$ is also false for all n , since when n ends with the digit 4, $n - 41$ ends with the digit 3 and, therefore, cannot be the square of a natural number.

Let us see whether there are any values of n for which the proposition $A(n) C(n)$ is true. If $A(n)$ and $C(n)$ are simultaneously true, then

$$\begin{cases} n+48=p^2 \\ n-41=q^2 \end{cases} \quad \text{where } p \in \mathbb{N}, \quad q \in \mathbb{N}$$

Subtracting the second equation from the first term-by-term, we get

$$89 = p^2 - q^2 = (p - q)(p + q).$$

But 89 is a prime number and, therefore, we must have

$$\begin{cases} p - q = 1, \\ p + q = 89, \end{cases}$$

whence we have $p = 45$ and $n = 45^2 - 48 = 1977$.

Consequently, only the value $n = 1977$ can be a solution of the problem. Verification shows that $n = 1977$ is indeed a solution of the problem.

9. The converse theorem is: if the discriminant of a quadratic equation is nonpositive, the equation does not possess two distinct real roots.

The opposite theorem is: if a quadratic equation possesses two distinct real roots, the discriminant of that equation is positive.

The theorem opposite to the converse is: if the discriminant of a quadratic equation is positive, the equation possesses two distinct real roots.

All four theorems are true.

10. (a) The statement is true. Let us prove this. Assume that n ($n \geq 5$) is a prime number. Then, $n - 1$ and $n + 1$ are two successive even numbers and, consequently, their product $n^2 - 1$ is divisible by 8. Besides, of the three successive numbers $n - 1$, $n + 1$, one must be divisible by 3. But n is not divisible by 3 and this means that $n^2 - 1$ must be necessarily divisible by 3.

(b) The statement is false. To prove this, it is sufficient to take $n = 25$.

11. It is easy to verify that the propositions $A(1)$, $A(2)$, . . . , $A(8)$ are false and $A(9)$ is true.

Suppose $A(n)$ is true for a certain value $n = k$. Using this assumption, we get

$$2^{k+1} = 2 \cdot 2^k > 2(4k^2 + 1).$$

Then it is easy to prove that

$$2(4k^2 + 1) > 4(k + 1)^2 + 1.$$

Consequently,

$$2^{k+1} > 4(k + 1)^2 + 1,$$

i.e. $A(k + 1)$ is true. This argument is correct for all values $k \geq 9$.

By mathematical induction, the proposition $A(n)$ is true for any $n \geq 9$.

12. In the case of one straight line the assertion is obvious. Assume that it is true for $n = k$. Suppose now that $k + 1$ straight lines are drawn on the plane in an arbitrary fashion. Let us prove that we can paint the plane black and white so that any two parts having a common side will be differently coloured. For that purpose, we shall choose one of the drawn lines and call it an auxiliary line for the sake of convenience. The rest k lines will be called the principal lines. We shall paint the plane as we need it, taking into account only the principal lines. By the inductive assumption it can be done. The auxiliary line partitions the now coloured plane into two half-planes. We shall

change the colour of one of them into the opposite colour. We shall show that after this change the entire plane, with all the $k - 1$ straight lines taken into account, is coloured as necessary. Let us take any two adjacent parts. They are partitioned either by any one of the principal lines or by the auxiliary line. In the first case, these parts had previously been coloured differently, and then either both changed their colours to the opposite colours or did not change colour at all. Thus, in the first case they are repainted different colours. In the second case, the parts had previously been painted the same colour and then the colour of one of them was changed to the opposite one, that is, in this case they are differently coloured. Thus, from the possibility of painting the plane in the required manner in the case of k straight lines it follows that such a colouring is possible in the case of $k + 1$ straight lines as well.

CHAPTER 3

1. Multiplying both sides of the given equation by $(x + 2) \times (x + 1)$, we obtain

$$3(x + 1) - (2x - 1)(2 + x) = 2x + 1,$$

which is, evidently, equivalent to the quadratic equation

$$x^2 + x - 2 = 0. \quad (1)$$

This means that only the roots of equation (1), $x = -2$ and $x = 1$, can serve as the solutions of the given equation. The number -2 is not a root of the given equation since for $x = -2$ both sides of the equation are not defined. Verification shows that the number 1 is the solution of the given equation.

Answer: $\{1\}$.

2. Here is the scheme of solution of the given equation:

$$\begin{aligned} \left(\sqrt{9-5x} = \sqrt{3-x} + \frac{6}{\sqrt{3-x}} \right) &\Rightarrow \left(9-5x = \left(\sqrt{3-x} + \frac{6}{\sqrt{3-x}} \right)^2 \right) \\ &\Leftrightarrow \left(9-5x = 3-x+12+\frac{36}{3-x} \right) \Leftrightarrow \left(6+4x = \frac{36}{x-3} \right) \\ &\Rightarrow ((3+2x)(x-3)-18=0) \Leftrightarrow (2x^2-3x-27=0) \\ &\Leftrightarrow (x=-3 \text{ or } x=9/2). \end{aligned}$$

Consequently, only $x = -3$ and $x = 9/2$ can be solutions of the given equation. For $x = 9/2$ both sides of the equation are not specified. Verification shows that $x = -3$ is the root of the given equation.

Answer: $\{-3\}$.

3. Squaring both sides of the equation, we obtain

$$16(x + 2) = (x + 1)^2 + 8|x + 1| + 16.$$

After requisite transformations, we get

$$x^2 - 14x - 15 + 8|x + 1| = 0. \quad (1)$$

If $x \geq -1$, then $|x + 1| = x + 1$ and equation (1) is equivalent to the quadratic equation $x^2 - 6x - 7 = 0$ whose roots are the num-

bers 7 and -1 . Note that these two numbers satisfy the condition $x \geq -1$.

Now if $x \leq -1$, then $|x + 1| = -x - 1$ and equation (1) is equivalent to the equation $x^2 - 22x - 23 = 0$ whose roots are the numbers 23 and -1 . Only the number -1 satisfies the condition $x \leq -1$.

Consequently, only the numbers -1 and 7 can be roots of the given equation. Verification shows that both these numbers are the roots of the given equation.

Answer: $\{-1, 7\}$.

4. We introduce a new variable

$$u = \frac{x-3}{x^2+4x+9}. \quad (1)$$

Then, with respect to u , we get the equation

$$u + \frac{1}{u} = -2,$$

which, after multiplication of both sides by u , reduces to the quadratic equation

$$u^2 + 2u + 1 = 0,$$

whose only root is $u = -1$. Substituting the value of u we have obtained into (1), we get the equation

$$\frac{x-3}{x^2+4x+9} = -1.$$

It can be easily transformed into the quadratic equation

$$x^2 + 5x + 6 = 0,$$

whose roots are the numbers -3 and -2 . It is easy to verify that these two numbers satisfy the given equation.

Answer: $\{-2; -3\}$.

5. We introduce a new variable

$$v = \sqrt{2x^2 + 3x + 9}.$$

Then we have

$$\begin{aligned} v^2 - 5v - 6 &= 0, \\ v &= \frac{5 \pm \sqrt{25+24}}{2} = \frac{5 \pm 7}{2}. \end{aligned}$$

Consequently, $v = 6$ and $2x^2 + 3x + 9 = 36$. The numbers 3 and $-9/2$ are the roots of the equation. Verification shows that these numbers are the roots of the given equation.

Answer: $\{3; -9/2\}$.

6. We subtract the first equation multiplied by 3 from the second equation and then subtract the first equation multiplied by 2 from the third equation. The result will be a system

$$\begin{cases} x + 2y + 3z = 8, \\ -5y - 8z = -18, \\ -3y - 4z = -10. \end{cases}$$

Now we add the second equation multiplied by $-3/5$ to the third equation term-by-term and obtain a system

$$\begin{cases} x + 2y + 3z = 8, \\ y + \frac{8}{5}z = \frac{18}{5}, \\ z = 1, \end{cases}$$

which is triangular in form and is, evidently, equivalent to the given system. From the system obtained we successively find $z = 1$, $y = 2$, $x = 1$.

Answer: $\{(1, 2, 1)\}$.

7. We subtract termwise the fourth equation multiplied, respectively, by 1, 2, and 3, from the first, second, and third equations. As a result we get a system

$$\begin{cases} x + y + z = 3, \\ y - 4z = -2, \\ y - 8z = -6, \\ -8z = -9. \end{cases}$$

Subtracting termwise the second equation of this system from the third, we get the equation $-4z = -4$, which, evidently, contradicts the last equation of the system. This means that the system has no solutions.

Answer: The system is inconsistent.

8. Adding term-by-term the first equation, multiplied by -2 , to the second equation, and, multiplied by -4 , to the third equation, we obtain a system

$$\begin{cases} x_1 + x_2 + x_3 + x_4 = 1, \\ -5x_2 - x_3 - 3x_4 = 0, \\ -5x_2 - x_3 - 3x_4 = 0. \end{cases}$$

The second and the third equation of the system coincide and, therefore, this system is equivalent to the system of equations

$$\begin{cases} x_1 + x_2 + x_3 + x_4 = 1, \\ -5x_2 - x_3 - 3x_4 = 0. \end{cases}$$

It follows from the second equation that

$$x_2 = -\frac{1}{5}x_3 - \frac{3}{5}x_4.$$

Substituting this expression for x_2 into the first equation, we get

$$\begin{aligned} x_1 = 1 - x_2 - x_3 - x_4 &= 1 - \left(-\frac{1}{5}x_3 - \frac{3}{5}x_4\right) \\ &\quad - x_3 - x_4 = 1 - \frac{4}{5}x_3 - \frac{2}{5}x_4. \end{aligned}$$

Consequently, the general solution of the given system has the form

$$\left(1 - \frac{4}{5}x_3 - \frac{2}{5}x_4; -\frac{1}{5}x_3 - \frac{3}{5}x_4; x_3; x_4\right),$$

where x_3 and x_4 are arbitrary. Thus, the given system has an infinite number of solutions.

Answer: $\left\{\left(1 - \frac{4}{5}x_3 - \frac{2}{5}x_4; -\frac{1}{5}x_3 - \frac{3}{5}x_4; x_3; x_4\right), x_3 \in \mathbb{R}, x_4 \in \mathbb{R}\right\}.$

9. To solve this system, we shall use the method of substitution. We find $x = 7 - 2y$ from the second equation of the system and substitute it into the first equation:

$$(7 - 2y)^2 + 2y^2 - 2y(7 - 2y) = 5.$$

Requisite transformations lead to the quadratic equation

$$5y^2 - 21y + 22 = 0.$$

It has the roots $y_1 = 2$ and $y_2 = 11/5$. Substituting these values for y into the equation $x = 7 - 2y$, we find $x_1 = 3$ and $x_2 = 13/5$.

Answer: $\{(3, 2); (13/5, 11/5)\}.$

10. Adding termwise the first equation, multiplied by 7, to the second equation, multiplied by -3 , we get the equation

$$14x^2 + 21y^2 - 28xy = 6x^2 - 3y^2,$$

which can be easily reduced to the equation

$$2x^2 - 7xy + 6y^2 = 0.$$

Solving it as a quadratic equation for x , we obtain

$$x = 2y \text{ and } x = 3y/2.$$

The given system is equivalent to the collection of two systems

$$\begin{cases} x = 2y, \\ 2x^2 - y^2 = 7 \end{cases} \quad \text{and} \quad \begin{cases} 2x = 3y, \\ 2x^2 - y^2 = 7. \end{cases}$$

Substituting $x = 2y$ into the second equation, we obtain

$$8y^2 - y^2 = 7, \quad y^2 = 1, \quad y = \pm 1.$$

Thus we see that the first system has the solutions $(2; 1)$ and $(-2; -1)$. Substituting $x = 3y/2$ into the second equation, we get

$$\frac{18}{4}y^2 - y^2 = 7, \quad y^2 = 2, \quad y = \pm \sqrt{2}.$$

Consequently, the second system has the solutions $(3/\sqrt{2}; \sqrt{2})$ and $(-3/\sqrt{2}; -\sqrt{2})$.

Answer: $\{(2; 1), (-2; -1), (3/\sqrt{2}; \sqrt{2}), (-3/\sqrt{2}; -\sqrt{2})\}.$

11. We introduce new variables $u = xy$ and $v = x^2 + y^2$, for which we have the system of equations

$$\begin{cases} 3u - v = 5, \\ -v^2 + 9u^2 = 155. \end{cases}$$

Substituting $v = 3u - 5$ into the second equation, we get $-9u^2 + 30u - 25 + 9u^2 = 155$. This yields $u = 6$ and $v = 13$. Then we solve the system

$$\begin{cases} xy = 6, \\ x^2 + y^2 = 13. \end{cases}$$

Substituting $y = 6/x$ into the second equation, we obtain

$$x^2 + \frac{36}{x^2} = 13, \quad x^4 - 13x^2 + 36 = 0.$$

By the formula for the roots of a biquadratic equation we find

$$x = \pm \sqrt[4]{\frac{13 \pm \sqrt{169 - 144}}{2}} = \pm \sqrt{\frac{13 \pm 5}{2}},$$

i.e. $x = \pm 3$ and $x = \pm 2$. Consequently, the system possesses 4 solutions: $(3; 2)$, $(-3; -2)$, $(2; 3)$ and $(-2; -3)$.

Answer: $\{(3; 2), (-3; -2), (2; 3), (-2; -3)\}$.

12. Adding term-by-term the first and the second equation to the third, we get the equation

$$x(y + z + x) - z(y + z + x) + x + y + z = 0,$$

which can be evidently reduced to the form

$$(x + y + z)(x - z + 1) = 0. \quad (1)$$

Let us now consider the system which can be obtained from the given system by replacing the third equation of the system by equation (1). The system obtained can be separated into two systems whose first two equations coincide with those of the given system, and the third equation of the first system is

$$x + y + z = 0, \quad (2)$$

and that of the second system is

$$x - z + 1 = 0. \quad (3)$$

Let us solve these systems.

(1) From (2) we find $z = -x - y$ and substitute it into the first two equations. As a result, we get the following system of two equations for x and y :

$$\begin{cases} 2xy + y^2 - x + y = 0, \\ 2x^2 + 3xy + y^2 - 3x + 3y = 0. \end{cases} \quad (4)$$

We subtract termwise the first equation, multiplied by 3, from the second equation and solve the resulting equation

$$-3xy + 2x^2 - 2y^2 = 0$$

as a quadratic equation in the variable x . Then we have

$$x = \frac{3y \pm \sqrt{9y^2 + 16y^2}}{4} = \frac{3y \pm 5y}{4},$$

i.e. $x = 2y$ or $x = -y/2$. We see that system (4) is equivalent to the collection of two systems in which the first equation is the same as

in (4) and the second equation is, respectively, $x = 2y$ or $x = -y/2$.

If $x = 2y$, then, from the first equation of system (4), we get, for y , the equation $5y^2 - y = 0$, which has two roots $y_1 = 0$ and $y_2 = 1/5$. Thus we see that in this case system (4) has two solutions $(0; 0)$ and $(2/5; 1/5)$.

If $y = -2x$, then the first equation yields $x = 0$. So in this case system (4) has one solution $(0; 0)$.

Taking into account the solutions of system (4) we have obtained, we find that the system under consideration has two solutions $(0; 0; 0)$ and $(2/5; 1/5; -3/5)$.

(2) From (3) we find $z = x + 1$ and substitute it into the first two equations of the given system. We obtain the following system of linear equations:

$$\begin{cases} 3x + y + 1 = 0, \\ 2x - 3y - 1 = 0, \end{cases}$$

which is easy to solve: $x = -2/11$, $y = -5/11$. Consequently, the system of three equations being discussed has the unique solution

$$x = -2/11, y = -5/11, z = 9/11.$$

Answer: $\{(0; 0; 0), (2/5; 1/5; -3/5), (-2/11; -5/11; 9/11)\}$.

13. We introduce new variables u and v by the formulas

$$x = u/z \text{ and } y = v/z. \quad (1)$$

Then, for the variables u , v , z we get the system of equations

$$\begin{cases} 27u^3 - v^3 - 13uv = 0, \\ 3u^2 - 4\frac{uv}{z} + 3 = 0, \\ 3u - v = 1. \end{cases} \quad (2)$$

Substituting $v = 3u - 1$ into the first equation of system (2), we get the quadratic equation $12u^2 - 4u - 1 = 0$, whose roots are the numbers $u_1 = 1/2$ and $u_2 = -1/6$. Substituting these numbers into the equation $v = 3u - 1$, we obtain $v_1 = 1/2$ and $v_2 = -3/2$. From the second equation of system (2) we find $z_1 = 4/15$ and $z_2 = 12/37$. Knowing now u , v , z , and using (1) we find, $x_1 = 15/8$, $y_1 = 15/8$ and $x_2 = -37/72$, $y_2 = -37/8$. Consequently, the given system has the solutions $(15/8; 15/8; 4/15)$ and $(-37/72; -37/8; 12/37)$.

Answer: $\{(15/8; 15/8; 4/15), (-37/72; -37/8; 12/37)\}$.

14. We determine y by the second equation of the system

$$y = \frac{1}{2} (3 + c - (6 + c)x),$$

and substitute it into the first equation of the system:

$$-4x + \frac{1}{2} c (3 + c - 6x - cx) = 1 + c.$$

After some transformations we get an equation

$$(c^2 + 6c + 8)x = c^2 + c - 2.$$

if $(c^2 + 6c + 8) \neq 0$, this equation has the unique solution

$$x = \frac{c^2 + c - 2}{c^2 + 6c + 8}.$$

Substituting this value of x into the second equation of the system, we determine the unique value of y . Thus, if $c^2 + 6c + 8 \neq 0$, the system has a unique solution. Assume now that c is such that $c^2 + 6c + 8 = 0$, i.e. $c = -4$ or $c = -2$. For $c = -4$ the given system assumes the form

$$\begin{cases} -4x - 4y = -3, \\ 2x + 2y = -1. \end{cases}$$

It is evident that no pair of numbers satisfies this system. Consequently, for $c = -4$ the given system is inconsistent. For $c = -2$ the system assumes the form

$$\begin{cases} -4x - 2y = -1, \\ 4x + 2y = 1. \end{cases}$$

The system has an infinite number of solutions.

It follows that only for $c = -4$ the given system does not possess any solutions.

Answer: $c = -4$.

15. We denote by v_0 the river flow velocity and by v the actual speed of the boat on its way from A to B . By the hypothesis, the actual speed of the boat on its way from B to C is equal to $2v/3$, and the boat's actual speed on its way from C to A is equal to $4v/3$. Consequently, the actual speed of the boat moving from A to C is less, on all the parts of the trip, than when it moves from C to A . However, the boat covered the distance from A to C in 7 hours and from C to A , in 8 hours. Hence it follows that the river flows from point A to point C . By the hypothesis, the boat covered the way from A to C in 7 hours, its speed on the parts AB and BC being equal, respectively, to $v + v_0$ and $\frac{2v}{3} + v_0$. Hence we have

$$\frac{|AB|}{v + v_0} + \frac{|BC|}{\frac{2v}{3} + v_0} = 7.$$

From the second condition it follows that the boat covered the way from C to A in 8 hours at the rate $\frac{4}{3}v - v_0$, and, therefore,

$$\frac{|AC|}{\frac{4}{3}v - v_0} = 8.$$

From the third condition of the problem we finally obtain

$$\frac{|AB|}{v - v_0} = 6.$$

We introduce the notation $a = |AB|/v_0$, $b = |BC|/v_0$, and $p = v/v_0$. Then, for a , b and p we get the following system of equations:

$$\begin{cases} \frac{a}{p+1} + \frac{b}{\frac{2}{3}p+1} = 7, \\ \frac{a+b}{\frac{4}{3}p-1} = 8, \\ \frac{a}{p-1} = 6. \end{cases}$$

From the third equation of the system it follows that $a = 6(p-1)$. Substituting the expression for a into the second equation of the system, we obtain $b = \frac{14}{3}p - 2$. Substituting the expressions for a and b into the first equation of the system, we obtain, for p , a quadratic equation

$$4p^2 - 7p - 15 = 0,$$

whose roots are the numbers 3 and $-5/4$. The number 3 alone satisfies the hypothesis. Then $b=12$, and the required time is equal to $\frac{12}{\frac{2}{3} \cdot 3 + 1}$,

that is, to 4 hours.

CHAPTER 4

1. They are not equivalent, since, for instance, $x = 7$ enters into the set of solutions of the second inequality but does not belong to the set of solutions of the first equation.

2. They are not equivalent, the value $x = 0$ satisfies the first inequality but does not satisfy the second.

3. They are equivalent. The set of solutions of the given inequalities is the same, it is an empty set.

4. They are not equivalent since $x = 0$ belongs to the set of solutions of the second inequality but does not belong to the set of solutions of the first inequality.

5. We reduce the inequality to the standard form:

$$\frac{x^2 + 2x - 15}{x + 2} > 0.$$

Next we find the critical points $x = -2$, $x = -5$, $x = 3$ and write the inequality as follows:

$$\frac{(x+5)(x-3)}{x+2} > 0.$$

The critical points partition the number axis into four subintervals (Fig. 129). We determine the sign of the left-hand side of the inequali-

ty on each subinterval and find that the rational function

$$\frac{(x+5)(x-3)}{x+2}$$

is positive on two subintervals, $(-5; -2)$ and $(3; +\infty)$. In our case, the critical points themselves do not, evidently, belong to the set of solutions.

6. The inequality is already in the standard form. We find the critical points:

$$x = -\sqrt{2}, x = -1, x = \sqrt{2}, x = 3, x = 4.$$

They partition the number axis into six subintervals (Fig. 130), on each of which the rational function retains sign. It is easy to see that



Fig. 129.

on the subintervals $(-\sqrt{2}; -1)$, $(-1; \sqrt{2})$, and $(3, 4)$, the rational function $\frac{(\sqrt{2}-x)(\sqrt{2}+x)(x-3)^3}{(x+1)^2(x-4)}$ is positive. We investigate the critical points. The points $x = \pm\sqrt{2}$, $x = 3$ are the zeros of the

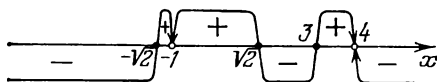


Fig. 130.

numerator and, the inequality being nonstrict, belong to the set of solutions. The zeros of the denominator $x = -1$, $x = 4$ do not belong to the set of solutions (the arrows in Fig. 130 indicate this).

Answer: $[-\sqrt{2}; -1) \cup (-1; \sqrt{2}] \cup [3; 4)$.

7. To find the domain of definition of the given function, we have to solve the rational inequality

$$\frac{2x+1}{x-2} - \frac{x+4}{2x+5} \geq 0,$$

which can be easily reduced to the standard form

$$\frac{3x^2+10x+13}{(x-2)(2x+5)} \geq 0.$$

The critical points $x = 2$ and $x = -5/2$ partition the number axis into three subintervals, two of which yield the set of solutions of the inequality and are, consequently, the domain of definition of the function.

Answer: $(-\infty; -5/2) \cup (2; +\infty)$.

8. The left-hand side of the inequality is meaningful if and only if $x^2 - 40x + 39 \geq 0$, that is, if $x \leq 1$ or $x \geq 39$. Now if $x < 1$, then the right-hand side of the inequality is negative whereas the left-hand side is positive. It follows that on the subinterval $(-\infty; 1)$ there are no solutions. The value $x = 1$ satisfies the inequality and, hence, belongs to the set of its solutions. If $x \geq 39$, then it is legitimate to square both sides of the inequality, i.e. $x^2 - 40x + 39 \leq x^2 - 2x + 1$, $-38x \leq -38$, or $x \geq 1$. Allowing for the restriction we get $x \geq 39$.

Answer: $\{1\} \cup [39; +\infty)$.

9. The square roots entering into the inequality exist simultaneously for those and only those values of x which satisfy the system of inequalities

$$\begin{cases} 3x - 4 \geq 0, \\ 2x - 13 \geq 0, \\ 13 - 2x \geq 0. \end{cases}$$

This system has a solution and this solution is unique: $x = 13/2$. Substituting it into the given inequality, we make sure that $x = 13/2$ is the solution.

10. We set $z = 9x^2 - 6x$ and then the inequality assumes the form

$$\sqrt{5z + 1} < 7 - z.$$

Evidently, the inequality can have solutions only when

$$-1/5 \leq z < 7,$$

but for these values of z we can square both sides of the inequality:

$$5z + 1 < 49 - 14z + z^2,$$

$$z^2 - 19z + 48 > 0,$$

$$z < 3 \text{ and } z > 16.$$

Consequently,

$$-1/5 \leq z < 3.$$

It now remains to solve the following system of quadratic inequalities:

$$-1/5 \leq 9x^2 - 6x < 3.$$

The inequality $9x^2 - 6x < 3$ is true for all values of x belonging to the interval

$$-1/3 < x < 1.$$

The inequality

$$-1/5 \leq 9x^2 - 6x$$

is satisfied by all $x \leq (5 - 2\sqrt{5})/15$ and all $x \geq (5 + 2\sqrt{5})/15$. Thus, both inequalities are valid if

$$-1/3 < x \leq (5 - 2\sqrt{5})/15 \text{ and } (5 + 2\sqrt{5})/15 \leq x < 1.$$

11. The given inequality is satisfied by those and only those values of the variable x which satisfy either the inequality

$$x - \frac{4}{x} - 2 \geq 1$$

or the inequality

$$x - \frac{4}{x} - 2 \leq -1.$$

Each of these rational inequalities can be easily solved by the method of intervals. We reduce the first inequality to the standard form:

$$\frac{x^2 - 3x - 4}{x} \geq 0 \quad \text{or} \quad \frac{(x+1)(x-4)}{x} \geq 0.$$

The critical points $x = -1$, $x = 0$, $x = 4$ partition the entire number axis into four subintervals on each of which we determine the sign of the rational function and obtain the solution

$$-1 \leq x < 0 \quad \text{and} \quad x \geq 4.$$

The second inequality is written in the standard form as

$$\frac{x^2 - x - 4}{x} \leq 0$$

or

$$\frac{\left(x - \frac{1 - \sqrt{17}}{2}\right) \left(x - \frac{1 + \sqrt{17}}{2}\right)}{x} \leq 0.$$

Of the four subintervals into which the number axis is partitioned by the three critical points $x = (1 - \sqrt{17})/2$, $x = 0$, $x = (1 + \sqrt{17})/2$, two subintervals, namely,

$$x < \frac{1 - \sqrt{17}}{2} \quad \text{and} \quad 0 < x < \frac{1 + \sqrt{17}}{2},$$

belong to the set of solutions. The right ends of these subintervals evidently also satisfy the inequality.

$$\text{Answer: } \left(-\infty; \frac{1 - \sqrt{17}}{2}\right] \cup [-1; 0) \cup \left(0; \frac{1 + \sqrt{17}}{2}\right] \cup [4; +\infty).$$

12. We immediately note that for $a = 3$ the inequality has no solutions. For all the other values of the parameter a the inequality is rational. Let us apply the method of intervals. We reduce the inequality to the standard form:

$$\frac{2a + 1 - x(a - 3)}{(a - 3)(x - 2)} \geq 0$$

or

$$\frac{x - \frac{2a + 1}{a - 3}}{x - 2} \leq 0.$$

The critical points $x = 2$ and $x = (2a + 1)/(a - 3)$ partition the number axis into three subintervals. The inequality is satisfied on the middle subinterval and at the point $x = (2a + 1)/(a - 3)$. Let us

determine which of the critical points is the left end of the middle subinterval. For that purpose we set up the difference

$$\frac{2a+1}{a-3} - 2 = \frac{7}{a-3}.$$

We can see now that for $a > 3$ the left end of the middle subinterval is the point $x = 2$; for $a < 3$, the point $x = (2a+1)/(a-3)$. Consequently, if $a > 3$, then the inequality is satisfied by all the values of x belonging to the subinterval

$$2 < x \leq (2a+1)/(a-3);$$

if $a < 3$, the inequality is satisfied by all the values of the variable belonging to the subinterval

$$\frac{2a+1}{a-3} \leq x < 2.$$

Answer: For $a > 3$, $x \in (2; (2a+1)/(a-3)]$,
for $a < 3$, $x \in [(2a+1)/(a-3); 2)$,
for $a = 3$, there are no solutions.

13. We transform the inequality as follows:

$$\begin{aligned} x + xy + y + yz + z + zx - 6\sqrt{xyz} &\geq 0, \\ (x - 2\sqrt{xy} + y) + (y - 2\sqrt{yz} + z) + (z - 2\sqrt{zx} + x) &\geq 0, \\ (\sqrt{x} - \sqrt{y})^2 + (\sqrt{y} - \sqrt{z})^2 + (\sqrt{z} - \sqrt{x})^2 &\geq 0. \end{aligned}$$

The validity of the last inequality is obvious. The transformation we have carried out have not violated the equivalence of the inequalities on the set of all nonnegative values of the variables. We have proved the inequality.

14. The left-hand side contains $2n+1$ summands. Each of them does not exceed $1/(n+1)$. Therefore,

$$\frac{1}{n+1} + \dots + \frac{1}{3n+1} \leq (2n+1) \frac{1}{n+1} < \frac{2n+2}{n+1} = 2.$$

15. In the inequality

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}$$

we set

$$x_1 = 1, x_2 = 2, x_3 = 2^2, \dots, x_n = 2^{n-1}.$$

We get

$$\begin{aligned} \frac{1+2+2^2+\dots+2^{n-1}}{n} &\geq \sqrt[n]{1 \cdot 2 \cdot 2^2 \dots 2^{n-1}}, \\ \frac{2^n-1}{n} &\geq \sqrt[n]{2^{1+2+\dots+(n-1)}}, \quad 2^n-1 \geq n \sqrt[n]{2^{(n-1)n/2}}, \end{aligned}$$

$$2^n-1 \geq n 2^{(n-1)/2}.$$

16. For any negative value of a the solution of the system is, for instance, $x = 1$. For $a = 0$ the system also has at least one solution since $x = 1$ is a solution of the system in this case as well. If $a > 0$,

then the system can be written as follows:

$$\begin{cases} x \leq 1/a, \\ x \geq 4a, \end{cases}$$

and for a solution to exist, it is necessary and sufficient that the inequality

$$4a \leq 1/a$$

should be satisfied, i.e. $a^2 \leq 1/4$, $a \leq 1/2$.

Answer: $(-\infty; 1/2]$.

17. Let us first consider the case when $a = 2$. Here $f(x) = 4x + 5$ and the only zero $x = -5/4$ of the function $f(x)$ belongs to the interval $(-2; 1)$. It follows that the value $a = 2$ satisfies the condition of the problem.

Suppose now that $a \neq 2$. Let us first make sure that the zeros of the function $f(x)$ do exist. For that it is necessary and sufficient that

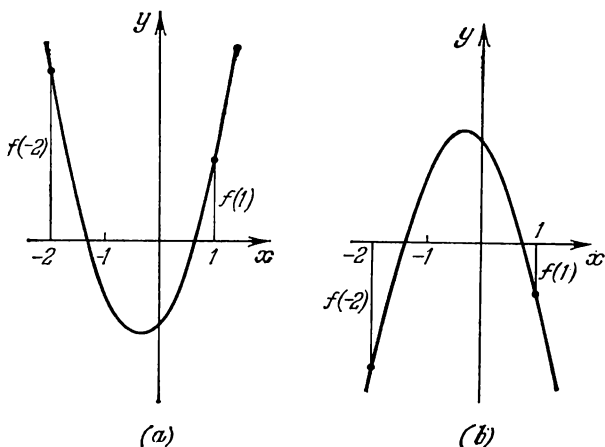


Fig. 131.

the discriminant of the quadratic trinomial $(a - 2)x^2 + 2ax + a + 3$ be nonnegative, i.e. $a^2 - (a - 2)(a + 3) \geq 0$, whence we have $a \leq 6$.

It is evident that the abscissa $x = \frac{-a}{a-2}$ of the vertex of the parabola $y = (a - 2)x^2 + 2ax + a + 3$ must belong to the interval $(-2; 1)$. Then, if $2 < a \leq 6$, then for the zeros of the function $f(x)$ to belong to the interval $(-2; 1)$, it is necessary and sufficient that the inequalities $f(-2) > 0$ and $f(1) > 0$ should be satisfied (see Fig. 131a). Now if $a < 2$, then for the zeros of the function $f(x)$ to lie on the interval $(-2; 1)$, it is necessary and sufficient that the inequalities $f(-2) < 0$ and $f(1) < 0$ be satisfied (Fig. 131b).

Thus it follows that in addition to the value $a = 2$, the conditions of the problems will be fulfilled by those and only those values

of a which satisfy at least one of the following two systems of equations:

$$\left\{ \begin{array}{l} 2 < a \leq 6, \\ -2 < \frac{-a}{a-2} < 1, \\ f(-2) = a-5 > 0, \\ f(1) = 4a+1 > 0, \end{array} \right. \quad \left\{ \begin{array}{l} a < 2, \\ -2 < \frac{-a}{a-2} < 1, \\ f(-2) = a-5 < 0, \\ f(1) = 4a+1 < 0. \end{array} \right.$$

The solution of the first system is $5 < a \leq 6$, and the second system is satisfied at $a < -1/4$.

Answer: $(-\infty; -1/4) \cup \{2\} \cup (5; 6]$.

18. We set $x = t - \frac{3}{2}$. Then we have

$$x(x+1)(x+2)(x+3) = \left(t^2 - \frac{9}{4}\right) \left(t^2 - \frac{1}{4}\right).$$

We have obtained a quadratic trinomial with respect to t^2 . Its least value is attained at

$$t^2 = \frac{\frac{9}{4} + \frac{1}{4}}{2} = \frac{5}{4}, \quad \text{i.e. } x = -\frac{3}{2} \pm \frac{\sqrt{5}}{2}.$$

The least value is equal to $\left(\frac{5}{4} - \frac{9}{4}\right) \left(\frac{5}{4} - \frac{1}{4}\right) = -1$.

Answer: For $x = (-3 \pm \sqrt{5})/2$ the polynomial assumes the least value equal to -1 .

19. Let us designate the length of the northern side as x and the length of the eastern side as y . Then $xy = 9000$ (sq m) and the cost S (in rubles) of the construction will be $S = 10x + 4y$. Let us find the least value of S :

$$S = 10x + 4y = 10x + \frac{4 \cdot 9000}{x} = 10 \left(x + \frac{3600}{x}\right).$$

Using the inequality

$$x + \frac{a}{x} \geq 2\sqrt{a}, \quad a > 0, \quad x \in (0; +\infty).$$

we find that

$$S \geq 10 \cdot 2 \sqrt{3600} = 1200,$$

the equality sign being legitimate only for

$$x = \sqrt{a} = \sqrt{3600} = 60.$$

Answer: The sum allocated for the construction will be enough only if the northern side of the sportsground is 60 metres long.

20. Let us designate the distance between points A and B as a . By the cosine theorem, we shall find from $\triangle LBM$ (Fig. 132) the distance $s(t)$ between the cars at the time moment t after the beginning of

product attains its greatest value when all the factors are equal, that is, if

$$a - 2x = a - 2x = 4x,$$

then $x = a/6$.

Answer: The side of the squares being cut off must equal $a/6$.

CHAPTER 5

1. Note that $\left| \frac{n^2-1}{n^2+1} - 1 \right| = \frac{2}{n^2+1} < \frac{2}{n^2}$, and the inequality $2/n^2 < \varepsilon$ is satisfied for all n complying with the condition $n > \sqrt{2/\varepsilon}$. For any $\varepsilon > 0$ we take $N = [\sqrt{2/\varepsilon}] + 1$ ($N > \sqrt{2/\varepsilon}$), then for all $n > N$ the following inequality is satisfied:

$$\left| \frac{n^2-1}{n^2+1} - 1 \right| < \frac{2}{n^2} < \frac{2}{N^2} < \varepsilon.$$

Taking into account the relationship between N and ε we have established, we fill in the following table:

ε	1/10	1/100	1/1000
N	5	15	45
	$\left \frac{n^2-1}{n^2+1} - 1 \right < \frac{1}{10}$	$\left \frac{n^2-1}{n^2+1} - 1 \right < \frac{1}{100}$	$\left \frac{n^2-1}{n^2+1} - 1 \right < \frac{1}{1000}$

2. For any positive number M we take $n = 2 ([M] + 1)$. Then, for the element of the sequence bearing this number, with due account of n being even, we obtain

$$|n^{(-1)^n}| = 2 ([M] + 1) > M.$$

3. Let us assume the contrary, that is, suppose that there is a number a which is a limit of the sequence (a_n) . By the definition of a limit, for $\varepsilon = 1/3$ there is a number N such that $|a_n - a| < 1/3$ for all $n > N$. Hence it follows that for any numbers n_1 and n_2 such that $n_1 > N$ and $n_2 > N$, the following inequality must hold:

$$|a_{n_1} - a_{n_2}| \leq |a_{n_1} - a| + |a_{n_2} - a| < 2/3.$$

Taking $n_1 = 2N$ and $n_2 = 2N + 1$, however, we get

$$|a_{n_1} - a_{n_2}| = \left| \frac{2N}{N+1} - 0 \right| = \frac{2N}{N+1} \geq 1.$$

The contradiction we have obtained proves that our assumption is false, i.e. the given sequence is divergent.

4. For any $\varepsilon > 0$ the inequality $1/n^\alpha < \varepsilon$ is satisfied for any natural n complying with the condition $n > \frac{1}{\varepsilon^{1/\alpha}}$. Thus, for any $\varepsilon > 0$

we can take $N = \left\lceil \frac{1}{\varepsilon^{1/\alpha}} \right\rceil + 1$, then at all $n > N$ we must have

$$\left| \frac{1}{n^\alpha} - 0 \right| = \frac{1}{n^\alpha} < \frac{1}{N^\alpha} < \varepsilon.$$

5. Assume $\lim_{n \rightarrow \infty} b_n = 0$. Then, for any $\varepsilon > 0$ there is an N such that $|b_n| < \varepsilon$ for all $n > N$. By the hypothesis, $|a_n| \leq |b_n|$ for $n = 1, 2, \dots$, then for $n > N$ the inequality $|a_n| < \varepsilon$ will be satisfied as well, which means that $\lim_{n \rightarrow \infty} a_n = 0$.

6. Since

$$\begin{aligned} |\sqrt{n+1} - \sqrt{n}| &= \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n+1} + \sqrt{n}} \\ &= \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{\sqrt{n}} \end{aligned}$$

and $\lim_{n \rightarrow \infty} (1/\sqrt{n}) = 0$ (this follows from problem 4 for $\alpha = 1/2$), this means, as follows from the preceding problem, that $\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0$.

7. Let us consider a new sequence (b_n) such that $b_n = a_n + x$ ($n = 1, 2, \dots$) and choose x such that the sequence (b_n) be defined as simply as possible. We substitute $a_n = b_n - x$ into the recurrence relation specifying a_n and get $b_{n+1} - x = \frac{1}{3}(b_n - x) + \frac{1}{3}$ or $b_{n+1} = \frac{1}{3}b_n + \frac{1}{3} + \frac{2}{3}x$. If we take $x = -1/2$, then $b_{n+1} = \frac{1}{3}b_n$, that is, the sequence (b_n) is a geometric progression with the common ratio $q = 1/3$ and the first term $b_1 = a_1 + x = 3/2$. Thus we have

$$b_n = b_1 q^{n-1} = \frac{1}{2} \left(\frac{1}{3} \right)^{n-2} \text{ and } a_n = \frac{1}{2} \left(\frac{1}{3} \right)^{n-2} + \frac{1}{2}.$$

We easily find that $\lim_{n \rightarrow \infty} a_n = \frac{1}{2}$ since $\lim_{n \rightarrow \infty} \left(\frac{1}{3} \right)^n = 0$.

8. By the formula for the sum of the first k terms of a geometric progression, we have, at $k = n + 1$,

$$1 + a + a^2 + \dots + a^n = \frac{1 - a^{n+1}}{1 - a}, \quad 1 + b + b^2 + \dots + b^n = \frac{1 - b^{n+1}}{1 - b}.$$

Therefore taking into account that $|a| < 1$ and $|b| < 1$, we get

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1 + a + a^2 + \dots + a^n}{1 + b + b^2 + \dots + b^n} &= \lim_{n \rightarrow \infty} \frac{(1 - b)(1 - a^{n+1})}{(1 - a)(1 - b^{n+1})} \\ &= \frac{1 - b}{1 - a} \lim_{n \rightarrow \infty} \frac{1 - a^{n+1}}{1 - b^{n+1}} = \frac{1 - b}{1 - a}. \end{aligned}$$

9. The radicand is the sum of the first n terms of the arithmetic progression with the first term $a_1 = 1$ and the difference $d = 2$. By the formula for the sum of an arithmetic progression, we have

$$1 + 3 + \dots + (2n-1) = \frac{1 + (2n-1)}{2} n = n^2,$$

and, therefore, $\frac{n \sqrt{1+3+\dots+(2n-1)}}{2n^2+n+1} = \frac{n^2}{2n^2+n+1}$ and

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{n \sqrt{1+3+\dots+(2n-1)}}{2n^2+n+1} &= \lim_{n \rightarrow \infty} \frac{n^2}{2n^2+n+1} \\ &= \lim_{n \rightarrow \infty} \frac{1}{2 + \frac{1}{n} + \frac{1}{n^2}} = \frac{1}{2}. \end{aligned}$$

10. It is evident that $\lim_{n \rightarrow \infty} \sqrt[n]{a} = 1$ for $a = 1$.

Suppose $a > 1$. Then, by the property of inequalities we have $\sqrt[n]{a} > 1$. Let us introduce the notation $h = \sqrt[n]{a} - 1$. The Bernoulli inequality $(1+h)^n \geq 1+nh$ implies that $((1+h)^n - 1)/n \geq h$. Substituting here $h = \sqrt[n]{a} - 1$, we get $(a-1)/n \geq \sqrt[n]{a} - 1$ and since $\sqrt[n]{a} - 1 > 0$, it follows that $|\sqrt[n]{a} - 1| \leq (a-1)/n$. For any $\varepsilon > 0$ we take $N = [(a-1)/\varepsilon] + 1$ and then, for all $n > N$, we shall have $|\sqrt[n]{a} - 1| \leq (a-1)/n < (a-1)/N < \varepsilon$ (since $N > (a-1)/\varepsilon$). And this means that $\lim_{n \rightarrow \infty} \sqrt[n]{a} = 1$.

For $0 < a < 1$ we have $1/a > 1$, and it follows from what has been proved that $\lim_{n \rightarrow \infty} \sqrt[n]{1/a} = 1$. Therefore we have $\lim_{n \rightarrow \infty} \sqrt[n]{a} =$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{1/a}} = \frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{1/a}} = 1.$$

11. It follows from the definition of a limit that for any $\varepsilon > 0$ there is an N_1 such that $|a_n - a| < \varepsilon$ at all $n > N_1$, i.e. $a - \varepsilon < a_n < a + \varepsilon$ and there can be found an N_2 such that $|b_n - a| < \varepsilon$ at all $n > N_2$, i.e. $a - \varepsilon < b_n < a + \varepsilon$. Then, if $N > N_1$ and $N > N_2$, the inequalities $a - \varepsilon < a_n$ and $b_n < a + \varepsilon$ hold true simultaneously. From the condition $a_n \leq x_n \leq b_n$ we find that $a - \varepsilon < a_n \leq x_n \leq b_n < a + \varepsilon$ for $n > N$, i.e. $|x_n - a| < \varepsilon$. This means that $\lim_{n \rightarrow \infty} x_n = a$.

It follows, in particular, from what we have proved that if for any n the inequalities $a \leq x_n \leq b_n$ are satisfied and $\lim_{n \rightarrow \infty} b_n = a$, then

$$\lim_{n \rightarrow \infty} x_n = a.$$

12. It is evident that $\sqrt[n]{3^n + 5^n} = 5 \sqrt[n]{(3/5)^n + 1}$. Furthermore, the inequalities $1 < (3/5)^n + 1 < 8/5$ hold for any n , whence it follows that $1 < \sqrt[n]{(3/5)^n + 1} < \sqrt[n]{8/5}$. The solution of problem 10 yields $\lim_{n \rightarrow \infty} \sqrt[n]{8/5} = 1$. Then it follows from the solution of problem 11

that $\lim_{n \rightarrow \infty} \sqrt[n]{(3/5)^n + 1} = 1$. Thus we have

$$\lim_{n \rightarrow \infty} \sqrt[n]{3^n + 5^n} = 5 \quad \lim_{n \rightarrow \infty} \sqrt[n]{(3/5)^n + 1} = 5.$$

13. Let us prove that for any n there holds the inequality

$$1 < a_n < 2. \quad (1)$$

This inequality is valid for $n = 1$ since $a_1 = \sqrt{2}$. Assume that inequalities (1) are also true for $n = k \geq 1$. From the recurrence formula $a_{k+1} = \sqrt{2 + a_k}$ and the condition $a_n > 1$ it follows that $a_{k+1} > \sqrt{2 + 1} > 1$ and from the condition $a_n < 2$ we get $a_{k+1} < \sqrt{2 + 2} = 2$. Thus we have $1 < a_{k+1} < 2$ and, in accordance with the principle of mathematical induction, inequality (1) is valid for any n .

Let us now show that $a_{n+1} > a_n$, i.e. $\sqrt{2 + a_n} > a_n$. Since $a_n > 0$, it follows that the given inequality is equivalent to the inequality $2 + a_n > a_n^2$ or $a_n^2 - a_n - 2 < 0$. The quadratic function $f(x) = x^2 - x - 2$ turns into zero for $x = -1$ and $x = 2$ and is negative on the interval $(-1; 2)$. Since $a_n \in (1; 2)$ for any n , we have $a_n^2 - a_n - 2 < 0$ and, consequently, $\sqrt{2 + a_n} > a_n$. Thus it follows that the sequence (a_n) is bounded and monotonic. According to the Weierstrass theorem it has a limit, which we shall denote by

The equation $a_{n+1} = \sqrt{2 + a_n}$ yields

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \sqrt{2 + a_n} = \sqrt{2 + \lim_{n \rightarrow \infty} a_n},$$

i.e. $a = \sqrt{2 + a}$. From this we find $a = 2$ ($a = -1$ does not, evidently, suit since $1 < a_n < 2$).

14. Assume that a is the initial term and d , the difference of an arithmetic progression, S_k is the sum of the first k terms of that progression, and n is the number of its terms. The conditions of the problem can be written as follows:

$$\frac{S_{13}}{S_n - S_{n-13}} = \frac{1}{2}, \quad \frac{S_n - S_3}{S_{n-3}} = \frac{4}{3}.$$

Using formula (6) of 5.4 for S_k we obtain from this

$$a + 19d = nd,$$

$$-2a + 22d = nd.$$

Multiplying the first equation by 2 and adding it to the second equation term-by-term, we obtain $60d = 3nd$, whence we find $n = 20$.

15. Assume that a is the initial term and d is the difference of an arithmetic progression, n being the number of its terms, and S_n its sum.

$$S_n = \frac{2a + d(n-1)}{2} n.$$

The sum $\tilde{S}_n$ of n terms of the arithmetic progression with the first term a and the difference $d + 3$ is

$$\tilde{S}_n = \frac{2a + (d+3)(n-1)}{2} n,$$

and the sum of n terms of the arithmetic progression with the first term $4a$ and the difference d is equal to

$$\tilde{\tilde{S}}_n = \frac{8a + d(n-1)}{2} n.$$

By the hypothesis, $\tilde{S}_n = 2S_n = \tilde{\tilde{S}}_n$. Hence we have

$$4a = d(n-1) \text{ and } 2a = (n-1)(3-d),$$

whence we find that $d = 2$ since $a \neq 0$.

16. Assume that a is the first term and q is the common ratio of an infinitely decreasing geometric progression and suppose that $a_n = 1/6$.

The sum of all terms of the progression preceding that term is equal to the sum of the first $n-1$ terms according to formula (5) in 5.5:

$$S_{n-1} = a \frac{1-q^{n-1}}{1-q}.$$

The sum of all terms succeeding the term a_n is equal to the sum of an infinitely decreasing geometric progression with the initial term aq^n

and the common ratio q , that is, to $aq^n \frac{1}{1-q}$. Thus we have

$$a \frac{1-q^{n-1}}{1-q} : a \frac{q^n}{1-q} = 30. \quad (1)$$

In addition, by the hypothesis, $a/(1-q) = 16/3$ and $aq^{n-1} = 1/6$. Removing a , we have from the last two equations

$$q^{n-1}(1-q) = \frac{1}{32} \quad \text{or} \quad q^{n-1} = \frac{1}{32} + q^n,$$

and formula (1) yields $30q^n = 1 - q^{n-1}$. Substituting here the expression for q^{n-1} , we get

$$30q^n = \frac{31}{32} - q^n, \quad \text{i.e.} \quad 31q^n = \frac{31}{32}, \quad q^n = \frac{1}{32}.$$

Then we have $q^{n-1} = \frac{1}{32} + q^n = \frac{1}{16}$. Now we easily find that $q = 1/2$ and $n = 5$.

17. By definition,

$$0.2(7) = \frac{2}{10} + \frac{7}{100} + \frac{7}{1000} + \dots = \frac{2}{10} + \frac{7}{100} \left(1 + \frac{1}{10} + \frac{1}{100} + \dots \right).$$

Using the formula for the sum of an infinitely decreasing geometric progression, we obtain

$$0.2(7) = \frac{2}{10} + \frac{7}{100} \cdot \frac{1}{1 - \frac{1}{10}} = \frac{2}{10} + \frac{7}{90} = \frac{5}{18}.$$

By analogy we get

$$\begin{aligned} 0.1(63) &= \frac{1}{10} + \frac{63}{1000} + \frac{63}{100000} + \dots \\ &= \frac{1}{10} + \frac{63}{1000} \left(1 + \frac{1}{100} + \frac{1}{10000} + \dots \right) \\ &= \frac{1}{10} + \frac{63}{1000} \cdot \frac{1}{1 - \frac{1}{100}} = \frac{1}{10} + \frac{63}{990} = \frac{9}{55}. \end{aligned}$$

Thus it follows that

$$0.2(7) \cdot 0.1(63) = 1/22.$$

18. The function $\sin(1/x)$ being defined for all $x \neq 0$ and $|\sin(1/x)| \leq 1$, it follows that

$$|x \sin(1/x) - 0| = |x \sin(1/x)| \leq |x|$$

and the inequality $|x \sin(1/x) - 0| < \varepsilon$, evidently holds for all x such that $0 < |x| < \varepsilon$. Thus, for any $\varepsilon > 0$ we take $\delta = \varepsilon$; then for all x satisfying the condition $0 < |x| < \delta$ the following equality holds true:

$$|x \sin(1/x) - 0| \leq |x| < \delta = \varepsilon.$$

$$19. \lim_{x \rightarrow 2} \frac{x+2}{\sqrt{x+3}-1} = \frac{\lim_{x \rightarrow 2} x+2}{\lim_{x \rightarrow 2} \sqrt{x+3}-1} = \frac{4}{\sqrt{5}-1}.$$

$$20. \lim_{x \rightarrow 1} \frac{x^3-1}{x-1} = \lim_{x \rightarrow 1} \frac{(x-1)(x^2+x+1)}{x-1} = \lim_{x \rightarrow 1} (x^2+x+1) = 3.$$

$$\begin{aligned} 21. \lim_{x \rightarrow -1} \frac{2x^2+x-1}{x^3-6x-7} &= \lim_{x \rightarrow -1} \frac{(2x-1)(x+1)}{(x-7)(x+1)} \\ &= \lim_{x \rightarrow -1} \frac{2x-1}{x-7} = \frac{3}{8}. \end{aligned}$$

22. Since $\frac{\cos 2x}{\sin x + \cos x} = \frac{(\cos^2 x - \sin^2 x)}{\sin^2 x + \cos^2 x} = \cos x - \sin x$ (provided that $\sin x + \cos x \neq 0$) and, as was mentioned, the functions $\cos x$ and $\sin x$ are continuous at any point, it follows that

$$\begin{aligned} \lim_{x \rightarrow -\frac{\pi}{4}} \frac{\cos 2x}{\sin x + \cos x} &= \lim_{x \rightarrow -\frac{\pi}{4}} (\cos x - \sin x) \\ &= \cos\left(-\frac{\pi}{4}\right) - \sin\left(-\frac{\pi}{4}\right) = \sqrt{2}. \end{aligned}$$

23. Since $|x| = \begin{cases} x & \text{for } x > 0, \\ -x & \text{for } x < 0, \end{cases}$ it follows that

$$\frac{x - |x|}{2x} = \begin{cases} 0 & \text{for } x > 0, \\ 1 & \text{for } x < 0. \end{cases}$$

It is evident that $\lim_{x \rightarrow +0} \frac{x - |x|}{2x} = 0$ and $\lim_{x \rightarrow -0} \frac{x - |x|}{2x} = 1$.

24. If $x > 1$, then $|x - 1| = x - 1$ and

$$f(x) = \frac{x^2 - 1}{x^2 - 2x + 1} = \frac{(x+1)(x-1)}{(x-1)} = \frac{x+1}{x-1}.$$

It is evident that $f(x) > 0$ for $x > 1$ and $\lim_{x \rightarrow 1+0} \frac{1}{f(x)} = \lim_{x \rightarrow 1} \frac{x-1}{x+1} = 0$ and, consequently,

$$\lim_{x \rightarrow 1+0} f(x) = +\infty.$$

Now if $x < 1$, then $|x - 1| = -x + 1$ and

$$f(x) = \frac{x^2 - 1}{x^2 + 2x - 3} = \frac{(x-1)(x+1)}{(x-1)(x+3)} = \frac{x+1}{x+3},$$

therefore

$$\lim_{x \rightarrow 1-0} f(x) = \lim_{x \rightarrow 1} \frac{x+1}{x+3} = \frac{1}{2}.$$

$$25. \text{ Since } \frac{(x-1)^2(7x+2)^2}{(2x+1)^4} = \frac{\left(1 - \frac{1}{x}\right)^2 \left(7 + \frac{2}{x}\right)^2}{\left(2 + \frac{1}{x}\right)^4} \text{ and } \lim_{x \rightarrow \infty} \frac{1}{x} =$$

0, it follows, as it is easy to see, that

$$\lim_{x \rightarrow \infty} \frac{(x-1)^2(7x+2)^2}{(2x+1)^4} = \frac{49}{16}.$$

26. Since

$$\begin{aligned} \sqrt{x + \sqrt{x}} - \sqrt{x} &= \frac{(\sqrt{x + \sqrt{x}} - \sqrt{x})(\sqrt{x + \sqrt{x}} + \sqrt{x})}{\sqrt{x + \sqrt{x}} + \sqrt{x}} \\ &= \frac{\sqrt{x}}{\sqrt{x + \sqrt{x}} + \sqrt{x}} = \frac{1}{\sqrt{1 + \sqrt{\frac{1}{x}}} + 1}, \end{aligned}$$

we can easily get

$$\lim (\sqrt{x + \sqrt{x}} - \sqrt{x}) = \frac{1}{2}.$$

27. The function $\sin x$ cannot, evidently, have an infinite limit since $|\sin x| \leq 1$. We shall show that this function does not possess a finite limit either as $x \rightarrow +\infty$. Let us suppose the contrary, that is, assume that $b = \lim_{n \rightarrow +\infty} \sin x$. By the definition of a limit at infinity,

for any $\varepsilon > 0$ and, consequently, for $\varepsilon = 1/2$, there is a number $M > 0$ such that $|\sin x - b| < 1/3$ for all x such that $x > M$. Hence it follows that if $x_1 > M$ and $x_2 > M$, then

$$|\sin x_1 - \sin x_2| \leq |\sin x_1 - b| + |\sin x_2 - b| < 2/3.$$

However, if we choose $n = [M] + 1$ ($n > M$) and take $x_1 = \pi n$ ($x_1 > M$) and $x_2 = \frac{\pi}{2} + 2\pi n$ ($x_2 > M$), then we have

$$|\sin x_1 - \sin x_2| = \left| \sin \pi n - \sin \left(\frac{\pi}{2} + 2\pi n \right) \right| = 1 > 2/3.$$

The contradiction obtained proves that the function $\sin x$ does not have a limit as $x \rightarrow +\infty$.

28. The function $z(x) = \frac{1}{\sqrt{x}}$ can be regarded as a composite function $z = u^{-1}$, where $u(x) = \sqrt{x}$. Knowing that $z' = -1/u^2$ and $u' = 1/(2\sqrt{x})$, we get, according to the rule of differentiation of a composite function, $(1/\sqrt{x})' = z'u' = -1/(2x\sqrt{x})$.

The derivative of the function $y(x) = (v(x))^{50}$, where $v(x) = \sqrt{x} + \frac{1}{\sqrt{x}}$ can also be found by the rule of differentiation of a composite function: $y'(x) = 50v^{49}v'$. Since

$$v' = (\sqrt{x})' + \left(\frac{1}{\sqrt{x}} \right)' = \frac{1}{2\sqrt{x}} - \frac{1}{2x\sqrt{x}} = \frac{1}{2\sqrt{x}} \left(1 - \frac{1}{x} \right),$$

it follows that

$$\begin{aligned} y'(x) &= 50 \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^{49} \cdot \frac{1}{2\sqrt{x}} \left(1 - \frac{1}{x} \right) \\ &= 25 \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^{48} \left(1 - \frac{1}{x^2} \right). \end{aligned}$$

29. If there exists the derivative $f'(x_0)$, then the equation of the tangent to the graph of the function $y = f(x)$ at the point with the abscissa x_0 has the form $y = f'(x_0)(x - x_0) + f(x_0)$. Since

$$f'(x) = \frac{4-2x}{2\sqrt{4x-3-x^2}}$$

we can substitute $x_0 = 3/2$, $f(x_0) = \sqrt{3}/2$, and $f'(x_0) = 1/\sqrt{3}$ into the equation of the tangent and obtain $y = \frac{1}{\sqrt{3}}x$.

30. Suppose that $f_1(x) = x^2 - 2x + 5$ and $f_2(x) = x^2 + 2x - 11$ and let the straight line $y = kx + l$ touch the graphs of the functions

$y = f_1(x)$ and $y = f_2(x)$ at the points x_1 and x_2 respectively. Then the equation of that line can be written in two forms:

$$y = f'_1(x_1)(x - x_1) + f_1(x_1),$$

$$y = f'_2(x_2)(x - x_2) + f_2(x_2).$$

Hence it follows that

$$k = f'_1(x_1) = f'_2(x_2),$$

$$l = f_1(x_1) - x_1 f'_1(x_1) = f_2(x_2) - x_2 f'_2(x_2).$$

For the given functions $f_1(x)$ and $f_2(x)$ we get the system

$$\begin{cases} 2x_1 - 2 = 2x_2 + 2 \\ 5 - x_1^2 = -11 - x_2^2 \end{cases} \quad \text{or} \quad \begin{cases} x_2 = x_1 - 2, \\ x_1^2 - x_2^2 = 16, \end{cases}$$

whose solution yields $x_1 = 5$ and $x_2 = 3$. Then we have $k = 8$, $l = -20$. The equation of the common tangent is $y = 8x - 20$.

CHAPTER 6

1. The function $f_1(x)$ is neither even nor odd since, for instance, $f(1) = \sqrt{2}$, $f(-1) = \sqrt{12}$.

The function $f_2(x)$ is neither even nor odd since it is defined only for $x \geq 2/3$.

The function $f_3(x)$ is defined throughout the real axis and is odd, being the sum of two odd functions x^3 and $3 \sin x$.

The function $f_4(x)$ is defined on the entire real axis and is even being the difference between two even functions x^4 and $5 \cos x$.

The function $f_5(x)$ is defined throughout the real axis. It is neither even nor odd since, for instance, $f_5(\ln 2) = -1/6$, $f_5(-\ln 2) = 2/3$.

The function $f_6(x)$ is defined for any $x \in \mathbb{R}$. Since

$$-x + \sqrt{x^2 + 1} = \frac{-x^2 + x^2 + 1}{x + \sqrt{x^2 + 1}} = \frac{1}{x + \sqrt{x^2 + 1}},$$

for any x , it follows that

$$f_6(-x) = \ln(-x + \sqrt{x^2 + 1}) = -\ln(x + \sqrt{x^2 + 1}) = -f_6(x).$$

Consequently, $f_6(x)$ is an odd function.

2. (a) The function $f(x)$ is periodic, its period being any positive number. Indeed, suppose T is a positive rational number. Then, if x is rational, then $x + T$ is also rational and, hence, $f(x) = 1 = f(x + T)$. Now if x is irrational, then $x + T$ is irrational and, therefore, $f(x) = 0 = f(x + T)$.

No irrational positive number T can be a period of that function since $f(-T) = 0$ but $f(-T + T) = f(0) = 1$.

Let us show that the function $f(x)$ does not have the least period. We shall assume that T_0 is the least positive rational number which is a period of the function $f(x)$. But the number $T_0/2 < T_0$ is also a period of that function. Consequently, our supposition concerning the existence of the least positive period is erroneous.

(b) The functions $\cos x$ and $\sin x$ being periodic with a period 2π , the function $f(x)$ is also periodic and the number 2π is its period.

Let us determine the least period of the function. We have

$$f(x) = \cos x \sin x (\cos^2 x - \sin^2 x) = \frac{1}{2} \sin 2x \cos 2x = \frac{1}{4} \sin 4x.$$

Consequently, the least period of the function $f(x)$ is equal to $\pi/2$.

3. (a) Since $f(x) \rightarrow \infty$ as $x \rightarrow -3$ and as $x \rightarrow +3$, it follows that the straight lines $x = -3$ and $x = 3$ are vertical asymptotes. Since

$$f(x) = \frac{x^2 - 9 + 13}{x^2 - 9} = 1 + \frac{13}{x^2 - 9} \rightarrow 1$$

as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$ the straight line $y = 1$ is an asymptote to the graph of the function $f(x)$ as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$.

(b) The line $x = 1$ is a vertical asymptote since $\lim_{x \rightarrow 1} f(x) = \infty$.

Furthermore, the function $f(x)$ can be represented in the form

$$f(x) = \frac{x^2 - 3}{x - 1} = \frac{x^2 - x + x - 1 - 2}{x - 1} = x + 1 - \frac{2}{x - 1}.$$

Since $\lim_{x \rightarrow \infty} \frac{2}{x - 1} = 0$, the straight line $y = x + 1$ is an asymptote as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$.

(c) The function $f(x)$ is defined for $x \geq -2$ and for $x \leq -4$. There are no vertical asymptotes to the graph of the given function. To find the asymptotes as $x \rightarrow +\infty$ and $x \rightarrow -\infty$, we shall make use of formulas (2) and (3) in 6.3. In accordance with formula (3),

$$k = \lim_{x \rightarrow +\infty} \frac{\sqrt{x^2 + 6x + 8}}{x} = \lim_{x \rightarrow +\infty} \sqrt{1 + \frac{6}{x} + \frac{8}{x^2}} = 1.$$

From formula (2) we have

$$\begin{aligned} b &= \lim_{x \rightarrow +\infty} (\sqrt{x^2 + 6x + 8} - x) \\ &= \lim_{x \rightarrow +\infty} \frac{(\sqrt{x^2 + 6x + 8} - x)(\sqrt{x^2 + 6x + 8} + x)}{\sqrt{x^2 + 6x + 8} + x} \\ &= \lim_{x \rightarrow +\infty} \frac{x^2 + 6x + 8 - x^2}{x \left(1 + \sqrt{1 + \frac{6}{x} + \frac{8}{x^2}} \right)} \\ &= \lim_{x \rightarrow +\infty} \frac{6 + \frac{8}{x}}{1 + \sqrt{1 + \frac{6}{x} + \frac{8}{x^2}}} = 3. \end{aligned}$$

It follows that the line $y = x + 3$ is an asymptote to the graph of the function $f(x)$ as $x \rightarrow +\infty$. Similar argument may be used to show that the line $y = -x - 3$ is an asymptote as $x \rightarrow -\infty$.

4. (a) The function $f(x)$ is defined on the entire real axis and is neither even nor odd. Nor is it periodic or have asymptotes. It follows

from the definition of the absolute value that

$$f(x) = \begin{cases} x \left(\frac{4}{3}x - 2 \right), & \text{if } x \geq 1, \\ -\frac{2}{3}x^2, & \text{if } x < 1. \end{cases}$$

The graph of the function $y = -\frac{2}{3}x^2$ is a parabola whose vertex lies at the origin and which is concave down. The function $f(x)$ increases

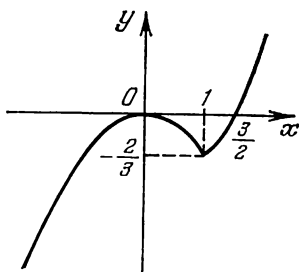


Fig. 133.

on the infinite interval $(-\infty; 0)$ and decreases on the interval $(0, 1)$. The point $x = 0$ is a point of maximum of the function $f(x)$. The graph of the function $y = x \left(\frac{4}{3}x - 2 \right)$ is a parabola which cuts the Ox axis at the points $x = 0$ and $x = 3/2$. Consequently, the abscissa x of the vertex of the parabola is equal to $3/4$ and the ordinate y , to $-3/4$. It follows that the function $f(x)$ increases for $x \geq 1$. The point $x = 1$ is a point of minimum because at that point the function $f(x)$

is continuous and decreases on the interval $(0; 1)$, while on the interval $(1; +\infty)$ it increases. The graph of the function $f(x)$ is shown in Fig. 133.

(b) The function $f(x) = \log(\sin x)$ is defined for those values of x for which $\sin x > 0$, i.e. for $x \in (2\pi n; 2\pi n + \pi)$, $n \in \mathbb{Z}$.

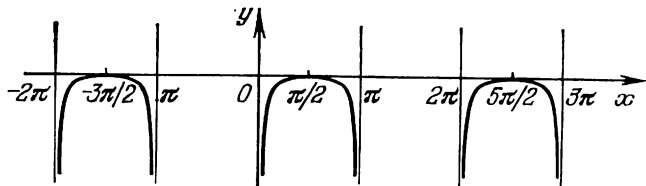


Fig. 134.

The function $\sin x$ being periodic with a period 2π , the function $\log(\sin x)$ is also periodic with the same period. Therefore, it suffices to construct the graph of the function on the interval $(0, \pi)$ and then complete the graph of the function at all the points at which the function is defined. The straight lines $x = 0$ and $x = \pi$ are vertical asymptotes because $f(x) \rightarrow -\infty$ as $x \rightarrow +0$ and as $x \rightarrow \pi - 0$. On the interval $(0; \pi/2)$ the function $\sin x$ increases from 0 to 1 and the function $\log(\sin x)$ increases from $-\infty$ to 0. On the interval $(\pi/2; \pi)$ the function $\sin x$ decreases from 1 to 0 and the function $\log(\sin x)$ decreases from 0 to $-\infty$. The point $\pi/2$ is a point of maximum for the given function. The graph of the function $f(x)$ is shown in Fig. 134.

(c) The function $f(x) = 2^{\cos x}$ is defined on the entire real axis. The function $\cos x$ being even, the function $f(x)$ is also even. The function $\cos x$ is periodic with a period 2π , therefore, the function $f(x)$ is also periodic with the same period. Consequently, it is sufficient to construct the graph of the function on the interval $[0; 2\pi]$.

Note that the graph of the function $f(x)$ possesses no vertical asymptotes.

On the interval $[0; \pi]$ the function $\cos x$ decreases from 1 to -1 and, therefore, the function $f(x)$ decreases from 2 to $1/2$. On the interval

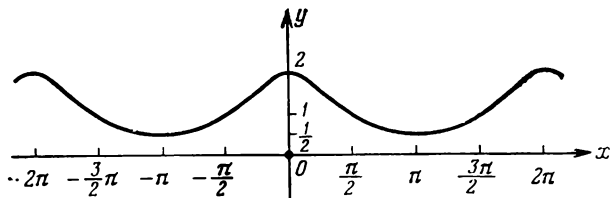


Fig. 135.

$[\pi; 2\pi]$ the function $\cos x$ increases from -1 to 1 and the function $f(x)$ increases from $1/2$ to 2. At the points $x = 2\pi n$, $n \in \mathbb{Z}$, the function $f(x)$ possesses maxima, equal to 2, and at the points $x = 2\pi n + \pi$, $n \in \mathbb{Z}$, the function $f(x)$ possesses minima, equal to $1/2$. The graph of the function $f(x) = 2^{\cos x}$ is shown in Fig. 135.

5. (a) The function $f(x) = \frac{(2-x)^3}{(x-3)^2}$ is defined for all $x \neq 3$.

It is neither even nor odd, nor periodic. Its graph cuts the Ox axis at the point $x = 2$ and the Oy axis at the point $y = 8/9$. Since $f(x) \rightarrow -\infty$ as $x \rightarrow 3$, the straight line $x = 3$ is a vertical asymptote to the graph of the function $f(x)$. Note that

$$f(x) = \frac{8 - 12x + 6x^2 - x^3}{x^2 - 6x + 9} = -x + \frac{8 - 3x}{x^2 - 6x + 9},$$

therefore, the line $y = -x$ is an inclined asymptote to the graph of the function as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$. Let us find the derivative

$$f'(x) = \frac{3(2-x)^2(-1)(x-3)^2 - (2-x)^3 2(x-3)}{(x-3)^4} = \frac{(2-x)^2(5-x)}{(x-3)^3}.$$

The points $x = 2$ and $x = 5$ are stationary. On the infinite interval $(-\infty; 2)$ the derivative $f'(x)$ is negative and, consequently, the function $f(x)$ decreases from $-\infty$ to 0. On the interval $(2; 3)$ the derivative is negative and, therefore, the function decreases from 0 to $-\infty$. On the interval $(3; 5)$ the derivative is positive and the function $f(x)$ increases from $-\infty$ to $-27/4$. And, finally, on the infinite interval $(5; +\infty)$ the derivative is negative and the function $f(x)$ decreases from $-27/4$ to $-\infty$. Hence it follows that the point $x = 5$ is a point of maximum. We tabulate all the data obtained as follows;

x	$(-\infty; 2)$	2	$(2; 3)$	$(3; 5)$	5	$(5; +\infty)$
y	$(+\infty; 0)$	0	$(0; -\infty)$	$(-\infty; -\frac{27}{4})$	$-\frac{27}{4}$	$(-\frac{27}{4}; -\infty)$
y'	-	0	-	+	0	-
	$\searrow$		$\searrow$	$\nearrow$	max	$\searrow$

The graph of the function $f(x)$ is shown in Fig. 136.

(b) The function $f(x) = x + e^{-x}$ is defined for all $x \in \mathbb{R}$. It is neither even nor odd, nor periodic. Its graph cuts the axis of ordinates at the point $y = 1$. It does not intersect the Ox axis. There are no vertical asymptotes. Since $\lim_{x \rightarrow +\infty} e^{-x} = 0$, the line $y = x$ is an inclined asymptote as $x \rightarrow +\infty$. As $x \rightarrow -\infty$ there is no inclined asymptote because $\frac{f(x)}{x} \rightarrow -\infty$ as $x \rightarrow -\infty$. Let us find the derivative

$$f'(x) = 1 - e^{-x}.$$

The point $x = 0$ is stationary and the derivative vanishes at that point. It is easy to see that $f'(x) < 0$ for $x < 0$ and $f'(x) > 0$ for $x > 0$.

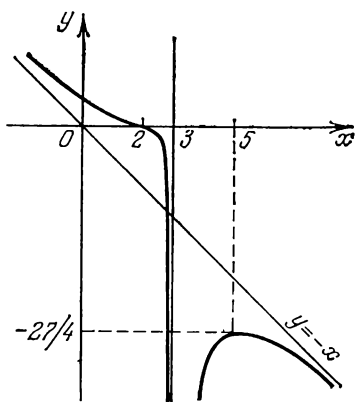


Fig. 136.

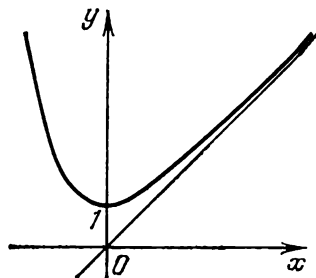


Fig. 137.

Therefore, the function $f(x)$ decreases on the infinite interval $(-\infty; 0)$ and increases on $(0; +\infty)$. Hence it follows that at the point $x = 0$ the function has a minimum. The graph of the function $f(x)$ is shown in Fig. 137.

(c) The function $f(x) = \frac{6 \sin x}{2 + \cos x}$ is defined for all $x \in \mathbb{R}$. It is an odd function, and periodic with a period 2π . It is, therefore, sufficient to construct its graph on the interval $[0; \pi]$. Since $2 + \cos x \geq 1$, the function possesses no vertical asymptotes. We find the derivative

$$f'(x) = 6 \frac{\cos x (2 + \cos x) - \sin x (-\sin x)}{(2 + \cos x)^2} = 6 \frac{2 \cos x + 1}{(2 + \cos x)^2}.$$

The points $x = 2\pi n \pm \frac{2}{3}\pi$, $n \in \mathbb{Z}$, are stationary. The interval $[0; \pi]$ contains the point $x = 2\pi/3$ in its interior. On the interval $(0; 2\pi/3)$ the derivative is positive and, consequently, the function $f(x)$ decreases. Therefore, the point $x = 2\pi/3$ is a point of maximum. The graph of the function $f(x)$ is shown in Fig. 138. Note that at all points of the kind $x = 2\pi n + \frac{2\pi}{3}$, $n \in \mathbb{Z}$, the function $f(x)$ possesses maxima and at all points of the kind $x = 2\pi n - \frac{2\pi}{3}$, $n \in \mathbb{Z}$, it possesses minima.

6. The function $f(x) = 2 \ln(x-2) - x^2 + 4x + 1$ is defined for $x > 2$. Let us find its derivative:

$$f'(x) = \frac{2}{x-2} - 2x + 4 = \frac{2(x-1)(3-x)}{x-2}.$$

The point $x = 3$ is a critical point of the given function. On the interval $(2; 3)$ the derivative $f'(x)$ is positive and, consequently, on that

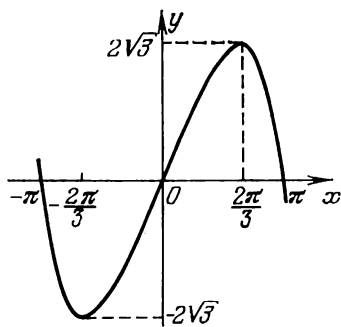


Fig. 138.

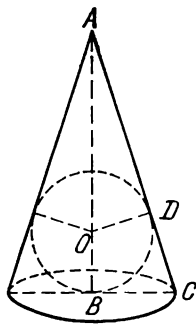


Fig. 139.

interval the function increases. And on the infinite interval $(3; +\infty)$ the derivative $f'(x)$ is negative and, therefore, the function $f(x)$ decreases on that interval. The point $x = 3$ is a point of maximum.

7. The function $f(x)$ is defined and has a derivative for all $x \in \mathbb{R}$. We find the derivative of the function $f(x)$:

$$f'(x) = x^2 - 5x + 6 = (x-2)(x-3).$$

The function $f(x)$ has two critical points $x=2$ and $x=3$. Both points belong to the interval $[1; 4]$. Consequently,

$$\begin{aligned}\min_{x \in [1; 4]} f(x) &= \min \{f(1), f(2), f(3), f(4)\} \\ &= \min \left\{ \frac{23}{6}, \frac{28}{6}, \frac{27}{6}, \frac{32}{6} \right\} = \frac{23}{6}, \\ \max_{x \in [1; 4]} f(x) &= \max \{f(1), f(2), f(3), f(4)\} = \frac{32}{6}.\end{aligned}$$

Answer: The greatest value is equal to $32/6$ and the least value, to $23/6$.

8. Suppose a certain cone is circumscribed about a given sphere of radius R (Fig. 139). It is evident that the centre of the sphere O lies on the altitude AB of the cone. Let us denote by r the radius of the base of the cone and by h , its altitude. Then the volume of the cone V is equal to $\frac{1}{3} \pi r^2 h$. It is easy to prove that $|BC| = |CD|$, i.e. $|CD| = r$. From the similarity of the triangles ABC and AOD it follows that

$$\frac{|BC|}{|OD|} = \frac{|AB|}{|AD|}, \text{ i.e. } \frac{r}{R} = \frac{h}{\sqrt{r^2 + h^2} - r}.$$

Hence we have

$$\begin{aligned}r \sqrt{r^2 + h^2} &= r^2 + hR, \\ r^2 (r^2 + h^2) &= (r^2 + hR)^2,\end{aligned}$$

and, finally,

$$r^2 = \frac{hR^2}{h - 2R}.$$

Consequently,

$$V = \frac{1}{3} \frac{\pi R^2 h^2}{h - 2R}.$$

Let us find the minimum of the function $V(h)$. We calculate the derivative

$$V'(h) = \frac{1}{3} \pi R^2 \frac{2h^2 - 4Rh - h^2}{(h - 2R)^2} = \frac{1}{3} \pi R^2 \frac{h(h - 4R)}{(h - 2R)^2}.$$

The derivative $V'(h)$ vanishes for $h = 4R$, changing its sign from minus to plus when it passes the point $h = 4R$. It follows that for $h = 4R$ the function $V(h)$ has a minimum. The least value of the volume is equal to

$$V(4R) = \frac{1}{3} \pi R^2 \frac{16R^2}{2R} = \frac{8}{3} \pi R^3.$$

CHAPTER 7

1. $\triangle BOC \sim \triangle AOD$ (Fig. 140), therefore, $\frac{|OC|}{|AO|} = \frac{|BC|}{|AD|} = n$, whence, taking into account that $|AC| = |AO| + |OC|$, we find

$|\vec{AO}| = \frac{1}{1+n} |\vec{AC}|$. Furthermore, $\vec{AO} \uparrow \vec{AC}$ and, therefore, $\vec{AO} = \frac{1}{1+n} \vec{AC}$, $\vec{BC} \uparrow \vec{AD}$, $|\vec{BC}| = n \cdot |\vec{AD}|$, consequently, $\vec{BC} = n\vec{AD}$, and since $\vec{AC} = \vec{AB} + \vec{BC}$, we finally have

$$\vec{AO} = \frac{1}{1+n} (\vec{AB} + n\vec{AD}).$$

2. Using the collinearity criterion, we can write $\mathbf{a} + \mathbf{b} = x\mathbf{c}$, and $\mathbf{b} + \mathbf{c} = y\mathbf{a}$. The vectors $\mathbf{b}$ and $\mathbf{c}$ are noncollinear and, consequently, $y \neq 0$ and $\mathbf{a} = \frac{1}{y} \mathbf{b} + \frac{1}{y} \mathbf{c}$. Substituting the expression into the first equality, we get $(1 + \frac{1}{y})\mathbf{b} = (x - \frac{1}{y})\mathbf{c}$. The vectors $\mathbf{b}$ and $\mathbf{c}$ are nonzero and noncollinear, therefore, the last equality can hold only under the condition $1 + \frac{1}{y} = 0$ and $x - \frac{1}{y} = 0$. Then $y = -1$, $x = -1$ and we easily find $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$.

3. Assume that D is the midpoint of the edge OA of the tetrahedron $OABC$ (Fig. 141). We have $\vec{BC} = \vec{DC} - \vec{DB}$, $\vec{MN} = \vec{DN} - \vec{DM}$.

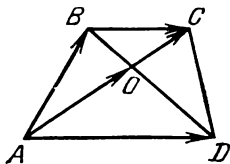


Fig. 140.

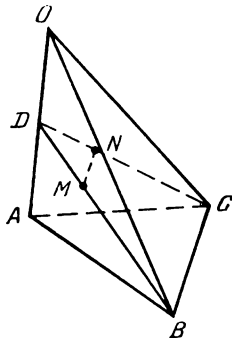


Fig. 141.

Since $\vec{DN} \uparrow \vec{DC}$ and $\vec{DM} \uparrow \vec{DB}$ and, besides, by the property of a median

$$\frac{|\vec{DN}|}{|\vec{DC}|} = \frac{1}{3} \text{ and } \frac{|\vec{DM}|}{|\vec{DB}|} = \frac{1}{3},$$

it follows that $\vec{DN} = \frac{1}{3} \vec{DC}$ and $\vec{DM} = \frac{1}{3} \vec{DB}$. Thus we have

$$\vec{MN} = \frac{1}{3} \vec{DC} - \frac{1}{3} \vec{DB} = \frac{1}{3} \vec{BC}.$$

By the collinearity criterion, the vector $\overrightarrow{MN}$ is collinear with the vector $\overrightarrow{BC}$ and

$$|\overrightarrow{MN}| : |\overrightarrow{BC}| = 1/3.$$

4. Suppose $m = |\overrightarrow{AM}| / |\overrightarrow{AB}|$, then (Fig. 142)

$$\overrightarrow{AM} = m\overrightarrow{AB}, \quad \overrightarrow{BN} = m\overrightarrow{BC} \text{ and } \overrightarrow{MB} = (1 - m)\overrightarrow{AB}.$$

We find $\overrightarrow{MN} = \overrightarrow{MB} + \overrightarrow{BN} = (1 - m)\overrightarrow{AB} + m\overrightarrow{BC}$. By the hypothesis we also have $\overrightarrow{PC} = m\overrightarrow{DC}$, $\overrightarrow{QD} = m\overrightarrow{AD}$ and, therefore,

$$\overrightarrow{DP} = (1 - m)\overrightarrow{DC} \text{ and } \overrightarrow{QP} = (1 - m)\overrightarrow{DC} + m\overrightarrow{AD}.$$

$ABCD$ being a parallelogram, we have $\overrightarrow{DC} = \overrightarrow{AB}$ and $\overrightarrow{BC} = \overrightarrow{AD}$.

Comparing the expressions for the vectors $\overrightarrow{MN}$ and $\overrightarrow{QP}$, we find that $\overrightarrow{MN} = \overrightarrow{QP}$. It follows from this equality that $|\overrightarrow{MN}| = |\overrightarrow{QP}|$ and $\overrightarrow{MN} \parallel \overrightarrow{QP}$. Thus the opposite sides of the quadrilateral $MNPQ$ are equal in length and are parallel, and, therefore, $MNPQ$ is a parallelogram.

5. Necessity. Suppose $\mathbf{a}$ and $\mathbf{b}$ are nonzero collinear vectors. By the collinearity criterion (7.4) there is a number α such that $\mathbf{b} = \alpha\mathbf{a}$. Then for $x = \alpha$ and $y = -1$ the following equality holds true:

$$x\mathbf{a} + y\mathbf{b} = 0. \quad (1)$$

Sufficiency. Suppose $\mathbf{a}$ and $\mathbf{b}$ are nonzero vectors for which equation (1) holds true, x and y being simultaneously nonzero. If, say,

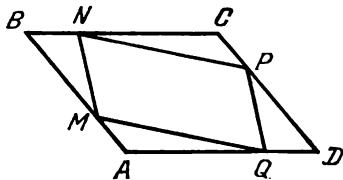


Fig 142.

$x \neq 0$, then equation (1) yields $\mathbf{a} = -\frac{y}{x}\mathbf{b}$, whence it follows (by the collinearity criterion) that $\mathbf{a}$ and $\mathbf{b}$ are collinear. The case $y \neq 0$ can be treated similarly.

6. Suppose $|\mathbf{a} + \mathbf{b}| = |\mathbf{a} - \mathbf{b}|$ then $|\mathbf{a} + \mathbf{b}|^2 = |\mathbf{a} - \mathbf{b}|^2$. Since $|\mathbf{a} + \mathbf{b}|^2 = (\mathbf{a} + \mathbf{b})(\mathbf{a} + \mathbf{b}) = |\mathbf{a}|^2 + 2\mathbf{a} \cdot \mathbf{b} + |\mathbf{b}|^2$ and $|\mathbf{a} - \mathbf{b}|^2 = (\mathbf{a} - \mathbf{b})(\mathbf{a} - \mathbf{b}) = |\mathbf{a}|^2 - 2\mathbf{a} \cdot \mathbf{b} + |\mathbf{b}|^2$, it follows from the equality $|\mathbf{a} + \mathbf{b}|^2 = |\mathbf{a} - \mathbf{b}|^2$ that $\mathbf{a} \cdot \mathbf{b} = 0$. The scalar

product of nonzero vectors is equal to zero if and only if they are orthogonal.

7. Adding termwise the equalities

$$|a + b|^2 = |a|^2 + 2a \cdot b + |b|^2,$$

$$|a - b|^2 = |a|^2 - 2a \cdot b + |b|^2,$$

we get

$$|a + b|^2 + |a - b|^2 = 2|a|^2 + 2|b|^2.$$

Hence, knowing that $|a| = 11$, $|b| = 23$ and $|a - b| = 30$, we find that $|a + b| = 20$.

8. If the vectors are orthogonal, then their scalar product is zero and, consequently,

$$(a + 3b)(7a - 5b) = 0, (a - 4b)(7a - 2b) = 0.$$

Using the properties of a scalar product, we find that

$$7|a|^2 + 16a \cdot b - 15|b|^2 = 0,$$

$$7|a|^2 - 30a \cdot b + 8|b|^2 = 0.$$

Removing $|a|^2$ from these equalities, we find that $a \cdot b = |b|^2/2$, and substituting this expression into the first equality, we get $|a| =$

$|b|$. Since $a \cdot b = |a| \cdot |b| \cos(\widehat{a, b})$, we find from the condition $a \cdot b = |b|^2/2$ for $|a| = |b|$ that $\cos(\widehat{a, b}) = 1/2$. Consequently, the angle between the vectors a and b is equal to 60° .

9. Provided that $a + b + c = 0$ and $|a| = |b| = |c| = 1$, the equation

$$(a + b + c)^2 = |a|^2 + |b|^2 + |c|^2 + 2(a \cdot b + b \cdot c + a \cdot c)$$

yields

$$a \cdot b + b \cdot c + a \cdot c = -3/2.$$

10. The vectors a and b are, evidently, noncollinear. The vector c , coplanar with a and b , can be resolved along these vectors: $c = xa + yb$. This vector equality is equivalent to a system of three scalar equalities:

$$m = x + 3y, 0 = -x + y, 2 = 2x.$$

We find: $x = 1$, $y = 1$, $m = 4$.

11. The vector b is collinear with the vector a , therefore, $b = ma$ and $|b| = |m| \cdot |a|$. Since $|b| = 3$ and $|a| = \sqrt{1 + 1 + \frac{1}{4}} = \frac{3}{2}$, it follows that $|m| = 2$. If $m = 2$, then $b = (2; 2; -1)$, and if $m = -2$, then $b = (-2; -2; 1)$. Assume that $b = (x; y; z)$, then $\cos(\widehat{b, k}) = z/|b|$. If the angle between the vectors b and k is acute, then the cosine of that angle is positive and the z -coordinate of the vector b is also positive. Thus we have $b = (-2; -2; 1)$.

12. Suppose $p = (x; y; z)$. The vector p is orthogonal to the vectors a and b , consequently, $a \cdot p = 0$ and $b \cdot p = 0$. Expressing the scalar

products in terms of the coordinates, we obtain

$$x + y + z = 0 \text{ and } x + 3y - z = 0,$$

whence we find that $z = y$ and $x = -2y$. The vector $\mathbf{p}$ is a unit vector, consequently, $x^2 + y^2 + z^2 = 6y^2 = 1$. The angle between the vectors $\mathbf{p}$ and $\mathbf{j}$ is obtuse and the cosine of that angle is negative, and since

$\widehat{(\mathbf{p}, \mathbf{j})} = y$, it follows that $y < 0$. Thus we have $y = -1/\sqrt{6}$ and $\mathbf{p} = (2/\sqrt{6}; -1/\sqrt{6}; -1/\sqrt{6})$.

13. The vector $\mathbf{b}$ is coplanar with the vectors $\mathbf{i}$ and $\mathbf{j}$, consequently, it can be resolved along these vectors: $\mathbf{b} = x\mathbf{i} + y\mathbf{j}$. Since $\mathbf{a} \perp \mathbf{b}$, we have $\mathbf{a} \cdot \mathbf{b} = 0$. Writing the scalar product in terms of the coordinates of the vectors, we get $4x - 3y = 0$, i.e. $y = \frac{4}{3}x$. The length of the vector $\mathbf{a}$ is equal to $\sqrt{4^2 + 3^2 + 5^2} = \sqrt{50}$ and is equal to the length of the vector $\mathbf{b}$ and, therefore, $x^2 + y^2 = 50$, whence we get $x^2 = 18$, $x = \pm 3\sqrt{2}$ and $y = \frac{4}{3}x$. Thus we have $\mathbf{b} = \pm (3\sqrt{2}; 4\sqrt{2})$.

14. If the point $M(x; y; z)$ belongs to the axis of ordinates, then $x = z = 0$. Thus it follows that the point M has the coordinates $(0; y; 0)$ and, therefore $|\overrightarrow{AM}| = \sqrt{2^2 + (y+1)^2 + 1^2}$ and $|\overrightarrow{BM}| = \sqrt{(y-1)^2 + 3^2}$. It follows from the condition $|\overrightarrow{AM}| = |\overrightarrow{BM}|$ that $(y+1)^2 + 5 = (y-1)^2 + 9$, whence we find $y = 1$ and $M(0; 1; 0)$.

15. Suppose $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. If the vector $\overrightarrow{OC}$ has a direction similar to that of the vector $\overrightarrow{OB}$ and $|\overrightarrow{OC}| = |\overrightarrow{OA}|$, then the vector $\overrightarrow{OD}$ is a diagonal of the rhombus $OADC$ and divides the angle AOB in half. Since $|\mathbf{a}| = 5$, $|\mathbf{b}| = 15$, it, evidently, follows that $\overrightarrow{OC} = \mathbf{b}/3 = (14/3; 2/3; -5/3)$, and, then, $\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{OC} = (2/3; 2/3; 4/3)$. The required vector $\mathbf{d}$ is of the same direction as $\overrightarrow{OD}$ and, therefore, $\mathbf{d} = x\overrightarrow{OD}$, $x > 0$. Since $|\overrightarrow{OD}| = \frac{2}{3}\sqrt{6}$ and $|\mathbf{d}| = \sqrt{6}$, we have $x = 3/2$. Thus, $\mathbf{d} = (1; 1; 2)$.

16. If A, B, C , and D are the consecutive vertices of a parallelogram, then $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{AD}$. Knowing the coordinates of the points A, B and D , we find $\overrightarrow{AB} = (2; -1)$, $\overrightarrow{AD} = (-2; -4)$. Then $\overrightarrow{AC} = (0; -5)$ and $|\overrightarrow{AC}| = 5$.

17. Assume that O is the origin and the point N belongs to the line segment BC with $|BN| : |BC| = n$. Then (see Example 2 in 7.5), $\overrightarrow{ON} = n \cdot \overrightarrow{OC} + (1-n) \cdot \overrightarrow{OB}$. This means that the coordinates of the point N can be expressed in terms of the coordinates of the points $C(x_C; y_C; z_C)$ and $B(x_B; y_B; z_B)$ as follows:

$$N(nx_C + (1-n)x_B; ny_C + (1-n)y_B; nz_C + (1-n)z_B).$$

If N is the midpoint of the segment BC , then $n = 1/2$. The coordinates of the midpoint of a segment are equal to the half-sum of the coordinates

of its end points. For the given points B and C we find $N(-1; 5; -2)$. Then

$$|MN| = \sqrt{(1+1)^2 + (-1-5)^2 + (1+2)^2} = 7.$$

In the case when $n = 1/3$, the coordinates of the point N are the following: $N(0; -3; 4)$ and

$$|MN| = \sqrt{1^2 + (-3+1)^2 + (4-1)^2} = \sqrt{14}.$$

18. If the plane α is parallel to the plane $ax + by + cz + d = 0$, then the vector $(a; b; c)$ is orthogonal to the plane α . Hence it follows that the equation of the plane passing through the point $A(1; -3; 2)$ parallel to the plane $4x - 2y - z + 7 = 0$ has the form $4(x-1) - 2(y+3) - (z-2) = 0$, i.e. $4x - 2y - z - 8 = 0$.

19. If O is the origin, then the vector $\vec{OA} = (-1; -1; 2)$ is orthogonal to the given plane and the equation of the plane is $-(x+1) - (y+1) + 2(z-2) = 0$, i.e. $x + y - 2z + 6 = 0$.

20. Assume that the point $B(x_0; y_0; z_0)$ is the foot of the perpendicular dropped from the point A on the plane. The point B lies on the plane, consequently, its coordinates satisfy the equation of the plane:

$$2x_0 + 2y_0 - z_0 + 3 = 0. \quad (1)$$

The vector $\vec{AB} = (x_0 + 2; y_0 - 3; z_0 + 4)$ is orthogonal to the plane and, therefore, it is collinear with the vector $\mathbf{n} = (2; 2; -1)$, that is, the coordinates of the vectors $\vec{AB}$ and $\mathbf{n}$ are proportional:

$$\frac{x_0+2}{2} = \frac{y_0-3}{2} = \frac{z_0+4}{-1}.$$

Hence we get $y_0 = x_0 + 5$, $z_0 = -\frac{1}{2}x_0 - 5$. Substituting these expressions into (1), we find $x_0 = -4$, and, then, $y_0 = 1$ and $z_0 = -3$. Thus we have $\vec{AB} = (-2; -2; 1)$ and $|\vec{AB}| = 3$.

21. Assume that β_1 and β_2 are, respectively, planes passing through the points A, B, C and A, B, D . Let us set up their equations (see example 4 in 7.8); we get

$$x - 2y + z = 0 \text{ and } x + y - 2z = 0.$$

The vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, orthogonal to those planes, are the following:

$$\mathbf{n}_1 = (1; -2; 1) \text{ and } \mathbf{n}_2 = (1; 1; -2).$$

The cosine of the angle between the planes β_1 and β_2 can be found by formula (5) of 7.8:

$$\cos \alpha = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1| |\mathbf{n}_2|} = \frac{1}{2}.$$

Thus we have $\alpha = 60^\circ$.

CHAPTER 8

$$\begin{aligned}
 1. \quad z &= \frac{(1+2i)^2 - (1-i)^3}{(3+2i)^3 - (2+i)^2} = \frac{1+4i+4i^2 - (1-i)(1-2i+i^2)}{(3+2i)(9+12i+4i^2) - (4+4i+i^2)} \\
 &= \frac{-3+4i+2i(1-i)}{(3+2i)(5+12i) - (3+4i)} = \frac{-1+6i}{-9+46i-(3+4i)} \\
 &= \frac{-1+6i}{-12+42i} = \frac{1-6i}{6(2-7i)} = \frac{(1-6i)(2+7i)}{6(2-7i)(2+7i)} = \frac{44-5i}{6(4+49)} = \frac{44-5i}{318}.
 \end{aligned}$$

$$\text{Answer: } \frac{22}{159} - \frac{5}{318}i.$$

2. (a) We successively find the moduli and the arguments of the numbers:

Number	Its modulus	One of its arguments
$1+i$	$\sqrt{2}$	$\pi/4$
$(1+i)^{13}$	$(\sqrt{2})^{13}$	$13\pi/4$
$1-i$	$\sqrt{2}$	$-\pi/4$
$(1-i)^7$	$(\sqrt{2})^7$	$-7\pi/4$
$\frac{(1+i)^{13}}{(1-i)^7}$	$(\sqrt{2})^6 = 8$	$\frac{13\pi}{4} - \left(-\frac{7\pi}{4}\right) = 5\pi$

Answer: The modulus of the number is equal to 8, the arguments, to $(2k+1)\pi$, $k \in \mathbb{Z}$.

(b) We find the modulus:

$$\begin{aligned}
 r &= \sqrt{(\sin(6\pi/5))^2 + (1 + \cos(6\pi/5))^2} = \sqrt{2 + 2\cos(6\pi/5)} \\
 &= 2\sqrt{\cos(3\pi/5)} = 2|\cos(3\pi/5)| = -2\cos(3\pi/5).
 \end{aligned}$$

The arguments φ of the given complex number must satisfy the equation

$$\tan \varphi = \frac{1 + \cos(6\pi/5)}{\sin(6\pi/5)} \quad (1)$$

or

$$\tan \varphi = \frac{2\cos^2(3\pi/5)}{2\sin(3\pi/5)\cos(3\pi/5)} = \cot \frac{3\pi}{5} = \tan\left(-\frac{\pi}{10}\right),$$

whence we find $\varphi = -\frac{\pi}{10} + \pi k$, $k \in \mathbb{Z}$. The given complex number lies in the second quadrant since $\sin \frac{6\pi}{5} < 0$ and $1 + \cos \frac{6\pi}{5} > 0$.

Therefore, the arguments are those and only those solutions of equation (1) which lie in the second quadrant, that is,

$$\varphi = -\frac{\pi}{10} + \pi + 2\pi n, \quad n \in \mathbb{Z}.$$

Answer: $r = -2 \cos(3\pi/5), \quad \varphi = \frac{9\pi}{10} + 2\pi n, \quad n \in \mathbb{Z}.$

3. (a) We first find the trigonometric form of the given complex number. Let us write the number $\sqrt{3} - i$ in trigonometric form. Since $|\sqrt{3} - i| = 2$, and one of the arguments $\varphi_1 = -\pi/6$, we have

$$\sqrt{3} - i = 2(\cos(-\pi/6) + i \sin(-\pi/6)).$$

It follows that

$$(\sqrt{3} - i)^{100} = 2^{100}(\cos(-100\pi/6) + i \sin(-100\pi/6)),$$

or

$$(\sqrt{3} - i)^{100} = 2^{100}(\cos(4\pi/3) + i \sin(4\pi/3)).$$

It is easy now to write the given number in algebraic form:

$$\begin{aligned} (\sqrt{3} - i)^{100} &= 2^{100}(\cos(4\pi/3) + i \sin(4\pi/3)) \\ &= 2^{100} \left(-\frac{1}{2} - i \frac{\sqrt{3}}{2} \right) = -2^{99} - i 2^{99} \sqrt{3}. \end{aligned}$$

Answer: $-2^{99} - i 2^{99} \sqrt{3}, \quad 2^{100}(\cos(4\pi/3) + i \sin(4\pi/3)).$

(b) The given number is written in algebraic form. To write it in trigonometric form, we first find the modulus of this number:

$$\begin{aligned} \left| 1 + \cos \frac{10\pi}{9} + i \sin \frac{10\pi}{9} \right| &= \sqrt{\left(1 + \cos \frac{10\pi}{9} \right)^2 + \left(\sin \frac{10\pi}{9} \right)^2} \\ &= \sqrt{2 + 2 \cos \frac{10\pi}{9}} = 2 \sqrt{\cos^2 \frac{5\pi}{9}} = 2 \left| \cos \frac{5\pi}{9} \right| = -2 \cos \frac{5\pi}{9}. \end{aligned}$$

To determine the argument, we solve the equation

$$\tan \varphi = \frac{\sin(10\pi/9)}{1 + \cos(10\pi/9)}.$$

Transforming the right-hand side, we get

$$\tan \varphi = \frac{\sin(10\pi/9)}{1 + \cos(10\pi/9)} = \frac{2 \sin(5\pi/9) \cos(5\pi/9)}{2 \cos^2(5\pi/9)} = \tan \frac{5\pi}{9},$$

whence it follows that $\varphi = \frac{5\pi}{9} + \pi k, \quad k \in \mathbb{Z}$. Since $1 + \cos(10\pi/9) > 0$, and $\sin(10\pi/9) < 0$, the number lies in the fourth quadrant of the complex plane, and one of its arguments is $\varphi_1 = 14\pi/9$.

Answer: $1 + \cos(10\pi/9) + i \sin(10\pi/9),$

$$-2 \cos(5\pi/9) (\cos(14\pi/9) + i \sin(14\pi/9)).$$

4. We find the modulus of the complex number $\tan 1 - i$:

$$|\tan 1 - i| = \sqrt{\tan^2 1 + 1} = 1/\cos 1.$$

The arguments φ of the number $\tan 1 - i$ satisfy the equation

$$\tan \varphi = -1/\tan 1.$$

Solving this equation, we get

$$\tan \varphi = -1/\tan 1 = -\cot 1 = \tan \left(\frac{\pi}{2} + 1 \right),$$

$$\varphi = \frac{\pi}{2} + 1 + \pi k, \quad k \in \mathbb{Z}.$$

The number $\tan 1 - i$ lies in the fourth quadrant of the complex plane and, therefore,

$$\varphi = \frac{\pi}{2} + 1 + \pi + 2\pi n, \quad n \in \mathbb{Z}.$$

We can now write the number $\tan 1 - i$ in trigonometric form:

$$\tan 1 - i = \frac{1}{\cos 1} \left(\cos \left(\frac{3\pi}{2} + 1 \right) + i \sin \left(\frac{3\pi}{2} + 1 \right) \right).$$

Applying formula (1) of 8.6, we finally have

$$(\tan 1 - i)^4 = \frac{1}{\cos^4 1} (\cos 4 + i \sin 4).$$

Answer: $z = \frac{1}{\cos^4 1} (\cos 4 + i \sin 4).$

5. Let us write z in algebraic form:

$$z = x + iy.$$

Then we can write the equation in the form

$$(x + iy)^2 + x^2 + y^2 = 0$$

or

$$x^2 + 2xyi - y^2 + x^2 + y^2 = 0.$$

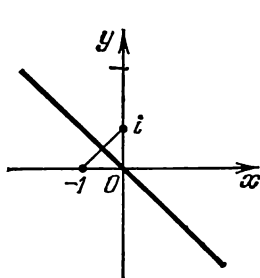


Fig. 143.

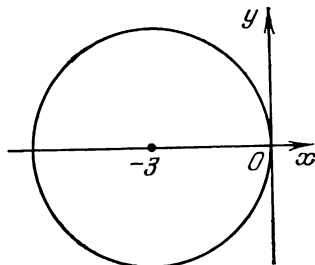


Fig. 144.

A complex number is equal to zero if and only if its real and imaginary parts are equal to zero and, therefore,

$$2x^2 = 0, \quad 2xy = 0.$$

The system obtained has an infinite number of solutions $(0; b)$, where b is an arbitrary real number. Hence it follows that the set of solutions of the original equation is infinite. It consists of all complex numbers corresponding to the points of the complex plane which lie on the axis of imaginaries.

Answer: $z = bi$ (b being an arbitrary real number).

6. (a) The condition $|z + 1| = |z - i|$ is satisfied by those and only those points of the complex plane which are equidistant from the points $z_1 = -1$ and $z_2 = i$. The set of points equidistant from the points z_1 and z_2 is a straight line perpendicular to the line segment connecting those points and passing through its midpoint. Figure 143 shows the line l giving the sought-for set.

(b) We set $z = x + iy$; then the given condition assumes the following form:

$$x^2 + y^2 + 3(x + iy) + 3(x - iy) = 0,$$

$$x^2 + 6x + y^2 = 0,$$

$$(x + 3)^2 + y^2 = 9.$$

This is an equation of a circle with centre at the point $(-3; 0)$, whose radius is equal to 3 (Fig. 144).

(c) The given inequality is equivalent to the inequalities

$$2\pi k < |z| < \pi + 2\pi k, \quad k = 0, 1, 2, \dots$$

The required set of points (hatched in Fig. 145) is an infinite system of concentric rings with centre at the point $z = 0$. The point $z = 0$ itself does not belong to the set.

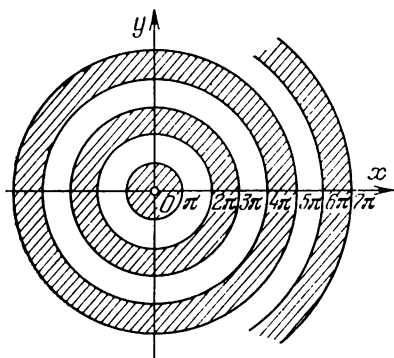


Fig. 145.

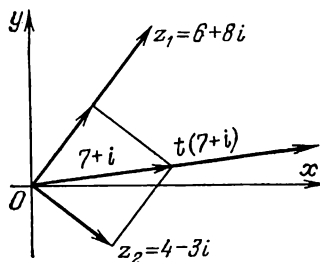


Fig. 146.

7. To obtain some vector passing along the bisector, it is sufficient to sum up any two vectors of the same length whose direction coincides with that of the vectors z_1 and z_2 (Fig. 146). Since the length of the vector z_1 is twice that of the vector z_2 , the vectors $z_1/2$ and z_2 have the same length and, therefore, the vector, which is equal to their sum,

$$\frac{1}{2} z_1 + z_2 = \frac{6+8i}{2} + 4-3i = 7+i,$$

lies on the bisector of the angle formed by the vectors z_1 and z_2 . The whole collection of vectors complying with the condition put in the problem is, evidently, given by the formula $t(7+i)$, where t is an arbitrary positive number.

Answer: $7t + it$, t being an arbitrary positive number.

8. The complex numbers satisfying the condition

$$|z + 3 - \sqrt{3}i| = \sqrt{3}$$

lie on the circle of radius $\sqrt{3}$ with centre at the point $z = -3 + \sqrt{3}i$ (Fig. 147). To find the point of the circle with the least positive argument, we draw a tangent line OT to the circle. The point of tangency P will, evidently, possess the required property. From the right-angled

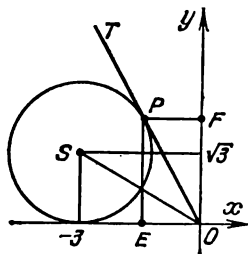


Fig. 147.

triangle OPE (the angle $\widehat{EOP} = 60^\circ$, $|OP| = 3$), we find

$$|PE| = |OP| \sin \widehat{EOP} = 3\sqrt{3}/2,$$

$$|OE| = |OP| \cos \widehat{EOP} = 3/2.$$

Answer: The number

$$z = -\frac{3}{2} + \frac{3\sqrt{3}}{2}i.$$

has the least positive argument.

9. The problem reduces to finding all values $\sqrt[4]{-8 - 8\sqrt{3}i}$. Let us write the number $w = -8 - 8\sqrt{3}i$ in trigonometric form:

$$|w| = \sqrt{64 + 3 \cdot 64} = 16,$$

$$\tan \varphi = \sqrt{3}, \varphi_1 = 240^\circ,$$

$$w = 16 (\cos 240^\circ + i \sin 240^\circ)$$

Applying formula (2) of §8.6 to seek all the values of $\sqrt[n]{w}$, we get

$$z_k = 2 (\cos (60^\circ + 90^\circ k) + i \sin (60^\circ + 90^\circ k)), \quad k = 0, 1, 2, 3.$$

Consequently,

$$z_0 = 2 (\cos 60^\circ + i \sin 60^\circ),$$

$$z_1 = 2 (\cos 150^\circ + i \sin 150^\circ),$$

$$z_2 = 2 (\cos 240^\circ + i \sin 240^\circ),$$

$$z_3 = 2 (\cos 330^\circ + i \sin 330^\circ).$$

Answer: $1 + i\sqrt{3}$, $-\sqrt{3} + i$, $-1 - i\sqrt{3}$, $\sqrt{3} - i$.

10. We set $\sqrt{5 + 12i} = x + iy$, then

$$5 + 12i = x^2 + 2xyi - y^2$$

and, consequently, x and y satisfy the system

$$x^2 - y^2 = 5, \quad xy = 6.$$

Solving the system, we obtain two solutions $(3; 2)$ and $(-3; -2)$. By the hypothesis, the real part $\sqrt{5 + 12i}$ is negative. Thus it follows that $\sqrt{5 + 12i} = -3 - 2i$. By analogy we find $\sqrt{5 - 12i} = -3 + 2i$. Now we easily find

$$\frac{\sqrt{5+12i} + \sqrt{5-12i}}{\sqrt{5+12i} - \sqrt{5-12i}} = \frac{-3-2i-3+2i}{-3-2i+3-2i} = \frac{-6}{-4i} = -\frac{3}{2}i.$$

Answer: $-\frac{3}{2}i$.

11. When dividing a polynomial of degree n by a polynomial of the first degree, we get a polynomial of degree $n - 1$ as a quotient and a polynomial of degree zero as a remainder. Denoting the quotient of the division of $P_n(z)$ by $z - z_0$ as $Q_{n-1}(z)$ and the remainder as w_0 , we can write the identity

$$P_n(z) = Q_{n-1}(z)(z - z_0) + w_0.$$

Substituting $z = z_0$ into this identity, we get

$$P_n(z_0) = Q_{n-1}(z_0)(z_0 - z_0) + w_0,$$

i.e. $w_0 = P_n(z_0)$.

Answers to Problems of Section II

CHAPTER 1

1. $A \cup B = \{1; 2; 3; 4; 5; 6; 12\}$, $B \cap C = \{5\}$, $(A \cup B) \cap C = \{3; 5\}$, $A \cap B \cap C = \emptyset$.

2. $A \cup B = \{1; 3; 4; 6; 7; 8; 21\}$, $B \cap C = \{3; 7\}$, $(A \cup B) \cap C = \{3; 7\}$, $A \cap B \cap C = \emptyset$.

3. For instance, (a) $A = (-\infty, a]$, $B = (a; +\infty)$, where a is an arbitrary real number, (b) B is a subset of any set A .

4. $A \cup B = [0; 5]$, $A \cap B = [1; 3]$, $A \cap C = \{0\}$, $B \cup C = (-2; 0] \cup (1; 5)$, $A \cap B \cap C = \emptyset$, $(A \cup B) \cap C = \{0\}$.

5. $A \cup B = (-\infty; +\infty)$, $A \cap B = \{1\}$, $A \cap C = (0; 1)$, $B \cup C = (0; +\infty)$, $A \cap B \cap C = \emptyset$, $(A \cup B) \cap C = (0; 1)$.

6. $A \cup B = [-3; 1] \cup [2; +\infty)$, $A \cap B = \emptyset$, $A \cap C = [-3; -2)$, $B \cup C = (-\infty; -2) \cup (2; +\infty)$, $(A \cup B) \cap C = [-3; -2)$, $A \cap B \cap C = \emptyset$.

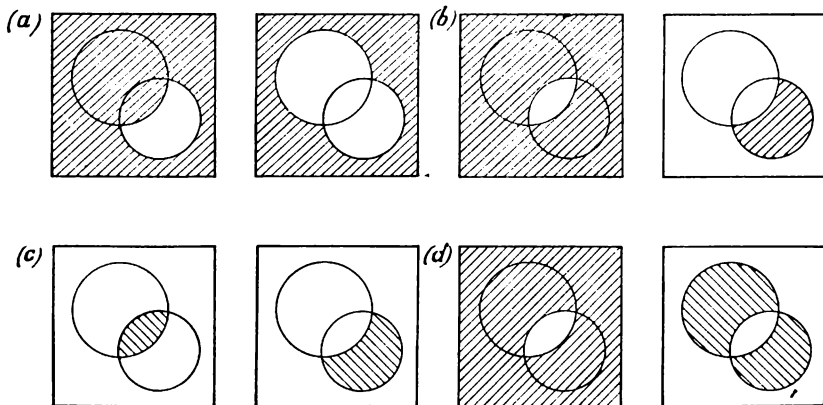
7. $A \cup B = (-1; 2)$, $A \cap B = \{0\}$, $A \cup \bar{B} = (-\infty; 0] \cup [2; +\infty)$, $A \cap B = (0; 2)$, $\bar{A} \cup \bar{B} = (-\infty; -1] \cup [2; +\infty)$.

8. $A \cup B = (-1; +\infty)$, $A \cap B = [0; 3]$, $A \cup \bar{B} = (-\infty; -1] \cup [0; 3]$, $\bar{A} \cap B = (-1; 0] \cup [3; +\infty)$, $\bar{A} \cup \bar{B} = (-\infty; -1]$.

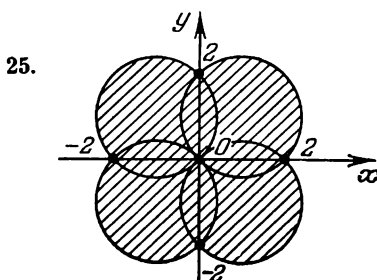
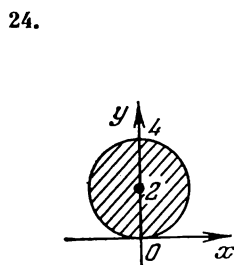
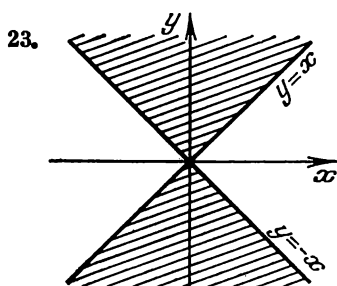
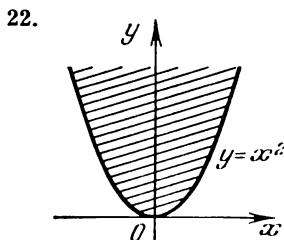
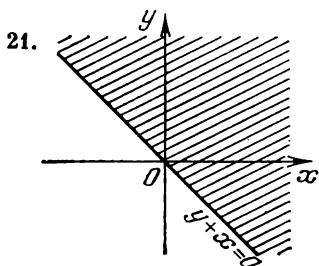
9. $A \cup B = (-\infty; 1]$, $A \cap B = (-\infty; 3)$, $A \cup \bar{B} = (-\infty; +\infty)$, $\bar{A} \cap B = \emptyset$, $\bar{A} \cup \bar{B} = (1; +\infty)$.

10. $A \cup B = [-1; +\infty)$, $A \cap B = (0; 1)$, $A \cup \bar{B} = (-\infty; -1) \cup (0; +\infty)$, $\bar{A} \cap B = [-1; 0]$, $\bar{A} \cup \bar{B} = (-\infty; -1)$.

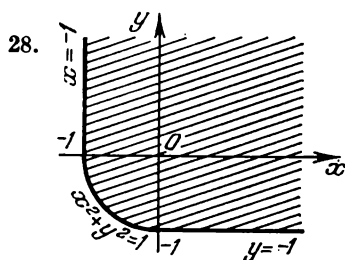
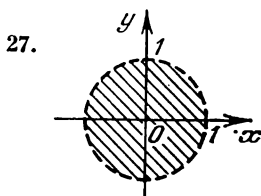
11.



12. 125. 13. 81. 14. 10%. 15. 30; 7; 18; 5. 16. 4. 17. 22.
 18. Six students only read book *A*, 5 only read book *B*, and 4 students only read book *C*. Three students read neither of the books.
 19. 94; 65. 20. $12n + 6$ ($n = 0, 1, 2, \dots$).



26. The straight line $y = x + \frac{1}{2}$.



29. $\left[-3; \frac{1}{2}\right]$. 30. $[0; 1)$.

31. $\left(-\frac{\pi}{2} + 2\pi n; 2\pi n\right) \cup \left(2\pi n; 2\pi n + \frac{\pi}{2}\right)$, $n \in \mathbb{Z}$.

32. $[1; 2)$. 33. $(-\infty; 1 - \sqrt{2}) \cup (1; 1 + \sqrt{2}]$.

34. $(0; 1/2)$. 35. $(-\infty; -2] \cup [2; +\infty)$.

36. $(-1; 0) \cup (0; 1) \cup (2; +\infty)$.

37. $\left(-6; -\frac{5}{3}\pi\right] \cup \left[-\frac{\pi}{3}; \frac{1}{6}\right]$. 38. $(0; 125]$.

39. $(-1; +1]$. 40. $\{1\}$. 41. $[-\sqrt{26}; +\sqrt{26}]$.

42. $(-\infty; -2] \cup [2; +\infty)$. 43. $[0; 2]$. 44. $\left[\frac{3}{4}; +\infty\right)$.

46. $-\sqrt{x+1}$. 47. $\frac{x+1}{x-1}$. 48. x^4 , $x \in [0; +\infty)$. 49. $\sqrt[3]{\frac{1-x}{x}}$.

50. $-1 + \sqrt{1+x}$ for $x \geq 0$, $1 - \sqrt{1-x}$ for $x < 0$.

51. Yes, $f^{-1}(x) = \frac{1}{2} + \sqrt{\frac{9}{4} - x}$.

52. No (for instance, $y(1) = y(-1) = 0$).

53. No (for instance, $y(0) = y(2) = -3$).

54. No (for instance, $y(0) = y(2) = -8$).

56. $d = -a$. 57. ± 1 .

CHAPTER 2

1. (a) There is at least one hit. (b) All the three bullets hit the target. (c) One and only one bullet hit the target.

2. Smith was.

3. The following persons should be chosen for the expedition: E as a biologist, B as a hydrologist, C as a radio operator, D as a doctor, F as a weather forecaster, and H as a mechanic.

4. (a) Yes, it can. Such a test will be that, for instance, in which the number of problems is equal to the number of students, with the first student having solved only the first problem, the second student, only the second problem, the third student, only the third problem, and so on.

(b) No, it cannot. From the fact that there is one student who solved all the three problems it follows that every problem was solved by at least one student (say, by that one who solved all the three problems).

5. Yes, there are. The second and fourth answers are equivalent.

6. $n = 45$. 7. $x = 2$, $x = (3 \pm \sqrt{5})/2$. 8. $[-1/2; 1/4]$.

9. (a) $[-36/5; 9]$, (b) $[-27/5; 0]$.

11. 1 and 4, 2 and 5 are mutually converse; 2 and 3, 4 and 6 are mutually opposite; 1 and 6, 3 and 5 are opposite to the converse.

Theorems 1 and 6 are true, the rest are false.

12. The direct theorem and the theorem opposite to the converse are false. The converse and the opposite theorem are true.

13. **The converse theorem.** If the function is continuous at the point x , then it is differentiable at that point.

The converse theorem. If the function is not differentiable at the point x , then it cannot be continuous at that point.

The theorem opposite to the converse. If the function is not continuous at the point x , then it is not differentiable at that point.

The direct theorem and that opposite to the converse are true. The converse and the opposite theorems are false (counter-example: $y(x) = |x|$, $x = 0$).

14. (a) Necessary; (b) sufficient, (c) sufficient; (d) necessary and sufficient; (e) necessary and sufficient.

21. (a) $n \geq 4$; (b) $n \geq 7$. 23. $n^2 - n + 2$.

CHAPTER 3

1. $\{0.5\}$ 2. $\{1; -0.75\}$. 3. $\{-0.8\}$.

4. $\{-6; 2; -2 - \sqrt{6}; -2 + \sqrt{6}\}$. 5. $\{0; \frac{4}{3}\}$. 6. $\emptyset$. 7. $\{4\}$.

8. $x \in [1; 2]$. 9. $\{(-7 + \sqrt{13})/3\}$. 10. $\{-1. 1\}$. 11. $\{-3; -4\}$.

12. 6.25. 13. $\{(1; 2; 1)\}$.

14. $\left\{\left(-\frac{1}{2} + \frac{5}{2}z; \frac{1}{2} - \frac{3}{2}z; z\right), z \in \mathbb{R}\right\}$.

15. The system is inconsistent. 16. $\{3; 1; 1\}$. 17. The system is inconsistent.

18. $\{(-1 - x_2 + x_4; x_2; 2 - x_4; x_4), x_2 \in \mathbb{R}, x_4 \in \mathbb{R}\}$.

19. $\{(4; 5), (5; 4)\}$. 20. $\{(7; -3), (-7; 3)\}$.

21. $\{(3; 1), (-3; -1), (-\sqrt{2}; 2\sqrt{2}), (\sqrt{2}; -2\sqrt{2})\}$.

22. $\{(0, 0), (1; 1), ((1 \pm \sqrt{5})/6; (1 \pm \sqrt{5})/6)\}$.

23. $\{(1/4; -3), (27/4; -1/3)\}$.

24. $\{(1; 2), (-1; -2), (\sqrt{2}; \sqrt{2}), (-\sqrt{2}; -\sqrt{2})\}$.

25. $\{(-3; -2), (3; 2), (-2; -3), (2; 3)\}$.

26. $\{(2; 3), (23/4; -9/2)\}$. 27. $\{(6; 4), (36/5; 5)\}$.

28. $\{(5; 4), (-0.5; -0.4)\}$. 29. $\{(4; 1)\}$. 30. $\{(5; 7)\}$.
 31. If $a = 1/2$, then it has 4 solutions, if $a = 1$, then it has 4 solutions, if $1/2 < a < 1$, then it has 8 solutions. For the rest of a there are no real solutions.
 32. $5/2$. 33. $\{(0; 0; 0); (-6/5; 8/5; -2/5), (0; -4/3; 2/3)\}$.
 34. $\{(2; -1; -1), (-3/2; 9/2; 3/2), (x; 0; 0), x \in \mathbb{R}\}$.
 35. $\{(0; 0; 0), (0; -1; 1)\}$.
 36. $\{(0; 0; 0), (7; 7; 7), (5/2; 5; 15/2)\}$.
 37. $\{(1; -1; 2), (-1; 1; -2), (3\sqrt{2}/5; 5\sqrt{2}/8; 4\sqrt{2}), (-3\sqrt{2}/5, -5\sqrt{2}/8, -4\sqrt{2})\}$.
 38. $\{(3/2; 2; 5/2), (-3/2; -2; -5/2)\}$.
 39. 63. 40. 30% and 60%. 41. $mn/(m+n)$ kg.
 42. 60 cu m; 2.4 cu m and 2 cu m. 43. 0.425.
 44. 336 and 280. 45. 16 h and 24 h.
 46. 28 days and 21 days. 47. 20 and 15. 48. $3 \cdot 10^4$ and $4 \cdot 10^4$.
 49. 56 s. 50. 21 m/s and 147 m.
 51. 14 min and 56/3 min.
 52. 0.5. 53. 1 h. 54. 4h. 55. 1.4. 56. 13 min.

CHAPTER 4

1. Not equivalent. 2. Not equivalent. 3. Equivalent. 4. Not equivalent. 5. Equivalent. 6. $(1/3; 4)$. 7. $\{-2\} \cup [2; 5/2]$.
 8. $(5 - \sqrt{5}; 3) \cup (4; 5 + \sqrt{5})$. 9. $(-\infty; -1) \cup (3; 7)$.
 10. $(-\infty; -3) \cup (-2; 0)$. 11. $(-1; 0) \cup (1; 4)$.
 12. $(-4; 0) \cup (0; +\infty)$. 13. $(-1; 2) \cup (2; 5)$.
 14. $(-\infty; -3) \cup (-1; 0) \cup (0; 1) \cup (7; +\infty)$.
 15. $(2; +\infty)$. 16. $(-\infty; -1) \cup (-1; 2]$.
 17. $(-2; 2) \cup (2; +\infty)$. 18. $(-\infty; -1) \cup (-1; 0) \cup (0; 1)$.
 19. $(-5; -3] \cup (3, 4]$. 20. $(-1; 2)$. 21. $[-5/2; 2)$.
 22. $(-\infty; 0] \cup (9/2; +\infty)$. 23. $(-\infty; -3] \cup (13; +\infty)$.
 24. $(-\infty; -1] \cup (8; +\infty)$. 25. $(-\infty; -2] \cup [5; 74/13)$.
 26. $(-\infty; (\sqrt{37} - 13)/6)$. 27. $[15/8; 2) \cup (3; +\infty)$.
 28. $(-\infty; -2/3] \cup [1/2; +\infty)$.
 29. $((5 - \sqrt{30})/10; (5 - \sqrt{20})/10) \cup (5 + \sqrt{20})/10; (5 + \sqrt{30})/10)$.
 30. $(2/3; 3/2) \cup (3/2; 4)$. 31. $[-1; -\sqrt{3}/2] \cup [\sqrt{3}/2; 1]$.
 32. $(-\infty; -2] \cup (0; +\infty)$. 33. $[-4; (1 - \sqrt{1431})/10] \cup (0; 4]$.
 34. $[2; (-7 + \sqrt{146})/2)$. 35. $\{2\}$. 36. $\{-1, 1\}$. 37. $[-2; 0) \cup (0; 8/5)$. 38. $(1; 2/\sqrt{3}]$. 39. $(-2; -1) \cup [-2/3; 1/3)$. 40. $\{7\}$.
 41. For $|a| > 3$ $(-\infty; \frac{1}{a-3})$, for $|a| < 3$ $(\frac{1}{a-3}; +\infty)$,
 for $a = -3$ there are no solutions, for $a = 3$ $(-\infty; +\infty)$.
 42. For $a < \frac{3}{2}$ $(-\infty; \frac{a^2 - 12a - 2}{2a - 3}]$, for $a = \frac{3}{2}$ $(-\infty; +\infty)$,
 for $a > \frac{3}{2}$ $[\frac{a^2 - 12a - 2}{2a - 3}; +\infty)$.

43. For $a > 3$ $\left(2; \frac{2a+1}{a-3}\right]$, for $a < 3$ $\left[\frac{2a+1}{a-3}; 2\right)$ for $a = 3$ there are no solutions.

44. For $a \leq 0$ $(-\infty; 0)$; for $a > 0$ $\left(-\frac{1}{\sqrt{a}}; 0\right) \cup \left(\frac{1}{\sqrt{a}}; +\infty\right)$.

45. If $a = 0$, then $x \in (-1; 0)$, if $a > 0$, then $x \in \left(\frac{-1 - \sqrt{1+4a}}{2}; a\right]$.

46. If $b < 2|a|$ there are no solutions, if $2|a| \leq b \leq 3|a|$, then $x \in [-b+2|a|, b-2|a|]$, if $b > 3|a|$, then $x \in [-b/3, b/3]$.

47. $a < 71/24$. 48. $-1 < a < 5$. 49. $a = 7/4$.

50. $a \geq 1/3$. 51. $3 \leq a \leq 18/5$. 52. $|a| < 1/2$.

59. 12; 24/5. 60. 1250 sq m. 61. $\frac{6 + \sqrt{3}}{132} p^3$.

62. The customer got more than two kg of grocery.

63. 2. 64. The distance between the cars will be least 8.4 minutes after midday. At that moment the first car will be 1.6 km from the crossing. The second car will be 1.2 km beyond the crossing.

65. The first tourist.

66. The speed of the first cyclist is faster than 10 km/h and slower than 40 km/h.

67. The speed of the boat must be faster than

$$\frac{a + \sqrt{a^2 + v^2 t^2}}{t}.$$

68. For $x = \pm \sqrt{\sqrt{2} - 1} y$ assumes the least value $2\sqrt{2} - 2$.

69. 18. 70. 4 boxes of the first kind and 25 boxes of the second kind.

CHAPTER 5

5. 10^5 . 6. 6. 7. 1. 8. $2/5$. 9. $1/2$. 10. 1. 11. 2.

12. 1 (use $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$). 13. 1.

14. 0 (use $\lim_{n \rightarrow \infty} \sqrt[n]{a} = 1$).

16. (a) $ad - bc > 0$, (b) $ad - bc < 0$.

18. (a) 1 (show that (a_n) is increasing and bounded); (b) 1.

19. (a) $a_n = 1 + \frac{1}{2^{n-2}}$ (see problem 7, section I), $\lim_{n \rightarrow \infty} a_n = 1$;

(b) $a_n = 4 - \frac{3}{2^{n-1}}$, $\lim_{n \rightarrow \infty} a_n = 4$; (c) $a_n = \frac{3}{2 - 4^{n-1}}$, $\lim_{n \rightarrow \infty} a_n = 0$.

20. (a) 2; (b) $7/33$; (c) $5/14$. 21. 40. 22. 38. 23. $1/3$. 24. -2.

25. -2. 26. 3; 6; 12; 18. 27. $1/5$. 28. 28. 29. $1/2$.

30. 4; 2; 1 and 1; 2; 4. 31. 6; 18; 54 and 26; 26; 26.
 32. 24. 33. $1/3$. 34. 64; $64/3$.
 35. $81/2$. 36. 9. 37. 26. 38. $4/3$.
 39. $2/3$. 41. 2. 42. 3. 43. $1/2$. 44. -1 . 45. $1/2$. 46. $1/2$.
 47. $3(x^8 - 1) = (x - 1)(x^7 + x^6 + \dots + 1)$, $x^4 - 1 = (x - 1)(x^3 + x^2 + x + 1)$. 48. 2. 49. $1/2$. 50. 0. 51. $1/2$. 52. 1; 0. 53. -1 ; 1.
 54. 1; 0. 55. (a) 3; $1/3$, (b) $1/3$; 3. 57. $+\infty$; 0. 58. $+\infty$; 1.
 59. $\frac{x+3}{2(x+1)\sqrt{x+1}}$.
 60. $(2-x)\sin x$. 61. $\frac{x+6\sqrt{x}-1}{2\sqrt{x}(x+1)^2}$. 62. $\frac{2}{(\sin x + \cos x)^2}$.
 63. $\ln(x^2 - 1) + \frac{2x^2}{x^2 - 1}$. 64. $2^x(1-x)\ln 2 - 2^x$.
 65. $-\frac{8x}{(x^2 - 4)^2}$. 66. $(1 - \ln 2)/2$. 67. $(-1; 0) \cup (0; +1)$.
 68. $y = 2x + 0.25$. 69. 135° . 70. $y = 8x + 4$.
 71. (a) $y = 2$; (b) $y + x = 5/2$.

CHAPTER 6

1. f_1, f_4, f_7 and f_8 are even, f_2 and f_6 are odd, and f_3 and f_5 are neither even nor odd.

2. The functions f_1, f_3, f_4 can be represented as $f_i = \varphi_i(x) - \psi_i(x)$, where $\varphi_i(x) = \frac{f_i(x) + f_i(-x)}{2}$ is an even function, $\psi_i(x) = \frac{f_i(x) - f_i(-x)}{2}$ is an odd function ($i = 1, 3, 4$); the function f_2 can be represented in such a form since its domain of definition is asymmetric about the origin.

4. (a) False; (b) false. 5. (a) No; (b) yes; (c) no.

6. (a) $e^{|x|}$; (b) $e^{|x|} \sin x$.

7. f_1, f_3 are nonperiodic; f_2 is periodic, $T = 6\pi$, f_4 is periodic, $T = \pi/2$.

8. It can.

10. (1) $x = 1$, $x = -1$, $y = 0$; (2) $x = 2/3$; (3) $x = 1$, $y = 0$;
 (4) $x = \frac{\pi}{2} + \pi n$, $n \in \mathbb{Z}$; (5) $x = 0$, $y = 0$; (6) $y = 0$.

11. (1) The function is periodic with period 2π ; the straight lines $x = \pi n$, $n \in \mathbb{Z}$, are vertical asymptotes; on the intervals $\left(2\pi n - \frac{\pi}{2}; 2\pi n\right)$ and $\left(2\pi n; 2\pi n + \frac{\pi}{2}\right)$ the function decreases; on the intervals $\left(2\pi n + \frac{\pi}{2}; 2\pi n + \pi\right)$ and $\left(2\pi n + \pi; 2\pi n + \frac{3\pi}{2}\right)$ it increases;

$x = \frac{3\pi}{2} + 2\pi n$ are the points of maximum, $x = \frac{\pi}{2} + 2\pi n$ are the points of minimum. (2) The function is defined for $x > 0$; $x = e^{2\pi n}$, $n \in \mathbb{Z}$, are the points of maximum; $x = e^{\pi + 2\pi n}$ are the points of minimum; $f_2(e^{2\pi n}) = 1$, $f_2(e^{\pi + 2\pi n}) = -1$; there are no asymptotes. (3) The function is defined on $\mathbb{R}$, it is odd, periodic with period 2π ; $f_3(x) = x$ on $(-\pi/2; \pi/2)$, $f_3(x) = \pi - x$ on $(\pi/2; 3\pi/2)$; $x = \frac{\pi}{2} +$

$2n\pi$, $n \in \mathbb{Z}$, are the points of maximum, $x = \frac{3\pi}{2} + 2\pi n$, $n \in \mathbb{Z}$, are the points of minimum. (4) The function is defined on $\mathbb{R}$, it is even; $x = \frac{\pi}{2} + 2n\pi$ and $x = -\frac{\pi}{2} - 2n\pi$, where $n = 0; 1; 2; \dots$, are

the points of maximum; $x = \frac{3\pi}{2} + 2n\pi$ and $x = -\frac{3\pi}{2} - 2n\pi$, where $n = 0; 1; 2; \dots$, are the points of minimum; the straight line $y = 0$ is a horizontal asymptote. (5) The function is even, periodic with period 1; $f_5(x) = x$ for $x \in [0; 1/2]$, $f_5(x) = 1 - x$ for $x \in [1/2; 1]$.

12. (1) The function is defined for all $x \neq -1$; $x = -1$ and $y = 1$ are asymptotes; $x = -3$ is a point of minimum; $(-\infty; -3)$ and $(-1; +\infty)$ are the intervals of decrease, $(-3; -1)$ is the interval of increase. (2) The function is defined for all $x \neq \pm\sqrt{3}$; $x = -\sqrt{3}$, $x = \sqrt{3}$ and $y = -x$ are asymptotes; $(-\infty; -3)$ and $(3; +\infty)$ are the intervals of decrease; $(-3; -\sqrt{3})$, $(-\sqrt{3}; \sqrt{3})$ and $(\sqrt{3}; 3)$ are the intervals of increase; $x = -3$ is a point of minimum, $x = 3$

is a point of maximum. (3) The function is defined for all x ; $(-\infty; -\frac{2}{5})$ and $(0; +\infty)$ are the intervals of increase, $(-\frac{2}{5}; 0)$

is the interval of decrease; $x = -\frac{2}{5}$ is a point of maximum; $x = 0$

is a point of minimum; for $x = 0$ the graph of the function has a vertical tangent. (4) The function is defined outside of the interval $(0; 4)$; the asymptotes are $y = 1$ as $x \rightarrow -\infty$ and $y = -1$ as $x \rightarrow +\infty$; the function is decreasing. (5) The function is periodic with period

$\pi/2$; $x = k\pi/2$, $k \in \mathbb{Z}$, are points of maximum, $x = \frac{\pi}{4} + \frac{k\pi}{2}$, $k \in \mathbb{Z}$,

are points of minimum. (6) The function is defined for $x > 0$; $x = 1$ is the point at which the graph of the function cuts the abscissa axis; $y = 0$ is an asymptote as $x \rightarrow +\infty$, $x = 0$ is a vertical asymptote; $(0, e^2)$ is the interval of increase, $(e^2; +\infty)$ is the interval of decrease; $x = e^2$ is a point of maximum.

13. For $a \leq -3$ and $a \geq 1$. 14. 2.

15. $\max f = f(6) = 132$, $\min f = f(3) = -57$.

16. $\max f = f(3) = -\log_3 10$, $\min f = f(6) = -\log_3 40$.

17. $x = -1$ is a point of maximum, $x = 2$ is a point of minimum; $\max f = f(4) = e^0$, $\min f = f(2) = -e^3$.

18. $\max f = 5$, $\min f = 2.75$. 19. $\sqrt[3]{V/2\pi}$. 20. 0.8.

CHAPTER 7

$$1. \vec{BC} = \vec{AB} + \vec{AF}, \vec{BD} = \vec{AB} + 2\vec{AF}. \quad 2. \vec{AM} = \frac{1}{2}(\vec{OB} + \vec{OC} - \vec{OA}).$$

$$3. \vec{BD} = \vec{BM} - \vec{MC}, \vec{AM} = \vec{BM} + 2\vec{MC}.$$

$$4. \vec{AA_1} = \vec{DA_1} - \vec{DB_1} + \vec{DC_1}, \vec{AC} = \vec{DC_1} - \vec{DA_1}, \vec{DB} = 2\vec{DB_1} - \vec{DC_1} - \vec{DA_1}.$$

$$5. \vec{AB} = -\frac{2}{3}\vec{CM} - \frac{4}{3}\vec{BN}, \vec{AC} = -\frac{4}{3}\vec{CM} - \frac{2}{3}\vec{BN}, \vec{MN} = (\vec{BN} - \vec{CM})/3.$$

$$6. \vec{AM} = \frac{1}{2}\vec{OB} - \vec{OA}, \vec{BN} = \frac{1}{2}\vec{OC} - \vec{OB}, \vec{MN} = \frac{1}{2}(\vec{OC} - \vec{OB}).$$

$$7. \vec{AM} = -\vec{BA} + \frac{1}{2}\vec{BB_1} + \frac{1}{2}\vec{BC}, \vec{A_1M} = -\vec{BA} - \frac{1}{2}\vec{BB_1} + \frac{1}{2}\vec{BC}.$$

8. (a) Yes, **a** and **b** are either any noncollinear vectors or nonzero vectors of the same direction; (b) yes, **a** and **b** are either oppositely directed or at least one of them is a zero vector, (c) no.

$$10. (a) x = 3; (b) x = -3. \quad 11. x = 3/2, y = 1/2.$$

$$12. x = 4; y = -2. \quad 13. x = 3 \text{ or } x = -2. \quad 14. = 1.$$

$$15. 2/5. \quad 16. x = 4, y = -1. \quad 17. (-6/7; 3/7; -2/7).$$

$$18. |\mathbf{a} + \mathbf{b}| = 6, |\mathbf{a} - \mathbf{b}| = 14. \quad 19. 24. \quad 20. 7.$$

$$21. |2\mathbf{a} - \mathbf{b}| = 5, |3\mathbf{b} - \mathbf{a}| = \sqrt{50}. \quad 22. 90^\circ. \quad 23. \arccos(11/15).$$

$$24. -3, 7, 19, 1. \quad 25. -13. \quad 26. (-4, 6). \quad 27. -25.$$

$$28. \pm(1/\sqrt{5}, -2/\sqrt{5}). \quad 29. -20. \quad 30. -3/2. \quad 31. 60^\circ.$$

$$32. (-2; 2; -4), 33. \pm(\sqrt{2}; -2/\sqrt{2}). \quad 34. m = 4; n = -3/2.$$

$$35. 1. \quad 36. \pm(1; 2). \quad 37. 2/3; -1/3; -2/3. \quad 38. -11/2.$$

$$39. \arccos(-3/\sqrt{11}). \quad 40. (6/7; 2/7; -3/7). \quad 41. 60^\circ. \quad 42. 45^\circ.$$

$$43. (-4; 3; -1); (8; -6; 2). \quad 44. M_1(1; 2; -5); M_2(1; -2; -5).$$

$$45. \arccos(-9/25). \quad 46. 5/2. \quad 47. \sqrt{5}. \quad 48. 13; -524. \quad 49. 90^\circ.$$

$$50. 15; \sqrt{593}. \quad 51. \arccos(63/65). \quad 52. (2/\sqrt{5}; \sqrt{5}; -5).$$

$$53. (48; -64; 60). \quad 54. \pm(2; 0; 3). \quad 55. \pm(1/\sqrt{3}; -1/\sqrt{3}; 1/\sqrt{3}).$$

$$56. \pm(1; 1; 1). \quad 57. (-24; -32; 30). \quad 58. m = -31/12; n = 41/12.$$

$$59. \pm(0; 1; 1). \quad 60. \pm(2/3; 2/3; -1/3). \quad 61. x - 2y + 5z = 0.$$

$$62. 2x + y + 2z - 3 = 0. \quad 63. (1) \text{ Perpendicular, } (2) \text{ parallel.}$$

$$64. -1. \quad 65. 2. \quad 66. 30^\circ. \quad 67. 60^\circ.$$

CHAPTER 8

$$1. (a) z_1 + z_2 = 3 - 2i, z_1 z_2 = 37 - 9i, z_1 - z_2 = 1 + 12i, \frac{z_1}{z_2} = -\frac{33}{50} + \frac{19}{50}i; (b) z_1 + z_2 = 2\sqrt{2}, z_1 z_2 = 5, z_1 - z_2 = -2\sqrt{3}i, \frac{z_1}{z_2} = -\frac{1}{5} - \frac{2\sqrt{6}}{5}i.$$

2. (a) i ; (b) $\frac{1}{2} + \frac{\sqrt{3}}{2}i$; (c) $-\frac{18}{25} + \frac{23}{50}i$; (d) $-117 + 44i$.

3. $z = -1 - i = \sqrt{2} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right)$.

4. $z_1 = 1 - i$, $z_2 = i$. 5. $(-2; -2)$; $(-2; 2)$.

6. Prove the equality for two numbers and then apply the principle of mathematical induction.

7. (a) $z_1 = 0$, $z_2 = -1$, $z_3 = \frac{1}{2} + \frac{\sqrt{3}}{2}i$, $z_4 = \frac{1}{2} - \frac{\sqrt{3}}{2}i$; (b) $z_1 = 0$, $z_2 = i$, $z_3 = -i$; (c) $z = 2 - \frac{3}{2}i$; (d) $z = 0$, $z_k = \cos \frac{2\pi k}{5} + i \sin \frac{2\pi k}{5}$, $k = 0, 1, 2, 3, 4$; (e) $z = c \pm \sqrt{3}ci$, where c is an arbitrary real nonpositive number.

8. $z = \frac{7}{6} + \frac{5}{6}i$. 10. $\frac{z_1 + z_2 + z_3}{3}$.

11. (a) The straight line passing through the points $z = 0$ and $z = 1 + i$. (b) The straight line passing through the points $z = 0$ and $z = -1 + i$. (c) The point $z = -1 - 2i$. (d) The real negative semi-axis, $z = x$, $x < 0$. (e) A half-plane (the boundary inclusive) bounded by the straight line passing through the points $z = 0$ and $z = 1 + i$ and containing the point $z = -i$. (f) A circle of radius $R = \sqrt{3}$ with centre at the point $z = -2 + i$. (g) A circle of radius $R = 10$ with the centre at the point $z = -i$, without the boundary of the circle and without the centre. (h) The angle (the boundary exclusive) with vertex at the point $z = -i$, whose sides pass through the points $z = 1$ and $z = 0$. (i) A circle of radius $R = 3$ with centre at the point $z = 0$.

12. (a) $z = i$; (b) $z = \frac{12}{5} + \frac{16}{5}i$.

13. $2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$; (b) $\cos \pi + i \sin \pi$; (c) $\cos \frac{13\pi}{12} + i \sin \frac{13\pi}{12}$; (d) $\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}$;
(e) $\frac{1}{\sqrt{2} \sin \frac{\pi}{5}} \left(\cos \frac{11\pi}{20} + i \sin \frac{11\pi}{20} \right)$.

14. (a) $z = -\sqrt{2}$; (b) $z = -\frac{\sqrt{3}}{2} + \frac{i}{2}$; (c) $z = (-1)^n$;
(d) $z = -2^7 i$.

15. (a) 1 , $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$, $-\frac{1}{2} - \frac{\sqrt{3}}{2}i$; (b) $\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$, $-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$, $-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$, $\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$; (c) $2 \left(\cos \frac{2\pi}{9} + i \sin \frac{2\pi}{9} \right)$,

$$2 \left(\cos \frac{8\pi}{9} + i \sin \frac{8\pi}{9} \right), \quad 2 \left(\cos \frac{14\pi}{9} + i \sin \frac{14\pi}{9} \right), \quad (d) \sqrt[16]{2} \times \\ \left(\cos \frac{8k+1}{32} \pi + i \sin \frac{8k+1}{32} \pi \right), \quad k=0, 1, 2, 3, 4, 5, 6, 7.$$

16. (a) $2n$; (b) n .

17. (a) $z_1 = 2 + i$, $z_2 = -2 + i$, (b) $z_1 = 13 - i$, $z_2 = 7 + i$;

(c) $z_k = \sqrt[5]{2} \left(\cos \frac{6k+1}{15} \pi + i \sin \frac{6k+1}{15} \pi \right)$, $k = 0, 1, 2, 3, 4$;

(d) $z_k = \cos \frac{\pi k}{3} + i \sin \frac{\pi k}{3}$, $k = 1, 2, 3, 4, 5$; (e) $z_1 = 1$, $z_2 = 3$,

$z_3 = 2i$, $z_4 = -2i$; (f) $z_1 = 1 + \sqrt{3}i$, $z_2 = -1 + \sqrt{3}i$, $z_3 = -2$; $z_4 = -1 - \sqrt{3}i$, $z_5 = 1 - \sqrt{3}i$.

18. Use the formulas of problem 6.

$$19. z_2 = 1 - i; z_3 = \frac{\sqrt{13}-1}{6}; z_4 = \frac{-\sqrt{13}-1}{6}.$$

$$20. z_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i; z_2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i. \quad 21. \frac{i}{2}z + \frac{1}{2} + i.$$

22. No, it cannot. *Hint.* Prove that all the vertices are in the interior of the circle of radius $R = \left| \frac{1}{1-z} \right|$ with centre at the point $\frac{1}{1-z}$.

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HIGH-SCHOOL MATHEMATICS

Part 1

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This two-part study aid in mathematics was written by the lecturers of the Department of Higher Mathematics at the Moscow Institute of Physics and Technology, and based on their long experience as promoters of mathematical knowledge among high-school students.

Part One includes eight chapters containing theoretical material and numerous examples and problems some of which are followed by solutions, others only by answers.

The book is intended for high-school students who wish to obtain deeper knowledge of mathematics. It can also be of use to those who want to learn how to solve problems in mathematics.

Contents of Part Two:

Trigonometric, Exponential, and Logarithmic Equations, Systems, and Inequalities. Combinatorics. Newton's Binomial Theorem. Random Events and Their Probabilities. The Integral. Solving Problems in Plane Geometry. Sets of Points on the Plane and in Space. Construction Problems. Solid Geometry. Solids of Revolution.